CHEMISTRY OF VARIOUS TYPES OF STEELS AND THEIR COMPARISON WITH INTERNATIONAL GRADES

BS:970		EQUIVALENT NATIONAL / INTERNATIONAL GRADES							CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)				CONDITION	TYPE OF STEEL AND APPLICATIONS
En No.	NUMERICAL	ISS (BIS)	ASTM	SAE	AISI	DIN	GOST	JIS	C	Mn	Si	S MAX.	P MAX.	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E %	HARDNESS (BHN) / IZOD(J)		
1A	220M07	11C10S25		---	1213	1213	9527	---	--- 0.15	0.07-1.20	0.80-MAX.	0.10 0.30	0.20-	0.07	---	---	---	---	---	40	24	14	140	CR/CD	FREECUTTING STEEL FOR LOW DUTYNUT, BOLTS, STUDS, PINS.
1B	240M07	---	---	---	---	---	---	---	0.07-0.15	1.00-1.40	0.10 MAX.	0.30-0.60	0.06	---	---	---	---	38	23	14	140	CR/CD	FREE CUTTINGSTEEL (HIGHER SULPHUR FOR LOWDUTY BOLTS NUTS, STUDS, PINS.
2	090A17	ST42S (IS:226)	---	1018	1018	---	6 XT 4 (GOST 380)	---	0.20 MAX.	0.80 MAX.	---	0.06	0.06	---	---	---	---	32	18	28	120	AR	GENERAL PURPOSE COLD FORMINGSTEEL FOR FASTENERS & ALLIED COMPONENTS
2A	040A12	---	---	1015	1015	---	5 XT 3 (GOST 380)	S 15 C	0.12 MAX.	0.50 MAX.	---	0.05	0.05	---	---	---	---	30	18	28	120	A/AR	SPECIAL PURPOSE COLD FORMING STEEL FOR TYPICAL HEAD CONFIGURATION
2A/1	040A10	---	---	1008	1008	---	08КП (GOST 1560)	---	0.10 MAX.	0.50 MAX.	---	0.04	0.04	---	---	---	---	30	18	28	120	A/AR	SPECIAL PURPOSE COLD FORMING STEEL FOR TYPICAL HEAD
2B	040A12	---	---	1012	1012	---	---	---	0.15 MAX.	0.50 MAX.	---	0.05	0.05	---	---	---	---	30	18	28	120	A/AR	SPECIAL PURPOSE COLD FORMING STEEL FOR TYPICAL HEAD CONFIGURATION
2C	050A20	---	---	1020	1020	Cq 22	---	---	0.15-0.25	0.40-0.60	---	0.05	0.05	---	---	---	---	32	18	28	120	A/AR	SPECIAL PURPOSE COLD FORMING STEEL FOR TYPICAL HEAD CONFIGURATION
2D	060A22	---	---	1025	1025	---	---	---	0.15-0.30	0.40 0.70	---	0.05	0.05	---	---	---	---	40	24	22	140	A/AR	SPECIAL PURPOSE COLD FORMING STEEL FOR TYPICAL HEAD CONFIGURATION
2E	050A15	---	---	1015	1015	Cq15 (0572)	—	S 15 C	0.15 MAX.	0.50 MAX.	0.10-0.35	0.05	0.05	---	---	---	---	40	24	22	140	A/AR	COLD FORMING STEEL (FULLY KILLED) FOR FASTENERS
3	080A20	ST42S (IS:226)	A194 Gr. 1	1022	1022	---	---	---	0.25 MAX.	1.00 MAX.	0.05-0.35	0.06	0.06	---	---	---	---	40-56	26	25	140	AR	20 CARBON STEEL FOR AIRCRAFT SOCKETS, LEVERS, PLUG ENDS, FASTENERS
3A	070A20	20C8 (C-20) IS:2073	---	1020	1020	---	---	---	0.15-0.25	0.40-0.90	0.05-0.35	0.06	0.06	---	---	---	---	44	25	25	140	AR/N	20 CARBON STEEL FOR AIRCRAFT SOCKETS, PLUG ENDS, LEVERS, FASTENERS

ABBREVATIONS USED: HT – HOT ROLLED; AC – AIR COOLED; OH – OIL HARDENED; AH – AIR HARDENED; H & T – HARDENED & TEMPERED; AR – AS ROLLED; CR – COLD ROLLED; CD – COLD DRAWN; N – NORMALISED; A – ANNEALED

HEAT TREATMENT CONDITION: Q, R, S, T, U, V, W, X, Y, Z

CONTD.

CHEMISTRY OF VARIOUS TYPES OF STEELS AND THEIR COMPARISON WITH INTERNATIONAL GRADES

BS:970		EQUIVALENT NATIONAL / INTERNATIONAL GRADES							CHEMICAL COMPOSITION, PERCENT									MECHANICAL PROPERTIES (MIN.)				CONDITION	TYPE OF STEEL AND APPLICATIONS
En No.	NUMERICAL	ISS (BIS)	ASTM	SAE	AISI	DIN	GOST	JIS	C	Mn	Si	S MAX.	P MAX.	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E %	HARDNESS (BHN) / IZOD(J)		
1A	220M07	11C10S25		---	1213	1213	9527	---	--- 0.15	0.07-1.20	0.80-MAX.	0.10 0.30	0.20-	0.07	---	---	---	---	40	24	14	140	CR/CD	FREECUTTING STEEL FOR LOW DUTYNUT, BOLTS, STUDS, PINS.
1B	240M07	---	---	---	---	---	---	---	0.07-0.15	1.00-1.40	0.10 MAX.	0.30-0.60	0.06	---	---	---	---	38	23	14	140	CR/CD	FREE CUTTINGSTEEL (HIGHER SULPHUR FOR LOWDUTY BOLTS NUTS, STUDS,PINS.
2	080A17	ST42S (IS:226)	---	1018	1018	---	6 XT 4 (GOST 380)	---	0.20 MAX.	0.80 MAX.	---	0.06	0.06	---	---	---	---	32	18	28	120	AR	GENERAL PURPOSE COLD FORMINGSTEEL FOR FASTENERS & ALLIED COMPONENTS
2A	040A12	---	---	1015	1015	---	5 XT 3 (GOST 380)	S 15 C	0.12 MAX.	0.50 MAX.	---	0.05	0.05	---	---	---	---	30	18	28	120	A/AR	SPECIAL PURPOSE COLD FORMING STEEL FOR TYPICAL HEAD CONFIGURATION
2A/1	040A10	---	---	1008	1008	---	08KП (GOST 1560)	---	0.10 MAX.	0.50 MAX.	---	0.04	0.04	---	---	---	---	30	18	28	120	A/AR	SPECIAL PURPOSE COLD FORMING STEEL FOR TYPICAL HEAD
2B	040A12	---	---	1012	1012	---	---	---	0.15 MAX.	0.50 MAX.	---	0.05	0.05	---	---	---	---	30	18	28	120	A/AR	SPECIAL PURPOSE COLD FORMING STEEL FOR TYPICAL HEAD CONFIGURATION
2C	050A20	---	---	1020	1020	Cq 22	---	---	0.15-0.25	0.40-0.60	---	0.05	0.05	---	---	---	---	32	18	28	120	A/AR	SPECIAL PURPOSE COLD FORMING STEEL FOR TYPICAL HEAD CONFIGURATION
2D	060A22	---	---	1025	1025	---	---	---	0.15-0.30	0.40 0.70	---	0.05	0.05	---	---	---	---	40	24	22	140	A/AR	SPECIAL PURPOSE COLD FORMING STEEL FOR TYPICAL HEAD CONFIGURATION
2E	050A15	---	---	1015	1015	Cq15 (0572)	---	S 15 C	0.15 MAX.	0.50 MAX.	0.10-0.35	0.05	0.05	---	---	---	---	40	24	22	140	A/AR	COLD FORMING STEEL (FULLY KILLED) FOR FASTENERS
3	080A20	ST42S (IS:226)	A194 Gr. 1	1022	1022	---	---	---	0.25 MAX.	1.00 MAX.	0.05-0.35	0.06	0.06	---	---	---	---	40-56	26	25	140	AR	20 CARBON STEEL FOR AIRCRAFT SOCKETS, LEVERS, PLUG ENDS, FASTENERS
3A	070A20	20C8 (C-20) IS:2073	---	1020	1020	---	---	---	0.15-0.25	0.40-0.90	0.05-0.35	0.06	0.06	---	---	---	---	44	25	25	140	AR/N	20 CARBON STEEL FOR AIRCRAFT SOCKETS, PLUG ENDS, LEVERS, FASTENERS

ABBREVATIONS USED: HT – HOT ROLLED; AC – AIR COOLED; OH – OIL HARDENED; AH – AIR HARDENED; H & T – HARDENED & TEMPERED; HEAT TREATMENT CONDITION: Q, R, S, T, U, V, W, X, Y, Z

AR – AS ROLLED; CR – COLD ROLLED; CD – COLD DRAWN; N – NORMALISED; A – ANNEALED

CONTD.

CHEMISTRY OF VARIOUS TYPES OF STEELS AND THEIR COMPARISON WITH INTERNATIONAL GRADES

BS:970		EQUIVALENT NATIONAL / INTERNATIONAL GRADES							CHEMICAL COMPOSITION, PERCENT									MECHANICAL PROPERTIES (MIN.)				CONDITION	TYPE OFSTEELAND APPLICATIONS
En No.	NUMERICAL	ISS (BIS)	ASTM	SAE	AISI	DIN	GOST	JIS	C	Mn	Si	S MAX.	P MAX.	Cr	Ni	Mo	OTHERS	UTS	YS Kg/mm²	E %	HARDNESS (BHN) / IZOD (J)		
3B	080A20	ST42S (IS:226)	---	1019	1019	---	---	---	0.25 MAX.	1.00 MAX.	0.35 MAX.	0.06	0.06	---	---	---	---	44	34	17	160	CD	20 CARBON STEEL COLD DRAWN FOR COMPONENTS MFD. ON AUTOMATS.
3C	070A20	20C8 (IS:1875)	---	1021	1021	---	GR.15Г (GOST 1050)	---	0.17-0.23	0.60-1.00	0.05 0.35	0.05	0.05	---	---	---	---	44	25	25	140	AR/N	20 CARBON STEEL FOR BOLTS,NUTS, PINS HAVINGHEAD WITH SPECIAL CONFIGURATION
3D	070A20	ST55 HTw (IS:961)	---	1018	1018	---	GR. 15Г (GOST 1050)	---	0.15-0.25	0.60-1.00	0.05 0.35	0.06 MAX.	0.06	---	---	---	---	52	34	15	160	CD	20 CARBON STEEL OR BOLTS,SCREWS, NUTS FOR MEDIUM CLASS HIGH TENSILE PROPERTIES
4	080A25	25 C8 (IS:1875)	A106 GR. B	1028	1028	---	---	---	0.30 MAX.	1.00 MAX.	0.05-0.35	0.06	0.06	---	---	---	---	44-60	28	25	140/28	N	25 CARBON STEEL FOR GEARS, MACHINERY PARTS, ETC.
4A	080A25	25 C8 (IS:1875)	A106 GR. B	1028	1028	---	---	---	0.30 MAX.	1.00 MAX.	0.05-0.35	0.06	0.06	---	---	---	---	50-66	34	12	170	CD	25 CARBON STEEL COLD DRAWN FOR TAPER/CYL. PINS, BOLTS, NUTS TO PROP. CLASS 5.6
5	080M30	C-30 (IS: 2073)	A106 GR. C	1028	1028	---	GR. 25Г (GOST 1050)	---	0.25-0.35	0.60-1.00	0.05-0.35	0.06	0.06	---	---	---	---	55 63 70 50	34 38 42 30	22 20 20 25	152-207/28 179-229/28 201-255/56 145-165/28	P Q R N	30 CARBON STEEL HARDENED AND TEMPERED FOR MANUFACTURE OF FASTENERS TO P-5.6, 6.6.
5K	080A30	30C8 (IS: 2073)	A106 GR. C	1028	1028	---	---	---	0.25-0.35	0.60-1.00	0.05-0.35	0.05	0.05	---	---	---	---	55 63 70	34 38 42	22 20 20	152-207/28 179-229/28 201-255/56	P Q R	30 CARBON STEEL FOR SPECIAL APPLICATION ONLY.
5A	080A27	25C8 (IS: 1875)	---	1028	1028	---	GR. 25 (GOST 1050)	---	0.25-	0.70-0.30	0.05-0.90	0.06 0.35	0.06	---	---	---	---	MECH. PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION.					27 CARBON STEEL FOR SPECIAL APPLICATION ONLY.
5B	080A30	---	---	1030	1030	---	---	---	0.25-0.33	0.70-0.90	0.05-0.35	0.06	0.06	---	---	---	---	MECH. PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION.					27 CARBON STEEL FOR SPECIAL APPLICATION ONLY.
5C	080A32	---	---	1036	1036	---	GR. 30 (GOST 1050)	---	0.30-0.35	0.70-0.90	0.05-0.35	0.06	0.06	---	---	---	---	MECH. PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION.					27 CARBON STEEL FOR SPECIAL APPLICATION ONLY.

ABBREVATIONS USED: HT – HOT ROLLED; AC – AIR COOLED; OH – OIL HARDENED; AH – AIR HARDENED; H & T – HARDENED & TEMPERED; HEAT TREATMENT CONDITION:

AR – AS ROLLED; CR – COLD ROLLED; CD – COLD DRAWN; N – NORMALISED; A – ANNEALED; Q, R, S, T, U, V, W, X, Y, Z

CONTD.

CHEMISTRY OF VARIOUS TYPES OF STEELS AND THEIR COMPARISON WITH INTERNATIONAL GRADES

BS:970		EQUIVALENT NATIONAL / INTERNATIONAL GRADES							CHEMICAL COMPOSITION, PERCENT									MECHANICAL PROPERTIES (MIN.)				CONDITION	TYPE OF STEEL AND APPLICATIONS
En No.	NUMERICAL	ISS (BIS)	ASTM	SAE	AISI	DIN	GOST	JIS	C	Mn	Si	S MAX.	P MAX.	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E%	HARDNESS (BHN) / IZOD (J)		
5D	080A30	30C8 (IS:1875)	A350 GRLF1	1030	1030	---	---	---	0.25-0.30	0.60-0.90	0.05-0.35	0.06	0.06	---	---	---	---	55	34	15	152-207 / 28	HT & CD	30 CARBON STEEL H & T AND BRIGHT DRAWN FOR BOLTS, SCREWS / STUDS.
6	---	---	---	---	---		---	---	0.40 MAX.	0.50-0.90	0.05-0.35	0.06	0.06	---	---	---	---	55-70	34	12	152-231 / 20	CD	BRIGHT CARBON STEEL FOR MACHINED PARTS.
6K	---	---	---	---	---		---	---	0.40 MAX.	0.50-0.90	0.05-0.35	0.05	0.05	---	---	---	---	55-70	34	12	152-231 / 20	CD	BRIGHT CARBON STEEL FOR SPECIAL APPLICATION ONLY.
6A	---	---	---	---	---		---	---	0.40 MAX.	0.50-0.90	0.05-0.35	0.06	0.06	---	---	---	---	55-70	34	12	152-231 / 20	CD	BRIGHT CARBON STEEL FOR SPECIAL APPLICATION ONLY.
7	---	40C10S18 (IS:4431)	---	---	---		---	---	0.10 0.30	1.00-1.30	0.25 MAX.	0.10-0.18	0.06	---	---	---	---	55-70	34	15	152-231 / 20	CD	BRIGHT CARBON SEMI FREE CUTTING STEEL FOR TURNED COMPONENTS. NOT RECOMMENDED FOR CASE HARDENING.
7A	---	14MnS14 (IS:2073)	---	1118	1118		---	---	0.12 0.18	1.00-1.50	0.25 MAX.	0.10-0.18	0.06	---	---	---	---	44-54	26	18	152-231 / 20	CD	15 CARBON SEMI FREE CUTTING STEEL IN COLD DRAWN CONDITION FOR TURNED COMPONENTS.
8	080M40	40C8 C-40 (IS:2073)	---	1040	1040	---	GR.40 (GOST 1050)	---	0.35-0.45	0.60-1.00	0.05-0.35	0.06	0.06	---	---	---	---	58-68 63 70 60-70 63 70	34 38 42 40 40 42	18 20 20 17 17 17	170-200 179-229 / 28 201-255 / 56 187-240 187-240 187-240	N Q R N & CD Q & CD R & CD	40 MEDIUM CARBON STEEL FOR MANUFACTURE OF BOLTS, NUTS, SCREWS, STUD, AND MACHINED PARTS IN GENERAL CONN. RODS, CRANK SHAFTS & PARTS REQUIRING STRENGTH & WEAR RESISTANCE (WITHOUT GRAIN SIZE CONTROL
8A	080A35	---	---	1037	1037	---	---	GR.40 (GOST 1050)	---	0.33 0.38	0.70-0.90	0.05-0.35	0.06	0.06	---	---	---	---	MECH. PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					MEDIUM CARBON STEEL FOR SPECIAL APPLICATIONS ONLY SUCH AS MACHINED PARTS, GEARS, SPROCKETS, SPL. TYPES OF FASTENERS.

ABBREVATIONS USED: HT – HOT ROLLED; AC – AIR COOLED; OH – OIL HARDENED; AH – AIR HARDENED; H & T – HARDENED & TEMPERED

AR – AS ROLLED; CR – COLD ROLLED; CD – COLD DRAWN; N – NORMALISED; A – ANNEALED

HEAT TREATMENT CONDITION: Q, R, S, T, U, V, W, X, Y, Z

CONTD.

CHEMISTRY OF VARIOUS TYPES OF STEELS AND THEIR COMPARISON WITH INTERNATIONAL GRADES

BS:970		EQUIVALENT NATIONAL / INTERNATIONAL GRADES							CHEMICAL COMPOSITION, PERCENT									MECHANICAL PROPERTIES (MIN.)				CONDITION	TYPE OF STEEL AND APPLICATIONS
En No.	NUMERICAL	ISS (BIS)	ASTM	SAE	AISI	DIN	GOST	JIS	C	Mn	Si	S MAX.	P MAX.	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E%	HARDNESS (BHN)/IZOD (J)		
8B	080A37	---	---	1038	1038	---	GR. 35 (GOST 1050)	---	0.35 0.40	0.70-0.90	0.05-0.35	0.06	0.06	---	---	---	---	MECH. PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					MEDIUM CARBON STEEL FOR SPECIAL APPLICATIONS ONLY SUCH AS MACHINED PARTS, GEARS, SPROCKETS, SPL. TYPES OF FASTENERS.
8C	080A40	---	---	1040	1040	---	GR. 35	---	0.38 0.43	0.70-0.90	0.05-0.35	0.06	0.06	---	---	---	---	MECH. PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					SAME AS ABOVE
8D	080A42	---	A194 2, 2H	1042	1042	---	GR. 35	---	0.40 0.45	0.70-0.90	0.05-0.35	0.06	0.06	---	---	---	---	MECH. PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					SAME AS ABOVE
8E	---	---	---	---	---	---	GR. 35	---	0.35 0.40	0.90-1.00	0.05-0.35	0.06	0.06	---	---	---	---	MECH. PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					SAME AS ABOVE
8K	080M40	C-40	---	1038	1038	---	GR. 35	---	0.35 0.40	0.60-1.00	0.05-0.35	0.05	0.05	---	---	---	---	MECH. PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					SAME AS ABOVE
8M	212M44	---	---	1140	1140	---	A40Г	---	0.35-0.45	1.00-1.30	0.25 MAX.	0.12-0.20	0.06	---	---	---	---	55 63	34 40	20 22	170-200 / 20 179-229 / 34	N Q	SAME AS ABOVE FOR SPECIAL APPLICATIONS
8AM	---	---	A105	---	---	---	---	---	0.33-0.38	0.90-1.30	0.25 MAX.	0.12-0.20	0.06	---	---	---	---	55 63 70 60	34 40 42 36	20 22 20 12	170-200 179-229 / 28 201-255 / 56 187-240	N Q R CD	40 MEDIUM CARBON FREE CUTTING STEEL
8BM	212A37	---	---	---	---	---	A40Г	---	0.35-0.45	0.90-1.30	0.25 MAX.	0.12-0.20	0.06	---	---	---	---	MECH. PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					40 MEDIUM CARBON FREE CUTTING STEEL FOR SPECIAL APPLICATIONS
8CM	212A37	---	---	---	---	---	---	---	0.38-0.43	0.90-1.30	0.25 MAX.	0.12-0.20	0.06	---	---	---	---	MECH. PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					40 MEDIUM CARBON FREE CUTTING STEEL FOR SPECIAL APPLICATIONS.
8DM	212A42	---	A194 2/2H	---	---	---	---	---	0.40-0.45	0.90-1.30	0.25 MAX.	0.12-0.20	0.06	---	---	---	---	MECH. PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					40 MEDIUM CARBON FREE CUTTING STEEL FOR SPECIAL APPLICATIONS.
9	070M55	55C8	A449 TYPE-1	1055	1055	CK55	GR. 50 (GOST 1050)	---	0.50-0.60	0.50-0.80	0.05-0.35	0.06	0.06	---	---	---	---	70 70 79 87 79-102	40 42 48 60 60	18 18 18 16 12	187-231 201-255 223-277 248-302 223-277	N R S T CD	55 CARBON STEEL FOR CYLINDERS, GEARS, MACHINE TOOLS, RIFLE BARREL AND HT FASTENERS

ABBREVATIONS USED: HT -- HOT ROLLED; AC -- AIR COOLED; OH -- OIL HARDENED; AH -- AIR HARDENED; H & T -- HARDENED & TEMPERED; HEAT TREATMENT CONDITION: Q, R, S, T, U, V, W, X, Y, Z

AR -- AS ROLLED; CR -- COLD ROLLED; CD -- COLD DRAWN; N -- NORMALISED; A -- ANNEALED

CONTD

CHEMISTRY OF VARIOUS TYPES OF STEELS AND THEIR COMPARISON WITH INTERNATIONAL GRADES

BS:970		EQUIVALENT NATIONAL / INTERNATIONAL GRADES							CHEMICAL COMPOSITION, PERCENT									MECHANICAL PROPERTIES (MIN.)				CONDITION	TYPE OF STEEL AND APPLICATIONS
En No.	NUMERICAL	ISS (BIS)	ASTM	SAE	AISI	DIN	GOST	JIS	C	Mn	Si	S MAX.	P MAX.	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS Kg/mm²	E%	HARDNESS (BHN)/IZOD (J)		
9K	070M55	55C8 C-55 IS:1875	---	1055	1055	---	GR.50	---	0.50 0.60	0.50-0.80	0.05-0.35	0.05	0.05	---	---	---	---	70 70 79 87 79-102	40 42 48 60 60	18 18 20 15 12	187-231 201-255 223-277 248-302 223-277	N R S T CD	55 CARBON STEEL FOR SPECIAL APPLICATIONS
10	---	---	---	---	---	---	---	---	0.50-0.60	0.50-0.80	0.05-0.35	0.06	0.06	---	0.50-0.80	---	---	MECH. PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					55 CARBON 3/4% Ni STEEL H&T FOR HIGH TENSILE FASTENERS
11	526M60	T55Cr3 (T55Cr70) IS:3749	---	---	---	---	---	---	0.50 0.70	0.50-0.80	0.10-0.35	0.05	0.05	0.50-0.80	---	---	---	87 102	60 80	15 12	248-302 / 34 293-341 / 16	T V	60 CARBON Cr. STEEL DIRECT HARDENING STEEL FOR STRENGTH & WEAR RESISTANCE (TOOL STEEL FOR COLD WORK)
12	503M40	---	---	---	---	---	---	---	0.30 0.45	1.50 MAX.	0.10-0.35	0.05	0.05	---	0.60-1.00	---	---	55 63	34 40	20 22	170-200 / 27 179-229 / 48	N Q	1% NICKEL STEEL FOR MOTOR AXLE SHAFTS, CONN. RODS, BOLTS, CRANKSHAFTS, STEERING LEVERS.
12A	503A32	---	---	---	---	---	---	---	0.30 0.35	0.70-0.90	0.10-0.35	0.05	0.05	---	0.60-1.00	---	---	MECH. PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					SAME AS ABOVE FOR SPECIAL APPLICATION
12B	503A37	---	---	---	---	---	---	---	0.35 0.40	0.70-0.90	0.10-0.35	0.05	0.05	---	0.60-1.00	---	---	MECH. PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					SAME AS ABOVE FOR SPECIAL APPLICATION
12C	503A42	---	---	---	---	---	---	---	0.40 0.45	0.70-0.90	0.10-0.35	0.05	0.05	---	0.60-1.00	---	---	MECH. PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					SAME AS ABOVE FOR SPECIAL APPLICATION
13	785M19	---	---	---	---	---	---	---	0.15 0.25	1.40-1.80	0.10-0.35	0.05	0.05	---	0.40-0.70	0.15-0.35	---	63	40	22	179-229 / 54	Q	Mn-Ni-Mo FOR USE SAME AS PER En-12
14A	150M19	20Mn2 (IS:1570)	---	1024	1024	21Mn6	---	---	0.15-0.25	1.30-1.70	0.10-0.35	0.06	0.06	0.25 MAX.	0.40 MAX.	---	---	55 63 70 70	34 40 42 45	20 20 20 15	170-220 / 20 179-229 / 41 201-255 / 34 187-220 / 20	N Q R CD	CARBON MANGANESE STEEL FOR WELDING QUALITY.
14A1	---	---	---	---	---	---	---	---	0.23 MAX.	1.20 MIN.	0.05-0.35	0.06	0.06	---	---	---	---	55-65	36	18	179-229	HR/N	SAME AS ABOVE FOR SPECIAL APPLICATION
14B	150M28	---	---	1027	1027	28Mn6	---	---	0.20-0.30	1.30-1.70	0.10-0.35	0.05	0.05	---	---	---	---	60 63 70 70	36 40 42 45	20 20 20 15	179-229 / 20 179-229 / 47 201-255 / 41 187-220 / 20	N Q R CD	CARBON MANGANESE STEEL FOR WELDING QUALITY.

ABBREVATIONS USED: HT – HOT ROLLED; AC – AIR COOLED; OH – OIL HARDENED; AH – AIR HARDENED; H & T – HARDENED & TEMPERED; HEAT TREATMENT CONDITION:

AR – AS ROLLED; CR – COLD ROLLED; CD – COLD DRAWN; N – NORMALISED; A – ANNEALED; Q, R, S, T, U, V, W, X, Y, Z

CONTD.

CHEMISTRY OF VARIOUS TYPES OF STEELS AND THEIR COMPARISON WITH INTERNATIONAL GRADES

BS:970	EQUIVALENT NATIONAL / INTERNATIONAL GRADES							CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)			CONDITION	TYPE OF STEEL AND APPLICATIONS
En No. NUMERICAL	ISS (BIS)	ASTM	SAE	AISI	DIN	GOST	JIS	C	Mn	Si	S MAX.	P MAX.	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E % HARDNESS (BHN) / I ZOD (J		
5 150M36	37Mn2 (IS:1570)	A325M TYPE-1	1041	1041	36Mn5	35Г2	---	0.30-0.40	1.30-1.70	0.10-0.35	0.05	0.05	---	---	---	---	63 70 79 89	40 42 48 68	22 179-229 / 34 20 201-255 / 41 20 223-277 / 41 14 248-302	Q R S T/P-8.8	CARBON MANGA-NESE STEEL FOR HIGH TENSILE FASTENERS ≤ M24
15A ---	37Mn2 (IS:1570)	---	---	---	---	35Г2	---	0.30-0.40	1.30-1.70	0.05-0.35	0.06	0.06	---	---	---	---	63 70 79 89	40 42 48 68	22 179-229 / 34 20 201-255 / 41 20 223-277 / 41 14 248-302 / 34	Q R S T/P-8.8	CARBON MANGA-NESE STEEL FOR HIGH TENSILE FASTENERS ≤ M24
15AM 216M36	40Mn15S12— (IS:1570)		1137	1137	---	---	---	0.30-0.40	1.30-1.70	0.25 MAX.	0.12 0.20	0.06	---	---	---	---	70	42	20 201-255 / 41	R	CARBON MANGA-NESE STEEL FREE CUTTING STEEL FOR SPECIAL APPLICATION
15B 120M36	---	---	---	---	---	40Г2	---	0.35-0.40	1.10-1.70	0.05-0.35	0.06	0.06	---	---	---	---	63 70 79 87	40 42 48 68	22 179-229 / 34 20 201-255 / 41 20 223-277 / 41 14 248-302 / 34	Q R S T/P-8.8	CARBON MANGA-NESE STEEL FOR HIGH TENSILE FASTENERS ≤ M24
16 605M36	35Mn2Mo28 (IS:4368)	A194 GR.4	---	---	---	---	---	0.30-0.40	1.30-1.80	0.10-0.35	0.05	0.05	---	---	0.20-0.35	---	70 79 87 95 102	42 48 68 70 80	22 201-255 / 54 20 223-277 / 54 16 248-302 / 54 16 269-321 / 47 14 293-341 / 47	R S T/P-8.8 U/P-8.8 V/P-8.8	CARBON MANGA-NESE Mo STEEL FOR HIGH TENSILE FASTENERS ≤ M24
16A ---	---	---	---	---	---	---	---	0.25-0.30	1.30-1.80	0.10-0.35	0.05	0.05	---	---	0.20-0.35	---	MECH. PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION				CARBON MANGA-NESE Mo STEEL FOR SPECIAL APPLICATION
16B 605A32	---	---	---	---	---	---	---	0.30-0.30	1.30-1.80	0.10-0.35	0.05	0.05	---	---	0.20-0.35	---	MECH. PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION				CARBON MANGA-NESE Mo STEEL FOR SPECIAL APPLICATION
16C 605A37	35Mn6Mo3--- (IS:3930)		---	---	---	---	---	0.35-0.40	1.30-1.80	0.10-0.35	0.05	0.05	---	---	0.20-0.35	---	MECH. PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION				CARBONMANGA-NESE Mo STEEL FOR SPECIAL APPLICATION
16M 606M36	---	---	---	---	---	---	---	0.30-0.40	1.30-1.80	0.25 MAX.	0.12-0.20	0.05	---	---	0.20-0.35	---	MECH. PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION				CARBON MANGA-NESE Mo FREE CUTTING STEEL FOR SPECIAL APPLICATION

ABBREVATIONS USED:
HT -- HOT ROLLED
AC -- AIR COOLED
OH -- OIL HARDENED
AH -- AIR HARDENED
H & T -- HARDENED & TEMPERED
AR -- AS ROLLED
CR -- COLD ROLLED
CD -- COLD DRAWN
N -- NORMALISED
A -- ANNEALED

HEAT TREATMENT CONDITION:
Q, R, S, T, U, V, W, X, Y, Z

CONTD.

CHEMISTRY OF VARIOUS TYPES OF STEELS AND THEIR COMPARISON WITH INTERNATIONAL GRADES

BS:970		EQUIVALENT NATIONAL / INTERNATIONAL GRADES							CHEMICAL COMPOSITION, PERCENT									MECHANICAL PROPERTIES (MIN.)				CONDITION	TYPE OF STEEL AND APPLICATIONS
En No.	NUMERICAL	ISS (BIS)	ASTM	SAE	AISI	DIN	GOST	JIS	C	Mn	Si	S MAX.	P MAX.	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E%	HARDNESS (BHN)/IZOD (J)		
17	608M38	35Mn6Mo4 (IS:3930)	---	---	---	---	---	---	0.30-0.40	1.30-1.80	0.10-0.35	0.05	0.05	---	---	0.35-0.55	---	70 79 87 95 102	42 48 68 70 80	22 20 16 16 14	201-255 / 54 223-277 / 54 248-302 / 54 269-321 / 47 293-341 / 47	R S T/P-8.8 U/P-8.8 V/P-8.8	CARBON MANGANESE Mo STEEL FOR HIGH TENSILE FASTENERS ≥ M27 (HIGHER Mo)
18	530M40	40Cr4 (IS:3930)	---	5140H	5140H	41Cr4	40X	---	0.35-0.45	0.60-0.95	0.10-0.35	0.05	0.05	0.85-1.15	---	---	---	70 79 87 95 102	42 48 68 70 80	22 20 16 16 14	201-255 / 54 223-277 / 54 248-302 / 54 269-321 / 47 293-341 / 47	R S T/P-8.8 U/P-8.8 V/P-8.8	1% Cr. STEEL FOR HIGH TENSILE FASTENERS GREATER THAN M36, SHAFTS, AXLES, CONN. RODS, ETC.
18A	530A30	---	A325M TYPE-2	5130	5130	---	30X	---	0.27-0.32	0.65-0.80	0.10-0.35	0.05	0.05	0.85-1.15	---	---	---	MECH. PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					AS ABOVE FOR SPECIAL APPLICATION
18B	530A32	---	---	5132	5132	34Cr4	30X	---	0.30-0.35	0.65-0.80	0.10-0.35	0.05	0.05	0.85-1.15	---	---	---	MECH. PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					AS ABOVE FOR SPECIAL APPLICATION
18C	530A36	---	---	5135	5135	34Cr4	35X	---	0.35-0.38	0.65-0.80	0.10-0.35	0.05	0.05	0.85-1.15	---	---	---	MECH. PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					AS ABOVE FOR SPECIAL APPLICATION
18D	530A40	---	---	5140	5140	41Cr4	40X	---	0.38-0.43	0.65-0.80	0.10-0.35	0.05	0.05	0.85-1.15	---	---	---	MECH. PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					AS ABOVE FOR SPECIAL APPLICATION
19	709M40	---	A490M TYPE-1	4140	4140	42Cr Mo4	35XM	---	0.35-0.45	0.50-0.80	0.10-0.35	0.05	0.05	0.90-1.50	---	0.20-0.40	---	70 79 87 95 102 110 126	42 48 68 70 80 95 100	22 20 16 16 14 14 10	201-255 / 54 223-277 / 54 248-302 / 54 269-321 / 47 293-341 / 47 345-440 / 40 385-490 / 27	R S T/P-8.8 U/P-8.8 V/P-8.8 W/P-10.9 Y/P-10.9	1% Cr. Mo STEEL FOR HIGH TENSILE FASTENERS FOR HIGHER TENSILE PROPERTIES
19A	708M40	40Cr4 Mo3	A193 B7/B7M	4140	4140	42Cr Mo4	35CM	---	0.35-0.45	0.50-0.80	0.10-0.35	0.05	0.05	0.90-1.20	---	0.20-0.40	---	70 79 87 95 102 110 126	42 48 68 70 80 95 100	22 20 16 16 14 14 10	201-255/54 223-277/54 248-302/54 269-321/47 293-341/47 345-440/40 385-490/27	R S T/P-8.8 U/P-8.8 V/P-8.8 W/P-10.9 Y/P-10.9	1% Cr. Mo STEEL FOR HIGH TENSILE FASTENERS FOR HIGHER TENSILE PROPERTIES
19B	708A37	---	A320 L-7	4137	4137	42Cr Mo4	---	---	0.35-0.40	0.50-0.80	0.10-0.35	0.05	0.05	0.90-1.20	---	0.20-0.35	---	MECH. PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					AS ABOVE FOR SPECIAL APPLICATION
19C	708A42	---	A320 L-7	4142	4142	42Cr Mo4	---	---	0.40-0.45	0.50-0.80	0.10-0.35	0.05	0.05	0.90-1.20	---	0.20-0.35	---	MECH. PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					AS ABOVE FOR SPECIAL APPLICATION

ABBREVATIONS USED: HT – HOT ROLLED; AC – AIR COOLED; OH – OIL HARDENED; AH – AIR HARDENED; H & T – HARDENED & TEMPERED; HEAT TREATMENT CONDITION:

AR – AS ROLLED; CR – COLD ROLLED; CD – COLD DRAWN; N – NORMALISED; A – ANNEALED; Q, R, S, T, U, V, W, X, Y, Z

CONTD.

CHEMISTRY OF VARIOUS TYPES OF STEELS AND THEIR COMPARISON WITH INTERNATIONAL GRADES

BS:970		EQUIVALENT NATIONAL / INTERNATIONAL GRADES							CHEMICAL COMPOSITION, PERCENT									MECHANICAL PROPERTIES (MIN.)				CONDITION	TYPE OF STEEL AND APPLICATIONS
En No.	NUMERICAL	ISS (BIS)	ASTM	SAE	AISI	DIN	GOST	JIS	C	Mn	Si	S MAX.	P MAX.	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E%	HARDNESS (BHN)/IZOD (J)		
20A	---	---	---	---	---	---	---	---	0.20-0.30	0.40-0.70	0.10-0.35	0.05	0.05	0.50-1.00	---	0.50-0.80	---	87 95 102	68 70 80	16 16 14	248-302 / 54 269-321 / 47 293-341 / 47	T/P-8.8 U/P-8.8 V/P-8.8	1% Cr Mo STEEL (HIGHER Mo) FOR HIGH TENSILE FASTENERS FOR SERVICE AT STEAM TEMPERATURE.
20B	---	---	---	---	---	---	---	---	0.35-0.45	0.40-0.70	0.10-0.35	0.05	0.05	1.00-1.50	---	0.50-0.90	---	87 95 102	68 70 80	16 16 14	248-302 / 54 269-321 / 47 293-341 / 47	T/P-8.8 U/P-8.8 V/P-8.8	1% Cr Mo STEEL (HIGHER Mo) FOR HIGH TENSILE FASTENERS FOR SERVICE AT STEAM TEMPERATURE.
21	---	---	---	---	---	---	---	---	0.25-0.35	0.35-0.75	0.10-0.35	0.05	0.05	0.30 MAX.	2.75-3.25	---	---	70 79	42 48	22 20	201-255 / 54 223-277 / 54	R S	3% Ni STEEL - SUITABLE FOR HT FASTENERS IN GRADE 8.8 OR EQUI.
21A	---	---	---	---	---	---	---	---	0.20-0.30	0.60 MAX.	0.10-0.35	0.05	0.05	---	2.50-3.50	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					3% Ni STEEL IN THE FORM OF PLATE, SHEET/STRIP.
22	---	T40Ni14 (IS:3749)	---	---	---	---	34Ni5	---	---	0.35-0.45	0.50-0.80	0.10-0.35	0.05	0.05	0.30 MAX.	3.25-3.75	---	---	79 87	48 68	20 16	223-277 / 54 248-302 / 54	S T/P-8.8	3.5% Ni STEEL - SUITABLE FOR HT FASTENERS IN GRADE P-8.8
23	653M31	---	---	4820	4820	---	---	---	0.25-0.35	0.45 0.70	0.10-0.35	0.05	0.05	0.50-1.00	2.75-3.50	0.65 MAX. (OPT.)	---	79 87 95 110	48 68 70 95	20 16 16 14	223-277 / 54 248-302 / 54 269-321 / 47 345 - 440 / 40	S T/P-8.8 U/P-8.8 V/P-10.9	3.5% Ni STEEL - SUITABLE FOR HT FASTENERS IN GRADE P-8.8/10.9
24	817M40	T40Ni6 Cr4Mo3 (IS:5517)	F568	4340	4340	34CrNi Mo6	34X H2M	---	0.35-0.45	0.45 0.70	0.10-0.35	0.05	0.05	0.90-1.40	1.30-1.80	0.20-0.35	---	79 87 95 110 115 120 126 157	48 68 70 95 100 110 115 125	20 16 16 14 12 10 9 8	223-277 / 54 248-302 / 54 269-321 / 47 345 - 440 / 40 365 - 450 / 35 395 - 470 / 34 410 - 490 / 30 430 - 500 / 25	S T/P-8.8 U/P-8.8 V/P-10.9 W/P-10.9 X/P-12.9 Y/P-12.9 Z/P-12.9	1.5% Ni-Cr-Mo STEEL - SUITABLE FOR HT FASTENERS IN GRADE P-10.9/12.9 ACCORDING TO RULING SECTION.
25	826M31	31Ni10 Cr3Mo6	F541	---	---	---	---	---	0.27-0.35	0.50-0.70	0.10-0.35	0.05	0.05	0.50-0.80	2.30-2.80	0.40-0.70	---	87 95 110 115 120 126 157	68 70 95 100 110 115 125	16 16 14 12 10 9 8	248-302 / 54 269-321 / 47 345 - 440 / 40 365 - 450 / 35 395 - 470 / 34 410 - 490 / 30 430 - 500 / 25	T/P-8.8 U/P-8.8 V/P-10.9 W/P-10.9 X/P-12.9 Y/P-12.9 Z/P-12.9	2.5% Ni-Cr-Mo STEEL (MEDIUM CARBON) SUITABLE FOR HT FASTENERS IN GRADE P-10.9/12.9

ABBREVATIONS USED: HT -- HOT ROLLED; AC -- AIR COOLED; OH -- OIL HARDENED; AH -- AIR HARDENED; H & T -- HARDENED & TEMPERED; HEAT TREATMENT CONDITION: Q, R, S, T, U, V, W, X, Y, Z

AR -- AS ROLLED; CR -- COLD ROLLED; CD -- COLD DRAWN; N -- NORMALISED; A -- ANNEALED

CONTD.

CHEMISTRY OF VARIOUS TYPES OF STEELS AND THEIR COMPARISON WITH INTERNATIONAL GRADES

BS:970		EQUIVALENT NATIONAL / INTERNATIONAL GRADES							CHEMICAL COMPOSITION, PERCENT									MECHANICAL PROPERTIES (MIN.)				CONDITION	TYPE OF STEEL AND APPLICATIONS
En No.	NUMERICAL	ISS (BIS)	ASTM	SAE	AISI	DIN	GOST	JIS	C	Mn	Si	S MAX.	P MAX.	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E%	HARDNESS (BHN)/IZOD(J)		
25	826M40	40Ni3 Cr65Mo55 (IS:4368)	---	---	---	---	---	--- 0.44	0.36-0.70	0.50-0.35	0.10-	0.05	0.05 0.80	0.50-2.80	2.30-0.70	0.40-	--- 110	95 95 115 120 126 157	70 14 100 110 115 125	16 12 10 9 8	269-321 / 47 345 - 440 / 40 365 - 450 / 35 395 - 470 / 34 410 - 490 / 30 430 - 500 / 25		U/P-8.8 V/P-10.9 W/P-10.9 X/P-12.9 Y/P-12.9 Z/P-12.9	2.5% Ni-Cr-Mo STEEL (HIGH CARBON) SUITABLE FOR TENSILE RANGE 100-160 Kg/mm²
27	830M31	---	---	---	---	---	---	---	0.20-0.35	0.70-MAX.	0.10-0.35	0.05	0.05	0.50-1.30	3.00-3.75	0.20-0.65	---	87 95 110 115	68 70 95 100	16 16 14 12	248-302 / 54 269-321 / 47 345 - 440 / 40 365 - 450 / 35		T/P-8.8 U/P-8.8 V/P-10.9 W/P-10.9	2.5% Ni-Cr-Mo STEEL (HIGH CARBON) SUITABLE FOR TENSILE RANGE 85-125 Kg/mm²
28	---	---	---	---	---	---	---	---	0.25-0.40	0.70-MAX.	0.10-0.35	0.05	0.05	0.75-1.50	3.00-4.50	0.20-0.65	---	95 110 115 126	70 95 100 115	16 14 12 9	269-321 / 47 345 - 440 / 40 365 - 450 / 35 410 - 490 / 30		U/P-8.8 V/P-10.9 W/P-10.9 Y/P-12.9	3.5% Ni-Cr-Mo STEEL SUITABLE FOR TENSILE RANGE 95-130 Kg/mm²
29A	---	---	---	---	---	---	---	---	0.15-0.25	0.65 MAX.	0.10-0.35	0.05	0.05	2.50-3.50	0.40 MAX.	0.30-0.70	---	70 79 87 95 110 115 157	42 48 68 70 95 100 125	22 20 16 16 14 12 8	201-255 / 54 223-277 / 54 248-302 / 54 269-321 / 47 345 - 440 / 40 365 - 450 / 35 430 - 500 / 25		R S T/P-8.8 U/P-8.8 V/P-10.9 W/P-10.9 Z/P-12.9	3% Cr- Mo STEEL SUITABLE FOR TENSILE RANGE 70-158 Kg/mm²
29B	---	25Cr3Mo55— (IS:4348)	---	---	---	---	---	---	0.25-0.35	0.65 MAX.	0.10-0.35	0.05	0.05	2.50-3.50	0.40 MAX.	0.30-0.70	---	70 79 87 95 110 115 157	42 48 68 70 95 100 125	22 20 16 16 14 12 8	201-255 / 54 223-277 / 54 248-302 / 54 269-321 / 47 345 - 440 / 40 365 - 450 / 35 430 - 500 / 25		R S T/P-8.8 U/P-8.8 V/P-10.9 W/P-10.9 Z/P-12.9	3% Cr- Mo STEEL SUITABLE FOR TENSILE RANGE 70-158 Kg/mm² (HIGHER CARBON)
30A	---	T30Ni16Cr5— (IS:3749)	---	---	---	---	---	---	0.26-0.34	0.40-0.60	0.10-0.35	0.05	0.05	1.00-1.40	3.90-4.30	---	---	157 157	125 125	10 10	430 - 500 /25 430 - 500 / 25		OH AH	4.25% Ni-Cr STEEL FOR TENSILE RANGE 158 Kg/mm2 GEARS, AERO ENGINE CONN. RODS, HEAVY DUTY FASTENERS, SPINDLES, ETC.
30B	835M30	---	---	---	---	---	---	---	0.26-0.34	0.40-0.60	0.10-0.35	0.05	0.05	1.00-1.40	3.90-4.30	0.20-0.40	---						4.25% Ni-Cr TOOL STEEL
31	534A99	103 Cr2	---	---	---	---	IIIX15	---	0.90-1.20	0.30-0.75	0.10-0.35	0.05	0.05	1.00-1.60	---	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					1% CARBON Cr STEEL FOR PARTS OF MAX. HARDNESS SUCH AS BALL RACES.
32A	--	C-14 (IS:2073)	---	---	---	---	MCT 2KJ1	---	0.15 MAX.	0.40-0.70	0.05-0.35	0.05	0.05	---	---	---	---	40	24	35	54	AR/N	CASE HARDENING STEEL-BARS FOR FORGINGS/MACHINING/BRIGHT BARS

ABBREVATIONS USED:
AR – AS ROLLED
HT – HOT ROLLED
CR – COLD ROLLED
AC – AIR COOLED
CD – COLD DRAWN
OH – OIL HARDENED
N – NORMALISED
AH – AIR HARDENED
A – ANNEALED
H & T – HARDENED & TEMPERED

HEAT TREATMENT CONDITION:
Q, R, S, T, U, V, W, X, Y, Z

CONTD.

CHEMISTRY OF VARIOUS TYPES OF STEELS AND THEIR COMPARISON WITH INTERNATIONAL GRADES

BS:970		EQUIVALENT NATIONAL / INTERNATIONAL GRADES							CHEMICAL COMPOSITION, PERCENT									MECHANICAL PROPERTIES (MIN.)				CONDITION	TYPE OF STEEL AND APPLICATIONS
En No.	NUMERICAL	ISS (BIS)	ASTM	SAE	AISI	DIN	GOST	JIS	C	Mn	Si	S MAX.	P MAX.	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E%	HARDNESS (BHN)/IZOD(J)		
32B	--	---	---	---	---	---	---	---	0.10-0.18	0.60-1.00	0.05-0.35	0.07	0.05	---	---	---	---	40	24	26	54	AR	CASE HARDENING STEEL-FOR SPECIAL APPLICATIONS
32C	--	C15Mn75 (IS:4432)	1016	1016	---	---	---	---	0.10-0.18	0.60-1.00	0.05-0.35	0.05	0.05	---	---	---	---	40	24	26	54	AR	CASE HARDENING STEEL-FOR SPECIAL APPLICATIONS
32M	--	10S11 (IS:4432)	1117	1117	---	---	---	---	0.10-0.18	0.90-1.20	0.05-0.35	0.10-0.15	0.05	---	---	---	---	40	24	26	50	AR	CASE HARDENING FREECUTTING STEEL
33	--	---	---	---	---	---	---	---	0.10-0.15	0.30-0.60	0.10-0.35	0.05	0.05	0.30 MAX.	2.75-3.00	---	---	70	45	18	54	HT	3% NiCASE HARDENING STEEL FOR GUDGEN PINS, CAM-SHAFTS, PINIONS.
34	665M17	20Ni2 Mo25 (IS:4423)	---	4615	4615	---	---	---	0.14-0.20	0.30-0.60	0.10-0.35	0.05	0.05	---	1.50-2.00	0.20-0.30	---	70	45	18	54	HT	2% Ni Mo CASE HARDENING STEEL (LOW CARBON)-USES SAME AS ABOVE
35	665M23	---	---	4620	4620	---	---	---	0.20-0.28	0.30-0.60	0.10-0.35	0.05	0.05	---	1.50-2.00	0.20-0.30	---	87	50	15	30	HT	2% Ni Mo CASE HARDENING STEEL (HIGH CARBON)-FOR SIMILAR PARTS TO En33 REQUIRING STRONGER AND TOUGHER CORE.
35A	665A22	20Ni2 Mo25 (IS:4432)	---	4620	4620	---	---	---	0.20-0.25	0.30-0.60	0.10-0.35	0.05	0.05	---	1.50-2.00	0.20-0.30	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					SAME AS ABOVE FOR SPECIAL APPLICATIONS.
35B	665A24	---	---	---	---	---	---	---	0.23-0.28	0.30-0.60	0.10-0.35	0.05	0.05	---	1.50-2.00	0.20-0.30	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					SAME AS ABOVE FOR SPECIAL APPLICATIONS.
36A	665M13	13Ni3Cr80— (IS:4432)	9310	---	---	12XH3A	---		0.15 MAX.	0.30-0.60	0.10-0.35	0.05	0.05	0.60-1.10	3.00-3.75	---	---	87	66	15	243-305/48	HT	3% Ni-Cr CASE HARDENING STEEL FOR SPECIAL APPLICATION.
36B	---	13Ni3Cr80 F541 (IS:4432)	---	---	---	12XH3A	---		0.12-0.18	0.30-0.60	0.10-0.35	0.05	0.05	0.60-1.10	3.00-3.75	---	---	103	92	13	325-355/40	HT	3% Ni-Cr CASE HARDENING STEEL FOR SPECIAL APPLICATION.
36C	832M13	---	---	---	---	---	---	---	0.12-0.18	0.30-0.60	0.10-0.35	0.05	0.05	0.60-1.10	3.00-3.75	0.10-0.25	---	103	92	13	325-355/40	HT	3% Ni-Cr-Mo CASE HARDENING STEEL FOR SPECIAL APPLICATION.

ABBREVATIONS USED:

AR -- AS ROLLED; HT -- HOT ROLLED; CR -- COLD ROLLED; AC -- AIR COOLED; CD -- COLD DRAWN; OH -- OIL HARDENED; N -- NORMALISED; AH -- AIR HARDENED; A -- ANNEALED; H & T -- HARDENED & TEMPERED

HEAT TREATMENT CONDITION: Q, R, S, T, U, V, W, X, Y, Z

CONTD

CHEMISTRY OF VARIOUS TYPES OF STEELS AND THEIR COMPARISON WITH INTERNATIONAL GRADES

BS:970		EQUIVALENT NATIONAL / INTERNATIONAL GRADES							CHEMICAL COMPOSITION, PERCENT									MECHANICAL PROPERTIES (MIN.)				CONDITION	TYPE OF STEEL AND APPLICATIONS
En No.	NUMERICAL	ISS (BIS)	ASTM	SAE	AISI	DIN	GOST	JIS	C	Mn	Si	S	P MAX.	Cr MAX.	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E%	HARDNESS (BHN) / IZOD (J)		
37	---	---	---	---	---	---	---	---	0.16 MAX.	0.45 MAX.	0.10-0.35	0.05	0.05	0.30-MAX.	4.50-5.20	---	---	63	40	20	213-297/68	HT	5% Ni-CASE HARDENING STEEL FOR GUDGEON PINS, CAM SHAFTS, GEARS, PUSH ROD HEADS, PINIONS.
38	---	---	---	---	---	---	---	---	0.16 MAX.	0.60 MAX.	0.10-0.35	0.05	0.05	0.30 MAX.	4.50-5.50	0.15-0.30	---	103	92	13	325-355/40	HT	5% Ni-Mo CASE HARDENING STEEL FOR ITEMS REQUIRING STRONGER AND TOUGHER CORE.
39A	659M15	15Ni4Cr1 (IS:4368)	---	---	---	---	12X2H4A	---	0.12-0.18	0.50 MAX.	0.10-0.35	0.05	0.05	1.00-1.40	3.80-4.50	---	---	134	110	12	363-440/34	HT	4.25% Ni-Cr CASE HARDENING STEEL FOR HIGH DUTY TRANSMISSION GEARS, REDUCTION GEARS, ETC.
39B	835M15	---	---	---	---	---	---	---	0.12-0.18	0.50 MAX.	0.10-0.35	0.05	0.05	1.00-1.40	3.80-4.50	0.15-0.35	---	134	110	12	363-440/34	HT	4.25% Ni-Cr-Mo CASE HARDENING STEEL FOR HIGH DUTY TRANSMISSION GEARS, REDUCTION GEARS, ETC.
40A	---	15Cr13 Mo6 (IS:5517)	---	---	---	---	---	---	0.10-0.20	0.40-0.65	0.10-0.35	0.05	0.05	2.90-3.50	0.40 MAX.	0.40-0.70	---	70 79 87 95	42 48 60 70	18 18 15 13	201-255/56 223-277/34 248-302/30 269-321/25	R S T U	3% Cr-Mo NITRIDING STEELFOR CYL. LINERS, CRANK-SHAFTS, AIR SCREW SHAFTS.
40B	722M24	25Cr13 Mo6 (IS:5517)	---	---	---	---	---	---	0.20-0.30	0.40-0.65	0.10-0.35	0.05	0.05	2.90-3.50	0.40 MAX.	0.40-0.70	---	70 79 87 95	42 48 60 70	18 18 15 13	201-255/56 223-277/34 248-302/30 269-321/25	R S T U	3% Cr-Mo NITRIDING STEELFOR CYL. LINERS, CRANK-SHAFTS, AIR SCREW SHAFTS(HIGH CARBON).
40C	897M39	40Cr13 Mo10V2 (IS:5517)	---	---	---	---	---	---	0.30-0.50	0.40-0.80	0.10-0.35	0.05	0.05	2.50-3.50	0.40 MAX.	0.70-1.20	---	134	120	10	363-470/20	HT	3% Cr-Mo-V NITRIDING STEEL FOR CYL. LINERS, CRANK SHAFTS, AIR SCREW SHAFTS
41A	905M31	40Cr7 A110Mo2 (IS:5517)	---	---	---	34CrA1 Mo2	---	---	0.25-0.35	0.65 MAX.	0.10-0.45	0.05	0.05	1.40-1.80	0.40 MAX.	0.10-0.25	0.90-1.30 Al.	70 79 87	42 48 60	18 18 15	201-255/56 223-277/34 248-302/30	R S T	1.5% Cr-Al-Mo NITRIDING STEEL
41B	905M39	40Cr7 A110Mo2 (IS:5517)	---	---	---	42CrMo4	---	---	0.35-0.45	0.65 MAX.	0.10-0.45	0.05	0.05	1.40-1.80	0.40 MAX.	0.10-0.25	0.90-1.30 Al.	70 79 87	42 48 60	18 18 15	201-255/56 223-277/34 248-302/30	R S T	1.5% Cr-Al-Mo NITRIDING STEEL

ABBREVATIONS USED:
HT -- HOT ROLLED AC -- AIR COOLED OH -- OIL HARDENED AH -- AIR HARDENED H & T -- HARDENED & TEMPERED
AR -- AS ROLLED CR -- COLD ROLLED CD -- COLD DRAWN N -- NORMALISED A -- ANNEALED

HEAT TREATMENT CONDITION:
Q, R, S, T, U, V, W, X, Y, Z

CONTD

CHEMISTRY OF VARIOUS TYPES OF STEELS AND THEIR COMPARISON WITH INTERNATIONAL GRADES

BS:970		EQUIVALENT NATIONAL / INTERNATIONAL GRADES							CHEMICAL COMPOSITION, PERCENT									MECHANICAL PROPERTIES (MIN.)				CONDITION	TYPE OF STEEL AND APPLICATIONS
En No.	NUMERICAL	ISS (BIS)	ASTM	SAE	AISI	DIN	GOST	JIS	C	Mn	Si	S MAX.	P MAX.	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E %	HARDNESS (BHN)/IZOD(J)		
42	070A72	80T6 (IS:3749)	---	1080	1080	C75	GR.80	---	0.70-0.85	0.55-0.75	0.10-0.40	0.05	0.05	---	---	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					CARBON SPRING STEEL OIL HARDENING AND TEMPERING QUALITY
42B	---	CL-6 (IS:1875)	---	1085	1085	C67	GR.65	---	0.60-0.70	0.55-0.80	0.35 MAX.	0.05	0.05	---	---	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					CARBON SPRING STEEL (ANNEALED WIRE)
42C	---	80T6 (IS:3749)	---	1074	1074	C75	GR.75	---	0.70-0.80	0.55-0.80	0.35 MAX.	0.05	0.05	---	---	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					CARBON SPRING STEEL (ANNEALED WIRE)
42D	---	85T6 (IS:3749)	---	1084	1084	---	GR.85	---	0.80-0.90	0.55-0.80	0.35 MAX.	0.05	0.05	---	---	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					CARBON SPRING STEEL (ANNEALED WIRE)
42E	---	70T6 (IS:3749)	---	1069	1069	---	---	---	0.60-0.75	0.30-0.60	0.35 MAX.	0.05	0.05	---	---	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					CARBON SPRING STEEL (COLD ROLLED STRIP)
42F	---	---	---	1070	1070	---	---	---	0.60-0.75	0.60-0.90	0.35 MAX.	0.05	0.05	---	---	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					CARBON SPRING STEEL (COLD ROLLED STRIP)
42G	---	---	---	1078	1078	M85	---	---	0.75-0.90	0.30-0.60	0.35 MAX.	0.05	0.05	---	---	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					CARBON SPRING STEEL (COLD ROLLED STRIP) OIL HARDENING & TEMPERING QUALITY.
42J	---	85T6 (IS:3749)	---	1080	1080	---	---	---	0.75-0.90	0.60-0.90	0.35 MAX.	0.05	0.05	---	---	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					CARBON SPRING STEEL (COLD ROLLED STRIP) OIL HARDENING & TEMPERING QUALITY.
43	080A52	50T8 (IS:3749)	---	1055	1055	C55	---	---	0.45-0.60	0.60-0.80	0.10 0.40	0.05	0.05	---	---	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					CARBON SPRING STEEL WATER HARDENING AND TEMPERING QUALITY.
43A	080M50	50T8 (IS:3749)	---	1050	1050	---	GR.50	---	0.45-0.55	0.70-1.00	0.05 0.35	0.06	0.06	---	---	---	---	63 --- 18 --- 71 --- 18 --- 79 --- 18 --- 71-95 --- 12 ---				N R S CD	50 CARBON STEEL BARS & BILLETS FOR FORGINGS, DROP FORGINGS, BRIGHT FINISH M/C
43B	080A47	CL-4 (IS:1875)	---	1045	1045	CK45	GR.45	---	0.45-0.50	0.70-1.00	0.05 0.35	0.05	0.05	---	---	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					45 CARBON STEEL BARS & BILLETS FOR FORGINGS, DROP FORGINGS, BRIGHT FINISH M/C

ABBREVATIONS USED: HT -- HOT ROLLED; AC -- AIR COOLED; OH -- OIL HARDENED; AH -- AIR HARDENED; H & T -- HARDENED & TEMPERED; AR -- AS ROLLED; CR -- COLD ROLLED; CD -- COLD DRAWN; N -- NORMALISED; A -- ANNEALED

HEAT TREATMENT CONDITION: Q, R, S, T, U, V, W, X, Y, Z

CONTD.

CHEMISTRY OF VARIOUS TYPES OF STEELS AND THEIR COMPARISON WITH INTERNATIONAL GRADES

BS:970		EQUIVALENT NATIONAL / INTERNATIONAL GRADES							CHEMICAL COMPOSITION, PERCENT									MECHANICAL PROPERTIES (MIN.)	CONDITION	TYPE OF STEEL AND APPLICATIONS
En No.	NUMERICAL	ISS (BIS)	ASTM	SAE	AISI	DIN	GOST	JIS	C	Mn	Si	S MAX.	P MAX.	Cr	Ni	Mo	OTHERS	UTS YS E% HARDNESS Kg/mm² (BHN)/IZOD(J)		
43C	080A52	50T8 (IS:3749)	---	1053	1053	---	---	---	0.50-0.55	0.70-1.00	0.05 0.35	0.05	0.05	---	---	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION		52 CARBON STEEL BARS & BILLETS FOR FORGINGS, DROP FORGINGS, BRIGHT FINISH M/C
43D	060A62	---	---	1060	1060	CK55	GR.60	---	0.60-0.65	0.40-0.60	0.05 0.35	0.06	0.06	---	---	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION		62 CARBON STEEL BARS & BILLETS FOR FORGINGS, DROP FORGINGS, BRIGHT FINISH M/C
43E	080A67	65T6 (IS:3749)	---	1070	1070	C67	GR.65	---	0.65-0.70	0.70-0.90	0.05 0.35	0.06	0.06	---	---	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION		67 CARBON STEEL BARS & BILLETS FOR FORGINGS, DROP FORGINGS, BRIGHT FINISH M/C
43G	---	---	---	---	---	---	---	---	0.45-0.60	0.30-0.60	0.35 MAX.	0.05	0.05	---	---	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION		CARBON SPRING STEEL FOR WATER HARDENING AND TEMPERING(COLD ROLLED SHEET AND FLAT.
43J	---	50T8 (IS:3749)	---	1055	1055	---	GR.55	---	0.45-0.60	0.60-0.90	0.35 MAX.	0.05	0.05	---	---	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION		CARBON SPRING STEEL FOR WATER HARDENING AND TEMPERING(COLD ROLLED SHEET AND FLAT.
44	060A96	---	---	---	---	---	---	---	0.90-1.20	0.45-0.70	0.30 MAX.	0.05	0.05	---	---	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION		CARBON SPRING STEEL FOR OIL HARDENING AND TEMPERING (HIGHER CARBON)
44B	---	---	---	---	---	---	---	---	0.90-1.00	0.40-0.70	0.35 MAX.	0.05	0.05	---	---	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION		CARBON SPRING STEEL FOR WIRES
44C	---	---	---	---	---	---	---	---	1.00-1.20	0.40-0.70	0.35 MAX.	0.05	0.05	---	---	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION		CARBON SPRING STEEL FOR WIRES
44D	---	---	---	1095	1095	MK101	---	---	0.90-1.05	0.30-0.70	0.35 MAX.	0.05	0.05	---	---	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION		CARBON SPRING STEEL FOR COLD ROLLED STRIPS.
44E	---	---	---	---	---	---	---	---	1.05-1.25	0.30-0.70	0.35 MAX.	0.05	0.05	---	---	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION		CARBON SPRING STEEL FOR COLD ROLLED STRIPS.
45	250A53	T55Si7 (IS:3749)	---	9255	9255	55Si7	55C2	---	0.55-0.60	0.70-1.00	1.50-2.00	0.05	0.05	---	---	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION		Si-Mn SPRING STEEL FOR OIL HARDENING AND TEMPERING.

ABBREVATIONS USED:

AR -- AS ROLLED | HT -- HOT ROLLED | CR -- COLD ROLLED | AC -- AIR COOLED | CD -- COLD DRAWN | OH -- OIL HARDENED | N -- NORMALISED | AH -- AIR HARDENED | A -- ANNEALED | H & T -- HARDENED & TEMPERED

HEAT TREATMENT CONDITION:

Q, R, S, T, U, V, W, X, Y, Z

CONTD.

CHEMISTRY OF VARIOUS TYPES OF STEELS AND THEIR COMPARISON WITH INTERNATIONAL GRADES

BS:970		EQUIVALENT NATIONAL / INTERNATIONAL GRADES							CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)	CONDITION	TYPE OF STEEL AND APPLICATIONS
En No.	NUMERICAL	ISS (BIS)	ASTM	SAE	AISI	DIN	GOST	JIS	C	Mn	Si	S MAX.	P MAX.	Cr	Ni	Mo	OTHERS	UTS Kg/mm² YS E% HARDNESS (BHN)/IZOD (J)		
45A	250A58	60Si7 (IS:3195)	---	9260	9260	65Si7	60C2	---	0.55-0.65	0.70-1.00	1.70-2.00	0.05	0.05	---	---	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION		Si-Mn SPRING STEEL FOR OIL HARDENING AND TEMPERING.
46	---	---	---	---	---	---	---	---	0.35-0.45	0.70-1.00	1.50-2.00	0.05	0.05	---	---	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION		Si-Mn SPRING STEEL FOR WATER HARDENING AND TEMPERING.
47	735A50	T50Cr4V2— (IS:3749)		6150	6150	50CrV4	50ΧΦΑ	---	0.45-0.55	0.50-0.80	0.50 MAX.	0.05	0.05	0.80-1.20	---	---	0.15V MIN.	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION		1%Cr-V SPRING STEEL FOR OIL HARDENING AND TEMPERING.
48	527A60	---	---	---	---	---	50ΧΓ	---	0.45-0.55	0.50-0.80	0.10-0.50	0.05	0.05	1.00-1.40	---	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION		1%Cr SPRING STEEL FOR OIL HARDENING AND TEMPERING.
48A	527A60	T55Cr3	--- (IS:3749)	---	---	---	---	---	0.50-	0.60-0.60	1.35-0.90	0.05 1.65	0.05	0.55-	--- 0.85	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION		Si-Cr SPRING STEEL FOR OIL HARDENING AND TEMPERING.
49A	---	80T3 (IS:3749)	---	---	---	---	---	---	0.40-0.85	1.00 MAX.	0.30 MAX.	0.05	0.05	---	---	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION		CARBON STEEL FOR HARD DRAWN WIRE
49B	---	80T3 (IS:3749)	---	---	---	---	---	---	0.45-0.85	1.00 MAX.	0.30 MAX.	0.05	0.05	---	---	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION		CARBON STEEL FOR HARD DRAWN WIRE
49C	---	80T3 (IS:3749)	---	---	---	---	---	---	0.55-0.85	0.75 MAX.	0.30 MAX.	0.04	0.04	---	---	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION		CARBON STEEL FOR HARD DRAWN WIRE
49D	---	80T3 (IS:3749)	---	---	---	---	---	---	0.65-0.85	0.75 MAX.	0.30 MAX.	0.04	0.04	---	---	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION		CARBON STEEL FOR HARD DRAWN WIRE
50	---	50Cr4V2 (IS:3930)	---	---	---	---	---	---	0.40-0.50	0.50-0.70	0.10-0.35	0.04	0.04	1.00-1.50	0.15V MIN.	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION		CR-V STEEL WIRE FOR VALVE SPRINGS.
51	---	---	---	---	---	---	30ХН3А	---	0.25-0.35	0.35-0.75	0.10-0.35	0.05	0.05	0.30 MAX.	2.75-3.25	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION		3% Ni VALVE STEEL
52	401S45	---	---	---	---	---	X9C2	---	0.40-0.50	0.30-0.60	3.00-3.75	0.04	0.04	7.50-9.50	0.50 MAX.	---	---	225-293	HT	Si-Cr VALVE STEEL FOR FORGINGS.
53	---	--	---	---	---	---	---	---	0.55-0.65	0.30-0.60	1.40-1.70	0.05	0.05	5.75-6.75	0.50 MAX.	---	---	235-285 / 16	HT	Si-Cr VALVE STEEL FOR FORGINGS.
54	331S40	--	---	---	---	---	---	---	0.35-0.50	1.50 MAX.	1.00-2.50	0.045	0.045	12.00-16.00	10.00 MIN.	---	2.00-4.00W	302 / 20 MAX.	HT	HIGH Cr-Ni TUNGSTEN VALVE STEEL

ABBREVATIONS USED: HT – HOT ROLLED AC – AIR COOLED OH – OIL HARDENED AH – AIR HARDENED H & T – HARDENED & TEMPERED HEAT TREATMENT CONDITION:

AR – AS ROLLED CR – COLD ROLLED CD – COLD DRAWN N – NORMALISED A – ANNEALED Q, R, S, T, U, V, W, X, Y, Z

CONTD.

CHEMISTRY OF VARIOUS TYPES OF STEELS AND THEIR COMPARISON WITH INTERNATIONAL GRADES

BS:970		EQUIVALENT NATIONAL / INTERNATIONAL GRADES							CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)				CONDITION	TYPE OF STEEL AND APPLICATIONS
En No.	NUMERICAL	ISS (BIS)	ASTM	SAE	AISI	DIN	GOST	JIS	C	Mn	Si	S MAX.	P MAX.	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E %	HARDNESS (BHN)/I ZOD (J)		
54A	331S42	--	---	---	---	---	---	---	0.37-0.47	0.50-0.80	1.00-2.00	0.045	0.045	13.00-15.00	13.00-15.00	0.40-0.60	2.20-3.30W				269/20 MAX.	HT	HIGHCr-Ni TUNGTEN VALVE STEEL
55	---	---	---	---	---	---	---	---	0.18-0.45	1.00 MAX.	1.00-2.50	0.045	0.045	17.00 MIN.	6.00-12.00	---	2.00-4.00W				302/27 MAX.	HT	HIGHCr-Ni TUNGTEN VALVE STEEL
56A	410S21	12Cr13 (IS:6529)	F738M GR.C1	51410	410	---	0X13	---	0.12 MAX.	1.00 MAX.	1.00 MAX.	0.045	0.045	12.00-14.00	1.00 MAX.	---	---	55 71 80 87 118	32 45 50 55 90	25 20 16 12 10	240- 302/60 280-345/34 290-363/27 302-390/14 350-445/10	P/C1-50 R/C1-70 S/C1-80 T X	Cr RUST RESISTING STEEL FOR TURBINE BLADES AND S.S FASTENERS.
56B	420S29	12Cr13 (IS:6529)	---	51420	420	---	2X13	---	0.12-0.18	1.00 MAX.	1.00 MAX.	0.045	0.045	12.00-14.00	1.00 MAX.	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					Cr RUST RESISTING LOW CARBON STAINLESS STEEL
56C	420S37	20Cr13 (IS:6529)	---	---	--	---	2X13	---	0.18-0.25	1.00 MAX.	1.00 MAX.	0.045	0.045	12.00-14.00	1.00 MAX.	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					Cr RUST RESISTING LOW CARBON STAINLESS STEEL
56D	420S45	30Cr13 (IS:6529)	---	---	---	---	3X13	---	0.25-0.35	1.00 MAX.	1.00 MAX.	0.045	0.045	12.00-14.00	1.00 MAX.	---	---	55 71 80 87 118	32 45 50 55 90	25 20 16 12 10	240-302/60 280-345/34 290-363/27 302-390/14 350-445/10	P R S T X	Cr RUST RESISTING STEEL FOR TURBINE BLADES AND S.S FASTENERS.
56AM	416S41	---	---	51416	416	---	---	---	0.12 MAX.	1.50 MAX.	1.00 MAX.	0.075	0.045	12.00-14.00	1.00 MAX.	0.60 MAX.	0.60Se., Zr. & Pb. MAX.	55 71 79	25 35 40	20 15 12	210- 280/33 280-345/27 ---	P R S	Cr RUST RESISTING FREE CUTTING STEEL FOR SPECIAL APPLICATION
56BM	416S19	---	---	---	---	---	---	---	0.12-0.18	1.50 MAX.	1.00 MAX.	0.075	0.045	12.00-14.00	1.00 MAX.	0.60 MAX.	0.60Se., Zr. & Pb. MAX.	55 71 79	25 35 40	20 15 12	210-280/33 280-345/27 ---	P R S	Cr RUST RESISTING FREE CUTTING STEEL FOR SPECIAL APPLICATION
56CM	416S37	---	---	---	---	---	---	---	0.18-0.25	1.50 MAX.	1.00 MAX.	0.075	0.045	12.00-14.00	1.00 MAX.	0.60 MAX.	0.60Se., Zr. & Pb. MAX.	55 71 79	25 35 40	20 15 12	210-280/33 280-345/27 ---	P R S	Cr RUST RESISTING FREE CUTTING STEEL FOR SPECIAL APPLICATION
56DM	---	---	---	---	---	---	---	---	0.25-0.35	1.50 MAX.	1.00 MAX.	0.075	0.045	12.00-14.00	1.00 MAX.	0.60 MAX.	0.60Se., Zr. & Pb. MAX.	55 71 79	25 35 40	20 15 12	210-280/33 280-345/27 ---	P R S	Cr RUST RESISTING FREE CUTTING STEEL FOR SPECIAL APPLICATION
57	431S29	15Cr16Ni2 (IS:6529)	---	51431	431	X22Cr Ni17	2X17H2	---	0.25 MAX.	1.00 MAX.	0.10-1.00	0.045	0.045	15.50-20.00	1.00-3.00	---	---	80 87	50 50	15 15	290-380/33 302-445/27	AH OH	MARTENSITIC Cr-Ni RUST RESISTING STEEL FOR SPECIAL APPLICATION
58A	302S25	---	---	30302	302	---	X18H9	---	0.16 MAX.	2.00 MAX.	0.20 MIN.	0.045	0.045	17.00-20.00 +	7.00-10.00 +	---	---	55	25	30	88	SOFENED	AUSTENITIC Cr-Ni RUST RESISTING STEEL (+ Ni & Cr 25% MINIMUM.)

ABBREVATIONS USED: HT -- HOT ROLLED; AC -- AIR COOLED; OH -- OIL HARDENED; AH -- AIR HARDENED; H & T -- HARDENED & TEMPERED; HEAT TREATMENT CONDITION:

AR -- AS ROLLED; CR -- COLD ROLLED; CD -- COLD DRAWN; N -- NORMALISED; A -- ANNEALED; Q, R, S, T, U, V, W, X, Y, Z

CONTD.

CHEMISTRY OF VARIOUS TYPES OF STEELS AND THEIR COMPARISON WITH INTERNATIONAL GRADES

BS:970		EQUIVALENT NATIONAL / INTERNATIONAL GRADES							CHEMICAL COMPOSITION, PERCENT									MECHANICAL PROPERTIES (MIN.)				CONDITION	TYPE OF STEEL AND APPLICATIONS
En No.	NUMERICAL	ISS (BIS)	ASTM	SAE	AISI	DIN	GOST	JIS	C	Mn	Si	S MAX.	P MAX.	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E %	HARDNESS BRINELL		
58B	321S12	04Cr18 Ni10Ti20 (IS:6529)	---	30321	321	04Cr18 Ni10Ti20	X18H10T	---	0.15 MAX.	2.00 MAX.	0.20 MIN.	0.045	0.045	17.00-20.00 +	7.00-10.00 +	---	Ti-4C MIN.	55	25	35	187	SOFTENED	AUSTENITIC CrNi RUST RESISTING STEEL (+ Ni & Cr 25% MINIMUM.)
58C	321S20	04Cr18 Ni10Ti20 (IS:6529)	---	30321	321	04Cr18 Ni10Ti20	X18H10T	---	0.15 MAX.	2.00 MAX.	0.20 MIN.	0.045	0.045	17.00-20.00	9.00-12.00	---	Ti-4C MIN.	55	25	35	187	SOFTENED	AUSTENITIC CrNi RUST RESISTING STEEL (+ Ni & Cr 25% MINIMUM.)
58D	---	---	---	---	---	---	---	---	0.16 MAX.	2.00 MAX.	0.20 MIN.	0.045	0.045	11.00-14.00 +	11.00-14.00 +	---	---	---	---	---	---	SOFTENED	AS ABOVE FOR DEEP DRAWING OR SPINNING (+ Ni & Cr 25% MINIMUM.)
58E	304S15	04Cr18 Ni10 (IS:6529)	A199-68	30304	304	X8CrNi 1910	0X18H9	---	0.08 MAX.	2.00 MAX.	0.20 MIN.	0.045	0.045	17.00-20.00	8.00-11.00	---	---	55	25	30	168	SOFTENED	AUSTENITIC CrNi RUST RESISTING STEEL (+ Ni & Cr 25% MINIMUM.)
58F	347S17	---	---	30347	347	---	---	---	0.15 MAX.	2.00 MAX.	0.20 MIN.	0.045	0.045	17.00-20.00	7.00-10.00	---	Nb. 8C MIN.	55	25	30	168	SOFTENED	AUSTENITIC CrNi RUST RESISTING STEEL (+ Ni & Cr 25% MINIMUM.)
58G	347S17	04Cr18 Ni10Nb40 (IS:6529)	---	30347	347	---	---	---	0.15 MAX.	2.00 MAX.	0.20 MIN.	0.045	0.045	17.00-20.00	9.00-10.00	---	Nb. 8C MIN.	55	25	30	168	SOFTENED	AUSTENITIC CrNi RUST RESISTING STEEL (+ Ni & Cr 25% MINIMUM.)
58H	316S16	---	---	---	---	---	---	---	0.12 MAX.	2.00 MAX.	0.20 MIN.	0.045	0.045	17.00-20.00	8.00-10.00	1.5-2.5	Ti & Nb OPTIONAL	55	25	30	168	SOFTENED	AUSTENITIC CrNiMO RUST RESISTINGSTEEL (+ Ni & Cr 25% MINIMUM.)
58J	316S16	04Cr17 Ni12--Mo2 (IS:6529)	---	30316	316	X5CrNi Mo18	X17H13 M2T	---	0.12 MAX.	2.00 MAX.	0.20 MAX.	0.045	0.045	17.00-20.00	8.00-12.00	2.50-3.50	---	55	25	30	168	SOFTENED	AUSTENITIC CrNiMO RUST RESISTINGSTEEL (+ Ni & Cr 25% MINIMUM.)
58M	303S21	---	---	30303	303	---	1X18H9	---	0.12 MAX.	1.00-2.00	0.20-1.00	0.15-3.00	0.045	17.00-19.00	8.00-11.00	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					AUSTENITIC CrNi RUST RESISTING STEEL
59	443S65	---	---	---	---	---	---	---	0.74-0.84	0.20-0.60	1.75-2.25	0.030	0.030	19.00-20.50	1.15-1.65	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					CrNi- Si VALVE STEEL
60	430S15	05Cr17 (IS:6529)	---	51430	430	---	X17	---	0.12 MAX.	1.00 MAX.	1.00 MAX.	0.045	0.045	16.00-18.00	0.50 MAX.	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					FERRITIC Cr RUST RESISTING STEEL
61	---	---	---	---	---	---	---	---	0.12 MAX.	1.00 MAX.	1.00 MAX.	0.045	0.045	20.00-22.00	0.50 MAX.	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					FERRITIC Cr RUST RESISTING STEEL

ABBREVATIONS USED:
HT -- HOT ROLLED
AR -- AS ROLLED
CR -- COLD ROLLED
AC -- AIR COOLED
CD -- COLD DRAWN
OH -- OIL HARDENED
N -- NORMALISED
AH -- AIR HARDENED
A -- ANNEALED
H & T -- HARDENED & TEMPERED

HEAT TREATMENT CONDITION:
Q, R, S, T, U, V, W, X, Y, Z

CONTD.

CHEMISTRY OF VARIOUS TYPES OF STEELS AND THEIR COMPARISON WITH INTERNATIONAL GRADES

BS:970		EQUIVALENT NATIONAL / INTERNATIONAL GRADES							CHEMICAL COMPOSITION, PERCENT									MECHANICAL PROPERTIES (MIN.)				CONDITION	TYPE OF STEEL AND APPLICATIONS
En No.	NUMERICAL	ISS (BIS)	ASTM	SAE	AISI	DIN	GOST	JIS	C	Mn	Si	S MAX.	P MAX.	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E %	HARDNESS (BHN/ZDD/Li)		
100	945M38	---	---	---	---	---	---	---	0.35-0.45	1.20-1.50	0.50 MAX.	0.05	0.05	0.30-0.60	0.50-1.00	0.15-0.25	---	70 79 87 95 102	42 48 60 70 80	22 20 18 17 16	201 - 255/56 223 - 277/56 248 - 302/56 269 - 321/50 293-341 / 50	R S T U V	LOW ALLOY STEEL FOR SPECIAL APPLICATIONS.
100A	---	---	---	---	---	---	---	---	0.25-0.30	1.20-1.50	0.10-0.50	0.05	0.05	0.50-0.60	0.50-1.00	0.15-0.25	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					LOW ALLOY STEEL FOR SPECIAL APPLICATIONS.
100B	---	---	---	---	---	---	---	---	0.30-0.35	1.20-1.50	0.10-0.50	0.05	0.05	0.50-0.60	0.50-1.00	0.15-0.25	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					LOW ALLOY STEEL FOR SPECIAL APPLICATIONS.
100C	945A40	---	---	---	---	---	---	---	0.35-0.40	1.20-1.50	0.10-0.50	0.05	0.05	0.50-0.60	0.50-1.00	0.15-0.25	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					LOW ALLOY STEEL FOR SPECIAL APPLICATIONS.
100D	---	---	---	---	---	---	---	---	0.40-0.45	1.20-1.50	0.10-0.50	0.05	0.05	0.50-0.60	0.50-1.00	0.15-0.25	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					LOW ALLOY STEEL FOR SPECIAL APPLICATIONS.
100E	---	---	---	---	---	---	---	---	0.25-0.35	1.20-1.50	0.50 MAX.	0.05	0.05	0.30-0.60	0.50-1.00	0.15-0.25	---	70 79 87 95 102	42 48 60 70 80	22 20 18 17 16	201 - 255/56 223 - 277/56 248 - 302/56 269 - 321/ 50 293-341 / 50	R S T U V	LOW ALLOY STEEL FOR SPECIAL APPLICATIONS.
110	---	40Ni6 Cr4Mo2 (IS:5517)	---	---	---	---	40XHMA	---	0.35-0.45	0.40-0.80	0.10-0.35	0.05	0.05	0.90-1.40	1.20-1.60	0.10-0.20	---	79 87 95 102 110	48 60 70 80 95	20 18 17 16 15	223 - 277 / 56 248 - 302/56 269 - 321 / 50 293-341 / 50 350 - 440 / 40	S T U V W	LOW Ni-Cr-Mo STEEL FOR SPECIAL APPLICATIONS.
111	640M40	35Ni5 Cr80 (IS:4368)	3140	3140	---	---	40XH	---	0.30-0.40	0.60-0.90	0.10-0.35	0.05	0.05	0.45-0.75	1.00-1.50	---	---	70 79 87 95	42 48 60 70	22 20 18 17	201 - 255/56 223 - 277/56 248 - 302/56 269 - 321/ 50	R S T U	LOW Ni-Cr STEEL FOR SPECIAL APPLICATIONS.
111A	640M35	---	--	---	---	---	---	---	0.33-0.38	0.60-0.90	0.10-0.35	0.05	0.05	0.45-0.75	1.00-1.50	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					LOW Ni-Cr STEEL FOR SPECIAL APPLICATIONS.
160	---	---	---	---	---	---	40HM	---	0.35-0.45	0.30-0.60	0.10-0.35	0.05	0.05	---	1.50-2.00	0.20-0.35	---	70 79 87 95	42 48 60 70	22 20 18 17	201 - 255/56 223 - 277/56 248 - 302/56 269 - 321/ 50	R S T U	2% Ni-Mo STEEL FOR SPECIAL APPLICATIONS.
160A	---	---	---	---	---	---	40HM	---	0.38-0.43	0.30-0.60	0.10-0.35	0.05	0.05	---	1.50-2.00	0.20-0.35	---	70 79 87 95	42 48 60 70	22 20 18 17	201 - 255/56 223 - 277/56 248 - 302/56 269 - 321/ 50	R S T U	2% Ni-Mo STEEL FOR SPECIAL APPLICATIONS.

ABBREVATIONS USED: HT – HOT ROLLED; AC – AIR COOLED; OH – OIL HARDENED; AH – AIR HARDENED; H & T – HARDENED & TEMPERED; HEAT TREATMENT CONDITION: Q, R, S, T, U, V, W, X, Y, Z

AR – AS ROLLED; CR – COLD ROLLED; CD – COLD DRAWN; N – NORMALISED; A – ANNEALED

CONTD.

CHEMISTRY OF VARIOUS TYPES OF STEELS AND THEIR COMPARISON WITH INTERNATIONAL GRADES

BS:970		EQUIVALENT NATIONAL / INTERNATIONAL GRADES							CHEMICAL COMPOSITION, PERCENT									MECHANICAL PROPERTIES (MIN.)				CONDITION	TYPE OF STEEL AND APPLICATIONS
En No.	NUMERICAL	ISS (BIS)	ASTM	SAE	AISI	DIN	GOST	JIS	C	Mn	Si	S MAX.	P MAX.	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E %	HARDNESS (BHN)/ IZOD (J)		
201	---	---	--	---	---	---	---	---	0.18 MAX.	1.10-1.50	0.05-0.35	0.05	0.05	---	---	---	---	63	32	20	54		C - Mn CASE HARDENING STEEL FOR SPECIAL APPLICATION
202	---	---	--	---	---	---	Gr. A12	---	0.18 MAX.	1.20-1.50	0.05-0.35	0.10-0.18	0.05	---	---	---	---	60	30	20	40		C - Mn SEMI FREE CUTTING STEEL FOR SPECIAL APPLICATION
206	532A14	15Cr65 (IS:4368)	---	5015	5015	15Cr3	12XT	---	0.12-0.17	0.30-0.50	0.10-0.35	0.05	0.05	0.30-0.50	---	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					LOW Cr CASE HARDENING STEEL
207	527A19	15Cr65 (IS:4368)	---	---	---	15Cr3	12XT	---	0.16-0.21	0.60-0.80	0.10-0.35	0.05	0.05	0.60-0.80	---	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					LOW Cr. CASE HARDENING STEEL
325	---	---	--	---	---	---	---	---	0.22 MAX.	0.45-0.65	0.10-0.35	0.05	0.05	0.40-0.60	1.50-2.00	0.20-0.30	---	87	45	15	40		LOW Ni - Cr CASE HARDENIN STEEL FOR SPECIAL APPLICATION
351	635M15	16Ni80 Cr60 (IS:4432)	---	---	---	---	---	---	0.20 MAX.	0.60-1.00	0.35 MAX.	0.05	0.05	0.40-0.80	0.60-1.00	0.10 MAX.	---	71	36	18	40		3/4% Ni - Cr. CASE HARDENING STEEL FOR SPL.APPLICATION
352	637M17	16Ni1 Cr80 (IS:4432)	---	---	---	15Cr Ni6	20XMA	---	0.20 MAX.	0.50-1.00	0.35 MAX.	0.05	0.05	0.60-1.00	0.85-1.25	0.10 MAX.	---	87	45	15	27		1 % Ni - Cr. CASE HARDENING STEEL FOR SPL.APPLICATION
353	815M17	16Ni1 Cr1 Mo12 (IS:4432)	---	---	---	---	---	---	0.20 MAX.	0.50-1.00	0.35 MAX.	0.05	0.05	0.75-1.25	0.10-1.50	0.08-0.15	---	102	55	12	27		1 -1/4% Ni - Cr. CASE HARDENING STEEL FOR SPL.APPLICATION
354	820M17	16Ni Cr1 Mo12 (IS:4432)	---	---	---	---	---	---	0.20 MAX.	0.50-1.00	0.35 MAX.	0.05	0.05	0.75-1.25	1.50-2.00	0.10-0.20	---	118	60	12	27		1 -3/4% Ni - Cr-Mo CASE HARDENING STEEL FOR SPL.APPLICATION
355	822M17	16Ni Cr2 Mo20 (IS:4432)	---	---	---	18CrNi18	---	---	0.20 MAX.	0.40-0.70	0.35 MAX.	0.05	0.05	1.40-1.70	1.80-2.20	0.15-0.25	---	134	70	12	27		2% Ni - - Cr. CASE HARDENING STEEL FOR SPL.APPLICATION
361	805M17	---	---	---	---	---	---	---	0.13-0.17	0.70-1.00	0.35 MAX.	0.05	0.05	0.55-0.80	0.40-0.70	0.08-0.15	---	71	36	18	40		15 CARBON LOW ALLOY CASE HARDENING STEEL FOR SPL. APPLICATION.
362	805M20	---	---	---	---	---	---	---	0.18-0.23	0.70-1.00	0.35 MAX.	0.05	0.05	0.55-0.80	0.40-0.70	0.08-0.15	---	87	45	15	27		20 CARBON LOW ALLOY CASE HARDENING STEEL FOR SPL. APPLICATION.
363	805M25	---	---	8620	8620	---	---	---	0.22-0.26	0.70-1.00	0.35 MAX.	0.05	0.05	0.55-0.80	0.40-0.70	0.08-0.15	---	102	55	12	27		25 CARBON LOW ALLOY CASE HARDENING STEEL FOR SPL. APPLICATION.

ABBREVATIONS USED

HT -- HOT ROLLED AC -- AIR COOLED OH -- OIL HARDENED AH -- AIR HARDENED H & T -- HARDENED & TEMPERED

AR -- AS ROLLED CR -- COLD ROLLED CD -- COLD DRAWN N -- NORMALISED A -- ANNEALED

HEAT TREATMENT CONDITION

Q, R, S, T, U, V, W, X, Y, Z

CHEMISTRY AND MECHANICAL PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES OF INDIAN STANDARDS SPECIFICATIONS AT A GLANCE

CHEMISTRY & MECHANICAL PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES-INDIAN STANDARD SPCN.

DESIGNATION / COLOUR CODE FOR IDENTIFICATION OF STEEL(IS:2049)			*CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
ISS NO.	NEW	OLD	C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	%HARDNESS (BHN) / IZOD(J)			HOT FORMING TEMP. °C	ANNEALING TEMP. °C	HARDENING TEMP. °C	TEMPERING TEMP. °C QUENCHANT	NORMALIZ-ING TEMP.°C
226 NO COLOUR CODE SHALL BE APPLIED	Fe410-S	St42-S	0.25	---	---	0.055	0.055	0.20-0.35 (WHERE SO REQD.)	---	---	---	42-54	26	23	126-179	N/6-20 N/>20-40 N/ OVER 40	STRUCTURAL STEEL, STANDARD QUALITY. PLATES, ANGLES,TEES,BEAMS CHANNELS,FLATS, ROUND,SQUARE,HEX. BARS. A GENERAL PURPOSE STEEL FOR WELDED OR RIVETTED STRUCTURES, FORG-ING & MACHINED PARTS.	1200	880-920 FURNACE COOL	CARBURIZING AT 880-930 HARDENING AFTER CARBURIZING AT 760-780 WATER QUENCH	150-200	880-930 AIR COOL
												42-54	24	23	126-179							
												42-54	23	23	126-179							
												BELOW 6, BEND TEST SHALL ONLY BE REQUIRED.										
			'C' WILL BE 0.23% MAX. ONLY FOR THICKNESS / DIA. 20 mm. AND BELOW.																			
	Fe410-Cu-S GREY DOT ON ONE END	St42-Cu-S	0.25	---	---	0.055	0.055	0.20-0.35 (WHERE SO REQD.)	---	---	0.20-0.35 Cu	42-54	26	23	126-179	N/6-20 N/>20-40 N/ OVER 40	STRUCTURAL STEEL, STANDARD QUALITY. PLATES, ANGLES,TEES,BEAMS CHANNELS,FLATS, ROUND,SQUARE,HEX. BARS. A GENERAL PURPOSE STEEL FOR WELDED OR RIVETTED STRUCTURES, FORG-ING & MACHINED PARTS.	1200	880-920 FURNACE COOL	CARBURIZING AT 880-930 HARDENING AFTER CARBURIZING AT 760-780 WATER QUENCH	150-200	880-930 AIR COOL
												42-54	24	23	126-179							
												42-54	23	23	126-179							
												BELOW 6, BEND TEST SHALL ONLY BE REQUIRED.										
			'C' WILL BE 0.23% MAX. ONLY FOR THICKNESS / DIA. 20 mm. AND BELOW.																			
432	Fe410(St42) GREY WITH BLUE DOT	M.S.GR.-I	0.25	---	---	0.055	0.055	---	---	---	---	42-54	26	23	126-179	N/6-20 N/>20-40 N/ OVER 40	MILD STEEL & MEDIUM TENSILE STEEL BARS FOR CONCRETE RE-INFORCEMENT	1200	880-920 FURNACE COOL	CARBURIZING AT 880-930 HARDENING AFTER CARBURIZING AT 760-780 WATER QUENCH	150-200	880-930 AIR COOL
			'C' WILL BE 0.20% MAX. ONLY FOR THICKNESS / DIA. 20 mm. AND OVER.									42-54	24	23	126-179							
												42-54	23	23	126-179							
												BELOW 6, BEND TEST SHALL ONLY BE REQUIRED.										
	Fe380(St39) AL. PAINT WITH ORANGE DOT.	M.S.GR.-II	0.23	---	---	0.055	0.055	---	---	---	---	38-54	23	20	126-179	N/6-20 N/>20-40 N/ OVER 40	MILD STEEL & MEDIUM TENSILE STEEL BARS FOR CONCRETE RE-INFORCEMENT	1200	880-920 FURNACE COOL	CARBURIZING AT 880-930 HARDENING AFTER CARBURIZING AT 760-780 WATER QUENCH	150-200	880-930 AIR COOL
			'C' WILL BE 0.20% MAX. ONLY FOR THICKNESS / DIA. 20 mm. AND OVER.									38-54	22	20	126-179							
												38-54	21	20	126-179							
												BELOW 6, BEND TEST SHALL ONLY BE REQUIRED.										
	Fe540WHT GREY WITH YELLOW DOT	MEDIUM TENSILE STEEL	0.27	---	---	0.055	0.055	---	---	---	---	58	38	20	140-179	N/6-20 N/>20-40 N/ OVER 40	MEDIUM TENSILE STEEL BARS FOR CONCRETE RE-INFORCEMENT	1200	880-920 FURNACE COOL	CARBURIZING AT 880-930 HARDENING AFTER CARBURIZING AT 760-780 WATER QUENCH	150-200	880-930 AIR COOL
												56	34	18	140-179							
												56	33	17	140-179							
												BELOW 6, BEND TEST SHALL ONLY BE REQUIRED.										
			*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED.																			

ABBREVATIONS USED: HR -- HOT ROLLED AC -- AIR COOLED OH -- OIL HARDENED AH -- AIR HARDENED H & T -- HARDENED & TEMPERED

AR -- AS ROLLED CR -- COLD ROLLED CD -- COLD DRAWN N -- NORMALISED A -- ANNEALED

CONTD.

CHEMISTRY & MECHANICAL PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - INDIAN STANDARD SPCN.

DESIGNATION / COLOUR CODE FOR IDENTIFICATION OF STEEL(IS:2049)			* CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
ISS NO.	NEW	OLD	C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J)			HOT FORMING TEMP. °C	ANNEALING TEMP. °C	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZ-ING TEMP °C
513	O	ORDINARY	0.15	---	---	0.06	0.06	---	---	---	---	28	---	---	---	SCALE FREE IMPROVED SURFACE. OR BEST SURFACE.	COLD ROLLED CARBON STEEL SHEET STP'P FOR COLD FORMED COMPO-NENTS.	---	---	---	---	---
	D	DRAWN	0.12	0.50	---	0.05	0.05	---	---	---	---	28	---	---	---							
	DD	DEEP DRAWN	0.10	0.50	---	0.04	0.04	---	---	---	---	27-40	---	23	---							
	EDD	EXTRA DD	0.10	0.50	---	0.035	0.035	---	---	---	---	27-39	---	25	---							
O - GREY WITH ORANGE BAND, D - GREY WITH BLACK BAND, DD - GREY WITH BROWN BAND, EDD - GREY WITH VIOLET BAND																						
961	Fe570HT	St58HT	0.27	---	---	0.055	0.055	---	---	---	---	58	36	20	140-179	N / 6-28 N / > 28-63 N / OVER 63	HIGH TENSILE STRUCTURAL STEEL. GENERAL PURPOSE H.T.STEEL FOR TIE RODS, SPROCKETS, LEVERS, CAMS, AUTOMOBILE AND BICYCLE PARTS.	1200	860-900 FURNACE COOL	860-890 OIL OR WATER QUENCH	550-600	860-930 AIR COOL
	GREY WITH WHITE DOT											58	33	20	140-179							
												55	30	20	140-179							
												BELOW 6, BEND TEST SHALL ONLY BE REQUIRED.										
961	Fe540WHT	St55HTW	0.20	---	---	0.055	0.055	---	---	---	---	55	36	20	140-179	N / 6-28 N / > 28-63 N / OVER 63	HIGH TENSILE WELDABLE STEEL. GENERAL PURPOSE H.T.STEEL FOR TIE RODS, SPROCKETS, LEVERS, CAMS, AUTOMOBILE AND BICYCLE PARTS.	1200	860-900 FURNACE COOL	860-890 OIL OR WATER QUENCH	550-600	860-930 AIR COOL
	GREY WITH YELLOW DOT											52	34	20	140-179							
												50	29	20	140-179							
												BELOW 6, BEND TEST SHALL ONLY BE REQUIRED.										
1079	O	ORDINARY	---	---	---	0.06	0.06	---	---	---	---	---	---	---	---	HOT ROLLED ANNEALED, NORMALI-ZED AND DESCALED	HOT ROLLED CARBON STEEL SHEET AND STRIP FOR COLD FORMED COMPO-NENTS AND FOR OTHER GENERAL ENGG. PURPOSES.	---	---	---	---	---
	D	DRAWN	0.12	0.50	---	0.05	0.05	---	---	---	---	---	---	---	---							
	DD	DEEP DRAWN	0.10	0.50	---	0.04	0.04	---	---	---	---	27-40	---	23	---							
	EDD	EXTRA DD	0.10	0.50	---	0.035	0.035	---	---	---	---	27-39	---	25	---							
	St-34	UTS 34 Kg/mm²	0.15	---	---	0.05	0.05	---	---	---	---	34-42	---	25	---							
	St-42	UTS 42 Kg/mm²	0.25	---	---	0.05	0.05	---	---	---	---	42-50	---	22	---							
	St-50	UTS 50 Kg/mm²	0.30	---	---	0.05	0.05	---	---	---	---	50-60	---	20	---							
	St-52	UTS 52 Kg/mm²	0.22	---	---	0.05	0.05	---	---	---	---	52-62	---	20	---							
	SAME AS IS: 513																					
1148	Fe410R	ST42R	0.23	---	---	0.05	0.05	---	---	---	---	42-54	27	23	---	AR/N	SUITABLE FOR HOT FORGED RIVETS FOR STRUCTURAL PURPOSES.	1200	860-900 FURNACE COOL	860-890 OIL OR WATER QUENCH	550-600	860-930 AIR COOL
	GREY WITH BLUE DOT																					
1149	Fe460R	ST47R	0.23	---	---	0.05	0.05	---	---	---	---	47	32	22	---	AR/N	SUITABLE FOR HOT FORGED HIGH TENSILE RIVETS FOR STRUCTURAL PURPOSES.	1200	860-900 FURNACE COOL	860-890 OIL OR WATER QUENCH	550-600	860-930 AIR COOL
	GREY WITH BLUE & RED DOT																					
1673	10C4	C10	0.17	0.30-0.65	---	0.055	0.055	---	---	---	---	47	32	18	--	CD	C.S.WIRE FOR MFR. OFWOOD SCREWS BY COLD HEADING PROCESS.	---	---	---	---	---
	BLACK WITH WHITE & GREY DOT											50	35	16	---	CD						
	15C4	C15	0.22	0.30-0.65	---	0.055	0.055	---	---	---	---	47	32	18	---	CD FREE CUTTING STEEL						
	BLACK WITH WHITE & ORANGE DOT																					
	10C8S11	10S11	0.17	0.60-0.95	---	0.08-0.15	0.055	---	---	---	---											
	BROWN WITH WHITE DOT																					

*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED.

ABBREVATIONS USED: HR – HOT ROLLED AC – AIR COOLED OH – OIL HARDENED AH – AIR HARDENED H & T – HARDENED & TEMPERED

AR – AS ROLLED CR – COLD ROLLED CD – COLD DRAWN N – NORMALISED A – ANNEALED

CONTD.

CHEMISTRY & MECHANICAL PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - INDIAN STANDARD SPCN.

DESIGNATION / COLOUR CODE FOR IDENTIFICATION OF STEEL(IS:2049)			* CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
ISS NO.	NEW	OLD	C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J)			HOT FORMING TEMP. °C	ANNEALING TEMP. °C	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZ-ING TEMP.°C
1875/ 2004	15C8	CL 1A BLACK WITH RED DOT	0.10-0.20	0.60-0.90	0.15-0.35	0.05	0.05	---	---	---	---	42	24	25	110	N	CARBON STEEL BILLETS,BLOOMS, SLABS AND BARS FOR FORGINGS OF PARTS FOR AUTO-MOBILE, BICYCLE ANG GENERAL ENGG. GOODS.	1200	880-920 FURNACE COOL	CARBURIZING AT 880-930 HARDENING AFTER CARBURIZING AT 760-780 WATER QUENCH	150-200	880-930 AIR COOL
	20C8	CL. 2 BLACK WITH YELLOW DOT	0.15-0.25	0.60-0.90	0.15-0.35	0.05	0.05	---	---	---	---	44	26	24	120	N	SAME AS ABOVE	1200	880-920 FURNACE COOL	SAME AS ABOVE	150-200	880-930 AIR COOL
	25C8	CL. 2A BLACK WITH VIOLET DOT	0.20-0.30	0.60-0.90	0.15-0.35	0.05	0.05	---	---	---	---	47	28	22	130	N	SAME AS ABOVE	1200	880-920 FURNACE COOL	840-880	550-600	880-930 AIR COOL
	30C8	CL.3 BLACK WITH GREEN DOT	0.25-0.35	0.60-0.90	0.15-0.35	0.05	0.05	---	---	---	---	50	30	21	140	N	SAME AS ABOVE	1200	880-920 FURNACE COOL	840-880	550-600	880-930 AIR COOL
	35C8	CL.3A BLACK WITH BROWN DOT	0.30-0.40	0.60-0.90	0.15-0.35	0.05	0.05	---	---	---	---	55	30	20	155	N	SAME AS ABOVE	1200	880-920 FURNACE COOL	840-880	550-600	880-930 AIR COOL
	45C8	CL. 4 BLACK	0.40-0.50	0.60-0.90	0.15-0.35	0.05	0.05	---	---	---	---	63	36	15	175	N	SAME AS ABOVE	1200	880-920 FURNACE COOL	840-880	550-600	880-930 AIR COOL
	55C8	CL.5 BLACK WITH BROWN & WHITE DOT	0.50-0.60	0.60-0.90	0.15-0.35	0.05	0.05	---	---	---	---	72	42	13	200	N	SAME AS ABOVE	1200	880-920 FURNACE COOL	840-880	550-600	880-930 AIR COOL
	65C8	CL.6 BLACK WITH AL. PAINT DOT	0.60-0.70	0.60-0.90	0.15-0.35	0.05	0.05	---	---	---	---	75	45	10	210	N	SAME AS ABOVE	1200	880-920 FURNACE COOL	840-880	550-600	880-930 AIR COOL
1976	GR.1	--- NO COLOUR SPECIFIED	0.15	0.30-0.65	---	0.065	0.055	---	---	---	---	44-55	30	20	--	CD	C.S.WIRE FOR MFR. OF MACINE SCREWS BY COLD HEADING PROCESS.	---	---	---	---	---
	GR.2	--- NO COLOUR SPECIFIED	0.15	0.30-0.65	---	0.065	0.055	---	---	---	---	55-71	38	18	---	CD						
2002	GR.1	--- GREY WITH PINK BAND	0.18	---	0.10-0.35	0.04	0.04	---	---	---	---	37-45	24	26	---	AR	BOILER QUALITY PLATES WHICH ARE REQD. TO BE EITHER WELDED, FLANGED 0R FLAME CUT	---	---	---	---	---
	GR. 2A	--- GREY WITH GREEN BAND	0.20	---	0.10-0.35	0.05	0.05	---	---	---	---	42-50	28	25	---	AR		---	---	---	---	---
	GR. 2B	--- GREY WITH RED BAND	0.22	---	0.10-0.35	0.05	0.05	---	---	---	---	52-62	30	20	---	AR		---	---	---	---	---
2062	FE410-W	S142-W GREEN DOT ON ONE END	0.20	---	---	0.055	0.055	---	---	---	---	42-54	26	23	126-179	N / 6-20	STRUCTURAL STEEL, STANDARD QUALITY. PLATES, ANGLES, TEES, BEAMS, CHANNELS,FLATS, ROUNDS, SQUARE, HEX BARS - FUSION WELDED QUALITY.	---	---	---	---	---
	FE410-WC	S142-WC GREEN DOT ON ONE END AND SURFACE	0.20	---	---	0.055	0.055	---	---	---	0.20-0.35 Cu.	42-54 42-54 BELOW 6, BEND TEST SHALL ONLY BE REQUIRED.	24 23	23 23	126-179 126-179	N / > 20-40 N/ OVER 40						
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DESIGNATION / COLOUR CODE FOR IDENTIFICATION OF STEEL(IS:2049)			*CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
ISS NO.	NEW	OLD	C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J)			HOT FORMING TEMP. °C	ANNEALING TEMP. °C	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZING TEMP. °C
2073	14C8	C-14 BLACK WITH WHITE & BLUE DOT	0.10-0.18	0.40-0.70	---	0.055	0.055	---	---	---	---	37-45 50	--- ---	26 13	110 140	N CD	CARBON STEEL BARS FOR MACHINED FOR GENERAL ENGG. PURPOSE.	1200	880-920 FURNACE COOL	CARBURIZING AT 880-930 HARDENING AFTER CARBURIZING AT 760-780 WATER QUENCH	150-200	880-930 AIR COOL
	20C8	C-20 BLACK WITH YELLOW DOT	0.15-0.25	0.60-0.90	0.05-0.35	0.055	0.055	---	---	---	---	44-52 55	--- ---	24 11	120 160	N CD	SAME AS ABOVE	1200	880-920 FURNACE COOL	SAME AS ABOVE	150-200	880-930 AIR COOL
	30C8	C-30 BLACK WITH GREEN DOT	0.25-0.35	0.60-0.90	0.05-0.35	0.055	0.055	---	---	---	---	50-60 60 62-77	--- --- 42	21 10 18	140 170 170-187/55	N CD H&T	SAME AS ABOVE	1200	880-920 FURNACE COOL	840-880	550-600	880-930 AIR COOL
	40C8	C-40 BLACK WITH WHITE AND BROWN DOT	0.35-0.45	0.60-0.90	0.05-0.35	0.055	0.055	---	---	---	---	58-68 65 70-85	--- --- 50	18 9 17	175 187 195	N CD H&T	SAME AS ABOVE	1200	880-920 FURNACE COOL	840-880	550-600	880-930 AIR COOL
	45C8	C-45 BLACK	0.40-0.50	0.60-0.90	0.05-0.35	0.055	0.055	---	---	---	---	63-71 70 70-85	--- --- 50	15 8 16	210 225 235	N CD H&T	SAME AS ABOVE	1200	880-920 FURNACE COOL	840-880	550-600	880-930 AIR COOL
	50C4	C-50 BLACK WITH GREEN AND WHITE DOT	0.45-0.55	0.30-0.60	0.05-0.35	0.055	0.055	---	---	---	---	67-80 75 81-96	--- --- 55	13 7 13	230 240 250	N CD H&T	SAME AS ABOVE	1200	880-920 FURNACE COOL	840-880	550-600	880-930 AIR COOL
	55C8	C-55Mn-75 BLACK WITH WHITE DOT	0.50-0.60	0.60-0.90	0.05-0.35	0.055	0.055	---	---	---	---	72 74 81-96	--- --- 55	13 7 13	220 240 250	N CD H&T	SAME AS ABOVE	1200	880-920 FURNACE	840-880	550-600	880-930 AIR COOL
	65C6	C-65 BLACK WITH AL. PAINT DOT	0.60-0.70	0.50-0.80	0.05-0.35	0.055	0.055	---	---	---	---	75 88-100	--- ---	10 8	250 280	N H&T	SAME AS ABOVE	1200	880-920 FURNACE COOL	840-880	550-600	880-930 AIR COOL
	14C14S14	14Mn1S14 BROWN WITH GREEN DOT	0.10-0.18	1.20-1.50	0.05-0.35	0.10-0.18	0.055	---	---	---	---	44-54 55	--- ---	22 11	120 160	N CD	CARBON STEEL FREE CUTTING STEEL FOR MACINED PARTS FOR GENERAL ENGG. PURPOSE	1200	880-920 FURNACE COOL	---	---	880-930 AIR COOL
	13C10S25	13S25 BROWN	0.08-0.18	0.80-1.20	0.10 MAX.	0.20-0.30	0.055	---	---	---	---	38-50 55	--- ---	22 11	110 160	N CD	SAME AS ABOVE	1200	880-920 FURNACE COOL	---	---	880-930 AIR COOL
2830/ 6914	Fe-410-SB1 GREY DOT ON ONE END	St42-SB1	0.10-0.18*	0.50-0.90	---	0.055	0.055	---	---	---	0.20-0.35**	---	---	---	---		CARBON STEEL BILLETS, BLOOMS AND SLABS FOR REROLLING INTO STRUCTURAL STEEL	---	---	---	---	---
	Fe-410-SB2 GREEN DOT ON ONE END	St42-SB2	0.17-0.23	0.50-0.90	---	0.055	0.055	---	---	---	0.20-0.35**	---	---	---	---			---	---	---	---	---
	Fe-410-SB3 ORANGE DOT ON ONE END	St42-SB3	0.19-0.23	0.50-0.90	---	0.055	0.055	---	---	---	0.20-0.35**	---	---	---	---			---	---	---	---	---

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**Cu- IF SO REQUIRED

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DESIGNATION / COLOUR CODE FOR IDENTIFICATION OF STEEL(IS:2049)			* CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)	CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
ISS NO.	NEW	OLD	C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm² YS E % HARDNESS (BHN) / IZOD(J)			HOT FORMING TEMP. °C	ANNEALING TEMP. °C	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZ-ING TEMP.°C
3195	98C6 BLACK WITH YELLOW AND RED DOT	C98	0.90-1.05	0.50-0.80	0.10-0.35	0.05	0.05	---	---	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION		STEEL FOR THE MANUFACTURE OF VALVE AND HELICAL SPRINGS. ALSO SUITABLE FOR MACINE KNIVES, CHISELS, CUTTING & HAND TOOLS.	1100	800-920 FURNACE COOL	780-820 OIL	400-500	800-850 AIR COOL
	113C6 BLACK WITH YELLOW AND PINK DOT	C113	1.05-1.02	0.50-0.80	0.10-0.35	0.05	0.05	---	---	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION		SAME AS ABOVE	1100	800-920 FURNACE COOL	780-820 OIL	400-500	800-850 AIR COOL
	55Si7 GREEN WITH WHITE DOT	55Si2Mn90	0.50-0.80	0.80-1.00	1.50-2.00	0.05	0.05	---	---	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION		SAME AS ABOVE	1100	800-920 FURNACE COOL	780-820 OIL	400-500	800-850 AIR COOL
	60Si7 NO COLOUR SPECIFIED	60SiMn90	0.55-0.65	0.80-1.00	1.50-2.00	0.05	0.05	---	---	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION		SAME AS ABOVE	1100	800-920 FURNACE COOL	780-820 OIL	400-500	800-850 AIR COOL
	50Cr4V2 GREEN WITH YELLOW DOT	50Cr1V23	0.45-0.55	0.50-0.80	0.10-0.35	0.05	0.05	0.90-1.20	---	---	0.15-0.30V	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION		SAME AS ABOVE	1100	800-920 FURNACE COOL	780-820 OIL	400-500	800-850 AIR COOL
	60Cr4V2 NO COLOUR SPECIFIED	60Cr1V23	0.55-0.65	0.80-1.00	0.10-0.35	0.05	0.05	0.90-1.20	---	---	0.15V MIN.	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION		SAME AS ABOVE	1100	800-920 FURNACE COOL	780-820 OIL	400-500	800-850 AIR COOL
3748	XT33W9Cr3V4 AL. PAINT WITH BROWN DOT	T33W9Cr3V38	0.25-0.40	0.20-0.40	0.10-0.35	0.035	0.035	2.80-3.30	---	---	0.25-0.50 V 8.0-10.0W	245 MAX. 425 MIN.	A H & T	HOT WORK DIE STEEL. SUITABLE FOR DIES FOR EXTRUSION, FORGING,STAMPING, HOT HEADING, RIVETTING	850-1150	850 FURNACE COOL	1100 OIL/AIR	600-650	850 AIR COOL
	XT35Cr5Mo1V3 AL. PAINT WITH GREEN AND PINK DOT	T35Cr5Mo1V30	0.30-0.40	0.25-0.50	0.80-1.20	0.035	0.035	4.75-5.25	---	1.20-1.60	0.20-0.40 V	235 MAX. 320 MIN.	A H & T	SAME AS ABOVE	850-1150	850 FURNACE COOL	1030 OIL/AIR	550-650	850 AIR COOL
	XT35Cr5MoV1 AL. PAINT WITH GREEN DOT	T35Cr5MoV1	0.30-0.40	0.25-0.50	0.80-1.20	0.035	0.035	4.75-5.25	---	1.20-1.60	1.00-1.20 V	235 MAX. 320 MIN.	A H & T	SAME AS ABOVE	850-1150	850 FURNACE COOL	1000 OIL/AIR	550-650	850 AIR COOL
	XT35Cr5MoW1V3 AL. PAINT WITH RED DOT	T35Cr5MoW1V30	0.30-0.40	0.25-0.50	0.80-1.20	0.035	0.035	4.75-5.25	---	1.20-1.60	0.20-0.40V 1.20-1.60 W	235 MAX. 320 MIN.	A H & T	SAME AS ABOVE	850-1150	850 FURNACE COOL	1010 OIL/AIR	550-650	850 AIR COOL
	XT55W15Cr3V AL. PAINT WITH BROWN AND WHITE DOT	T55W14Cr3V45	0.50-0.60	0.20-0.40	0.10-0.35	0.035	0.035	2.80-3.30	---	---	0.30-0.60 V 13.0-15.0 W	248 MAX. 684 MIN.	A H & T	SAME AS ABOVE AND ALSO FOR MILLING CUTTERS, REAMERS,DRILLS	850-1150	850 FURNACE COOL	1260 OIL/AIR	570 (DOUBLE TEMPERING)	850 AIR COOL

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DESIGNATION / COLOUR CODE FOR IDENTIFICATION OF STEEL(IS:2049)			* CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)		CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
ISS NO.	NEW	OLD	C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS YS Kg/mm²	E % HARDNESS (BHN) / IZOD(J			HOT FORMING TEMP. °C	ANNEALING TEMP. °C	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP °C	NORMALIZ-ING TEMP. °C
3748	50T8	T50 ALUMINIUM PAINT	0.45-0.55	0.60-0.90	0.10-0.35	0.04	0.04	---	---	---	---		240 MAX. 465 MIN	A H & T	COLD WORK DIE STEEL SUITABLE FOR PRESS TOOLS, PUNCHES, SHEARING BLADES, RAZORS	850-1050	750 FURNACE COOL	840 OIL/AIR	150-400	850 AIR COOL
	55T8	T55 ALUMINIUM PAINT WITH WHITE DOT	0.50-0.60	0.60-0.90	0.10-0.35	0.04	0.04	---	---	---	---		240 MAX. 465 MIN	A H & T	SAME AS ABOVE	850-1050	750 FURNACE COOL	830 OIL/AIR	150-400	850 AIR COOL
	60T6	T60 ALUMINIUM PAINT WITH WHITE & BLUE DOT	0.55-0.65	0.50-0.80	0.10-0.35	0.04	0.04	---	---	---	---		240 MAX. 465 MIN	A H & T	SAME AS ABOVE	850-1050	750 FURNACE COOL	820-850 OIL/AIR	150-400	850 AIR COOL
	65T6	T65 ALUMINIUM PAINT WITH WHITE & GREEN DOT	0.60-0.70	0.50-0.80	0.10-0.35	0.04	0.04	---	---	---	---		240 MAX. 485 MIN	A H & T	SAME AS ABOVE	850-1050	750 FURNACE COOL	820-850 OIL/AIR	150-400	850 AIR COOL
	70T6	T70Mn65 ALUMINIUM PAINT WITH WATER &RED DOT	0.65-0.75	0.50-0.80	0.10-0.35	0.04	0.04	---	---	---	---		240 MAX. 485 MIN	A H & T	SAME AS ABOVE	850-1050	750 FURNACE COOL	820-850 OIL/AIR	150-400	850 AIR COOL
	75T6	T75 ALUMINIUM PAINT WITH WHITE & YELLOW DOT	0.70-0.80	0.50-0.80	0.10-0.35	0.04	0.04	---	---	---	---		240 MAX. 485 MIN.	A H & T	SAME AS ABOVE	850-1050	750 FURNACE COOL	820-850 OIL/AIR	150-400	850 AIR COOL
	80T6	T80Mn65 ALUMINIUM PAINT WITH WHITE & VIOLET DOT	0.75-0.85	0.50-0.80	0.10-0.35	0.04	0.04	---	---	---	---		240 MAX. 485 MIN.	A H & T	SAME AS ABOVE	850-1050	750 FURNACE COOL	820 OIL/AIR	150-400	850 AIR COOL
	85T6	T85 ALUMINIUM PAINT WITH WHITE & ORANGE DOT	0.80-0.90	0.50-0.80	0.10-0.35	0.04	0.04	---	---	---	---		240 MAX. 485 MIN.	A H & T	SAME AS ABOVE	850-1050	750 FURNACE COOL	820 OIL/AIR	150-400	850 AIR COOL
	70T3	T70 ALUMINIUM PAINT WITH WHITE & BLACK DOT	0.65-0.75	0.40	0.10-0.35	0.04	0.04	---	---	---	---		200 MAX. 445 MIN.	A H & T	SAME AS ABOVE	850-1050	750 FURNACE COOL	820 WATER	150-400	850 AIR COOL
	80T3	T80 ALUMINIUM PAINT WITH WHITE & GREY DOT	0.75-0.85	0.40	0.10-0.35	0.04	0.04	---	---	---	---		200 MAX. 445 MIN	A H & T	SAME AS ABOVE	850-1050	750 FURNACE COOL	820 WATER	150-400	850 AIR COOL
	90T3	T90 ALUMINIUM PAINT WITH ORANGE DOT	0.85-0.95	0.40	0.10-0.35	0.04	0.04	---	---	---	---		200 MAX. 485 MIN.	A H & T	SAME AS ABOVE	850-1050	750 FURNACE COOL	800 WATER	150-400	850 AIR COOL
	103T3	T103 ALUMINIUM PAINT WITH ORANGE & WHITE DOT	0.95-1.10	0.40	0.10-0.35	0.04	0.04	---	---	---	---		210 MAX. 485 MIN.	A H & T	SAME AS ABOVE	850-1050	750 FURNACE COOL	770 WATER	150-400	850 AIR COOL

*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED

ABBREVATIONS USED:
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HR – HOT ROLLED
CR – COLD ROLLED
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CC – COLD DRAWN
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H & T – HARDENED & TEMPERED

CONTD.

CHEMISTRY & MECHANICAL PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - INDIAN STANDARD SPCN.

DESIGNATION / COLOUR CODE FOR IDENTIFICATION OF STEEL(IS:2049)			* CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)			CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
ISS NO.	NEW	OLD	C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E %	HARDNESS (BHN) / IZOD(J			HOT FORMING TEMP. °C	ANNEALING TEMP. °C	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZ-ING TEMP.°C
3749 (CONTD.)	118T3 ALUMINIUM PAINT WITH ORANGE & BLUE DOT	T103	1.10-1.20	0.40 MAX.	0.10-0.35	0.04	0.04	--	---	---	---				210 MAX. 485 MIN	A H & T	COLD WORK DIE STEEL SUITABLE FOR PRESS TOOLS, PUNCHES, SHEARING BLADES, RAZORS	850-1050	750 FURNACE COOL	770 WATER	150-400	850 AIR COOL
	133T3 ALUMINIUM PAINT WITH RED & WHITE DOT	T133	1.25-1.40	0.40 MAX.	0.10-0.35	0.04	0.04	---	---	--	---				210 MAX. 485 MIN.	A H & T	SAME AS ABOVE	850-1050	750 FURNACE COOL	770 WATER	150-400	850 AIR COOL
	T80V2 ALUMINIUM PAINT WITH RED & GREEN DOT	T80V23	0.75-0.85	0.40 MAX.	0.10-0.35	0.04	0.04	---	---	—	0.15-0.30 V				220 MAX. 485 MIN.	A H & T	SAME AS ABOVE WITH IMPROVED TOUGHNESS	850-1050	750 FURNACE COOL	820 WATER	150-400	850 AIR COOL
	T90V2 ALUMINIUM PAINT WITH RED & GREEN DOT	T90V23	0.85-0.95	0.40 MAX.	0.10-0.35	0.04	0.04	---	---	—	0.15-0.30 V				220 MAX. 485 MIN.	A H & T	SAME AS ABOVE WITH IMPROVED TOUGHNESS	850-1050	750 FURNACE COOL	820 WATER	150-400	850 AIR COOL
	T103V2 ALUMINIUM PAINT WITH ORANGE & GREEN DOT	T103V23	0.95-1.10	0.40 MAX.	0.10-0.35	0.04	0.04	---	---	---	0.15-0.30 V				220 MAX. 485 MIN.	A H & T	SAME AS ABOVE WITH IMPROVED TOUGHNESS	850-1050	750 FURNACE COOL	810 WATER	150-400	850 AIR COOL
	T118Cr4 ALUMINIUM PAINT WITH ORANGE & GREY DOT	T118Cr45	1.10-1.25	0.40 MAX.	0.10-0.35	0.04	0.04	0.30-0.60	---	---	0.30V MAX.				220 MAX. 485 MIN.	A H & T	SAME AS ABOVE WITH IMPROVED TOUGHNESS AND WEAR RESISTANCE	850-1050	750 FURNACE COOL	810 WATER	150-400	850 AIR COOL
	T133Cr4 ALUMINIUM PAINT WITH ORANGE & BLUE DOT	T133Cr45	1.25-1.40	0.40 MAX.	0.10-0.35	0.04	0.04	0.30-0.60	---	---	0.30V MAX.				220 MAX. 485 MIN.	A H & T	SAME AS ABOVE	850-1050	750 FURNACE COOL	800 WATER	150-400	850 AIR COOL
	T105Cr5 ALUMINIUM PAINT WITH PINK & BROWN DOT	T105Cr1	0.90-1.20	0.20-0.40	0.10-0.35	0.04	0.04	1.00-1.60	---	---	---				230 MAX. 485 MIN.	A H & T	SAME AS ABOVE	850-1050	750 FURNACE COOL	800 WATER	150-400	850 AIR COOL
	T105Cr5Mn ALUMINIUM PAINT WITH PINK & YELLOW DOT	T105Cr1Mn60	0.90-1.20	0.40-0.80	0.10-0.35	0.04	0.04	1.00-1.60	---	---	---				230 MAX. 485 MIN.	A H & T	SAME AS ABOVE	850-1050	750 FURNACE COOL	800 WATER	150-400	850 AIR COOL
	T140W15Cr2 ALUMINIUM PAINT WITH BLACK & GREEN DOT	T140W4Cr50	1.30-1.50	0.25-0.50	0.10-0.35	0.04	0.04	0.30-0.70	---	---	3.50-4.20 W				250 MAX. 585 MIN.	A H & T	SAME AS ABOVE	850-1050	820 FURNACE COOL	800 WATER	150-400	850 AIR COOL
	T60Ni5 ALUMINIUM PAINT WITH BLUE & YELLOW DOT	T60Ni1	0.55-0.65	0.50-0.80	0.10-0.35	0.04	0.04	0.30 MAX.	1.00-1.50	---	---				255 MAX. 485 MIN.	A H & T	SAME AS ABOVE	850-1100	830 FURNACE COOL	820 OIL/AIR	150-400	850 AIR COOL
	T40Ni14 ALUMINIUM PAINT WITH BLUE & RED DOT	T40Ni3	0.35-0.45	0.50-0.80	0.10-0.35	0.04	0.04	0.30 MAX	3.20-3.60	---	---				255 MAX. 485 MIN.	A H & T	SAME AS ABOVE	800-1100	830 FURNACE COOL	840 OIL/AIR	150-400	850 AIR COOL
	T30Ni16Cr5 ALUMINIUM PAINT WITH BLUE & BROWN DOT	T30Ni4Cr1	0.26-0.34	0.40-0.70	0.10-0.35	0.04	0.04	1.10-1.40	3.90-4.30	---	---				255 MAX. 485 MIN.	A H & T	SAME AS ABOVE	800-1100	830 FURNACE COOL	840 OIL/AIR	150-400	850 AIR COOL
	T55Ni6Cr Mo3 ALUMINIUM PAINT WITH GREEN & WHITE DOT	T55Ni2Cr65 Mo30	0.50-0.60	0.50-0.80	0.10-0.35	0.04	0.04	0.50-0.80	1.25-1.75	0.25-0.35	---				255 MAX. 485 MIN	A H & T	SAME AS ABOVE	800-1100	830 FURNACE COOL	820 OIL/AIR	150-400	850 AIR COOL
			*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED.																			

ABBREVATIONS USED: AR – AS ROLLED; HR – HOT ROLLED; CR – COLD ROLLED; AC – AIR COOLED; CD – COLD DRAWN; OH – OIL HARDENED; N – NORMALISED; AH – AIR HARDENED; A – ANNEALED; H & T – HARDENED & TEMPERED

CONTD.

CHEMISTRY & MECHANICAL PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - INDIAN STANDARD SPCN.

DESIGNATION / COLOUR CODE FOR IDENTIFICATION OF STEEL(IS:2048)			* CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
ISS NO.	NEW	OLD	C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J)			HOT FORMING TEMP. °C	ANNEALING TEMP. °C	HARDENING TEMP °C QUENCHANT	TEMPERING TEMP °C	NORMALIZING TEMP °C
3749 (CONTD.)	T40Ni6 Cr4Mo3 ALUMINIUM PAINT WITH GREEN & YELLOW DOT.	T40Ni2 Cr1Mo28	0.35-0.45	0.40-0.70	0.10-0.35	0.04	0.04	0.90-1.30	1.25-1.75	0.20-0.35	---				255 MAX. 425 MIN.	A H & T	COLD WORK DIE STEEL SUITABLE FOR PRESS TOOLS, PUNCHES, SHEARING BLADES, DIES.	800-1100	800 FURNACE COOL	840 OIL/AIR	150-400	850 AIR COOL
	T31Ni10 Cr3Mo6 ALUMINIUM PAINT WITH GREEN & BLUE DOT.	T31Ni3 Cr65Mo55	0.27-0.35	0.40-0.70	0.10-0.35	0.04	0.04	0.50-0.80	2.25-2.75	0.40-0.70	---				255 MAX. 425 MIN.	A H & T	SAME AS ABOVE	800-1100	800 FURNACE COOL	840 OIL/AIR	150-400	850 AIR COOL
	T40Ni10 Cr3Mo6 ALUMINIUM PAINT WITH GREEN & BLUE DOT.	T40Ni3 Cr65Mo55	0.36-0.44	0.40-0.70	0.10-0.35	0.04	0.04	0.50-0.80	2.25-2.75	0.40-0.70	---				255 MAX. 425 MIN.	A H & T	SAME AS ABOVE	800-1100	800 FURNACE COOL	840 OIL/AIR	150-400	850 AIR COOL
	T105W6 CrV2 ALUMINIUM PAINT WITH BLACK DOT.	T105W2 Cr60V25	0.90-1.20	0.40	0.10-0.35	0.04	0.04	0.40-0.80	---	0.25 MAX.	0.20-0.30 V 1.25-1.75W				230 MAX. 425 MIN.	A H & T	SAME AS ABOVE	850-1050	780 FURNACE COOL	850 OIL/AIR	150-400	850 AIR COOL
	T110W6Cr4 ALUMINIUM PAINT WITH BLACK & WHITE DOT.	T110W2Cr1	1.00-1.20	0.25-0.50	0.10-0.35	0.04	0.04	0.90-1.30	---	---	1.25-1.75W				230 MAX. 425 MIN.	A H & T	SAME AS ABOVE	850-1050	780 FURNACE COOL	850 OIL/AIR	150-400	850 AIR COOL
	T90Mn6 WCr2 ALUMINIUM PAINT WITH BLUE DOT.	T90Mn2 W50Cr45	0.85-0.95	1.25-1.75	0.10-0.35	0.04	0.04	0.30-0.60	---	---	--**0.25 MAX. V 0.40-0.60 W **OPTIONAL				230 MAX. 500 MIN.	A H & T	SAME AS ABOVE	850-1050	780 FURNACE	790 OIL/AIR COOL	150-400	850 AIR COOL
	XT160Cr12 ALUMINIUM PAINT WITH BLACK & BLUE DOT.	T160Cr12	1.50-1.70	0.25-0.55	0.10-0.35	0.04	0.04	11.00-13.00	---	**0.80 MAX.	**0.80V MAX. **OPTIONAL				260 MAX. 590 MIN.	A H & T	SAME AS ABOVE	900-1050	850 FURNACE COOL	1020 OIL/AIR	150-400	850 AIR COOL
	XT215Cr12 ALUMINIUM PAINT WITH BLACK & RED DOT.	T215Cr12	2.00-2.30	0.25-0.55	0.10-0.35	0.04	0.04	11.00-13.00	---	**0.80 MAX.	**0.80V MAX. **OPTIONAL				260 MAX. 590 MIN.	A H & T	SAME AS ABOVE	900-1050	850 FURNACE COOL	980 OIL/AIR	150-400	950 AIR COOL
	T55Cr3 ALUMINIUM PAINT WITH ORANGE & VIOLET DOT	T55Cr70	0.50-0.60	0.60-0.80	0.10-0.35	0.04	0.04	0.60-0.80	---	---	---				230 MAX. 485 MIN.	A H & T	SAME AS ABOVE	850-1100	830 FURNACE COOL	820 OIL/AIR	150-400	850 AIR COOL
	T45Cr5Si3 ALUMINIUM PAINT WITH PINK DOT	T45Cr1Si95	0.40-0.50	0.55-0.75	0.80-1.10	0.04	0.04	1.20-1.60	---	---	---				230 MAX. 485 MIN.	A H & T	SAME AS ABOVE	850-1100	830 FURNACE COOL	820 OIL/AIR	150-400	850 AIR COOL
	T55Cr3V2 ALUMINIUM PAINT WITH PINK AND GREEN DOT	T55Cr70V15	0.50-0.60	0.60-0.80	0.10-0.35	0.04	0.04	0.60-0.80	---	---	0.10-0.20V				230 MAX. 485 MIN.	A H & T	SAME AS ABOVE	850-1100	830 FURNACE COOL	820 OIL/AIR	150-400	850 AIR COOL
	T50Cr4V2 ALUMINIUM PAINT WITH PINK AND RED DOT	T50Cr1V23	0.45-0.55	0.50-0.80	0.10-0.35	0.04	0.04	0.90-1.20	---	---	0.15-0.30V				230 MAX. 485 MIN.	A H & T	SAME AS ABOVE	850-1100	830 FURNACE COOL	820 OIL/AIR	150-400	850 AIR COOL
	T55Si7 ALUMINIUM PAINT WITH BLUE AND WHITE DOT	T55Si2Mn90	0.50-0.60	0.80-1.00	1.50-2.00	0.04	0.04	0.90-1.20	---	---	---				230 MAX. 485 MIN.	A H & T	SAME AS ABOVE	850-1100	740 FURNACE COOL	880 OIL/AIR	150-400	800 AIR COOL

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H & T -- HARDENED & TEMPERED

CONTD.

CHEMISTRY & MECHANICAL PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - INDIAN STANDARD SPCN.

DESIGNATION / COLOUR CODE FOR IDENTIFICATION OF STEEL(IS:2049)			* CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
ISS NO.	NEW	OLD	C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J			HOT FORMING TEMP. °C	ANNEALING TEMP. °C	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZ-ING TEMP.°C
3749 (CONTD.)	T55Si7Mo3 ALUMINIUM PAINT WITH GREEN & BLUE DOT.	T55Si2Mn90 Mo33	0.50-0.60	0.80-1.00	1.50-2.00	0.04	0.04	---	---	0.25-0.40	**0.12-0.20 V **OPTIONAL				230 MAX. 450 MIN.	A H & T	COLD WORK DIE STEEL. SUITABLE FOR PRESS TOOLS, PUNCHES, SHEARING BLADES, DIES.	800-1100	740 FURNACE COOL	880 OIL/AIR	150-400	810 AIR COOL
	T40W8 Cr5V2 ALUMINIUM PAINT WITH RED & ORANGE DOT.	T40W2 Cr1V18	0.35-0.45	0.20-0.40	0.50-1.00	0.04	0.04	1.00-1.50	---	---	0.10-0.25V 1.75-2.25W				230 MAX. 450 MIN.	A H & T	SAME AS ABOVE	850-1100	850 FURNACE COOL	1100 OIL/AIR	600-650	850 AIR COOL
	T50W8 Cr5V2 ALUMINIUM PAINT WITH RED & YELLOW DOT.	T50W2 Cr1V18	0.45-0.50	0.20-0.40	0.50-1.00	0.04	0.04	1.00-1.50	---	---	0.10-0.25V 1.75-2.25W				230 MAX. 450 MIN.	A H & T	SAME AS ABOVE	850-1100	850 FURNACE COOL	1100 OIL/AIR	600-650	850 AIR COOL
3930	30C8 BLACK WITH GREEN DOT	C30	0.27-0.35	0.60-0.90	0.10-0.40	0.05	0.05	---	---	---	---	50 55	--- 37	21 20	140 170/55	N H & T	F& I HARDENING STEEL. SUITABLE FOR SPROCKETS, CAMS, TIE RODS,ETC.	1200	870 FURNACE COOL	840-880 OIL/WATER	550-600	870 AIR COOL
	35C8 BLACK WITH BROWN DOT	C35Mn75	0.32-0.40	0.60-0.90	0.10-0.40	0.05	0.05	---	---	---	---	55 60	--- 37	20 18	160 187/55	N H & T	F & I HARDENING STEEL. SUITABLE FOR HAND TOOLS, COUPLINGS,FLANGES.	1200	870 FURNACE COOL	840-880 OIL/WATER	550-600	870 AIR COOL
	45C8 BLACK	C45	0.42-0.50	0.50-0.80	0.10-0.40	0.035	0.035	---	---	---	---	60 70	--- 37	15 15	180 200/41	N H & T	F & I HARDENING STEEL. SUITABLE FOR SPINDLES OF MACHINE TOOLS, GEARS,SPROCKETS.	1200	870 FURNACE COOL	840-880 OIL/WATER	550-600	870 AIR COOL
	55C6 BLACK WITH BROWN AND WHITE DOT	C55	0.50-0.57	0.50-0.65	0.10-0.40	0.035	0.035	---	---	---	---	71 78	--- 40	15 15	200 220/28	N H & T	F & I HARDENING STEEL. SUITABLE FOR CAMS, KEYS, CRANK SHAFTS, GEARS,SPROCKETS.	1150	840 FURNACE COOL	810-840 OIL/WATER	550-650	840 AIR COOL
	37Mn6 RED WITH BLUE DOT	37Mn2	0.32-0.40	1.30-1.70	0.10-0.35	0.035	0.035	---	---	---	---	65 80	--- 64	16 13	180 230/48	N H & T	F& I HARDENING STEEL. SUITABLE FOR KEYS, AXLES, CRANK SHAFTS, GEARS, H.T. NUTS, BOLTS, STUDS, ETC.	1200	840 FURNACE COOL	840-880 OIL/WATER	550-66	840 AIR COOL
	47Mn6 RED WITH VIOLET DOT	47Mn2	0.42-0.50	1.30-1.70	0.10-0.35	0.035	0.035	---	---	---	---	70 90	--- 72	15 12	200 250/48	N H & T	F& I HARDENING STEEL. SUITABLE FOR KEYS, AXLES, TRAM RAILS, GEARS, H.T. NUTS, BOLTS, STUDS, ETC.	1100	820-840 FURNACE COOL	820-840 5 OIL/WATER	50-66	820-840 AIR COOL
	40Mn15S12 RED WITH VIOLET DOT	40Mn2S12	0.35-0.45	1.30-1.70	0.25	0.08-0.12	0.06	---	---	---	---	65 80	--- 64	16 13	180 230	N H & T	SAME AS ABOVE	1100	820-840 FURNACE COOL	820-840 OIL/WATER	550-66	820-840 AIR COOL

*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED.

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H & T -- HARDENED & TEMPERED
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CONTD.

CHEMISTRY & MECHANICAL PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - INDIAN STANDARD SPCN.

DESIGNATION / COLOUR CODE FOR IDENTIFICATION OF STEEL(IS:2949)			* CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
ISS NO.	NEW	OLD	C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J)			HOT FORMING TEMP. °C	ANNEALING TEMP. °C	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZ-ING TEMP.°C
3930 (CONTD.)	35Mn6Mo3 RED WITH YELLOW DOT	35Mn2Mo28	0.32-0.40	1.30-1.80	0.10-0.35	0.035	0.035	---	---	0.20-0.35	---	65 80	37 64	20 15	180 230/55	N H & T	F &I HARDENING STEEL. SUITABLE FOR AXLE SHAFTS, CONN.RODS, CRANK-SHAFTS,GEARS,NUT-BOLTS, STUDS, ETC.	1200	830-860 FURNACE COOL	840-870 OIL/WATER	550-660	830-860 AIR COOL
	35Mn6Mo4 REDWITH GREEN DOT	35Mn2Mo45	0.32-0.40	1.30-1.80	0.10-0.35	0.035	0.035	---	---	0.35-0.55	---	67 87	40 74	18 14	200 250/55	N H & T	SAME AS ABOVE	1200	850-880 FURNACE COOL	850-880 OIL/WATER	550-700	850-880 AIR COOL
	40C4 BLUE	40Cr1	0.38-0.44	0.60-0.90	0.10-0.35	0.035	0.035	0.90-1.20	---	---	---	67 80	40 64	18 16	200 210/55	N H & T	SAME AS ABOVE	1200	850-880 FURNACE COOL	850-880 OIL/WATER	550-700	850-880 AIR COOL
	50C4 GREEN WITH BLUE DOT	50Cr1	0.45-0.55	0.60-0.90	0.10-0.35	0.035	0.035	0.90-1.20	---	---	---	67 80	40 64	18 16	200 210/28	N H & T	OIL HARDENING STEEL, USED FOR SPRINGS	1150	830-880 FURNACE COOL	830-880 OIL	400-550	830-880 AIR COOL
	50C4V2 GREEN WITH BLUE DOT	50Cr1V23	0.46-0.54	0.50-0.80	0.10-0.35	0.035	0.035	0.90-1.20	---	---	0.15-0.30V	67 80	40 64	18 16	200 210/28	N H & T	OIL HARDENING STEEL, USED FOR VOLUTE SPRINGS, H.D.LEAF SPRINGS.	1150	830-880 FURNACE COOL	830-880 OIL	400-550	830-880 AIR COOL
	40C4Mo3 YELLOW	40Cr1Mo28	0.38-0.45	0.50-0.80	0.10-0.35	0.035	0.035	0.90-1.20	---	0.20-0.35	---	67 80	40 64	18 16	200 225/55	N H & T	OIL HARDENING STEEL, USED FOR AXLE SHAFTS, CONN.RODS, CRANK-SHAFTS,GEARS,NUT-BOLTS, STUDS, ETC.	1200	850-880 FURNACE COOL	850-880 OIL	550-720	850-880 AIR COOL
	40Ni14 YELLOW WITH VIOLET DOT	40Ni3	0.38-0.45	0.50-0.80	0.10-0.35	0.035	0.035	0.30 MAX.	3.20-3.60	---	---	67 80	40 64	18 16	200 225/55	N H & T	SAME AS ABOVE	1200	830-860 FURNACE COOL	830-860 OIL	550-650	830-860 AIR COOL
	35Ni5Cr2 BLUE WITH RED DOT	35Ni1Cr60	0.32-0.40	0.60-0.90	0.10-0.35	0.035	0.035	0.45-0.75	1.00-1.50	---	---	67 80	40 64	18 16	200 225/55	N H & T	SAME AS ABOVE	1200	830-860 FURNACE COOL	830-860 OIL	550-650	830-860 AIR COOL
	40Ni6 Cr4Mo2 BLUE WITH BROWN DOT	40NiCr1 Mo15	0.38-0.45	0.40-0.70	0.10-0.35	0.035	0.035	0.90-1.30	1.20-1.50	0.10-0.20	---	67 87	40 70	18 15	200 250/55	N H & T	SAME AS ABOE	1200	850-880 FURNACE COOL	850-880 OIL	550-720	850-880 AIR COOL
	40Ni6 Cr4Mo3 BLUE WITH BROWN DOT	40Ni2Cr1 Mo28	0.36-0.45	0.40-0.70	0.10-0.35	0.035	0.035	0.90-1.30	1.25-1.75	0.20-0.35	---	70 100	40 90	17 12	220 285/48	N H & T	SAME AS ABOE	1200	850-880 FURNACE COOL	850-880 OIL	550-720	850-880 AIR COOL
	31Ni10 Cr3Mo6 YELLOW WITH BROWN DOT	31Ni3Cr65 Mo55	0.27-0.33	0.40-0.70	0.10-0.35	0.035	0.035	0.50-0.80	2.25-2.75	0.40-0.70	---	75 110	45 90	16 12	230 310/41	N H & T	SAME AS ABOE	1200	850-880 FURNACE COOL	850-880 OIL	550-720	850-880 AIR COOL

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CONTD.

CHEMISTRY & MECHANICAL PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - INDIAN STANDARD SPCN.

DESIGNATION / COLOUR CODE FOR IDENTIFICATION OF STEEL(IS:2049)			* CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
ISS NO.	NEW	OLD	C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J)			HOT FORMING TEMP. °C	ANNEALING TEMP. °C	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZING TEMP.°C
4368	20Mn6 RED WITH WHITE DOT	20Mn2	0.16-0.24	1.30-1.70	0.10-0.35	0.05	0.05	---	---	---	---	50-55 60	--- ---	20 17	241 MAX. APR/55	N CH & T	ALLOY STEEL BILLETS, BLOOMS & SLABS FOR FOR GENERAL ENGG. PURPOSES. SUITABLE IN CASE HARDENED CONDITION FOR GUDGEON PINS, RIFLES, ROPE SOCKETS, ETC.	1200	870 FURNACE COOL	880-930 CARBURIZING FOLLOWED BY HARDENING AT 770-790 OIL/WATER	150-220	870-930 AIR COOL
	15Cr3 WHITE WITH BLUE DOT	15Cr65	0.12-0.18	0.40-0.60	0.10-0.35	0.05	0.05	0.50-0.80	---	---	---	50-55 60	--- ---	20 13	241 MAX. APR/48	N CH & T	SAME AS ABOVE	1200	870 FURNACE COOL	900-950 CARBURIZING FOLLOWED BY HARDENING AT 770-790 WATER	150-220	870-930 AIR COOL
	17Mn5Cr4 WHITE WITH BLACK DOT	17Mn1Cr95	0.14-0.19	1.00-1.30	0.10-0.35	0.05	0.05	0.80-1.10	---	---	---	50-60 80	--- ---	18 10	241 MAX. APR/35	N CH & T	SAME AS ABOVE SUITABLE IN CASE HARDENED CONDITION FOR SMALL GEARS,SHAFTS, STEERING REGULATORS.	1200	870 FURNACE COOL	900-950 CARBURIZING FOLLOWED BY HARDENING AT 770-790 OIL/WATER	150-220	870-930 AIR COOL
	20MnCr5 WHITE	20MnCr1	0.17-0.22	1.00-1.40	0.10-0.35	0.05	0.05	1.00-1.30	---	---	---	60-70 100	--- ---	16 8	241 MAX. APR/38	N CH & T	SAME AS ABOVE SUITABLE IN CASE HARDENED CONDITION FOR MEDIUM SIZE GEARS,SHAFTS FOR VEHICLES.	1150	850-880 FURNACE COOL	900-930 CARBURIZING FOLLOWED BY HARDENING AT 780-820 OIL/WATER	150-220	850-880 AIR COOL
	21Cr4Mo3 ORANGE WITH RED DOT	21Cr1Mo28	0.26 MAX.	0.50-0.80	0.10-0.35	0.05	0.05	0.90-1.20	---	0.20-0.35	---	50-60 68	--- 40	18 16	241 MAX. 228-302	N H & T	SAME AS ABOVE CREEP RESISTING STEEL IN H & T CONDITION FOR MEDIUM SIZE GEARS,SHAFTS, STEERING ARMS.	1150	850-880 FURNACE COOL	860-880 OIL/WATER	500-550	850-880 AIR COOL
	07Cr4Mo6 ORANGE WITH WHITE DOT	07Cr90Mo55	0.12 MAX.	0.40-0.70	0.10-0.35	0.05	0.05	0.70-1.10	0.30	0.45-0.55	---	40-52	22	22	241 MAX.	N	SAME AS ABOVE CREEP RESISTING STEEL IN NORMALIZED CONDITION FOR BOILER & SUPERHEATER TUBES AND WELDING ELECTRODE	1150	850-880 FURNACE COOL	---	---	850-880 AIR COOL
	10Cr9Mo10 ORANGE WITH GREEN DOT	10Cr2Mo1	0.15 MAX.	0.40-0.70	0.50	0.05	0.05	2.00-2.50	0.30	0.90-1.10	---	50	32	20	241 MAX. 55 J	N	SAME AS ABOVE CREEP RESISTING STEEL IN NORMALIZED CONDITION FOR TUBES, PIPE FITTING FOR HIGH TEMP. SERVICE.	1150	850-880 FURNACE COOL	---	---	850-880 AIR COOL
	13Ni13Cr3 WHITE WITH RED DOT	13Ni3Cr80	0.10-0.15	0.40-0.70	0.10-0.35	0.05	0.05	0.60-1.00	3.00-3.50	---	---	70 150	45 125	16 8	241 MAX. 445/14	N H & T	SAME AS ABOVE GEARS, CONN. RODS, AIRCRAFTS PARTS, TRANSMISSION COMP.	1200	810-830 FURNACE COOL	810-830 OIL/AIR	250	810-830 AIR COOL

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CHEMISTRY & MECHANICAL PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - INDIAN STANDARD SPCN.

DESIGNATION / COLOUR CODE FOR IDENTIFICATION OF STEEL(IS:2049)			*CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
ISS NO.	NEW	OLD	C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J			HOT FORMING TEMP. °C	ANNEALING TEMP °C	HARDENING TEMP °C QUENCHANT	TEMPERING TEMP °C	NORMALIZING TEMP °C
4360 (CONTD.)	15Ni16Cr5 WHITE WITH RED DOT	15Ni4Cr1 BLUE AND	0.12-0.18	0.40-0.70	0.10-0.35	0.05	0.05	1.00-1.40	3.80-4.30	---	---	70 157	45 130	16 8	241 MAX. 445/14	N H & T	ALLOY STEEL FOR GEARS, CONN. RODS, AIRCRAFTS PARTS, TRANSMISSION COMP	1200	810-830 FURNACE COOL	810-830 OIL/AIR	200	810-830 AIR COOL
	15Ni5Cr4 Mo1 WHITE WITH	15NiCr1 Mo12 BROWN DOT	0.12-0.18	0.60-1.00	0.10-0.35	0.05	0.05	0.75-1.25	1.00-1.50	0.08-0.15	---	55 80	32 64	18 15	241 MAX. 250-300/55	N H & T	ALLOY STEEL FOR GEARS, CONN. RODS, AXLES,NUTS, BOLTS, SHAFTS, ETC.	1200	820-850 FURNACE COOL	820-850 OIL/AIR	550-660	820-850 AIR COOL
	15Ni7Cr4 Mo2 WHITE WITH	15Ni2Cr1 Mo15 ORANGE DOT	0.12-0.18	0.60-1.00	0.10-0.35	0.05	0.05	0.75-1.25	1.50-2.00	0.10-0.20	---	55 85	32 65	18 15	241 MAX. 250-300/55	N H & T	ALLOY STEEL FOR GEARS, CONN. RODS, AXLES,NUTS, BOLTS, SHAFTS, ETC.	1200	820-850 FURNACE COOL	820-850 OIL/AIR	550-660	820-850 AIR COOL
	16Ni6Cr6 Mo2 WHITE WITH	16NiCr2 Mo20 VIOLET DOT	0.12-0.20	0.40-0.70	0.10-0.35	0.05	0.05	1.40-1.70	1.80-2.20	0.15-0.25	---	60 87	36 68	16 14	241 MAX. 270-320/50	N H & T	ALLOY STEEL FOR GEARS, CONN. RODS, AXLES,NUTS, BOLTS, SHAFTS, ETC.	1200	820-850 FURNACE COOL	820-850 OIL/AIR	550-660	820-850 AIR COOL
	36S7 GREEN	37Si2Mn90	0.33-0.40	0.80-1.00	1.50-2.00	0.05	0.05	---	---	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					ALLOY STEEL FOR ALL TYPES OF LAMINATED SPRINGS FOR ROADAND RAIL VEHICLES	1150	860-920 FURNACE COOL	860-920 WATER	400-650	860-920 AIR COOL
	37Mn6 RED WITH	37Mn2 BLUE DOT	0.32-0.40	1.30-1.70	0.10-0.35	0.035	0.035	---	---	---	---	65 80	--- 64	16 13	277 MAX. 230/48	N H & T	F& I HARDENING STEEL SUITABLE FOR KEYS, AXLES, CRANK SHAFTS, GEARS, H.T. NUTS, BOLTS, STUDS, ETC.	1200	840 FURNACE COOL	840-880 OIL/WATER	550-66	840 AIR COOL
	35Mo6Mo3 RED WITH	35Mn2Mo28 YELLOW DOT	0.32-0.40	1.30-1.80	0.10-0.35	0.035	0.035	---	---	0.20-0.35	---	65 80	37 64	20 15	277 MAX. 230/55	N H & T	F &I HARDENING STEEL. SUITABLE FOR AXLE SHAFTS, CONN.RODS, CRANK-SHAFTS,GEARS,NUT-BOLTS, STUDS, ETC.	1200	830-860 FURNACE COOL	840-870 OIL/WATER	550-660 AIR COOL	830-860
	40Cr4 BLUE	40Cr1	0.38-0.44	0.60-0.90	0.10-0.35	0.035	0.035	0.90-1.20	---	---	---	67 80	40 64	18 16	277 MAX. 210/55	N H & T	SAME AS ABOVE	1200	850-880 FURNACE COOL	850-880 OIL/WATER	550-700	850-880 AIR COOL
	40Cr4Mo3 YELLOW	40Cr1Mo28	0.38-0.45	0.50-0.80	0.10-0.35	0.035	0.035	0.90-1.20	---	0.20-0.35	---	67 80	40 64	18 16	277 MAX. 225/55	N H & T	OIL HARDENING STEEL, USED FOR AXLE SHAFTS, CONN.RODS, CRANK-SHAFTS,GEARS,NUT-BOLTS, STUDS, ETC.	1200	850-880 FURNACE COOL	850-880 OIL	550-720	850-880 AIR COOL
	35Ni5Cr2 BLUE WITH	35Ni1Cr60 RED DOT	0.32-0.40	0.60-0.90	0.10-0.35	0.035	0.035	0.45-0.75	1.00-1.50	---	---	67 80	40 64	18 16	277 MAX. 225/55	N H & T	SAME AS ABOVE	1200	830-860 FURNACE COOL	830-860 OIL	550-650	830-860 AIR COOL
	40Ni6 Cr4Mo3 BLUE WITH	40Ni2Cr1 Mo28 BROWN DOT	0.36-0.45	0.40-0.70	0.10-0.35	0.035	0.035	0.90-1.30	1.25-1.75	0.20-0.35	---	70 100	40 90	17 12	277 MAX. 285/48	N H & T	SAME AS ABOE	1200	850-880 FURNACE COOL	850-880 OIL	550-720	850-880 AIR COOL

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																		HOT FORMING TEMP. °C	ANNEALING TEMP. °C	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZ-ING TEMP.°C
ISS NO.	NEW	OLD	C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J)							
4368 (CONTD.)	40Ni10Cr3 Mo6 YELLOW WITH WHITE DOT	40Ni3Cr65 M055	0.36-0.44	0.40-0.70	0.10-0.35	0.05	0.05	0.50-0.80	2.25-2.75	0.40-0.70	---	60 127	35 110	18 9	277 MAX. 390-490	N H & T	ALLOY STEEL FOR GEARS, CONN. RODS, AIRCRAFTS PARTS, NUTS, BOLTS.ETC.	1200	810-830 FURNACE COOL	810-830 OIL	550-660	810-830 AIR COOL
	25Cr20Mo6 ORANGE WITH BROWN DOT	25Cr3Mo55	0.20-0.30	0.40-0.70	0.10-0.35	0.05	0.05	2.90-3.40	0.30	0.45-0.65	---	60 80	35 64	18 12	277 MAX. 240-350 / 55	N H & T	ALLOY STEEL FOR GEARS, CONN. RODS, PLASTIC MOULDS, NUTS, BOLTS.ETC.	1200	840-880 FURNACE COOL	890-910 OIL	550-660	840-880 AIR COOL
	55Si7 GREEN WITH WHITE DOT	55Si2Mn90	0.50-0.80	0.80-1.00	1.50-2.00	0.05	0.05	---	---	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					ALLOY STEEL FOR HAND TOOLS, KNIVES, VALVES AND HELI-CAL SPRINGS.	1100	800-920 FURNACE	780-820 OIL COOL	400-500	800-850 AIR COOL
	50Cr4V2 GREEN WITH YELLOW DOT	50Cr1V23	0.45-0.55	0.50-0.80	0.10-0.35	0.05	0.05	0.90-1.20	---	---	0.15-0.30 V	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					SAME AS ABOVE	1100	800-920 FURNACE COOL	780-820 OIL	400-500	800-850 AIR COOL
4398	103Cr4 GREY	103Cr1	0.95-1.10	0.25-0.45	0.15-0.35	0.025	0.025	0.90-1.20	---	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					ALLOY STEEL FOR BALL , ROLLER AND BEARING RACES, BUSHES, COLLETS, CAMS, LATHE CENTRES, ETC.	1100	820-860 FURNACE COOL	820-860 OIL/WATER	150	820-860 AIR COOL
	103Cr6 GREY WITH GREEN DOT	103Cr2	0.95-1.10	0.25-0.45	0.15-0.35	0.025	0.025	1.40-1.60	---	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					SAME AS ABOVE	1100	820-860 FURNACE COOL	820-860 OIL/WATER	150	820-860 AIR COOL
	105Cr5 GREY WITH YELLOW DOT	105Cr1Mn60	0.95-1.10	0.60-0.80	0.15-0.35	0.025	0.025	1.50-1.70	---	---	---	MECHANICAL PROPERTIES ARE NOT PART OF CONTRACTUAL OBLIGATION					SAME AS ABOVE BUT WITH HIGHER Mn CONTENT FOR IMPROVED HARD-NESS.	1050	800-840 FURNACE COOL	800-840 OIL/WATER	130-180	800-840 AIR COOL
4430	T90Mn6 WCr2 ALUMINIUM PAINT WITH BLUE DOT.	T90Mn2 W50Cr45	0.85-0.95	1.25-1.75	0.10-0.35	0.04	0.04	0.30-0.60	---	---	--0.25 MAX. V 0.40-0.60 W **OPTIONAL				230 MAX. 500 MIN.	A H & T	MOULD STEEL AND ALSO USABLE FOR TOOLS, DRILLS, CUTTERS, PUNCHES.	850-1050	780 FURNACE COOL	790 OIL/AIR	150-400	850 AIR COOL
	T30Ni16Cr5 ALUMINIUM PAINT WITH BLUE & BROWN DOT	T30Ni4Cr1	0.26-0.34	0.40-0.70	0.10-0.35	0.04	0.04	1.10-1.40	3.90-4.30	---	---				255 MAX. 485 MIN.	A H & T	SAME AS ABOVE	800-1100	830 FURNACE COOL	840 OIL/AIR	150-400	850 AIR COOL
	T55Ni6Cr Mo3 ALUMINIUM PAINT WITH GREEN & WHITE DOT	T55Ni2Cr65 Mo30	0.50-0.60	0.50-0.80	0.10-0.35	0.04	0.04	0.50-0.80	1.25-1.75	0.25-0.35	---				255 MAX. 485 MIN.	A H & T	SAME AS ABOVE	800-1100	830 FURNACE COOL	820 OIL/AIR	150-400	850 AIR COOL
	XT215Cr12 ALUMINIUM PAINT WITH BLACK & RED DOT.	T215Cr12	2.00 2.30	0.25-0.55	0.10-0.35	0.04	0.04	11.00-13.00	---	---	**0.80 MAX. **0.80V MAX. **OPTIONAL				260 MAX. 590 MIN.	A H & T	SAME AS ABOVE	900-1050	850 FURNACE COOL	980 OIL/AIR	150-400	850 AIR COOL
	10T4 ALUMINIUM PAINT WITH YELLOW & GREEN DOT	T10	0.15	0.30-0.80	0.10-0.35	0.04	0.04	---	---	---	---				240 MAX. 465 MIN.	A H & T	SAME AS ABOVE	850-1050	810 FURNACE COOL	880-930 1 CARBURIZING AND HARDEN-INGAT810.O/W	50-400	810 AIR COOL

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ISS NO.	NEW	OLD	C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J)			HOT FORMING TEMP. °C	ANNEALING TEMP. °C	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZING TEMP. °C
4430 (CONTD.)	T15Cr3 ALUMINIUM PAINT WITH YELLOW & BLUE DOT.	T15Cr65	0..12-0.18	0.40-0.60	0.10-0.35	0.04	0.04	0.50-0.80	---	---	---				230 MAX. 500 MIN.	A H & T	MOULD STEEL AND ALSO USABLE FOR TOOLS, DRILLS, CUTTERS, PUNCHES.	850-1050	780 FURNACE COOL	880-930 CARBURIZING FOLLOWED BY HARDENING AT 770-790 OIL/WATER	150-400	850 AIR COOL
	T10Cr20 Mo6V2 ALUMINIUM PAINT WITH YELLOW & BLACK DOT	T10Cr5 Mo75V23	0..15	0.25-0.50	0.10-0.35	0.04	0.04	4.75-5.25	---	0.50-1.00	0.15-0.30V				230 MAX. 500 MIN.	A H & T	SAME AS ABOVE	850-1050	850 FURNACE COOL	SAME AS ABOVE	150-400	850 AIR COOL
	T16Ni3 Cr2 ALUMINIUM PAINT WITH YELLOW & PINK DOT	T16Ni80 Cr60	0..12-0.20	0.60 1.00	0.10-0.35	0.04	0.04	0.40-0.80	0.60-1.00	---					230 MAX. 500 MIN.	A H & T	SAME AS ABOVE	850-1050	850 FURNACE COOL	SAME AS ABOVE	150-400	850 AIR COOL
	T15Ni5 Cr4Mo ALUMINIUM PAINT WITH YELLOW & RED DOT	T15Ni Cr1Mo12	0..12-0.18	0.60-1.00	0.10-0.35	0.04	0.04	0.70-1.25	1.00-1.50	0.08-0.15					230 MAX. 500 MIN.	A H & T	SAME AS ABOVE	850-1050	850 FURNACE COOL	SAME AS ABOVE	150-400	850 AIR COOL
	T16Ni8 Cr6Mo2 ALUMINIUM PAINT WITH YELLOW & VIOLET DOT	T16Ni Cr2Mo20	0..12-0.20	0.40-0.70	0.10-0.35	0.04	0.04	1.40-1.70	1.80-2.20	0.15-0.25					230 MAX. 500 MIN.	A H & T	SAME AS ABOVE	850-1050	850 FURNACE COOL	SAME AS ABOVE	150-400	850 AIR COOL
	15Ni16Cr5 WHITE WITH BLUE AND RED DOT	15Ni4Cr1	0.12-0.18	0.40-0.70	0.10-0.35	0.05	0.05	1.00-1.40	3.80-4.30	---	---	70 157	45 130	16 8	241 MAX. 445/14	N H & T	ALLOY STEEL FOR GEARS, CONN. RODS, AIRCRAFTS PARTS, TRANSMISSION COMP.	1200	810-830 FURNACE COOL	SAME AS ABOVE	250	810-830 AIR COOL
4431	10C8S11 BROWN WITH WHITE DOT	10S11	0.17	0.60-0.95	---	0.08-0.15	0.055	---	---	---	---	47	32	18	---	CD FREE CUTTING STEEL	C.S.WIRE FOR MFR. OF WOOD SCREWS BY COLD HEADING PROCESS.	---	---	---	---	---
	14C14S14 BROWN WITH GREEN DOT	14Mn1S14	0.10-0.18	1.20-1.50	0.05-0.35	0.10-0.18	0.055	---	---	---	---	44-54 55	--- ---	22 11	120 160	N CD	CARBON STEEL FREE CUTTING STEEL FOR MACINED PARTS FOR GENERAL ENGG. PURPOSE.	1200	880-920 FURNACE COOL	---	---	880-930 AIR COOL
	25C12S14 BROWN WITH BLUE DOT	25Mn1S14	0.20-0.30	1.00-1.50	0.25	0.10-0.18	0.055	---	---	---	---	50-60 65	--- ---	20 9	150 187	N CD	SAME AS ABOVE	1200	880-920 FURNACE COOL	---	---	880-930 AIR COOL
	40C10S18 BROWN WITH GREY DOT	40S18	0.35-0.45	0.80-1.20	0.25	0.14-0.22	0.055	---	---	---	---	55-80 70	--- ---	17 8	170 200	N CD	SAME AS ABOVE	1200	880-920 FURNACE COOL	---	---	880-930 AIR COOL
	13C10S25 BROWN	13S25	0.08-0.18	0.80-1.20	0.10	0.20-0.30	0.055	---	---	---	---	38-50 55	--- ---	22 11	120 160	N CD	SAME AS ABOVE	1200	880-920 FURNACE COOL	---	---	880-930 AIR COOL
	40C15S12 BROWN WITH YELLOW DOT	40Mn2S12	0.35-0.45	1.30-1.70	0.25	0.08-0.15	0.055	---	---	---	---	65-84 70	--- ---	15 8	180 200	N CD	SAME AS ABOVE	1200	880-920 FURNACE COOL	---	---	880-930 AIR COOL

*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED.

ABBREVATIONS USED:
HR – HOT ROLLED; AC – AIR COOLED; OH – OIL HARDENED; AH – AIR HARDENED; H & T – HARDENED & TEMPERED; CH & T – CASE HARDENED & TEMPERED
AR – AS ROLLED; CR – COLD ROLLED; CD – COLD DRAWN; N – NORMALISED; A – ANNEALED; F & I – FLAME & INDUCTION HARDENING; APR – AS PER REQUIREMENTS

CONTD.

CHEMISTRY & MECHANICAL PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - INDIAN STANDARD SPCN.

DESIGNATION / COLOUR CODE FOR IDENTIFICATION OF STEEL(IS:2049)			* CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
ISS NO.	NEW	OLD	C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J			HOT FORMING TEMP. °C	ANNEALING TEMP. °C	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZ-ING TEMP.°C
4432	10C4 BLACK WITH WHITE & GREY DOT	C10	0.15	0.30-0.60	0.05-0.35	0.05	0.035	---	---	---	---	34 50	22 ---	28 17	130 150/55	CD & A/N CH & T***	CASE HARDENING STEEL SUITABLE FOR CAMS,GEARS, GUDGEON PINS, WORMS, COLLETS.	1200	880-930 FURNACE COOL	CARBURIZING 880-930 FOLLOWED BY HARDENING 760-780 WATER	150-200	880-930 AIR COOL
	14C6 BLACK WITH WHITE & BLUE DOT	C14	0.10-0.18	0.40-0.70	0.05-0.35	0.05	0.035	---	---	---	---	34 50	24 ---	26 17	130 150/55	CD & A/N CH & T***	CASE HARDENING STEEL SUITABLE FOR CAMS,GEARS, RATCHETS,SHAFTS, WORMS, COLLETS.	1200	880-930 FURNACE COOL	SAME AS ABOVE	150-200	880-930 AIR COOL
	15C8 BLACK WITH RED DOT	C15Mn75	0.15	0.60-0.90	0.05-0.35	0.05	0.035	---	---	---	---	40 50	24 ---	25 17	140 150/55	N CH & T***	CASE HARDENING STEEL SUITABLE FOR STAY BOLTS, AUTO WHEEL HUBS, BRAKE LEVERS.	1200	880-920 FURNACE COOL	SAME AS ABOVE	150-250	880-930 AIR COOL
	10C8S11 BROWN WITH WHITE DOT	10S11	0.15	0.60-0.90	0.05-0.30	0.08-0.13	0.035	---	---	---	---	40 50	24 ---	25 17	140 150/55	N CH & T***	CASE HARDENING STEEL SUITABLE FOR SMALL MACH-INED PARTS (FREE CUTTING STEEL).	1150	880-950 FURNACE COOL	SAME AS ABOVE (CYANIDING CARBO-NITRIDING AT 650-920)	150-250	880-930 AIR COOL
	14C14S14 BROWN WITH GREEN DOT	14Mn1S14	0.10-0.18	1.20-1.50	0.05-0.30	0.10-0.18	0.035	---	---	---	---	44 60	26 ---	21 17	140 165/40	N CH & T***	CASE HARDENING STEEL SUITABLE FOR SMALL MACH-INED PARTS (FREE CUTTING STEEL).	1150	880-950 FURNACE COOL	CARBURIZING 880-930 FOLLOWED BY HARDENING 760-780 WATER	150-250	880-930 AIR COOL
	11Mn6 WHITE WITH YELLOW AND RED DOT	11Mn2	0.16	1.30-1.70	0.10-0.30	0.05	0.035	---	---	---	---	44 60	26 ---	21 17	140 165/55	N CH & T***	CASE HARDENING STEEL SUITABLE FOR SMALL MACH-INED PARTS, RIFLES, COUPLINGS, ETC.	1200	880-950 FURNACE COOL	CARBURIZING 880-930 FOLLOWED BY HARDENING 760-780 OIL/WATER	150-250	880-930 AIR COOL
	15Cr3 WHITE WITH BLUE DOT	15Cr65	0.12-0.18	0.40-0.60	0.10-0.30	0.05	0.035	0.50-0.80	---	---	---	50 60	30 ---	18 13	150 165/48	N CH & T***	CASE HARDENING STEEL SUITABLE FOR GUDGEON PINS, ROLLER BEARINGS, MEASURING INSTRU-MENTS, ETC.	1200	900-950 FURNACE COOL	CARBURIZING 880-930 FOLLOWED BY HARDENING 760-780 WATER	150-250	880-930 AIR COOL
	17Mn5Cr4 WHITE WITH BLACK DOT	17Mn1Cr95	0.14-0.19	1.00-1.30	0.10-0.35	0.05	0.035	0.80-1.10	---	---	---	50-60 80	--- ---	18 10	241 MAX. 210/35	N CH & T***	SAME AS ABOVE SUITABLE IN CASE HARDENED CONDI-TION FOR SMALL GEARS,SHAFTS, STEE-RING REGULATORS.	1200	850-870 FURNACE COOL	CARBURIZING 900-950 FOLLOWED BY HARDENING AT 770-790 OIL/WATER	150-220	870-930 AIR COOL
	20MnCr5 WHITE	20MnCr1	0.17-0.22	1.00-1.40	0.10-0.35	0.05	0.035	1.00-1.30	---	---	---	60-70 100	--- ---	16 8	241 MAX. 240/38	N CH & T***	SAME AS ABOVE	1200	850-870 FURNACE COOL	SAME AS ABOVE	150-220	850-880 AIR COOL

*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED.
*** CORE PROPERTIES

ABBREVATIONS USED
AR -- AS ROLLED
HR -- HOT ROLLED
CR -- COLD ROLLED
AC -- AIR COOLED
CD -- COLD DRAWN
OH -- OIL HARDENED
N -- NORMALISED
AH -- AIR HARDENED
A -- ANNEALED
H & T -- HARDENED & TEMPERED
F & I -- FLAME & INDUCTION HARDENING
CH & T -- CASE HARDENED & TEMPERED
APR -- AS PER REQUIREMENTS

CONTD.

CHEMISTRY & MECHANICAL PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - INDIAN STANDARD SPCN.

DESIGNATION / COLOUR CODE FOR IDENTIFICATION OF STEEL(IS:2049)			* CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
ISS NO.	NEW	OLD	C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J)			HOT FORMING TEMP. °C	ANNEALING TEMP. °C	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZING TEMP.°C
4432 (CONTD.)	16Ni3Cr2 WHITE WITH GREEN DOT	16Ni80Cr60	0.12-0.20	0.60-1.00	0.10-0.35	0.05	0.035	0.40-0.80	0.60-1.00	---	---	50-60 70	-- ---	18 15	241 MAX. 180/41	N CH & T***	CASE HARDENING STEEL SUITABLE FOR CLUTCH PLATES, GUDGEON PINS, CAM-SHAFTS, GEARS, LEVERS, ETC.	1150	850-880 FURNACE COOL	CARBURIZING 880-930 FOLLOWED BY HARDENING AT 780-820 OIL	200-250	850-880 AIR COOL
	16Ni4Cr3 WHITE WITH PINK DOT	16Ni1Cr80	0.12-0.20	0.60-1.00	0.10-0.35	0.05	0.035	0.60-1.00	0.80-1.20	---	---	60-70 85	--- ---	16 12	241 MAX. 250/41	N CH & T***	CASE HARDENING STEEL SUITABLE FOR CAM, PINIONS, GUDGEON PINS, CAM-SHAFTS, GEARS, LEVERS, ETC.	1150	850-880 FURNACE COOL	CARBURIZING 880-930 FOLLOWED BY HARDENING AT 780-820 OIL	150-200	850-880 AIR COOL
	13Ni13Cr3 WHITE WITH RED DOT	13Ni3Cr80	0.10-0.15	0.40-0.70	0.10-0.35	0.05	0.035	0.60-1.00	3.00-3.50	---	---	70 85	45 ---	16 12	241 MAX. 250/48	N CH & T***	SAME AS ABOVE GEARS, CONN. RODS, AIRCRAFTS PARTS, TRANSMISSION COMPONENTS.	1150	850-880 FURNACE COOL	SAME AS ABOVE	150-250	850-880 AIR COOL
	15Ni16Cr5 WHITE WITH BLUE AND RED DOT	15Ni4Cr1	0.12-0.18	0.40-0.70	0.10-0.35	0.05	0.035	1.00-1.40	3.80-4.30	---	---	70 135	45 ---	16 9	241 MAX. 290/35	N CH & T***	SAME AS ABOVE	1150	850-880 FURNACE COOL	SAME AS ABOVE	150-250	850-880 AIR COOL
	20Ni7Mo2 WHITE WITH YELLOW DOT	26Mn2Mo25	0.17-0.22	0.40-0.65	0.15-0.35	0.05	0.035	---	1.65-2.00	0.20-0.30	---	70 85	45 ---	16 12	241 MAX. 290/62	N CH & T***	SAME AS ABOVE	1150	850-880 FURNACE COOL	SAME AS ABOVE	150-250	850-880 AIR COOL
	20NiCrMo2 WHITE WITH GREY DOT	20Ni55Cr50 Mn20	0.18-0.23	0.70-0.90	0.20-0.35	0.05	0.035	0.40-0.60	0.40-0.70	0.15-0.25	---	70 90	45 ---	16 11	241 MAX. 290/41	N CH & T***	SAME AS ABOVE	1150	850-880 FURNACE COOL	SAME AS ABOVE	150-250	850-880 AIR COOL
	15Ni5Cr4 Mo1 WHITE WITH BROWN DOT	15NiCr1 Mo12	0.12-0.18	0.60-1.00	0.10-0.35	0.05	0.035	0.75-1.25	1.00-1.50	0.08-0.15	---	75 100	50 ---	15 9	241 MAX. 300/41	N CH & T***	SAME AS ABOVE	1200	820-850 FURNACE COOL	SAME AS ABOVE	150-250	820-850 AIR COOL
	15Ni7Cr4 Mo2 WHITE WITH ORANGE DOT	15Ni2Cr1 Mo15	0.12-0.18	0.60-1.00	0.10-0.35	0.05	0.035	0.75-1.25	1.50-2.00	0.10-0.20	---	75 110	50 ---	15 9	241 MAX. 325/35	N CH & T***	SAME AS ABOVE	1200	820-850 FURNACE COOL	SAME AS ABOVE	150-250	820-850 AIR COOL
	16Ni8Cr6 Mo2 WHITE WITH VIOLET DOT	16NiCr2 Mo20	0.12-0.20	0.40-0.70	0.10-0.35	0.05	0.05	1.40-1.70	1.80-2.20	0.15-0.25	---	75 135	50 ---	15 9	241 MAX. 350/35	N CH & T***	SAME AS ABOVE	1200	820-850 FURNACE COOL	SAME AS ABOVE	150-250	820-850 AIR COOL
5489	14C6 BLACK WITH WHITE & BLUE DOT	C14	0.10-0.18	0.40-0.70	0.05-0.35	0.025	0.025	---	---	---	---	34 50	24 ---	26 17	130 150/55	CD & A/N CH & T***	CASE HARDENING STEEL SUITABLE FOR CAMS, GEARS, RATCHETS, SHAFTS, BEARING RACES.	1200	880-930 FURNACE COOL	SAME AS ABOVE	150-200	880-930 AIR COOL
	28C6 BLACK WITH WHITE AND GREEN DOT	C28	0.25-0.32	0.40-0.70	0.05-0.35	0.025	0.025	---	---	---	---	50 55-60	30 ---	24 18	160 180/48	N CH & T***	SAME AS ABOVE	1200	880-920 FURNACE COOL	SAME AS ABOVE	150-200	880-930 AIRCOOL
	17Mn5Cr4 WHITE WITH BLACK DOT	17Mn1Cr95	0.14-0.19	1.00-1.30	0.10-0.35	0.025	0.025	0.80-1.10	---	---	---	50-60 80	--- ---	18 10	241 MAX. APR/35	N CH & T***	SAME AS ABOVE	1200	870 FURNACE COOL	SAME AS ABOVE	150-220	870-930 AIR COOL

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*** CORE PROPERTIES

ABBREVATIONS USED:
HR – HOT ROLLED, CR – COLD ROLLED, AR – AS ROLLED, AC – AIR COOLED, CD – COLD DRAWN, OH – OIL HARDENED, N – NORMALISED, AH – AIR HARDENED, A – ANNEALED, H & T – HARDENED & TEMPERED, F & I – FLAME & INDUCTION HARDENING, CH & T – CASE HARDENED & TEMPERED, APR – AS PER REQUIREMENTS

CONTD.

CHEMISTRY & MECHANICAL PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - INDIAN STANDARD SPCN.

DESIGNATION / COLOUR CODE FOR IDENTIFICATION OF STEEL(IS:2049)			* CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
ISS NO.	NEW	OLD	C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J)			HOT FORMING TEMP. °C	ANNEALING TEMP. °C	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZ-ING TEMP.°C
5489 (CONTD.)	20Ni7Mo2 WHITE WITH YELLOW DOT	26Mn2Mo25	0.17-0.22	0.40-0.65	0.15-0.35	0.05	0.035	---	1.65-2.00	0.20-0.30	---	70 85	45 ---	16 12	241 MAX. 290/62	N CH & T***	CASE HARDENING STEEL SUITABLE FOR CAMS,GEARS, RATCHETS,SHAFTS, BEARING RACES.	1200	880-930 FURNACE COOL	CARBURIZING 880-930 FOLLOWED BY HARDENING AT 780-820 OIL	150-200	880-930 AIR COOL
	20NiCrMo2 WHITE WITH GREY DOT	20Ni55Cr50 Mn20	0.18-0.23	0.70-0.90	0.20-0.35	0.05	0.035	0.40-0.60	0.40-0.70	0.15-0.25	---	70 90	45 ---	16 11	241 MAX. 290/41	N CH & T***	SAME AS ABOVE	1150	850-880 FURNACE COOL	SAME AS ABOVE	150-250	850-880 AIR COOL
5517	30C8 BLACK WITH GREEN DOT	C30	0.27-0.35	0.60-0.90	0.10-0.40	0.05	0.05	---	---	---	---	50 55	--- 37	21 20	140 170/55	N H & T	H & T CONSTRUCTIO-NAL STEEL SUITABLE FOR SPROCKETS, CAMS, TIE RODS,ETC.	1200	870 FURNACE COOL	840-880 OIL/WATER	550-600	870 AIR COOL
	35C8 BLACK WITH BROWN DOT	C35Mn75	0.32-0.40	0.60-0.90	0.10-0.40	0.05	0.05	---	---	---	---	55 60	--- 37	20 18	160 187/55	N H & T	H & T CONSTRUCTIO-NAL STEEL SUITABLE FOR HAND TOOLS, COUPLINGS,FLANGES.	1200	870 FURNACE COOL	840-880 OIL/WATER	550-600	870 AIR COOL
	40C8 BLACK WITH WHITE AND BROWN DOT	C-40	0.35-0.45	0.60-0.90	0.05-0.35	0.055	0.055	---	---	---	---	58-68 65 70-85	--- --- 50	18 9 17	175 187 195	N CD H & T	SAME AS ABOVE	1200	880-920 FURNACE COOL	840-880 OIL/WATER	550-600	880-930 AIR COOL
	45C8 BLACK	C45	0.42-0.50	0.60-0.90	0.10-0.40	0.035	0.035	---	---	---	---	60 70	--- 37	15 15	180 200/41	N H & T	H & T CONSTRUCTIO-NAL STEEL SUITABLE FOR SPINDLES OF MACHINE TOOLS, GEARS,SPROCKETS.	1200	870 FURNACE COOL	840-880 OIL/WATER	550-600	870 AIR COOL
	50C8 BLACK WITH GREEN AND WHITE DOT	C-50	0.45-0.55	0.60-0.90	0.05-0.35	0.055	0.055	---	---	---	---	67-80 75 81-96	--- --- 55	13 7 13	230 240 250	N CD H & T	SAME AS ABOVE	1200	880-920 FURNACE COOL	840-880 OIL/WATER	550-600	880-930 AIR COOL
	55C8 BLACK WITH BROWN AND WHITE DOT	C55	0.50-0.60	0.60-0.90	0.10-0.40	0.035	0.035	---	---	---	---	71 78	--- 40	15 15	200 220/28	N H & T	H & T CONSTRUCTIO-NAL STEEL SUITABLE FOR CAMS, KEYS, CRANK SHAFTS, GEARS,SPROCKETS.	1150	840 FURNACE COOL	810-840 OIL/WATER	550-650	840 AIR COOL
	40C10S18 BROWN WITH GREY DOT	40S18	0.35-0.45	0.80-1.20	0.25	0.14-0.22	0.055	---	---	---	---	55-80 70 60-75	--- --- 40	17 8 18	170 200 180/41	N CD H & T	SAME AS ABOVE FREE CUTTING STEEL	1200	880-920 FURNACE COOL	810-840 OIL/WATER	550-650	880-930 AIR COOL
	40C15S12 BROWN WITH YELLOW DOT	40Mn2S12	0.35-0.45	1.30-1.70	0.25	0.08-0.15	0.055	---	---	---	---	65-84 70 70-85	--- --- 50	15 8 18	180 200 200/35	N CD H & T	SAME AS ABOVE FREE CUTTING STEEL	1200	880-920 FURNACE COOL	810-840 OIL/WATER	550-650	880-930 AIR COOL
	20C15 RED WITH WHITE DOT	20Mn2	0.16-0.24	1.30-1.70	0.10-0.35	0.05	0.05	---	---	---	---	50-55 60-75	--- 45	20 17	241 MAX. 200/48	N H & T	SAME AS ABOVE	1200	870 FURNACE COOL	840-870 OIL/WATER	550-650	870-930 AIR COOL

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ABBREVATIONS USED: HR -- HOT ROLLED; AR -- AS ROLLED; CR -- COLD ROLLED; AC -- AIR COOLED; CD -- COLD DRAWN; OH -- OIL HARDENED; N -- NORMALISED; AH -- AIR HARDENED; A -- ANNEALED; H & T -- HARDENED & TEMPERED; F & I -- FLAME & INDUCTION HARDENING; CH & T -- CASE HARDENED & TEMPERED; APR -- AS PER REQUIREMENTS

CONTD.

CHEMISTRY & MECHANICAL PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - INDIAN STANDARD SPCN.

DESIGNATION / COLOUR CODE FOR IDENTIFICATION OF STEELS(IS:2949)			* CHEMICAL COMPOSITION, PERCENT									MECHANICAL PROPERTIES (MIN.)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
ISS NO.	NEW	OLD	C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J			HOT FORMING TEMP. °C	ANNEALING TEMP. °C	HARDENING TEMP °C QUENCHANT	TEMPERING TEMP °C	NORMALIZING TEMP °C
5517 (CONTD.)	27Mn6 RED WITH BROWN DOT	27Mn2	0.22-0.32	1.30-1.70	0.10-0.35	0.045	0.045	---	---	---	---	60 71	36 50	20 16	180 210/41	N H & T	H & T LOW STEEL STEEL SUITABLE FOR KEYS, AXLES, CRANK SHAFTS, GEARS, H.T. NUTS, BOLTS, STUDS, ETC.	1200	840 FURNACE COOL	840-880 OIL/WATER	550-660	840 AIR COOL
	37Mn6 RED WITH BLUE DOT	37Mn2	0.32-0.40	1.30-1.70	0.10-0.35	0.045	0.045	---	---	---	---	65 80	40 64	16 13	200 230/38	N H & T	SAME AS ABOVE	1200	840 FURNACE COOL	840-880 OIL/WATER	550-660	840 AIR COOL
	35Mn6Mo3 RED WITH YELLOW DOT	35Mn2Mo28	0.30-0.40	1.30-1.80	0.10-0.35	0.045	0.045	---	---	0.20-0.35	---	65 80	37 64	20 15	180 230/55	N H & T	H & T LOW ALLOY STEEL SUITABLE FOR AXLE SHAFTS, CONN.RODS, CRANK-SHAFTS,GEARS,NUT-BOLTS, STUDS, ETC.	1200	830-880 FURNACE COOL	840-870 OIL	550-660	830-880 AIR COOL
	35Mn6Mo4 REDWITH GREEN DOT	35Mn2Mo45	0.30-0.40	1.30-1.80	0.10-0.35	0.045	0.045	---	---	0.35-0.55	---	67 87	40 70	18 15	200 250/55	N H & T	SAME AS ABOVE	1200	850-880 FURNACE COOL	850-880 OIL	550-700	850-880 AIR COOL
	55Cr3 BLUE WITH WHITE DOT	55Cr70	0.50-0.60	0.60-0.80	0.10-0.35	0.045	0.045	0.60-0.80	---	---	---	70 90	42 70	18 12	230 250/35	N H & T	SAME AS ABOVE	1150	800-850 FURNACE COOL	800-850 OIL	550-700	800-850 AIR COOL
	40Cr4 BLUE	40Cr1	0.35-0.45	0.60-0.90	0.10-0.35	0.045	0.045	0.90-1.20	---	---	---	67 80	40 64	18 16	217 MAX. 230/55	N H & T	SAME AS ABOVE	1200	850-880 FURNACE COOL	850-880 OIL	550-700	850-880 AIR COOL
	40Cr4Mo3 YELLOW	40Cr1Mo28	0.35-0.45	0.60-0.90	0.10-0.35	0.045	0.045	0.90-1.20	---	0.20-0.35	---	67 80	40 64	18 16	277 MAX. 225/55	N H & T	OIL HARDENING STEEL, USED FOR AXLE SHAFTS, CONN.RODS, CRANK-SHAFTS,GEARS,NUT-BOLTS, STUDS, ETC.	1200	850-880 FURNACE COOL	850-880 OIL	550-720	850-880 AIR COOL
	15Cr13Mo6 ORANGE WITH YELLOW DOT	15Cr3Mo55	0.10-0.20	0.40-0.70	0.10-0.35	0.045	0.045	2.90-3.40	---	0.45-0.65	---	67 80	40 64	18 16	277 MAX. 230/55	N H & T	SAME AS ABOVE CAN ALSO BE USED AS NITRIDING STTEL	1200	850-880 FURNACE COOL	850-880 OIL	550-700	850-880 AIR COOL
	25Cr13Mo6 ORANGE WITH BROWN DOT	25Cr3Mo55	0.20-0.30	0.40-0.70	0.10-0.35	0.045	0.045	2.90-3.40	---	0.45-0.65	---	67 80	40 64	18 16	277 MAX. 230/55	N H & T	SAME AS ABOVE CAN ALSO BE USED AS NITRIDING STTEL	1200	850-880 FURNACE COOL	850-880 OIL	550-700	850-880 AIR COOL
	40Cr13 Mo10V2 YELLOW WITH AL. PAINT DOT	40Cr3 Mo1V20	0.35-0.45	0.40-0.70	0.10-0.35	0.045	0.045	3.00-3.50	---	0.90-1.10	0.15-0.25V	80 134	50 113	14 8	277 MAX. 375/21	N H & T	SAME AS ABOVE CAN ALSO BE USED AS NITRIDING STTEL	1200	850-880 FURNACE COOL	900-940 OIL	550-700	850-880 AIR COOL
	40Cr7 A110Mo2 YELLOW WITH BLACK DOT	40Cr2 A11Mo18	0.35-0.45	0.40-0.70	0.10-0.45	0.045	0.045	1.50-1.80	---	0.10-0.25	0.90-1.30 Al	60 80	36 60	18 16	277 MAX. 225/55	N H & T	SAME AS ABOVE CAN ALSO BE USED AS NITRIDING STTEL AND USABLE FOR COLLETS,ROLLERS, CLUTCH PLATES.	1200	850-880 FURNACE COOL	880-910 OIL	550-700	850-880 AIR COOL

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CONTD.

CHEMISTRY & MECHANICAL PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - INDIAN STANDARD SPCN.

DESIGNATION / COLOUR CODE FOR IDENTIFICATION OF STEEL(IS:2049)			* CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
ISS NO.	NEW	OLD	C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J)			HOT FORMING TEMP. °C	ANNEALING TEMP. °C	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZ-ING TEMP.°C
5517 (CONTD.)	40Ni14 YELLOW WITH VIOLET DOT	40Ni3	0.35-0.45	0.50-0.80	0.10-0.35	0.045	0.045	0.30 MAX.	3.20-3.60	---	---	67 80	40 64	18 16	200 225/55	N H & T	MEDIUM ALLOY STEEL USABLE FOR VALVE ROCKERS, TURBINE SHAFTS, CONN. RODS,AXLES, PINIONS, HIGH TEMP. BOLTS & NUTS.	1200	830-860 FURNACE COOL	830-860 OIL	550-650	830-860 AIR COOL
	35Ni5Cr2 BLUE WITH RED DOT	35Ni1Cr60	0.30-0.40	0.60-0.90	0.10-0.35	0.045	0.045	0.45-0.75	1.00-1.50	---	---	67 80	40 64	18 16	200 225/55	N H & T	SAME AS ABOVE	1200	830-860 FURNACE COOL	830-860 OIL	550-650	830-860 AIR COOL
	30Ni13Cr5 YELLOW WITH GREEN DOT	30Ni4Cr1	0.26-0.34	0.40-0.70	0.10-0.35	0.045	0.045	1.10-1.40	3.90-4.30	---	---	80 157	60 130	16 8	240 445/14	N H & T	SAME AS ABOVE AND PARTS OF AERO ENGINES.	1200	810-830 FURNACE COOL	810-830 OIL	250	810-830 AIR COOL
	40Ni6 Cr4Mo2 BLUE WITH BROWN DOT	40NiCr1 Mo15	0.35-0.45	0.40-0.70	0.10-0.35	0.045	0.045	0.90-1.30	1.20-1.60	0.10-0.20	---	67 87	40 70	18 15	200 250/55	N H & T	SAME AS ABOE	1200	850-880 FURNACE COOL	850-880 OIL	550-720	850-880 AIR COOL
	40Ni6 Cr4Mo3 BLUE WITH BROWN DOT	40Ni2Cr1 Mo28	0.35-0.45	0.40-0.70	0.10-0.35	0.045	0.045	0.90-1.30	1.25-1.75	0.20-0.35	---	70 100	40 90	17 12	220 285/48	N H & T	SAME AS ABOE	1200	850-880 FURNACE COOL	850-880 OIL	550-720	850-880 AIR COOL
	31Ni10 Cr3Mo6 YELLOW WITH BROWN DOT	31Ni3Cr65 Mo55	0.27-0.35	0.40-0.70	0.10-0.35	0.045	0.045	0.50-0.80	2.25-2.75	0.40-0.70	---	75 110	45 90	16 12	230 310/41	N H & T	SAME AS ABOE	1200	850-880 FURNACE COOL	850-880 OIL	550-720	850-880 AIR COOL
	40Ni10Cr3 Mo6 YELLOW WITH WHITE DOT	40Ni3Cr65 M055	0.36-0.44	0.40-0.70	0.10-0.35	0.05	0.05	0.50-0.80	2.25-2.75	0.40-0.70	---	60 127	35 110	18 9	277 MAX. 390-490	N H & T	ALLOY STEEL FOR GEARS, CONN. RODS, AIRCRAFTS PARTS, NUTS, BOLTS.ETC.	1200	810-830 FURNACE COOL	810-830 OIL	550-660	810-830 AIR COOL
6529	X04Cr13 PINK WITH BLUE DOT	04Cr13	0.08	1.00	1.00	0.03	0.04	11.50-14.50	---	---	0.10-0.30 AL.	45 60-80	--- 44	20 18	187 MAX. 225	A H & T	FERITIC (MAGNETIC) STAINLESS STEEL SUITABLE FOR TURBINE BLADES, HOUSE HOLD ARTI-CLES, BOLTS, NUTS, BUILDING FITTINGS.	1000	750-800 FURNACE/ AIRCOOL	950-1000 OIL/AIR	680-780	750-800 AIR COOL
	X12Cr13 PINK WITH ORANGE DOT	12Cr13	0.09-0.15	1.00	1.00	0.03	0.04	11.50-14.50	1.00	---	---	75 65-80	--- 46	18 14	230 MAX. 255	A H & T	MARTENSITIC (MAGNETIC) STAINLESS STE EL SUITABLE FOR TURBINE BLADES, CUTLERY, DENTAL AND SURGINCAL INSTRUMENTS, BOLTS, NUTS, ETC.	1000	730-800 FURNACE COOL	980-1050 OIL/AIR	600-700	750-800 AIR COOL
	X20Cr13 PINK WITH GREEN DOT	20Cr13	0.16-0.25	1.00	1.00	0.03	0.04	12.00-14.00	1.00	---	---	75 70-85	--- 50	18 12	230 MAX. 275	A H & T	SAME AS ABOVE AND ALSO FOR BALL BEARING RACES, SPRINGS.	1000	730-800 FURNACE COOL	980-1050 OIL/AIR	600-700	750-800 AIR COOL

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DESIGNATION / COLOUR CODE FOR IDENTIFICATION OF STEEL (IS:2049)			* CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
ISS NO.	NEW	OLD	C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J)			HOT FORMING TEMP. °C	ANNEALING TEMP °C	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZ-ING TEMP °C
6529 (CONTD.)	X30Cr13 PINK WITH RED DOT	30Cr13	0.26-0.35	1.00	1.00	0.03	0.04	12.00-14.00	1.00	---	---	75 70-85	--- 50	18 12	230 MAX. 275	A H & T	SAME AS ABOVE	1000	730-800 AIR COOL	980-1050 OIL/AIR	600-700	750-800 AIR COOL
	X40Cr13 PINK WITH GREY DOT	40Cr13	0.36-0.45	1.00	1.00	0.03	0.04	12.50-14.50	1.00	---	---	80 90-100	--- 70	16 8	230 MAX. 475	A H & T	SAME AS ABOVE AND ALSO FOR GLASS MOULDS. HAND TOOLS, VALVES, KNIVES.	1000	750-850 FURNACE COOL	1000-1050 OIL/AIR	100-250	750-800 AIR COOL
	X05Cr17 PINK WITH BLACK DOT	05Cr17	0.10	1.00	1.00	0.03	0.04	16.00-18.00	0.50	---	---	45	---	18	192 MAX.	A	FERRITIC (MAGNETIC) CORROSION AND HEAT RESISTANT STEEL FOR NITRIC ACID PLANT & EQUIP-MENT, DOMESTIC & CATERING APPLIA-NCES. CAR TRIMS.	1000	700-780 AIR COOL	---	---	---
	X15Cr16Ni2 PINK WITH VIOLET DOT	15Cr16Ni2	0.10-0.20	1.00	1.00	0.03	0.04	15.00-18.00	1.50-3.00	---	---	80 75-95	--- 60	16 12	295 MAX. 250-310	A H & T	MARTENSITIC (MAGNETIC) STAINLESS STEEL FOR AIRCRAFT & GENERAL ENGG. FITTINGS, HIGH TENSILE NUT, BOLTS, PAPER MACHINE PARTS, ETC.	1120	650-750 FURNACE OR AIR COOL	980-1030 OIL/AIR	620-720 AIR COOL	---
	X105Cr18Mo PINK WITH BROWN DOT	105Cr18Mo50	0.90-1.20	1.00	1.00	0.03	0.04	16.00-19.00	0.50	0.75	---	--- ---	--- ---	--- ---	265 MAX. 580-650	A H & T	MARTENSITIC (MAGNETIC) STAINLESS STEEL FOR VALVES, RACES, NOZZLES, VALVE PARTS, SURGICAL TOOLS, ETC.	1120	650-750 FURNACE OR AIR COOL	980-1030 OIL/AIR	620-720 AIR COOL	---
	X02Cr18Ni11 PINK WITH BLUE AND WHITE DOT	02Cr18Ni11	0.03	2.00	1.00	0.03	0.045	17.00-20.00	9.00-13.00	---	---	45 85	18 50	45 12	192 MAX. 290-340	S CD	AUSTENITIC (NON-MAGNETIC) STAINLESS STEEL FOR KITCHEN WARES, CHEMICAL TEXTILE, PETROLEUM PLANT, WASHING M/C, DOOR BARS, BOLTS, NUTS, WAHSERS, ETC	1200	950-1150 S QUICKLY AIR/WATER/ OIL COOL	---	---	---
	X04Cr18 Ni10 PINK	02Cr18 Ni10	0.08	2.00	1.00	0.03	0.045	17.00-20.00	8.00-12.00	---	---	50 85	21 50	40 12	192 MAX. 290-340	S CD	AUSTENITIC (NON-MAGNETIC) STAINLESS STEEL FOR KITCHEN WARES, CHEMICAL HANDLING EQUIP-MENT, WASHING M/C, DOOR BARS, BOLTS, NUTS, WAHSERS, ETC	1200	950-1150 S QUICKLY AIR/WATER/ OIL COOL	---	---	---

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ISS NO.	NEW	OLD	C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J)			HOT FORMING TEMP. °C	ANNEALING TEMP. °C	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZ-ING TEMP.°C
6529 (CONTD.)	X07Cr18 Ni9 PINK WITH AL. PAINT DOT AND	07Cr18 Ni9	0.12	2.00	1.00	0.03	0.045	17.00-19.00	8.00-10.00	---	---	55 85	22 50	40 12	212 MAX. 290-340	S CD	AUSTENITIC (NON-MAGNETIC) STAINLESS STEEL FOR AIRCRAFT, AUTOMOBLE AND REFRIGERATOR PARTS, RADAR & MICROWAVE ANTENNAS AND HOUSEHOLD APPLIANCES.	1200	950-1150 S QUICKLY AIR/WATER/ OIL COOL	---	---	---
	X04Cr18 Ni9 PINK WITH AL. PAINT DOT	04Cr18 Ni9	0.08	2.00	1.00	0.03	0.045	17.00-19.00	8.00-10.00	---	---	55 85	22 50	40 12	212 MAX. 290-340	S CD						
	X10Cr17 Ni7 PINK WITH BLUE AND ORANGE DOT	10Cr17 Ni7	0.15	2.00	1.00	0.03	0.045	16.00-18.00	6.00-8.00	---	---	60 85	24 50	40 12	212 MAX. 290-340	S CD	AUSTENITIC (NON-MAGNETIC) STAINLESS STEEL FOR AIRCRAFT STRUCTURALS, RAILROAD CARS. FASTENERS, AUTOMOBILE PARTS, TRIMS, ETC.	1200	950-1150 S QUICKLY AIR/WATER/ OIL COOL	---	---	---
	X04Cr18 Ni10Ti PINK WITH WHITE DOT	04Cr18 Ni10Ti20	0.08	2.00	1.00	0.03	0.045	17.00-19.00	9.00-12.00	---	5C-0.8 Ti	50 85	21 50	35 12	192 MAX. 290-340	S CD	AUSTENITIC (NON-MAGNETIC) STAINLESS STEEL FOR AIRCRAFT ENGINE EXHAUST MANIFOLDS, BOILER SHIELDS, HIGH TEMP. CHEMICAL PLANTS, FASTENERS, ETC.	1200	950-1150 S QUICKLY AIR/WATER/ OIL COOL	---	---	---
	X04Cr18 Ni10Nb PINK WITH BLACK AND WHITE DOT	04Cr18 Ni10Nb40	0.08	2.00	1.00	0.03	0.045	17.00-19.00	9.00-12.00	---	10C-1.00 Nb	50 85	21 50	35 12	192 MAX. 290-340	S CD	SAME AS ABOVE	1200	950-1150 S QUICKLY AIR/WATER/ OIL COOL	---	---	---
	X04Cr17 Ni12Mo2 PINK WITH YELLOW DOT	04Cr17 Ni12Mo2	0.08	2.00	1.00	0.03	0.045	16.00-18.50	10.00-14.00	2.00-3.00	---	45 85	18 50	40 12	192 MAX. 290-340	S CD	AUSTENITIC (NON-MAGNETIC) STAINLESS STEEL FOR PARTS OF CHEMICAL, PAPER, TEXTILE AND PRINTING INDUST-RIES, DYE VATS, MARINE PARTS, ETC.	1200	950-1150 S QUICKLY AIR/WATER/ OIL COOL	---	---	---
	X02Cr17 Ni12Mo2 PINK WITH YELLOW DOT	02Cr17 Ni12Mo2	0.03	2.00	1.00	0.03	0.045	16.00-18.50	10.50-14.00	2.00-3.00	---	45 85	20 50	40 12	192 MAX. 290-340	S CD	AUSTENITIC (NON-MAGNETIC) STAINLESS STEEL FOR PARTS OF CHEMICAL. RAYON, RUBBER, MARINE, PULP HANDLING EQUIPMENTS, STEAM JACKETED KETTLES, ETC.	1200	950-1150 S QUICKLY AIR/WATER/ OIL COOL	---	---	---

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ISS NO.	NEW	OLD	C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J			HOT FORMING TEMP. °C	ANNEALING TEMP. °C	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZING TEMP °C
6529 (CONTD.)	X04Cr17 Ni12Mo2 PINK WITH WHITE DOT	04Cr17 Ni12Mo2Ti20 ORANGE AND	0.08	2.00	1.00	0.03	0.045	16.00-18.50	10.50-14.00	2.00-3.00	5C-0.8 Ti	50 65	21 50	40 12	192 MAX. 290-340	S CD	AUSTENITIC (NON-MAGNETIC) STAINLESS STEEL FOR AIRCRAFT ENGINE EXHAUST MANIFOLDS, BOILER SHIELDS, HIGH TEMP. CHEMICAL PLANTS, FASTENERS, ETC.	1200	950-1150 S QUICKLY AIR/WATER/ OIL COOL	---	---	---
	X10Cr17 Mn6Ni4N PINK WITH WHITE DOT	10Cr17 Mn6Ni4 RED AND	0.15	5.50-7.50	1.00	0.03	0.045	16.00-18.00	3.50-5.50	---	0.25 N	65 85	33 50	40 12	217 MAX. 290-340	S CD	AUSTENITIC (NON-MAGNETIC) STAINLESS STEEL FOR FLATWARE, AUTOMOBILE WHEEL COVER, TRIM, ETC.	1200	950-1150 S QUICKLY AIR/WATER/ OIL COOL	---	---	---
	X15Cr25N PINK WITH WHITE DOT	15Cr25N20 GREY AND	0.20	1.50	1.00	0.03	0.045	23.00-27.00	---	---	0.25 N	50	---	20	217 MAX.	A	AUSTENITIC (NON-MAGNETIC) STAINLESS STEEL-HIGH CHROMIUM CONTENT. SUITABLE FOR FURNACE PARTS, NOZZLES, COMBUSTION CHAMBERS, ETC.	1100	800-850 FURNACE/ AIR COOL	---	---	---
	X15Cr25Ni13 PINK WITH WHITE DOT	15Cr25Ni13 VIOLET AND	0.20	1.50	1.00	0.03	0.045	23.00-27.00	11.00-15.00	---	---	50	21	20	217 MAX.	S	AUSTENITIC (NON-MAGNETIC) STAINLESS STEEL-HIGH CHROMIUM USES SAME AS ABOVE.	1200	950-1150 S QUICKLY AIR/WATER OIL COOL	---	---	---
	X20Cr25 Ni20 PINK WITH WHITE DOT	15Cr25Ni20 BROWN AND	0.25	2.00	2.50	0.03	0.045	24.00-26.00	18.00-21.00	---	---	50	21	20	217 MAX.	S	AUSTENITIC (NON-MAGNETIC) STAINLESS STEEL-HIGH CHROMIUM USES SAME AS ABOVE.	1200	950-1150 S QUICKLY AIR/WATER OIL COOL	---	---	---
	X45Cr9Si4 PINK WITH BLUE DOT	45Cr9Si4 GREEN AND	0.40-0.50	0.30-0.60	3.25-3.75	0.03	0.045	7.50-9.50	0.50	---	---	70	40	14	225-295	H & T	SILICON CHROMIUM VALVE STEEL BEST SUITED FOR HIGH TEMP. APPLICATIONS SUCH AS OUTLET VALVES IN OIL ENGINES, LORRIES AND CARS, HIGH TEMP. NUTS & BOLTS.	1150	800-850 FURNACE COOL	1000-1050 OIL/AIR	650-850 AIR COOL/ OIL QUENCH	---
	X80Cr20Si2 Ni1 PINK WITH YELLOW DOT	80Cr20Si2 Ni1 RED AND	0.75-0.85	0.20-0.60	1.75-2.25	0.03	0.045	19.00-21.00	1.20-1.70	---	---	---	---	---	460	H & T	Cr-Si-Ni VALVE STEEL BEST SUITED FOR HIGHLY STRESSED OUTLET VALVES IN HIGH SPEED CARBU-RETTORS, FURNACE PARTS & OIL ENGINES.	1100 AIR	750 S OIL/AIR COOL	1020-1080 AIR COOL/ OIL QUENCH	700-750	---

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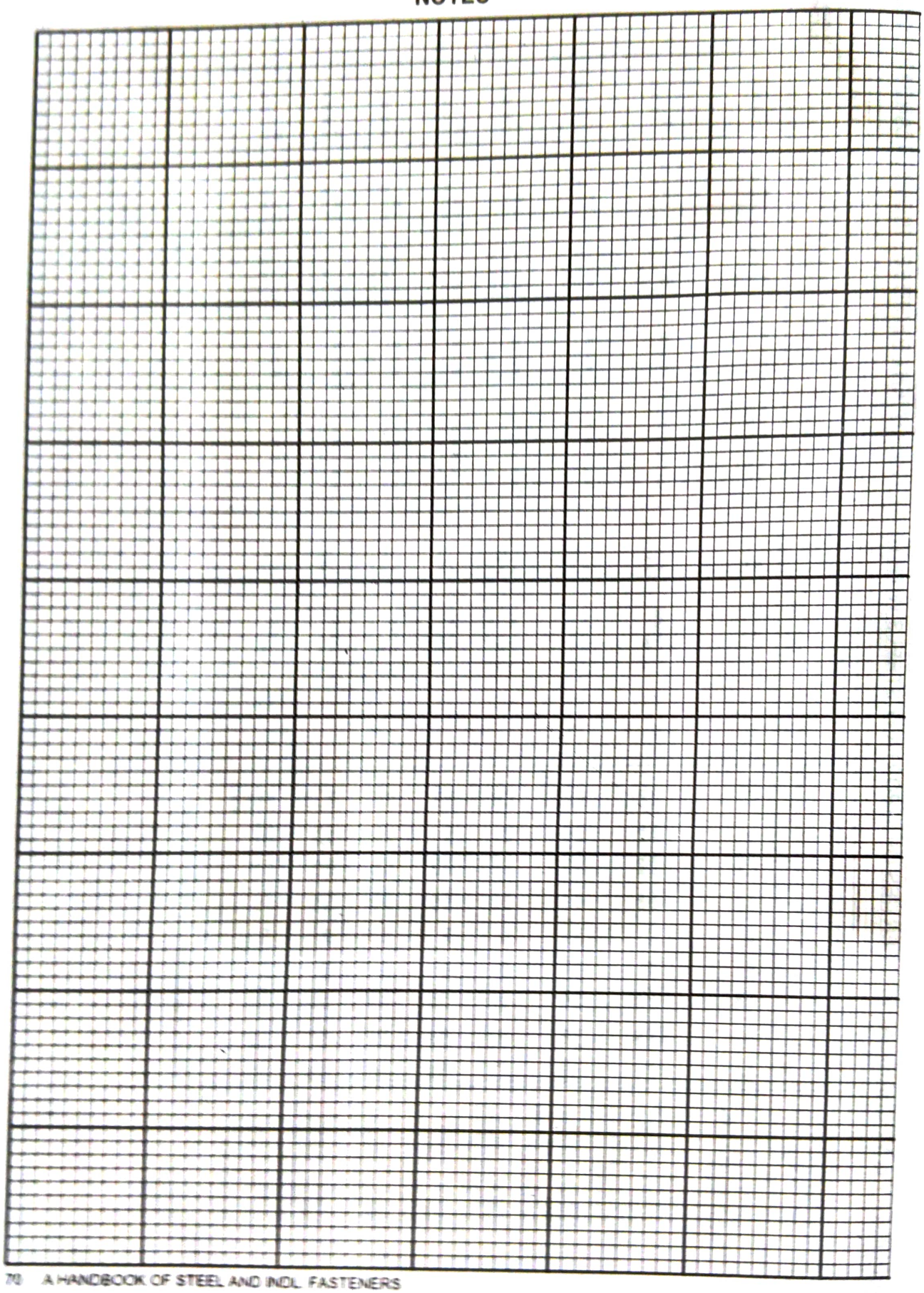
ISS NO.	NEW	OLD	C	Mn	Si	S	P	Cr	W	Mo	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J)			HOT FORMING TEMP. °C	ANNEALING TEMP. °C	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZ-ING TEMP °C
7291	XT70W18 Cr4V1 AL. PAINT WITH BROWN AND RED DOT	T70W18 Cr4V1	0.65-0.80	0.20-0.40	0.10-0.35	0.03	0.03	4.00-4.50	17.25-18.75	---	0.90-1.30V	--- ---	--- ---	--- ---	270 MAX. 684	A H & T	HIGH SPPED TOOL STEEL SUITABLE FOR MILLING AND WORM CUTTERS, REAMERS, SLITTING SAWS, DRILLS, HACKSAWS, ETC.	950-1150	850 FURNACE COOL	1260 OIL/AIR	570 (DOUBLE)	---
	XT75W18 Co6Cr4 VMo1 AL. PAINT WITH BROWN AND PINK DOT	T75W18 Co6Cr5V1 Mo75	0.70-0.80	0.20-0.40	0.10-0.35	0.03	0.03	4.00-4.50	17.50-19.00	0.50-1.00	1.00-1.50V 5.00-6.00 Co	--- ---	--- ---	--- ---	270 MAX. 684	A H & T	SAME AS ABOVE	950-1150	850 FURNACE COOL	1300 OIL/AIR	570 (DOUBLE)	---
	XT75W18 Co10Cr4 V2Mo1 AL. PAINT WITH BROWN AND VIOLET DOT	T75W18 Co10Cr4V2 Mo75	0.70-0.80	0.20-0.40	0.10-0.35	0.03	0.03	4.00-4.50	17.50-19.00	0.50-1.00	1.50-2.00V 9.00-10.00 Co	--- ---	--- ---	--- ---	270 MAX. 684	A H & T	SAME AS ABOVE	950-1150	850 FURNACE COOL	1310 OIL/AIR	570 (DOUBLE)	---
	XT123W14 Co5V4 AL. PAINT WITH BROWN AND GREEN DOT	T123W14 Co5V4	1.15-1.30	0.20-0.40	0.10-0.35	0.03	0.03	4.00-4.50	13.00-14.50	0.60	3.70-4.20V 4.75-5.25 Co	--- ---	--- ---	--- ---	270 MAX. 684	A H & T	SAME AS ABOVE	950-1150	850 FURNACE COOL	1250 OIL/AIR	570 (DOUBLE)	---
	XT70W14 Cr4V AL. PAINT WITH BROWN AND BLUE DOT	T70W14 Cr4V75	0.65-0.75	0.20-0.40	0.10-0.35	0.03	0.03	4.00-4.50	13.50-15.00	0.60	0.50-1.00V	--- ---	--- ---	--- ---	270 MAX. 684	A H & T	SAME AS ABOVE	950-1150	850 FURNACE COOL	1280 OIL/AIR	570 (DOUBLE)	---
	XT83Mo W6Cr4V2 AL. PAINT WITH YELLOW DOT	T83Mo W6Cr4V2	0.75-0.90	0.20-0.40	0.10-0.35	0.03	0.03	3.75-4.50	5.50-6.50	5.50-6.50	1.75-2.00V	--- ---	--- ---	--- ---	270 MAX. 684	A H & T	SAME AS ABOVE	950-1150	850 FURNACE COOL	1210 OIL/AIR	570 (DOUBLE)	---
	XT81W6Co Mo5Cr4V2 AL. PAINT WITH YELLOW AND RED DOT	T81W6Co Mo5Cr4V2	0.77-0.85	0.20-0.40	0.10-0.35	0.03	0.03	3.75-4.50	5.50-6.75	4.00-5.50	1.90-2.30V 4.75-5.25 Co	--- ---	--- ---	--- ---	270 MAX. 684	A H & T	SAME AS ABOVE	950-1150	850 FURNACE COOL	1210 OIL/AIR	570 (DOUBLE)	---
	XT110Mo10 Co8Cr4W1 AL. PAINT WITH YELLOW AND GREEN DOT	T110Mo10 Co8Cr4W1	1.05-1.15	0.15-0.40	0.10-0.35	0.03	0.03	3.50-4.50	1.15-1.85	9.00-10.00	0.95-1.35V 7.75-8.75 Co	--- ---	--- ---	--- ---	270 MAX. 684	A H & T	SAME AS ABOVE	950-1150	850 FURNACE COOL	1210 OIL/AIR	570 (DOUBLE)	---

*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED.

ABBREVATIONS USED
AR – AS ROLLED
HR – HOT ROLLED
CR – COLD ROLLED
AC – AIR COOLED
CD – COLD DRAWN
OH – OIL HARDENED
N – NORMALISED
AH – AIR HARDENED
A – ANNEALED
H & T – HARDENED & TEMPERED
F & I – FLAME & INDUCTION HARDENING
S – SOFENED

NOTES



CHEMISTRY AND MECHANICAL PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES OF SPECIAL PURPOSE STEELS AT A GLANCE

CHEMISTRY & MECH.PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - SPECIAL PURPOSE ALLOYS

DESCRIPTION OF SPECIAL PURPOSE ALLOY STEEL	* CHEMICAL COMPOSITION, PERCENT								MECH ANICAL PROPERTIES (MIN.)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
	Ni	Ti	Al	Nb	Co	Mo	REST	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J			HOT FORMING TEMP. °C	ANNEALING TEMP. °C	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZ-ING TEMP. °C
MARAGING STEELS	18.00-20.00	1.30-1.60	0.15-0.35	0.30-0.50	---	---	---	---	173.2-196.0	168.6-190.6	10-13	---	AGED	THESE STEELS ARE USED FOR NAVAL PURPOSES ,SPACE VEHICLES WHERE HIGH STRENGTH, FORMABILITY AND MODERATE CORRO-SION RESISTANCE ARE OF PRIME IMPORTANCE.	---	---	---	---	---
	25.00-26.00	1.30-1.60	0.15-0.35	0.30-0.50	---	---	---	---	182.6-204.8	168.6-192.2	10-15	---	AGED	SAME AS ABOVE	---	---	---	---	---
	17.00-19.00	0.15-0.25	0.05-0.15	---	8.90	3.00-3.50	---	---	140.2-154.4	133.8-148.0	14-16	---	AGED	SAME AS ABOVE	---	---	---	---	---
	18.00-19.00	0.50-0.70	0.05-0.15	---	8.50-9.50	4.70-5.20	---	---	211.0-217.4	207.8-214.2	12	---	AGED	SAME AS ABOVE	---	---	---	---	---

*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED.

DESCRIPTION OF SPECIAL PURPOSE ALLOY STEEL	* CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
	C	Cr	Ni	Mo	Al	Mn	Si	Cu	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J			HOT FORMING TEMP. °C	ANNEALING TEMP. °C	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZ-ING TEMP. °C
PH STAINLESS STEELS 17-4 PH ALLOY TYPE- MARTENSITIC	0.04	16.50	4.00	---	---	1.00	1.00	2.75	---	PLEASE SEE BELOW										
17-7 PH ALLOY TYPE- SEMI AUSTENITIC	0.07	17.00	7.00	---	1.15	0.60	0.40	---	---	PLEASE SEE BELOW										
PH 15-7 Mo ALLOY TYPE- SEMI AUSTENITIC	0.09	15.00	7.00	2.50	1.00	1.00	1.00	---	---	PLEASE SEE BELOW										
AM 350 ALLOY TYPE- SEMI AUSTENITIC	0.10	16.50	4.30	2.75	---	0.80	0.25	---	0.1 N	PLEASE SEE BELOW										
AM 355 ALLOY TYPE- SEMI AUSTENITIC	0.13	15.50	4.30	2.75	---	0.95	0.25	---	0.1 N	PLEASE SEE BELOW										

*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED.

APPLICATION AND PROPERTIES:

THESE ARE HIGH STRENGTH STAINLESS MATERIALS FOR APPLICATIONS AT TEMP. UPTO 540° C. HIGHER HARDNESS IS DEVELOPED DUE TO THE PRECIPITATION EFFECT. THESE STEELS FIND APPLICATIONS IN AIRCRAFTS AND MISSILE INDUSTRIES AS MATERIAL FOR SKINS, NIBS, BULKHEADS AND OTHER STRUCTURAL COMPONENTS.

HEAT TREATMENT AND MECHANICAL PROPERTIES:

ALLOYS OF MARTENSITIC GROUP ARE NORMALLY SOLUTION ANNEALED AT 1040° C AND AIR COOLED WITH A RESULTANT TRANSFORMATION OF AUSTENITE TO MARTENSITE. SUBSEQUENT AGING AT TEMP. OF 450 °C TO 560 °C CAUSES PRECIPITATION. AFTER AGING AT 480 °C, YIELD STRENGTH IS 125 Kg/mm² AND TENSILE STRENGTH IS 136 Kg/mm² WITH 13% ELONGATION. THESE ALLOYS ARE MORE COLD FORMING IN ANNEALED CONDITION.

THE SEMI-AUSTENITIC ALLOYS ARE SUBDIVIDED INTO TWO GROUPS: 17-7 PH & PH 15-7 Mo WHICH RESPOND TO PRECIPITATION HARDENING AND AM 350 & AM 355 WHICH, THOUGH CLASSIFIED IN THIS GROUP, CANNOT BE ACTUALLY HARDENED BY PRECIPITATION MECHANISM. ALL THE ALLOYS OF THIS GROUP ARE SOLUTION ANNEALED AT 1060 ° C AND PROVIDE MAXIMUM FORMABILITY. THE SUBSEQUENT HARDENING IS DONE BY SUB-ZERO COOLING TREATMENT RESULTING IN HIGHER STRENGTH OR A DOUBLE AGING TREATMENT THAT PRODUCES IMPROVED DUCTILITY.

CHEMISTRY & MECH.PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - SPECIAL PURPOSE ALLOYS

DESCRIPTION OF SPECIAL PURPOSE ALLOY STEEL		* CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
		C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J)			HOT FORMING TEMP. °C	ANNEALING TEMP. °C	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZING TEMP. °C
HADFIELDSTEELS		1.00-1.50	11.00-14.00	---	0.05	0.05	---	---	---	---	90	34.50	40	170	H & T	PLEASE SEE BELOW.	---	---	1065 WATER	---	---
																THIS AUSTENITIC STEEL HAS ABILITY TO WORK HARDEN AND PROBABLY HAS NO EQUAL IN THIS RESPECT. AFTER WATER QUENCHING, IT IS AUSTENITIC AND WORK HARDENABLE BUT BECOMES MARTENSITIC AND BRITTLE IF TEMPERED ABOVE 300 °C. REPEATED IMPACT ON THE WATER QUENCHED PARTS INCREASE THE HARDNESS CONSIDERABLY AND, HENCE, SUCH STEELS ARE USED FOR ALL PARTS SUBJECT TO EXCESSIVE WEAR AND ABRASION. TYPICAL USES BEING: JAWS AND PARTS OF CRUSHING MACHINES, POINTS AND CROSSINGS OF RAILS, BULLET PROOF HELMETS.					
HEAT RESISTANT ALLOYS	INCONEL X-750	---	---	---	---	---	15.00	74.00	---	2.50 Ti 0.60 Al 0.80 Nb 7.10 Fe	113.90	64.70	24	281	FULLY HEAT TREATED (AGE HARDENED)	PLEASE SEE BELOW.	---	---	---	---	---
																AGE HARDENED ALLOY HAVING HIGH STRENGTH, RESISTANCE TO CORROSION UPTO ABOUT 800 °C AND GOOD CREEP STRENGTH; FINDS ITS APPLICATIONS IN GAS TURBINE BLADES, COMBUSTION LINERS IN JET ENGINES, TAIL PIPES, MESH BELT CONVEYERS FOR FURNACES, ETC.					
	HASTELLOY B	---	1.00	1.00	---	---	---	65.00	28.00	5.00 Fe	55.10 95.30	39.40 43.30	8 42	210 230	CAST ROLLED	THIS ALLOY POSSESS HIGH STRENGTH , OXIDATION AND CORROSION RESISTANCE AT HIGH TEMP.	---	---	---	---	---
	HASTELLOY C	---	1.00	1.00	---	---	16.00	75.00	17.00	6.00 Fe 4.00 W	53.50 87.40	32.30 41.70	12 35	195 185	CAST ROLLED	THIS ALLOY POSSESS HIGH STRENGTH , OXIDATION AND CORROSION RESISTANCE AT HIGH TEMP.	---	---	---	---	---
	HASTELLOY D	---	1.00	9.00	---	---	1.00	84.00	---	1.00 Fe 4.00 Cu	26.80	26.80	---	525	CAST	THIS ALLOY POSSESS HIGH STRENGTH , OXIDATION AND CORROSION RESISTANCE AT HIGH TEMP.	---	---	---	---	---
	NIMONIC - 75	---	---	---	---	---	20.00	79.20	---	0.40 Ti	81.90	36.20	44	200	HOT ROLLED AND ANNEALED	AGE HARDENING ALLOY : CAN BE USED UPTO 750 °C AND SUITABLE FOR GAS TURBINE FRAMES, TUBES, FURNACE COMBUSTION CHAMBERS, ETC.	---	---	---	---	---
	NIMONIC - 80A	---	---	---	---	---	20.00	76.00	---	2.50 Ti 1.50 Al	108.70	61.40	39	245	FULLY HEAT TREATED	AGE HARDENING ALLOY : CAN BE USED UPTO 850 °C AND SUITABLE FOR GAS TURBINE FRAMES, TUBES, FURNACE COMBUSTION CHAMBERS, ETC.	---	---	---	---	---
	NIMONIC - 90	---	---	---	---	---	20.00	55.50	---	2.70 Ti 1.80 Al 20.00 Co	126.00	81.90	33	245	FULLY HEAT TREATED	AGE HARDENING ALLOY : CAN BE USED UPTO 870 °C AND SUITABLE FOR GAS TURBINE FRAMES, TUBES, FURNACE COMBUSTION CHAMBERS, ETC.	---	---	---	---	---

*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED.

CHEMISTRY AND MECHANICAL PROPERTIES

OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES

OF U. S. STANDARD GRADES STEELS AT A GLANCE

CHEMISTRY & MECH.PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - US STANDARD GRADES

UNS NO.	DESIGNATION SAE/AISI NO.	* CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
UNALLOYED STEELS		C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J)			HOT FORMING TEMP. °C	ANNEALING TEMP. °C (FC)	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZ-ING TEMP. °C
G10050	1005	0.06	0.35	0.10-0.35	0.05	0.04	---	---	---	---	35	---	28	130	A & N	MOSTLY USED IN ELECTRICAL IND.	1250	900-930	---	---	900-930
G10060	1006	0.08	0.25-0.40	0.10-0.35	0.05	0.04	---	---	---	---	35	---	28	130	A & N	MOSTLY USED IN ELECTRICAL IND.	1250	900-930	---	---	900-930
G10080	1008	0.10	0.50	0.10-0.35	0.05	0.04	---	---	---	---	35	---	28	135	A & N	MOSTLY USED FOR GENERAL ENGG. CONSTRUCTION SUBJECT TO LIGHT STRESSES.	1250	900-930	---	---	900-930
G10090	1009	0.15	0.60	0.15-0.35	0.05	0.04	---	---	---	---	35	---	28	130	A & N	MOSTLY USED IN AUTOMOBILE BODY AND TENDER STOCK, HOODS, FRAMES, PANELS, ETC.	1250	900-930	---	---	900-930
G10100	1010	0.08-0.13	0.30-0.60	0.15-0.35	0.05	0.04	---	---	---	---	35	---	28	130	A & N	SAME AS ABOVE.	1250	900-930	---	---	900-930
G10120	1012	0.10-0.15	0.30-0.60	0.15-0.35	0.05	0.04	---	---	---	---	35 32	--- ---	26 28	130 130	A & N CD & A	MOSTLY USED FOR CAMS, LIGHT DUTY GEARS, GUDGEON PINS, FORKS, WORMS COLLETS, ETC.	1200	880-930	CARBURI-ZING AT 880-930 FOLLOWED BY HARDEN-ING AT 760-780 WATER	150-200	900-930
G10130	1013	0.11-0.18	0.50-0.80	0.15-0.35	0.05	0.04	---	---	---	---	35 32	--- ---	26 28	130 130	A & N CD & A	SAME AS ABOVE	1200	880-930	SAME AS ABOVE	150-200	900-930
G10150	1015	0.12-0.18	0.30-0.60.	0.15-0.35	0.05	0.04	---	---	---	---	35 32	--- ---	26 28	130 130	A & N CD & A	MOSTLY USED FOR SHAFTS, CAM SHAFTS, LIGHT DUTY GEARS, WORMS, SPINDLES, RATCHETS, ETC.	1200	880-930	SAME AS ABOVE	150-200	900-930
G10160	1016	0.12-0.18	0.60-0.90	0.15-0.35	0.05	0.04	---	---	---	---	35 32	--- ---	26 28	130 130	A & N CD & A	SAME AS ABOVE	1200	880-930	SAME AS ABOVE	150-200	900-930
G10170	1017	0.14-0.20	0.30-0.60	0.15-0.35	0.05	0.04	---	---	---	---	37 32	--- ---	25 25	135 135	A & N CD & A	MOSTLY USED FOR SHELLS FOR NON-FERROUS BEARINGS, PRESSINGS, WELDING AND FABRICATED SECTIONS, FRAMES.	1200	880-930	SAME AS ABOVE	150-200	900-930
G10180	1018	0.14-0.20	0.60-0.90	0.15-0.35	0.05	0.04	---	---	---	---	40 32	--- ---	25 25	140 135	HR & N CD & A	MOSTLY USED FOR STAY BOLTS, BRAKE HOUSING, AUTO WHEEL HUBS, BRAKE PEDAL LEVERS, ETC.	1200	880-920	SAME AS ABOVE	150-200	880-920

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ABBREVATIONS USED:
AR -- AS ROLLED; HR -- HOT ROLLED; CR -- COLD ROLLED; AC -- AIR COOLED; CD -- COLD DRAWN; OH -- OIL HARDENED; N -- NORMALISED; AH -- AIR HARDENED; A -- ANNEALED; H & T -- HARDENED & TEMPERED; F & I -- FLAME & INDUCTION HARDENING; S -- SOFTENED; FC -- FURNACE COOL

CONTD.

CHEMISTRY & MECH.PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - US STANDARD GRADES

DESIGNATION UNS NO.	SAE/AISI NO.	* CHEMICAL COMPOSITION, PERCENT									MECHANICAL PROPERTIES (MIN.)				CONDITION/ NOMINAL THK./DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
UNALLOYED STEELS (CONTD.)		C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E%	HARDNESS (BHN)/IZOD(J)			HOT FORMING TEMP. °C	ANNEALING TEMP. °C (FC)	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZING TEMP. °C
G10190	1019	0.14-0.20	0.70-1.00	0.15-0.35	0.05	0.04	---	---	---	---	40 32	--- ---	25 25	140 135	HR & N CD & A	SUITABLE FOR SHAFTS, STAY BOLTS, WHEEL HUBS, GEAR SELECTORS, CLUTCH & BRAKE HOUSINGS, ETC.	1200	880-920	CARBURIZING AT 880-930 FOLLOWED BY HARDENING AT 760-780 WATER	150-200	880-920 AIR COOL
G10200	1020	0.17-0.23	0.30-0.60	0.15-0.35	0.05	0.04	---	---	---	---	40 32	--- ---	25 25	140 135	HR & N CD & A	SAME AS ABOVE	1200	880-920	SAME AS ABOVE	150-200	880-920 AIR COOL
G10210	1021	0.17-0.23	0.60-0.90	0.15-0.35	0.05	0.04	---	---	---	---	44 40	--- ---	24 25	140 135	HR & N CD & A	SAME AS ABOVE	1200	880-920	SAME AS ABOVE	150-200	880-920 AIR COOL
G10220	1022	0.17-0.23	0.70-1.00	0.15-0.35	0.05	0.04	---	---	---	---	44 40	--- ---	24 25	140 140	HR & N CD & A	SAME AS ABOVE	1200	880-920	SAME AS ABOVE	150-200	880-920 AIR COOL
G10230	1023	0.19-0.25	0.30-0.60	0.15-0.35	0.05	0.04	---	---	---	---	44 40	--- ---	25 25	140 135	HR & N CD & A	SAME AS ABOVE	1200	880-920	SAME AS ABOVE	150-200	880-920 AIR COOL
G10250	1025	0.22-0.28	0.30-0.60	0.15-0.35	0.05	0.04	---	---	---	---	44	---	23	140	HR & N	SAME AS ABOVE	1200	860-900	SAME AS ABOVE	150-200	860-900 AIR COOL
G10260	1026	0.22-0.28	0.60-0.90	0.15-0.35	0.05	0.04	---	---	---	---	44	---	23	140	HR & N	SAME AS ABOVE	1200	860-900	SAME AS ABOVE	150-200	860-900 AIR COOL
G10290	1029	0.25-0.31	0.60-0.90	0.15-0.35	0.05	0.04	---	---	---	---	50	---	21	160	HR & N	SAME AS ABOVE	1200	860-900	---	---	860-900 AIR COOL
G10300	1030	0.27-0.34	0.60-0.90	0.15-0.35	0.05	0.04	---	---	---	---	50 55	--- ---	21 20	160 165	HR & N H & T	GENERALLY USED FOR SOCKETS, TIE RODS, SPROCKETS, LEVERS, CAMS, PIPES FOR AUTOMOBILES, BICYCLE AND FURNITURES.	1200	860-890	860-890 OIL OR WATER	550-660	860-890 COOL IN AIR
G10330	1033	0.29-0.36	0.70-1.00	0.15-0.35	0.05	0.04	---	---	---	---	50 55	--- ---	21 20	160 165	HR & N H & T	SAME AS ABOVE	1200	860-890	860-890 OIL OR WATER	550-660	860-890 COOL IN AIR
G10350	1035	0.31-0.38	0.60-0.90	0.15-0.35	0.05	0.04	---	---	---	---	50 55	--- ---	21 20	160 165	HR & N H & T	SAME AS ABOVE	1200	860-890	860-890 OIL OR WATER	550-660	860-890 COOL IN AIR
G10370	1037	0.31-0.38	0.70-1.00	0.15-0.35	0.05	0.04	---	---	---	---	55 60	--- ---	20 18	170 175	HR & N H & T	SAME AS ABOVE AND ALSO FOR RAILWAY COUPLING, HAND TOOLS, ETC. CAN BE FLAME OR INDUCTION HARDNESS.	1200	860-890	860-890 OIL OR WATER	550-660	860-890 COOL IN AIR

*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED.

CONTD.

ABBREVATIONS USED:
AR -- AS ROLLED
HR -- HOT ROLLED
CR -- COLD ROLLED
AC -- AIR COOLED
CD -- COLD DRAWN
OH -- OIL HARDENED
N -- NORMALISED
AH -- AIR HARDENED
A -- ANNEALED
H & T -- HARDENED & TEMPERED
F & I -- FLAME & INDUCTION HARDENING
S -- SOFTENED
FC -- FURNACE COOL

CHEMISTRY & MECH.PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - US STANDARD GRADES

UNS NO.	DESIGNATION SAE/AISI NO.	* CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
		C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J)			HOT FORMING TEMP. °C	ANNEALING TEMP. °C (FC)	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZING TEMP. °C
UNALLOYED STEELS (CONTD.)																					
G10390	1039	0.34-0.42	0.60-0.90	0.15-0.35	0.05	0.04	---	---	---	---	55 60	--- ---	20 18	170 175	HR & N H & T	SAME AS AISI-1030 AND ALSO FOR RAILWAY COUPLING, HAND TOOLS, ETC. CAN BE FLAME OR INDUCTION HARDNESS.	1200	840-880	840-880 OIL OR WATER	550-660	840-880 COOL IN AIR
G10400	1040	0.36-0.44	0.70-1.00	0.15-0.35	0.05	0.04	---	---	---	---	58 63	--- ---	18 18	180 187	HR & N H & T	SAME AS AISI-1038 AND ALSO FOR CRANK SHAFTS, AXLES, SPINDLES, LIGHTLY STRESSED GEARS, ETC.	1200	840-880	840-880 OIL OR WATER	550-660	840-880 COOL IN AIR
G10410	1041	0.36-0.44	0.60-0.90	0.15-0.35	0.05	0.04	---	---	---	---	58 63	--- ---	18 18	180 187	HR & N H & T	SAME AS ABOVE	1200	840-880	840-880 OIL OR WATER	550-660	840-880 COOL IN AIR
G10420	1042	0.39-0.47	0.60-0.90	0.15-0.35	0.05	0.04	---	---	---	---	58 63	--- ---	18 18	180 187	HR & N H & T	SAME AS ABOVE	1200	840-880	840-880 OIL OR WATER	550-660	840-880 COOL IN AIR
G10430	1043	0.39-0.47	0.70-1.00	0.15-0.35	0.05	0.04	---	---	---	---	60 66	--- ---	17 17	190 200	HR & N H & T	SAME AS ABOVE	1200	840-880	840-880 OIL OR WATER	550-660	840-880 COOL IN AIR
G10440	1044	0.43-0.50	0.30-0.60	0.15-0.35	0.05	0.04	---	---	---	---	60 66	--- ---	17 17	190 200	HR & N H & T	SAME AS ABOVE	1200	840-880	840-880 OIL OR WATER	550-660	840-880 COOL IN AIR
G10450	1045	0.42-0.50	0.60-0.90	0.15-0.35	0.05	0.04	---	---	---	---	60 70	--- ---	15 15	200 210	HR & N H & T	SUITABLE FOR SPINDLES , LARGER GEARS, BOLTS, NUTS, SHAFTS, SPROCKETS.	1200	840-880	840-880 OIL OR WATER	550-660	840-880 COOL IN AIR
G10460	1046	0.42-0.50	0.70-1.00	0.15-0.35	0.05	0.04	---	---	---	---	60 70	--- ---	15 15	200 210	HR & N H & T	SUITABLE FOR SPINDLES , LARGER GEARS, BOLTS, NUTS, SHAFTS, SPROCKETS.	1200	840-880	840-880 OIL OR WATER	550-660	840-880 COOL IN AIR
G10490	1049	0.46-0.56	0.60-0.90	0.15-0.35	0.05	0.04	---	---	---	---	63 70	--- ---	13 13	230 230	HR & N H & T	SUITABLE FOR CYLINDERS, SMALL GEARS, BOLTS, NUTS, CAMS, SPROCKETS, KEYS, ETC.	1150	810-840	810-840 OIL OR WATER	550-660	810-840 COOL IN AIR
G10500	1050	0.47-0.55	0.60-0.90	0.15-0.35	0.05	0.04	---	---	---	---	63 70	--- ---	13 13	230 230	HR & N H & T	SAME AS ABOVE	1150	810-840	810-840 OIL OR WATER	550-660	810-840 COOL IN AIR
G10520	1052	0.46-0.56	1.10-1.40	0.15-0.35	0.05	0.04	---	---	---	---	72 65	--- ---	15 13	230 230	HR & N H & T	MOSTLY USED AS RAIL STEEL. CAN BE USED FOR CYLINDER LINERS, GEAR SHAFTS, ETC.	1100	810-840	810-840 OIL OR WATER	550-660	810-840 COOL IN AIR

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HR -- HOT ROLLED
CR -- COLD ROLLED
AC -- AIR COOLED
CD -- COLD DRAWN
OH -- OIL HARDENED
N -- NORMALISED
AH -- AIR HARDENED
A -- ANNEALED
H & T -- HARDENED & TEMPERED
F & I -- FLAME & INDUCTION HARDENING
S -- SOFTENED
FC -- FURNACE COOL

CONTD.

CHEMISTRY & MECH.PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - US STANDARD GRADES

UNS NO.	DESIGNATION SAE/AISI NO.	* CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
UNALLOYED STEELS (CONTD.)		C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J)			HOT FORMING TEMP. °C	ANNEALING TEMP. °C (FC)	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZ-ING TEMP. °C
G10520	1052	0.45-0.55	1.10-1.40	0.15-0.35	0.05	0.04	---	---	---	---	72 85	--- ---	15 13	230 230	HR & N H & T	MOSTLY USED AS RAIL STEEL. CAN BE USED FOR CYLINDER LINERS, CRANSHAFTS, ETC.	1100	810-840	810-840 OIL OR WATER	550-660	810-840 COOL IN AIR
G10530	1053	0.48-0.55	0.70-1.00	0.15-0.35	0.05	0.04	---	---	---	---	72 85	--- ---	15 13	230 230	HR & N H & T	SAME AS ABOVE	1100	810-840	810-840 OIL OR WATER	550-660	810-840 COOL IN AIR
G10550	1055	0.52-0.60	0.60-0.90	0.15-0.35	0.05	0.04	---	---	---	---	71 78	--- ---	15 15	230 230	HR & N H & T	MOSTLY USED FOR SPROCKETS, CAMS, SPRINGS, CYLINDERS, SMALL GEARS, KEYS. CAN BE SURFACE HARDENED BY FLAME OR INDUC-TION HEATING	1150	810-840	810-840 OIL OR WATER	550-660	810-840 COOL IN AIR
G10590	1059	0.55-0.65	0.50-0.80	0.15-0.35	0.05	0.04	---	---	---	---	72 78	--- ---	13 13	230 230	HR & N H & T	MOSTLY USED FOR AXLES, PINIONS, SPINDLES OF CRANK SHAFTS, SPRINGS. CAN BE SURFACE HARDENED BY FLAME OR INDUC-TION HEATING	1150	810-840	810-840 5 OIL OR WATER	50-660	810-840 COOL IN AIR
G10600	1060	0.55-0.66	0.60-0.90	0.15-0.35	0.05	0.04	---	---	---	---	72 78	--- ---	13 13	230 230	HR & N H & T	SAME AS ABOVE	1150	810-840	810-840 OIL OR WATER	550-660	810-840 COOL IN AIR
G10640	1064	0.59-0.70	0.50-0.80	0.15-0.35	0.05	0.04	---	---	---	---	MECH. PROPERTIES NOT SPECIFIED					USED IN HIGH TENSILE STRUCTURES, SPRINGS, WASHERS, STAMPED PARTS	1150	810-840	810-840 OIL OR WATER	550-660	810-840 COOL IN AIR
G10650	1065	0.59-0.70	0.60-0.90	0.15-0.35	0.05	0.04	---	---	---	---	MECH. PROPERTIES NOT SPECIFIED					SAME AS ABOVE	1150	810-840 OIL OR WATER	810-840	400-500	810-840 COOL IN AIR
G10690	1069	0.65-0.75	0.40-0.70	0.15-0.35	0.05	0.04	---	---	---	---	MECH. PROPERTIES NOT SPECIFIED					SAME AS ABOVE USED AS SPRING STEEL	1150	810-840	810-840 OIL OR WATER	400-500	810-840 COOL IN AIR
G10700	1070	0.65-0.76	0.60-0.90	0.15-0.35	0.05	0.04	---	---	---	---	MECH. PROPERTIES NOT SPECIFIED					SAME AS ABOVE USED AS SPRING STEEL	1150	810-840	810-840 OIL OR WATER	400-500	810-840 COOL IN AIR
G10740	1074	0.69-0.80	0.50-0.80	0.15-0.35	0.05	0.04	---	---	---	---	MECH. PROPERTIES NOT SPECIFIED					USED AS SPRING STEEL FOR FLAT SPRINGS, PLOUGH SHARES, RAKE TEETH, SCRAPERS AND CULTIVATOR SHOVELS.	1150	810-840	810-840 OIL OR WATER	400-500	810-840 COOL IN AIR

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A -- ANNEALED
H & T -- HARDENED & TEMPERED
F & I -- FLAME & INDUCTION HARDENING
S -- SOFTENED
FC -- FURNACE COOL

CONTD.

CHEMISTRY & MECH.PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - US STANDARD GRADES

DESIGNATION UNS NO.	SAE/AISI NO.	* CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)	CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
UNALLOYED STEELS (CONTD.)		C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS YS E % HARDNESS Kg/mm² (BHN) / IZOD(J			HOT FORMING TEMP. °C	ANNEALING TEMP. °C (FC)	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZ-ING TEMP °C
G10750	1075	0.69-0.80	0.40-0.70	0.15-0.35	0.05	0.04	---	---	---	---	MECH. PROPERTIES NOT SPECIFIED		USED AS SPRING STEEL FOR FLAT SPRINGS, PLOUGH SHARES, RAKE TEETH, SCRAPERS AND CULTIVATOR SHOVELS.	1150	810-840	810-840 OIL OR WATER	400-500	810-840 COOL IN AIR
G10780	1078	0.72-0.86	0.30-0.60	0.15-0.35	0.05	0.04	---	---	---	---	MECH. PROPERTIES NOT SPECIFIED		SAME AS ABOVE	1150	810-840	810-840 OIL OR WATER	400-500	810-840 COOL IN AIR
G10800	1080	0.74-0.88	0.60-0.90	0.15-0.35	0.05	0.04	---	---	---	---	MECH. PROPERTIES NOT SPECIFIED		SAME AS ABOVE ALSO USED FOR HAND TOOLS, SAWS, KNIVES, CLIPS,ETC.	1150	810-840	810-840 OIL OR WATER	400-500	810-840 COOL IN AIR
G10840	1084	0.80-0.94	0.60-0.90	0.15-0.35	0.05	0.04	---	---	---	---	MECH. PROPERTIES NOT SPECIFIED		SAME AS ABOVE ALSO USED FOR HAND TOOLS, SAWS, KNIVES, CLIPS, MUSICWIRES, ETC.	1150	810-840 8	10-840 OIL OR WATER	400-500	810-840 COOL IN AIR
G10850	1085	0.80-0.94	0.70-1.00	0.15-0.35	0.05	0.04	---	---	---	---	MECH. PROPERTIES NOT SPECIFIED		SAME AS ABOVE	1150	760-840	760-840 OIL OR WATER	400-500	810-840 COOL IN AIR
G10860	1086	0.80-0.94	0.30-0.50	0.15-0.35	0.05	0.04	---	---	---	---	MECH. PROPERTIES NOT SPECIFIED		SAME AS ABOVE	1150	760-840	760-840 OIL OR WATER	400-500	810-840 COOL IN AIR
G10900	1090	0.84-0.98	0.60-0.90	0.15-0.35	0.05	0.04	---	---	---	---	MECH. PROPERTIES NOT SPECIFIED		SAME AS ABOVE ALSO USED FOR FEELER GAUGES, MACHINE KNIVES, CHISELS, CUTTING AND HAND TOOLS	1150	810-920	780-820 OIL OR WATER	400-500	810-840 COOL IN AIR
G10950	1095	0.90-1.04	0.30-0.50	0.15-0.35	0.05	0.04	---	---	---	---	MECH. PROPERTIES NOT SPECIFIED		SAME AS ABOVE ALSO USED FOR FEELER GAUGES, MACHINE KNIVES, CHISELS, CUTTING AND HAND TOOLS	1150	810-920	780-820 OIL OR WATER	400-500	810-840 COOL IN AIR

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CONTD.

CHEMISTRY & MECH.PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - US STANDARD GRADES

UNS NO.	DESIGNATION SAE/AISI NO.	*CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
UNALLOYED FREE CUTTING STEELS		C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J)			HOT FORMING TEMP. °C	ANNEALING TEMP °C (FC)	HARDENING TEMP °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZING TEMP °C
G11080	1108	0.08-0.13	0.50-0.80	0.10-0.35	0.08-0.13	0.04	---	---	---	---	37	---	24	130	HR & N	MOSTLY USED FOR SMALL PARTS TO BE CYANIDED OR CARBONITRIDED	1150	950-950	CARBURIZING AT 950-950 FOLLOWED BY HARDENING AT 700-780 OR CYANIDING/ CARBONITRIDING AT 650-820	150-200	950-950 AIR COOL
G11090	1109	0.08-0.13	0.60-0.90	0.10-0.35	0.08-0.13	0.04	---	---	---	---	37	---	24	130	HR & N	SAME AS ABOVE	1150	950-950	SAME AS ABOVE	150-200	950-950 AIR COOL
G11100	1110	0.08-0.13	0.30-0.60	0.10-0.35	0.08-0.13	0.04	---	---	---	---	35	---	28	130	HR & N	SAME AS ABOVE	1150	950-950	SAME AS ABOVE	150-200	950-950 AIR COOL
G11170	1117	0.14-0.20	1.00-1.30	0.10-0.35	0.08-0.13	0.04	---	---	---	---	44	---	21	140	HR & N	SAME AS ABOVE ALSO FOR LIGHT DUTY STUDS AND BICYCLE PARTS.	1200	870-930	SAME AS ABOVE	150-200	950-950 AIR COOL
G11180	1118	0.14-0.20	1.30-1.60	0.10-0.35	0.08-0.13	0.04	---	---	---	---	44	---	21	140	HR & N	SAME AS ABOVE	1200	870-930	SAME AS ABOVE	150-200	950-950 AIR COOL
G11260	1126	0.20-0.30	1.00-1.50	0.25	0.10-0.18	0.04	---	---	---	---	46	---	20	150	HR & N	SAME AS ABOVE	1200	870-930	SAME AS ABOVE	150-200	950-950 AIR COOL
G11370	1137	0.32-0.38	1.35-1.65	0.10-0.35	0.08-0.13	0.04	---	---	---	---	50 55	--- 30	19 18	160 160	HR & N H & T	SAME AS ABOVE ALSO FOR SHAFTS, CONNECTING RODS, AXLES, BRACKETS, HOUSINGS, ETC.	1200	850-900	850-870 OIL OR WATER	550-660	850-900 COOL IN AIR
G11390	1139	0.35-0.43	1.35-1.65	0.10-0.35	0.13-0.20	0.04	---	---	---	---	60 75	--- 50	18 14	170 150	HR & N H & T	SAME AS ABOVE	1200	850-900	850-870 OIL OR WATER	550-660	850-900 COOL IN AIR
G11400	1140	0.37-0.44	0.70-1.00	0.10-0.35	0.08-0.13	0.04	---	---	---	---	55 60	--- 35	17 16	160 170	HR & N H & T	SAME AS ABOVE	1200	850-900	850-870 OIL OR WATER	550-660	850-900 COOL IN AIR
G11410	1141	0.37-0.45	1.35-1.65	0.10-0.35	0.08-0.13	0.04	---	---	---	---	60 75	--- 50	18 14	170 180	HR & N H & T	SAME AS ABOVE	1200	850-900	850-870 OIL OR WATER	550-660	850-900 COOL IN AIR
G11440	1144	0.40-0.48	1.35-1.65	0.10-0.35	0.24-0.33	0.04	---	---	---	---	68 85	--- 65	16 12	160 230	HR & N H & T	SAME AS ABOVE ALSO SUITABLE FOR HIGH TENSILE FASTENERS (P-8.8).	1200	850-900	850-870 OIL OR WATER	550-660	850-900 COOL IN AIR

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CONTD.

CHEMISTRY & MECH.PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - US STANDARD GRADES

DESIGNATION UNS NO.	SAE/AISI NO.	* CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
		C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J)			HOT FORMING TEMP. °C	ANNEALING TEMP. °C (FC)	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZING TEMP. °C
UNALLOYED FREE CUTTINGSTEELS (CONTD.)																					
G11460	1146	0.42-0.49	0.70-1.00	0.10-0.35	0.08-0.13	0.04	---	---	---	---	70 80	--- 50	16 14	180 200	HR & N H & T	SAME AS ABOVE ALSO FOR SHAFTS, CONNECTING RODS, AXLES, BRACKETS, HOUSINGS, NUTS, BOLTS(HIGH TENSILE)	1200	850-900	850-870 OIL OR WATER	550-660	850-900 COOL IN AIR
G11510	1151	0.48-0.55	0.70-1.00	0.10-0.35	0.08-0.13	0.04	---	---	---	---	75 85	--- 65	15 12	190 230	HR & N H & T	SAME AS ABOVE	1200	850-900	850-870 OIL OR WATER	550-660	850-900 COOL IN AIR
G12110	1211	0.13	0.60-0.90	0.10-0.35	0.10-0.15	0.07-0.12	---	---	---	---	35	---	26	130	HR & N	MOSTLY USED FOR SMALL PARTS TO BE CYANIDED OR CARBONITRIDED	1150	880-950	CARBURIZING AT 880-950 FOLLOWED BY HARDENING AT 700-780 OR CYANIDING/ CARBONITRIDING AT 650-920	150-200	880-950 AIR COOL
G12120	1212	0.13	0.70-1.00	0.10-0.35	0.16-0.23	0.07-0.12	---	---	---	---	35	---	26	130	HR & N	SAME AS ABOVE	1150	880-950	SAME AS ABOVE	150-200	880-950 AIR COOL
G12130	1213	0.13	0.70-1.00	0.10-0.35	0.24-0.33	0.07-0.12	---	---	---	---	35	---	26	130	HR & N	SAME AS ABOVE	1150	880-950	SAME AS ABOVE	150-200	880-950 AIR COOL
G12150	1215	0.09	0.75-1.05	0.10-0.35	0.26-0.35	0.04-0.09	---	---	---	---	35	---	26	130	HR & N	SAME AS ABOVE	1150	880-950	SAME AS ABOVE	150-200	880-950 AIR COOL
G12144	12L14	0.15	0.85-1.15	0.10-0.35	0.26-0.35	0.04-0.09	---	---	---	0.15-0.35 Pb	40	---	24	140	HR & N	SAME AS ABOVE	1150	880-950	SAME AS ABOVE	150-200	880-950 AIR COOL
UNALLOYED STEELS WITH HIGH MANGANESE CONTENT																					
G15130	1513	0.10-0.16	1.10-1.40	0.10-0.35	0.05	0.04	---	---	---	---	35	---	26	130	HR & N	SAME AS ABOVE	1150	880-950	SAME AS ABOVE	150-200	880-950 AIR COOL
G15220	1522	0.18-0.24	1.10-1.40	0.10-0.35	0.05	0.04	---	---	---	---	50	---	20	150	HR & N	MOSTLY USED FOR AUTOMOBILE AND BICYCLE PARTS, BOLTS, NUTS OF P-4.6/4.8	1150	880-950	---	---	880-950 AIR COOL
G15240	1524	0.19-0.25	1.35-1.65	0.10-0.35	0.05	0.04	---	---	---	---	50	---	20	150	HR & N	SAME AS ABOVE	1150	880-950	---	---	880-950 AIR COOL
G15260	1526	0.22-0.29	1.10-1.40	0.10-0.35	0.05	0.04	---	---	---	---	55	---	18	160	HR & N	SAME AS ABOVE BUT BOLTS & NUTS OF P-5.6.	1150	880-950	---	---	880-950 AIR COOL

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CONTD.

CHEMISTRY & MECH.PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - US STANDARD GRADES

DESIGNATION UNS NO.	SAE/AISI NO.	* CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
UNALLOYED STEELS WITH HIGH MANGANESE CONTENT (CONTD.)		C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J)			HOT FORMING TEMP. °C	ANNEALING TEMP. °C (FC)	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZING TEMP °C
G15270	1527	0.22-0.29	1.20-1.50	0.10-0.35	0.05	0.04	---	---	---	---	60	---	18	160	HR & N	MOSTLY USED FOR AUTOMOBILE AND BICYCLE PARTS, BOLTS, NUTS OF P-5,6	1150	880-950	---	---	880-950 COOL IN AIR
G15280	1528	0.30-0.37	1.20-1.50	0.10-0.35	0.05	0.04	---	---	---	---	65 68	--- 50	17 16	165 165	HR & N H & T	MOSTLY USED FOR AUTOMOBILE AND BICYCLE PARTS, BOLTS, NUTS OF P-6.8	1150	880-950	860-890 OIL OR WATER	550-660	880-950 COOL IN AIR
G15410	1541	0.36-0.44	1.35-1.65	0.10-0.35	0.05	0.04	---	---	---	---	70 82	--- 65	16 12	175 230	HR & N H & T	MOSTLY USED FOR AUTOMOBILE AND BICYCLE PARTS, BOLTS, NUTS OF P-8.8	1150	880-950	860-890 OIL OR WATER	550-660	880-950 COOL IN AIR
G15480	1548	0.44-0.52	1.10-1.40	0.10-0.35	0.05	0.04	---	---	---	---	70 82	--- 65	16 12	175 230	HR & N H & T	MOSTLY USED FOR AUTOMOBILE AND BICYCLE PARTS, BOLTS, NUTS OF P-8.8	1150	880-950	860-890 OIL OR WATER	550-660	880-950 COOL IN AIR
G15510	1551	0.45-0.56	0.85-1.15	0.10-0.35	0.05	0.04	---	---	---	---	70 82	--- 65	16 12	175 230	HR & N H & T	MOSTLY USED FOR AUTOMOBILE AND BICYCLE PARTS, BOLTS, NUTS OF P-8.8	1150	880-950	860-890 OIL OR WATER	550-660	880-950 COOL IN AIR
G15520	1552	0.47-0.55	1.20-1.50	0.10-0.35	0.05	0.04	---	---	---	---	75 102	--- 92	14 10	210 330	HR & N H & T	MOSTLY USED FOR AUTOMOBILE AND BICYCLE PARTS, BOLTS, NUTS OF P-10.9	1150	880-950	860-890 OIL OR WATER	550-660	880-950 COOL IN AIR
G15610	1561	0.55-0.65	0.75-1.05	0.10-0.35	0.05	0.04	---	---	---	---	75 102	--- 92	14 10	210 330	HR & N H & T	MOSTLY USED FOR AUTOMOBILE AND BICYCLE PARTS, BOLTS, NUTS OF P-10.9	1150	840-880	830-850 OIL OR WATER	550-660	840-880 COOL IN AIR
G15660	1566	0.60-0.71	0.85-1.15	0.10-0.35	0.05	0.04	---	---	---	---	80 112	--- 98	12 10	270 350	HR & N H & T	MOSTLY USED FOR AUTOMOBILE AND BICYCLE PARTS, BOLTS, NUTS OF P-10.9, SPRINGS.	1150	840-880	830-850 OIL OR WATER	550-660	840-880 COOL IN AIR

*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED.

ABBREVATIONS USED:
HR – HOT ROLLED
AR – AS ROLLED
CR – COLD ROLLED
AC – AIR COOLED
CD – COLD DRAWN
OH – OIL HARDENED
N – NORMALISED
AH – AIR HARDENED
A – ANNEALED
H & T – HARDENED & TEMPERED
F & I – FLAME & INDUCTION HARDENING
S – SOFTENED
FC – FURNACE COOL

CONTD.

CHEMISTRY & MECH.PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - US STANDARD GRADES

DESIGNATION UNS NO.	SAE/AISI NO.	*CHEMICAL COMPOSITION, PERCENT									MECHANICAL PROPERTIES (MIN.)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
ALLOYED STEELS		C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E%	HARDNESS (BHN) / IZOD(J)			HOT FORMING TEMP. °C	ANNEALING TEMP. °C (FC)	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZING TEMP. °C
G13300	1330	0.28-0.33	1.60-1.90	0.15-0.35	0.04	0.035	---	---	---	---	55 64	32 45	20 18	180 187/45	HR & N H & T	SUITABLE FOR LOW DUTY PINIONS, AUTOMOBILE AND BICYCLE PARTS, STUDS, BOLTS, SCREWS, NUTS TO P-6.8 AND CONNECTING RODS, ETC.	1200	840-880	840-880 OIL OR WATER	550-660	840-880 COOL IN AIR
G13350	1335	0.33-0.38	1.60-1.90	0.15-0.35	0.04	0.035	---	---	---	---	55 64	32 45	20 18	180 187/45	HR & N H & T	SAME AS ABOVE	1200	840-880	840-880 OIL OR WATER	550-660	840-880 COOL IN AIR
G13400	1340	0.38-0.43	1.60-1.90	0.15-0.35	0.04	0.035	---	---	---	---	65 82	36 65	16 14	187 230/43	HR & N H & T	SAME AS ABOVE BUT BOLTS, NUTS TO P-8.8, GR. S	1200	840-880	840-880 OIL OR WATER	550-660	840-880 COOL IN AIR
G13450	1345	0.43-0.48	1.60-1.90	0.15-0.35	0.04	0.035	---	---	---	---	68 88	38 68	16 13	197 250/43	HR & N H & T	SAME AS ABOVE BUT BOLTS, NUTS TO P-8.8, GR. S	1200	840-880	840-880 OIL OR WATER	550-660	840-880 COOL IN AIR
G40230	4023	0.20-0.25	0.70-0.90	0.15-0.35	0.04	0.035	---	---	0.20-0.30	---	65 74	40 52	18 16	180 210/43	HR & N H & T	SAME AS ABOVE BUT BOLTS, NUTS TO P-6.8 AND CONNECTING RODS, ETC.	1200	840-880	840-880 OIL OR WATER	550-660	840-880 COOL IN AIR
G40240	4024	0.20-0.25	0.70-0.90	0.15-0.35	0.035-0.050	0.035	---	---	0.20-0.30	---	65 74	40 52	18 16	180 210/43	HR & N H & T	SAME AS ABOVE BUT BOLTS, NUTS TO P-6.8 AND CONNECTING RODS, ETC.	1200	840-880	840-880 OIL OR WATER	550-660	840-880 COOL IN AIR
G40270	4027	0.25-0.30	0.70-0.90	0.15-0.35	0.04	0.035	---	---	0.20-0.30	---	65 74	40 52	18 16	180 210/43	HR & N H & T	SAME AS ABOVE	1200	840-880	840-880 OIL OR WATER	550-660	840-880 COOL IN AIR
G40280	4028	0.25-0.30	0.70-0.90	0.15-0.35	0.035-0.050	0.035	---	---	0.20-0.30	---	65 74	40 52	18 16	180 210/43	HR & N H & T	SAME AS ABOVE	1200	840-880	840-880 OIL OR WATER	550-660	840-880 COOL IN AIR
G40320	4032	0.30-0.35	0.70-0.90	0.15-0.35	0.04	0.035	---	---	0.20-0.30	---	68 78	42 55	17 14	190 220/41	HR & N H & T	SAME AS ABOVE	1200	840-880	840-880 OIL OR WATER	550-660	840-880 COOL IN AIR
G40370	4037	0.35-0.40	0.70-0.90	0.15-0.35	0.04	0.035	---	---	0.20-0.30	---	70 82	45 65	16 14	200 230/35	HR & N H & T	SAME AS ABOVE BUT BOLTS, NUTS TO P-8.8/GR. S	1200	840-880	840-880 OIL OR WATER	550-660	840-880 COOL IN AIR
G40420	4042	0.40-0.45	0.70-0.90	0.15-0.35	0.04	0.035	---	---	0.20-0.30	---	72 85	48 66	16 13	210 235/35	HR & N H & T	SAME AS ABOVE BUT BOLTS, NUTS TO P-8.8/GR. S	1200	840-880	840-880 OIL OR WATER	550-660	840-880 COOL IN AIR
G40470	4047	0.45-0.50	0.70-0.90	0.15-0.35	0.04	0.035	---	---	0.20-0.30	---	75 90	50 70	14 12	230 245/31	HR & N H & T	SAME AS ABOVE BUT BOLTS, NUTS TO P-8.8/GR. S,T	1200	840-880	840-880 OIL OR WATER	550-660	840-880 COOL IN AIR

*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED.

ABBREVATIONS USED:
AR -- AS ROLLED
HR -- HOT ROLLED
CR -- COLD ROLLED
AC -- AIR COOLED
CD -- COLD DRAWN
OH -- OIL HARDENED
N -- NORMALISED
AH -- AIR HARDENED
A -- ANNEALED
H & T -- HARDENED & TEMPERED
S -- SOFTENED
F & I -- FLAME & INDUCTION HARDENING
FC -- FURNACE COOL

CONTD.

CHEMISTRY & MECH.PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - US STANDARD GRADES

UNS NO.	DESIGNATION SAE/AISI NO.	*CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
ALLOYED STEELS (CONTD.)		C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J			HOT FORMING TEMP °C	ANNEALING TEMP °C (FC)	HARDENING TEMP °C QUENCHANT	TEMPERING TEMP °C	NORMALIZING TEMP °C
G41180	4118	0.18 0.23	0.70- 0.90	0.15- 0.35	0.04	0.035	0.40- 0.60	---	0.08- 0.15	---	65 68	40 50	18 16	180 200/43	HR & N H & T	MOSTLY USED FOR AUTOMOBILE PARTS SUCH AS GUDGEON PINS, ROLLER BEARINGS, MEASURING INSTRUMENTS, ETC.	1200	890-940	870-930 OIL OR WATER	550-660	890-940 COOL IN AIR
G41300	4130	0.28 0.33	0.40- 0.60	0.15- 0.35	0.04	0.035	0.80- 1.10	---	0.15- 0.25	---	70 82	42 65	16 14	200 230/41	HR & N H & T	SAME AS ABOVE AND ALSO FOR BOLTS, NUTS TO P-8.8, GR.S	1200	890-940	870-930 OIL OR WATER	550-660	890-940 COOL IN AIR
G41350	4135	0.33 0.38	0.70- 0.90	0.15- 0.35	0.04	0.035	0.80- 1.10	---	0.15- 0.25	---	75 92	45 68	15 13	210 250/37	HR & N H & T	SAME AS ABOVE AND ALSO FOR BOLTS, NUTS TO P-8.8, GR.S	1200	890-940	870-930 OIL OR WATER	550-660	890-940 COOL IN AIR
G41370	4137	0.35 0.40	0.70- 0.90	0.15- 0.35	0.04	0.035	0.80- 1.10	---	0.15- 0.25	---	80 102	50 92	14 11	240 350	HR & N H & T	SAME AS ABOVE AND ALSO FOR BOLTS, NUTS TO P-10.9/GR.T	1200	890-940	870-930 OIL OR WATER	550-660	890-940 COOL IN AIR
G41400	4140	0.38 0.43	0.75- 1.00	0.15- 0.35	0.04	0.035	0.80- 1.10	---	0.15- 0.25	---	80 102	50 92	14 11	240 350	HR & N H & T	MOSTLY USED FOR TRACK PINS, CRANK-SHAFTS, CONNEC-TING RODS, AXLES, SHAFTS, GEARS, BOLTS, NUTS, STUDS, TO P-10.9, GR.T, A193-B7, ETC.	1200	850-880	850-880 OIL OR WATER	550-720	850-880 COOL IN AIR
G41420	4142	0.40- 0.45	0.75- 1.00	0.15- 0.35	0.04	0.035	0.80- 1.10	---	0.15- 0.25	---	80 102	50 92	14 11	240 350	HR & N H & T	SAME AS ABOVE	1200	850-880	850-880 OIL OR WATER	550-720	850-880 COOL IN AIR
G41450	4145	0.43- 0.48	0.75- 1.00	0.15- 0.35	0.04	0.035	0.80- 1.10	---	0.15- 0.25	---	80 112	50 96	14 10	240 360	HR & N H & T	SAME AS ABOVE	1200	850-880	850-880 OIL OR WATER	550-720	850-880 COOL IN AIR
G41470	4147	0.45- 0.50	0.75- 1.00	0.15- 0.35	0.04	0.035	0.80- 1.10	---	0.15- 0.25	---	80 118	50 100	14 10	240 370	HR & N H & T	SAME AS ABOVE	1200	850-880	850-880 OIL OR WATER	550-720	850-880 COOL IN AIR
G41500	4150	0.48- 0.53	0.75- 1.00	0.15- 0.35	0.04	0.035	0.80- 1.10	---	0.15- 0.25	---	80 118	50 100	14 10	240 370	HR & N H & T	SAME AS ABOVE	1200	850-880	850-880 OIL OR WATER	550-720	850-880 COOL IN AIR
G41610	4161	0.56- 0.64	0.75- 1.00	0.15- 0.35	0.04	0.035	0.70- 0.90	---	0.25- 0.35	---	85 122	55 109	14 9	250 400	HR & N H & T	SAME AS ABOVE AND ALSO FOR BOLTS, NUTS TO P-12.9 GR.U,V	1200	850-880	850-880 OIL OR WATER	550-720	850-880 COOL IN AIR
G43200	4320	0.17- 0.22	0.45- 0.65	0.15- 0.35	0.04	0.035	0.40- 0.60	1.65- 2.00	0.20- 0.30	---	70 102	50 91	16 11	210 350	HR & N H & T	SAME AS ABOVE AND ALSO FOR BOLTS, NUTS TO P-10.9 ,GR.T, ETC.	1200	850-880	850-880 OIL OR WATER	550-720	850-880 COOL IN AIR

*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED.

ABBREVATIONS USED:
AR -- AS ROLLED
HR -- HOT ROLLED
CR -- COLD ROLLED
AC -- AIR COOLED
CD -- COLD DRAWN
OH -- OIL HARDENED
N -- NORMALISED
AH -- AIR HARDENED
A -- ANNEALED
H & T -- HARDENED & TEMPERED
F & I -- FLAME & INDUCTION HARDENING
S -- SOFTENED
FC -- FURNACE COOL
CS - CASE HARDENED

CONTD.

CHEMISTRY & MECH.PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - US STANDARD GRADES

DESIGNATION UNS NO.	SAE/AISI NO.	*CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
ALLOYED STEELS (CONTD.)		C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J)			HOT FORMING TEMP. °C	ANNEALING TEMP. °C (FC)	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZ-ING TEMP.°C
G43400	4340	0.38-0.43	0.60-0.80	0.15-0.35	0.04	0.035	0.70-0.90	1.85-2.00	0.20-0.30	---	80 122	55 109	14 9	250 400	HR & N H & T	MOSTLY USED FOR GEARS, AXLES, SHAFTS, CONNECT-ING RODS,BOLTS, NUTS TO P-12.9, GR. U,V.	1200	820-850	820-850 OIL	550-660	920-850 COOL IN AIR
G43406	E4340	0.38-0.43	0.65-0.85	0.15-0.35	0.04	0.035	0.70-0.90	1.85-2.00	0.20-0.30	---	80 122	55 109	14 9	250 400	HR & N H & T	SAME AS ABOVE	1200	820-850	820-850 OIL	550-660	920-850 COOL IN AIR
E - ELECTRIC FURNACE STEEL																					
G44220	4422	0.20-0.25	0.70-0.90	0.15-0.35	0.04	0.035	---	---	0.35-0.45	---	65 74	40 52	18 16	180 210/43	HR & N H & T	MOSTLY USED FOR AXLES, SHAFTS, CRANK SHAFTS, SPINDLES, STEERING LEVERS, BOLTS, NUTS TO P-6.8	1200	840-880	840-880 OIL OR WATER	550-660	840-880 COOL IN AIR
G44270	4427	0.24-0.29	0.70-0.90	0.15-0.35	0.04	0.035	---	---	0.35-0.45	---	65 74	40 52	18 16	180 210/43	HR & N H & T	SAME AS ABOVE	1200	840-880	840-880 OIL OR WATER	550-660	840-880 COOL IN AIR
G46150	4615	0.13-0.18	0.45-0.65	0.15-0.35	0.04	0.035	---	1.65-2.00	0.20-0.30	---	75 134	45 ---	16 9	200 400/35	HR & N CH & T	CASE HARDENING STEEL SUI ABLE FOR HEAVY DUTY GEARS FOR ALL TYPES OF MACHINERY, PINIONS, CAMSHAFTS, CLUTCH PLATES, ETC.	1200	840-880	CARBURI-ZING AT 880-930 FOLLOWED BY HARDEN-ING AT 790-820	200-250	840-880 COOL IN AIR
G46170	4617	0.15-0.20	0.45-0.65	0.15-0.35	0.04	0.035	---	1.65-2.00	0.20-0.30	---	75 134	45 ---	16 9	200 400/35	HR & N CH & T	SAME AS ABOVE	1200	840-880	SAME AS ABOVE	200-250	840-880 COOL IN AIR
G46200	4620	0.17-0.22	0.45-0.65	0.15-0.35	0.04	0.035	---	1.65-2.00	0.20-0.30	---	75 134	45 ---	16 9	200 400/35	HR & N CH & T	SAME AS ABOVE	1200	840-880	SAME AS ABOVE	200-250	840-880 COOL IN AIR
G46260	4626	0.24-0.29	0.45-0.65	0.15-0.35	0.04	0.035	---	0.70-1.00	0.15-0.25	---	70 124	40 ---	18 10	180 380/37	HR & N CH & T	SAME AS ABOVE	1200	840-880	SAME AS ABOVE	200-250	840-880 COOL IN AIR
G47180	4718	0.16-0.21	0.70-0.90	0.15-0.35	0.04	0.035	0.35-0.55	0.90-1.20	0.30-0.40	---	70 124	40 ---	18 10	180 380/37	HR & N CH & T	SAME AS ABOVE	1200	840-880	SAME AS ABOVE	200-250	840-880 COOL IN AIR
G47200	4720	0.17-0.22	0.50-0.70	0.15-0.35	0.04	0.035	0.35-0.55	0.90-1.20	0.15-0.25	---	70 124	40 ---	18 10	180 380/37	HR & N CH & T	SAME AS ABOVE	1200	840-880	SAME AS ABOVE	200-250	840-880 COOL IN AIR
G48150	4815	0.13-0.18	0.40-0.60	0.15-0.35	0.04	0.035	---	3.25-3.75	0.20-0.30	---	70 134	40 ---	18 9	180 400/35	HR & N CH & T	SAME AS ABOVE	1200	840-880	SAME AS ABOVE	200-250	840-880 COOL IN AIR

*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED

ABBREVATIONS USED: HR -- HOT ROLLED; AC -- AIR COOLED; OH -- OIL HARDENED; AH -- AIR HARDENED; H & T -- HARDENED & TEMPERED; S -- SOFTENED; CS - CASE HARDENED; CONTD
AR -- AS ROLLED; CR -- COLD ROLLED; CD -- COLD DRAWN; N -- NORMALISED; A -- ANNEALED; F & I -- FLAME & INDUCTION HARDENING; FC -- FURNACE COOL

CHEMISTRY & MECH.PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - US STANDARD GRADES

DESIGNATION UNS NO.	SAE/AISI NO.	* CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
ALLOYED STEELS (CONTD.)		C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J			HOT FORMING TEMP. °C	ANNEALING TEMP. °C (FC)	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZING TEMP. °C
G48170	4817	0.15-0.20	0.40-0.60	0.15-0.35	0.04	0.035	---	3.25-3.75	0.20-0.30	---	70 134	40 ---	18 9	180 400/35	HR & N CH & T	CASE HARDENING STEEL SUITABLE FOR HEAVY DUTY GEARS FOR ALL TYPES OF MACHINERY, PINIONS, CAMSHAFTS, CLUTCH PLATES, ETC.	1200	840-880	CARBURIZING AT 880-930 FOLLOWED BY HARDENING AT 780-820	200-250	840-880 COOL IN AIR
G48200	4820	0.18-0.23	0.50-0.70	0.15-0.35	0.04	0.035	---	3.25-3.75	0.20-0.30	---	70 134	40 ---	18 9	180 400/35	HR & N CH & T	SAME AS ABOVE	1200	840-880	SAME AS ABOVE	200-250	840-880 COOL IN AIR
G50401	50B40	0.38-0.43	0.75-1.00	0.15-0.35	0.04	0.035	0.40-0.60	---	---	0.0005-0.003 BORON	70 100	40 90	18 10	180 340	HR & N H & T	HIGH QUALITY ALLOY STEEL SUITABLE FOR GEARS, CONN. RODS, SPINDLES, AXLES, CRANK SHAFTS, HIGH TENSILE FASTENERS OF P-8.8 AND ABOVE.	1200	840-880	840-880 OIL	550-700	840-880 COOL IN AIR
G50441	50B44	0.43-0.48	0.75-1.00	0.15-0.35	0.04	0.035	0.40-0.60	---	---	0.0005-0.003 BORON	70 100	40 90	18 10	180 340	HR & N H & T	SAME AS ABOVE BUT FOR HIGH TENSILE FASTENERS OF P-10.9 & ABOVE	1200	840-880	840-880 OIL	550-700	840-880 COOL IN AIR
G50460	5046	0.43-0.48	0.75-1.00	0.15-0.35	0.04	0.035	0.20-0.35	---	---	---	65 80	36 64	18 14	160 230	HR & N H & T	SAME AS ABOVE BUT FOR HIGH TENSILE FASTENERS OF P-8.8	1200	840-880	840-880 OIL	550-700	840-880 COOL IN AIR
G50461	50B46	0.44-0.49	0.75-1.00	0.15-0.35	0.04	0.035	0.20-0.35	---	---	0.0005-0.003 BORON	70 90	40 70	18 14	180 250	HR & N H & T	SAME AS ABOVE BUT FOR HIGH TENSILE FASTENERS OF P-8.8 & ABOVE	1200	840-880	840-880 OIL	550-700	840-880 COOL IN AIR
G50501	50B50	0.48-0.53	0.75-1.00	0.15-0.35	0.04	0.035	0.40-0.60	---	---	0.0005-0.003 BORON	70 100	40 90	18 10	180 340	HR & N H & T	SAME AS ABOVE BUT FOR HIGH TENSILE FASTENERS OF P-10.9 & ABOVE	1200	840-880	840-880 OIL	550-700	840-880 COOL IN AIR
G50600	5060	0.56-0.64	0.75-1.00	0.15-0.35	0.04	0.035	0.40-0.60	---	---	---	65 80	36 64	18 14	160 230	HR & N H & T	SAME AS ABOVE BUT FOR HIGH TENSILE FASTENERS OF P-8.8	1200	840-880	840-880 OIL	550-700	840-880 COOL IN AIR
G50601	50B60	0.56-0.64	0.75-1.00	0.15-0.35	0.04	0.035	0.40-0.60	---	---	0.0005-0.003 BORON	70 100	40 90	18 10	180 340	HR & N H & T	SAME AS ABOVE BUT FOR HIGH TENSILE FASTENERS OF P-10.9 & ABOVE	1200	840-880	840-880 OIL	550-700	840-880 COOL IN AIR
G51150	5115	0.13-0.18	0.70-0.90	0.15-0.35	0.04	0.04	0.70-0.90	---	---	---	50 80	30 ---	22 9	160 400	HR & N CH & T	CASE HARDENING STEEL SUITABLE FOR CAMSHAFTS, BUSHINGS, CONNECTING RODS, ETC.	1200	840-880	CARBURIZING AT 880-930 FOLLOWED BY HARDENING AT 780-820	200-250	840-880 COOL IN AIR

*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED.

ABBREVATIONS USED:
AR -- AS ROLLED
HR -- HOT ROLLED
CR -- COLD ROLLED
AC -- AIR COOLED
CD -- COLD DRAWN
OH -- OIL HARDENED
N -- NORMALISED
AH -- AIR HARDENED
A -- ANNEALED
H & T -- HARDENED & TEMPERED
F & I -- FLAME & INDUCTION HARDENING
S -- SOFTENED
CS -- CASE HARDENED
FC -- FURNACE COOL

CONTD

CHEMISTRY & MECH.PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - US STANDARD GRADES

UNS NO.	DESIGNATION SAE/AISI NO.	* CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
		C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J)			HOT FORMING TEMP. °C	ANNEALING TEMP. °C (FC)	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZING TEMP. °C
ALLOYED STEELS (CONTD.)																					
G51170	5117	0.15-0.20	0.70-0.90	0.15-0.35	0.04	0.04	0.70-0.90	---	---	---	50 80	30 ---	22 9	160 400	HR & N CH & T	CASE HARDENING STEEL SUITABLE FOR CAMSHAFTS, BUSHINGS, CONNECTING RODS, ETC.	1200	840-880	CARBURIZING AT 880-930 FOLLOWED BY HARDENING AT 780-820	200-250	840-880 COOL IN AIR
G51200	5120	0.17-0.22	0.70-0.90	0.15-0.35	0.04	0.035	0.70-0.90	---	---	---	50 80	30 ---	22 9	160 400	HR & N CH & T	SAME AS ABOVE	1200	840-880	SAME AS ABOVE	200-250	840-880 COOL IN AIR
G51300	5130	0.28-0.33	0.70-0.90	0.15-0.35	0.04	0.035	0.80-1.10	---	---	---	70 82	42 65	16 14	200 230/41	HR & N H & T	SAME AS ABOVE AND ALSO FOR BOLTS, NUTS TO P-8.8, GR.S	1200	890-940	870-930 OIL OR WATER	550-660	890-940 COOL IN AIR
G51320	5132	0.30-0.35	0.60-0.80	0.15-0.35	0.04	0.035	0.75-1.00	---	---	---	70 82	42 65	16 14	200 230/41	HR & N H & T	SAME AS ABOVE AND ALSO FOR BOLTS, NUTS TO P-8.8, GR.S	1200	890-940	870-930 OIL OR WATER	550-660	890-940 COOL IN AIR
G51350	5135	0.33-0.38	0.60-0.80	0.15-0.35	0.04	0.035	0.80-1.05	---	---	---	75 92	45 68	15 13	210 250/37	HR & N H & T	SAME AS ABOVE AND ALSO FOR BOLTS, NUTS TO P-8.8, GR.S	1200	890-940	870-930 OIL OR WATER	550-660	890-940 COOL IN AIR
G51400	5140	0.38-0.43	0.70-0.90	0.15-0.35	0.04	0.035	0.70-0.90	---	---	---	75 92	45 68	15 13	210 250/37	HR & N H & T	SAME AS ABOVE AND ALSO FOR BOLTS, NUTS TO P-8.8, GR.S	1200	890-940	870-930 OIL OR WATER	550-660	890-940 COOL IN AIR
G51470	5147	0.46-0.51	0.70-0.95	0.15-0.35	0.04	0.035	0.85-1.15	---	---	---	80 102	50 92	14 11	240 350	HR & N H & T	SAME AS ABOVE AND ALSO FOR BOLTS, NUTS TO P-10.9/GR. T	1200	890-940	870-930 OIL OR WATER	550-660	890-940 COOL IN AIR
G51500	5150	0.48-0.53	0.70-0.90	0.15-0.35	0.04	0.035	0.70-0.90	---	---	---	80 102	50 92	14 11	240 350	HR & N H & T	MOSTLY USED FOR TRACK PINS, CRANKSHAFTS, CONNECTING RODS, AXLES, SHAFTS, GEARS, BOLTS, NUTS, STUDS, TO P-10.9, GR.T, ETC.	1200	830-880	830-880 OIL	550-720	830-880 COOL IN AIR
G51550	5155	0.51-0.59	0.70-0.90	0.15-0.35	0.04	0.035	0.70-0.90	---	---	---	85 112	55 100	14 10	250 380	HR & N H & T	SAME AS ABOVE	1200	830-880	830-880 OIL	550-720	830-880 COOL IN AIR
G51600	5160	0.56-0.64	0.75-1.00	0.15-0.35	0.04	0.035	0.70-0.90	---	---	---	85 122	55 110	14 9	250 400	HR & N H & T	SAME AS ABOVE AND ALSO FOR FASTENERS TO P-12.9 AND LAMINATED SPRINGS, ETC.	1200	830-880	830-880 OIL	550-720	830-880 COOL IN AIR

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ABBREVATIONS USED:
AR -- AS ROLLED; HR -- HOT ROLLED; CR -- COLD ROLLED; AC -- AIR COOLED; CD -- COLD DRAWN; OH -- OIL HARDENED; N -- NORMALISED; AH -- AIR HARDENED; A -- ANNEALED; H & T -- HARDENED & TEMPERED; S -- SOFTENED; CS - CASE HARDENED; F & I -- FLAME & INDUCTION HARDENING; FC -- FURNACE COOL

CONTD.

CHEMISTRY & MECH.PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - US STANDARD GRADES

DESIGNATION UNS NO. SAE/AISI NO.	* CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
ALLOYED STEELS (CONTD.)	C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	%HARDNESS (BHN) / IZOD(J)			HOT FORMING TEMP. °C	ANNEALING TEMP. °C (FC)	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZING TEMP °C
G51601 51B60	0.56-0.64	0.75-1.00	0.15-0.35	0.04	0.035	0.70-0.90	---	---	0.0005-0.003 BORON	80 120	50 108	14 9	240 400	HR & N H & T	MOSTLY USED FOR TRACK PINS, CRANK-SHAFTS, CONNEC-TING RODS, AXLES, SHAFTS, GEARS, BOLTS, NUTS, STUDS, TO P-10.9, GR.T AND ABOVE.	1200	830-880	830-880 OIL	550-720	830-880 COOL IN AIR
G50986 E50100	0.98-1.10	0.25-0.45	0.15-0.35	0.025	0.025	0.40-0.60	---	---	---	NOT SPECIFIED					USABLE FOR BALL & ROLLER BEARINGS, BEARING RINGS, BUSHES, COLLETS, CAMS, LATHE CENTRES, ETC.	1100	830-860	830-860 OIL	150-250	830-860 COOL IN AIR
G51986 E51100	0.98-1.10	0.25-0.45	0.15-0.35	0.025	0.025	0.90-1.15	---	---	---	NOT SPECIFIED					SAME AS ABOVE	1100	830-860	830-860 OIL	150-250	830-860 COOL IN AIR
G52986 E52100 E - ELECTRIC FURNACE STEEL	0.98-1.10	0.25-0.45	0.15-0.35	0.025	0.025	1.30-1.60	---	---	---	NOT SPECIFIED					SAME AS ABOVE	1100	830-860	830-860 OIL	150-250	830-860 COOL IN AIR
G51601 51B60	0.56-0.64	0.75-1.00	0.15-0.35	0.04	0.035	0.70-0.90	---	---	0.0005-0.003 BORON	80 120	50 108	14 9	240 400	HR & N H & T	MOSTLY USED FOR TRACK PINS, CRANK-SHAFTS, CONNEC-TING RODS, AXLES, SHAFTS, GEARS, BOLTS, NUTS, STUDS, TO P-10.9, GR.T AND ABOVE.	1200	830-880	830-880 OIL	550-720	830-880 COOL IN AIR
G81150 8115	0.13-0.18	0.70-0.90	0.15-0.35	0.04	0.035	0.30-0.50	0.15-0.35	0.08-0.15	---	50 80	30 ---	22 9	160 400	HR & N CH & T	CASE HARDENING STEEL SUITABLE FOR CAMSHAFTS, BUSH-INGS, AUTOMOBILE TRANSMISSION COMPONENTS, GUDGEON PINS.	1150	840-880	CARBURI-ZING AT 880-930 FOLLOWED BY HARDEN-ING AT 780-820 OIL	200-250	840-880 COOL IN AIR
G81451 81B45	0.43-0.48	0.75-1.00	0.15-0.35	0.04	0.035	0.20-0.40	0.35-0.55	0.08-0.15	0.0005-0.003 BORON	80 120	50 108	14 9	240 400	HR & N H & T	MOSTLY USED FOR TRACK PINS, CRANK-SHAFTS, AXLES, BOLTS, NUTS, STUDS, TO P-10.9, GR.T AND ABOVE.	1200	830-880	830-880 OIL	550-720	830-880 COOL IN AIR

*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED.

ABBREVATIONS USED:
AR – AS ROLLED HR – HOT ROLLED CR – COLD ROLLED AC – AIR COOLED CD – COLD DRAWN OH – OIL HARDENED N – NORMALISED AH – AIR HARDENED A – ANNEALED H & T – HARDENED & TEMPERED F & I – FLAME & INDUCTION HARDENING S – SOFTENED FC – FURNACE COOL CS - CASE HARDENED

CONTD.

CHEMISTRY & MECH.PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - US STANDARD GRADES

DESIGNATION UNS NO.	SAE/AISI NO.	*CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
		C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	%HARDNESS (BHN) / IZOD(J)			HOT FORMING TEMP. °C	ANNEALING TEMP. °C (FC)	HARDENING TEMP °C QUENCHANT	TEMPERING TEMP °C	NORMALIZING TEMP °C
ALLOYED STEELS (CONTD.)																					
G86150	8615	0.13-0.18	0.70-0.90	0.15-0.35	0.04	0.035	0.40-0.60	0.40-0.70	0.15-0.25	---	60 90	36 ---	20 8	170 450	HR & N CH & T	CASE HARDENING STEEL SUITABLE FOR CAMSHAFTS, BUSHINGS, AUTOMOBILE TRANSMISSION COMPONENTS, GUDGEON PINS.	1150	840-880	CARBURIZING AT 880-930 FOLLOWED BY HARDENING AT 780-820 OIL	200-250	840-880 COOL IN AIR
G86170	8617	0.15-0.20	0.70-0.90	0.15-0.35	0.04	0.035	0.40-0.60	0.40-0.70	0.15-0.25	---	62 90	36 ---	19 8	180 450	HR & N CH & T	SAME AS ABOVE	1150	840-880	SAME AS ABOVE	200-250	840-880 COOL IN AIR
G86200	8620	0.18-0.23	0.70-0.90	0.15-0.35	0.04	0.035	0.40-0.60	0.40-0.70	0.15-0.25	---	65 100	40 ---	18 8	190 480	HR & N CH & T	SAME AS ABOVE	1150	840-880	SAME AS ABOVE	200-250	840-880 COOL IN AIR
G86220	8622	0.20-0.25	0.70-0.90	0.15-0.35	0.04	0.035	0.40-0.60	0.40-0.70	0.15-0.25	---	70 82	42 65	16 14	200 230/41	HR & N H & T	LOW ALLOY STEEL SUITABLE FOR BOLTS, NUTS TO P-8.8, GR.S	1200	890-940	870-930 OIL OR WATER	550-660	890-940 COOL IN AIR
G86250	8625	0.23-0.28	0.70-0.90	0.15-0.35	0.04	0.035	0.40-0.60	0.40-0.70	0.15-0.25	---	70 86	42 68	16 14	200 250/41	HR & N H & T	LOW ALLOY STEEL SUITABLE FOR BOLTS, NUTS TO P-8.8, GR.S	1200	890-940	870-930 OIL OR WATER	550-660	890-940 COOL IN AIR
G86270	8627	0.25-0.30	0.70-0.90	0.15-0.35	0.04	0.035	0.40-0.60	0.40-0.70	0.15-0.25	---	70 90	42 70	16 12	200 270/38	HR & N H & T	LOW ALLOY STEEL SUITABLE FOR BOLTS, NUTS TO P-8.8, GR.S, AND HIGH TEMP. PIPE FITTINGS, ETC.	1200	870-930	870-930 OIL OR WATER	550-660	870-930 COOL IN AIR
G86300	8630	0.28-0.33	0.70-0.90	0.15-0.35	0.04	0.035	0.40-0.60	0.40-0.70	0.15-0.25	---	72 95	42 72	15 12	210 280/36	HR & N H & T	SAME AS ABOVE	1200	870-930	870-930 OIL OR WATER	550-660	870-930 COOL IN AIR
G86370	8637	0.35-0.40	0.75-1.00	0.15-0.35	0.04	0.035	0.40-0.60	0.40-0.70	0.15-0.25	---	72 95	42 72	15 12	210 280/36	HR & N H & T	SUITABLE FOR HIGH TEMP. BOLTING MAT.	1200	870-930	870-930 OIL	550-660	870-930 COOL IN AIR
G86400	8640	0.38-0.43	0.75-1.00	0.15-0.35	0.04	0.035	0.40-0.60	0.40-0.70	0.15-0.25	---	75 100	45 90	14 10	220 320/31	HR & N H & T	SUITABLE FOR HIGH TEMP. BOLTING MAT.	1150	820-870	830-870 OIL	550-660	830-870 COOL IN AIR
G86420	8642	0.40-0.45	0.75-1.00	0.15-0.35	0.04	0.035	0.40-0.60	0.40-0.70	0.15-0.25	---	75 105	45 92	14 10	220 330/31	HR & N H & T	SUITABLE FOR HIGH TEMP. BOLTING MAT. AND P-10.9	1150	820-870	830-870 OIL	550-660	830-870 COOL IN AIR
G86450	8645	0.43-0.48	0.75-1.00	0.15-0.35	0.04	0.035	0.40-0.60	0.40-0.70	0.15-0.25	---	75 105	45 92	14 10	220 330/31	HR & N H & T	SAME AS ABOVE	1150	820-870	830-870 OIL	550-660	830-870 COOL IN AIR

*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED

ABBREVATIONS USED: HR = HOT ROLLED; AC = AIR COOLED; OH = OIL HARDENED; AH = AIR HARDENED; H & T = HARDENED & TEMPERED; S = SOFTENED; CS = CASE HARDENED; CONTD.
AR = AS ROLLED; CR = COLD ROLLED; CD = COLD DRAWN; N = NORMALISED; A = ANNEALED; F & I = FLAME & INDUCTION HARDENING; FC = FURNACE COOL

CHEMISTRY & MECH. PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - US STANDARD GRADES

UNS NO.	DESIGNATION SAE/AISI NO.	*CHEMICAL COMPOSITION, PERCENT									MECHANICAL PROPERTIES (MIN.)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
		C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J)			HOT FORMING TEMP. °C	ANNEALING TEMP. °C (FC)	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZING TEMP. °C
ALLOYED STEELS (CONTD.)																					
G86451	8645	0.43-0.48	0.75-1.00	0.15-0.35	0.04	0.035	0.40-0.60	0.40-0.70	0.15-0.25	0.0005-0.003 BORON	80 125	50 110	12 9	250 430/25	HR & N H & T	SUITABLE FOR HIGH TEMP. BOLTING MAT. AND P-12.9	1150	820-870	830-870 OIL	550-660	830-870 COOL IN AIR
G86500	8650	0.48-0.53	0.75-1.00	0.15-0.35	0.04	0.035	0.40-0.60	0.40-0.70	0.15-0.25	---	80 120	48 108	13 9	240 410/28	HR & N H & T	SAME AS ABOVE AND ALSO FOR CAM AND GEAR SHAFTS, ETC.	1150	820-870	830-870 OIL	550-660	830-870 COOL IN AIR
G86550	8655	0.51-0.59	0.75-1.00	0.15-0.35	0.04	0.035	0.40-0.60	0.40-0.70	0.15-0.25	---	80 125	48 110	13 9	240 440/24	HR & N H & T	SAME AS ABOVE AND ALSO FOR AUTOMOTIVE SPRINGS, ETC.	1150	820-870	830-870 OIL	350-660	830-870 COOL IN AIR
G86600	8660	0.56-0.64	0.75-1.00	0.15-0.35	0.04	0.035	0.40-0.60	0.40-0.70	0.15-0.25	---	80 125	48 110	13 9	240 440/24	HR & N H & T	SAME AS ABOVE AND ALSO FOR AUTOMOTIVE SPRINGS, ETC.	1150	820-870	830-870 OIL	350-660	830-870 COOL IN AIR
G87200	8720	0.18-0.23	0.70-0.90	0.15-0.35	0.04	0.035	0.40-0.60	0.40-0.70	0.20-0.30	---	60 90	36 ---	20 8	170 450	HR & N CH & T	CASE HARDENING STEEL SUITABLE FOR CAMSHAFTS, BUSHINGS, AUTOMOBILE TRANSMISSION COMPONENTS, GUDGEON PINS.	1150	840-880	CARBURIZING AT 880-930 FOLLOWED BY HARDENING AT 780-820 OIL	200-250	840-880 COOL IN AIR
G87400	8740	0.38-0.43	0.75-1.00	0.15-0.35	0.04	0.035	0.40-0.60	0.40-0.70	0.20-0.25		75 100	45 90	14 10	220 320/31	HR & N H & T	SUITABLE FOR HIGH TEMP. BOLTING MAT.	1150	820-870	830-870 OIL	550-660	830-870 COOL IN AIR
G86220	8622	0.20-0.25	0.75-1.00	0.15-0.35	0.04	0.035	0.40-0.60	0.40-0.70	0.30-0.40	---	75 85	45 70	16 12	200 240/40	HR & N H & T	LOW ALOY STEEL SUITABLE FOR BOLTS, NUTS TO P-8.8, GR.S	1200	890-940	870-930 OIL OR WATER	550-660	890-940 COOL IN AIR
G92540	9254	0.51-0.59	0.60-0.80	1.20-1.60	0.04	0.035	0.60-0.80	---	---	---	80 125	48 110	13 9	240 440/24	HR & N H & T	SAME AS ABOVE AND ALSO FOR AUTOMOTIVE AND HEAVY DUTY LEAF SPRINGS, ETC.	1150	820-870	830-870 OIL	350-660	830-870 COOL IN AIR
G92600	9260	0.56-0.64	0.75-1.00	1.80-2.20	0.04	0.035	---	---	---	---	--- ---	--- ---	--- ---	240MAX. 500	A H & T	COLD WORK TOOL FOR CHIESELS, PUNCHES, SHEAR BLADES, SCARFING TOOLS, ETC.	850-110	740-800	860-880 OIL/AIR	150-400	740-800 COOL IN AIR
G93106 E - ELECTRIC FURNACE STEEL	E9310	0.08-0.13	0.45-0.65	0.15-0.35	0.025	0.025	1.00-1.40	3.00-3.50	0.08-0.15	---	60 90	36 ---	20 12	170 400/48	HR & N CH & T	CASE HARDENING STEEL SUITABLE FOR CAMSHAFTS, BUSHINGS, AUTOMOBILE TRANSMISSION COMPONENTS, GUDGEON PINS	1150	840-880	CARBURIZING AT 880-930 FOLLOWED BY HARDENING AT 760-780 OIL	200-250	840-880 COOL IN AIR

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ABBREVATIONS USED:
AR – AS ROLLED; HR – HOT ROLLED; CR – COLD ROLLED; AC – AIR COOLED; CD – COLD DRAWN; OH – OIL HARDENED; N – NORMALISED; AH – AIR HARDENED; A – ANNEALED; H & T – HARDENED & TEMPERED; F & I – FLAME & INDUCTION HARDENING; S – SOFTENED; FC – FURNACE COOL; CS - CASE HARDENED

CONTD.

CHEMISTRY & MECH.PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - US STANDARD GRADES

UNS NO.	DESIGNATION SAE/AISI NO.	* CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
		C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J)			HOT FORMING TEMP. °C	ANNEALING TEMP. °C (FC)	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZING TEMP. °C
ALLOYED STEELS (CONTD.)																					
G94151	94B15	0.13-0.18	0.75-1.00	0.15-0.35	0.040	0.035	0.30-0.50	0.30-0.60	0.08-0.15	---	55 80	30 ---	20 14	170 400/48	HR & N CH & T	CASE HARDENING STEEL SUITABLE FOR CAMSHAFTS, BUSHINGS, AUTOMOBILE TRANSMISSION COMPONENTS, GUDGEON PINS, CLUTCH PLATES.	1150	840-880	CARBURIZING AT 880-930 FOLLOWED BY HARDENING AT 780-820 OIL	200-250	840-880 COOL IN AIR
G94171	94B17	0.15-0.20	0.75-1.00	0.15-0.35	0.040	0.035	0.30-0.50	0.30-0.60	0.08-0.15	---	55 85	30 ---	20 14	170 400/48	HR & N CH & T	SAME AS ABOVE	1150	840-880	SAME AS ABOVE	200-250	840-880 COOL IN AIR
G94301	94B30	0.28-0.33	0.75-1.00	0.15-0.35	0.040	0.035	0.30-0.50	0.30-0.60	0.08-0.15	---	55 85	30 ---	20 14	170 400/48	HR & N CH & T	SUITABLE FOR HIGH TEMP. BOLTING MAT.	1200	840-880	830-880 OIL	550-650	840-880 COOL IN AIR
UNALLOYED "H" STEELS (STEELS WITH SCATTER BAND OF HARDENABILITY)																					
H10380	1038H	0.34-0.43	0.50-1.00	0.15-0.35	0.05	0.04	---	---	---	---	55 60	--- ---	20 18	170 175	HR & N H & T	DIRECT HARDENING STEEL FOR RAILWAY COUPLING, HAND TOOLS, ETC. CAN BE FLAME OR INDUCTION HARDNESS.	1200	840-880	840-880 OIL OR WATER	550-660	840-880 COOL IN AIR
H10450	1045H	0.42-0.51	0.50-1.00	0.15-0.35	0.05	0.04	---	---	---	---	60 70	--- ---	18 16	170 185	HR & N H & T	SAME AS ABOVE	1200	840-880	840-880 OIL OR WATER	550-660	840-880 COOL IN AIR
H15220	1522H	0.17-0.25	1.00-1.50	0.15-0.35	0.05	0.04	---	---	---	---	50	32	20	150	HR & N	MOSTLY USED FOR AUTOMOBILE AND BICYCLE PARTS, BOLTS, NUTS OF P-4.6/4.8	1150	880-950	---	---	880-950 COOL IN AIR
H15240	1524H	0.18-0.26	1.25-1.75	0.15-0.35	0.05	0.04	---	---	---	---	50	32	20	150	HR & N	SAME AS ABOVE	1150	880-950	---	---	880-950 COOL IN AIR
H15260	1526H	0.21-0.30	1.00-1.50	0.15-0.35	0.05	0.04	---	---	---	---	55	32	18	160	HR & N	SAME AS ABOVE BUT BOLTS & NUTS OF P-5.6.	1150	880-950	---	---	880-950 COOL IN AIR
H15211	15B21H	0.17-0.24	0.70-1.20	0.15-0.35	0.05	0.04	---	---	---	0.0005-0.003 BORON	65	48	18	180	HR & N	SAME AS ABOVE BUT BOLTS & NUTS OF P-6.8.	1150	880-950	---	---	880-950 COOL IN AIR
H15351	15B35H	0.31-0.39	0.70-1.20	0.15-0.35	0.05	0.04	---	---	---	0.0005-0.003 BORON	70 85	50 68	16 12	210 240	HR & N H& T	SAME AS ABOVE BUT BOLTS & NUTS OF P-8.8/ GR.S	1200	840-880	830-870 OIL	550-650	840-880 COOL IN AIR
H15371	15B37H	0.30-0.39	1.00-1.50	0.15-0.35	0.05	0.04	---	---	---	0.0005-0.003 BORON	70 90	50 70	16 12	210 260	HR & N H& T	SAME AS ABOVE BUT BOLTS & NUTS OF P-8.8 AND ABOVE.	1200	840-880	830-870 OIL	550-650	840-880 COOL IN AIR

*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED.

ABBREVATIONS USED: HR -- HOT ROLLED; AC -- AIR COOLED; OH -- OIL HARDENED; AH -- AIR HARDENED; H & T -- HARDENED & TEMPERED; S -- SOFTENED; CS - CASE HARDENED; CONTD
AR -- AS ROLLED; CR -- COLD ROLLED; CD -- COLD DRAWN; N -- NORMALISED; A -- ANNEALED; F & I -- FLAME & INDUCTION HARDENING; FC -- FURNACE COOL

CHEMISTRY & MECH.PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - US STANDARD GRADES

UNS NO.	DESIGNATION SAE/AISI NO.	* CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
UNALLOYED "H" STEELS (STEELS WITH SCATTER BAND OF HARDENABILITY) - CONTD.		C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J)			HOT FORMING TEMP. °C	ANNEALING TEMP. °C (FC)	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZ-ING TEMP. °C
H15411	15B41H	0.35-0.45	1.25-1.75	0.15-0.35	0.05	0.04	---	---	---	0.0005-0.003 BORON	75 102	55 82	15 10	230 350	HR & N H& T	MOSTLY USED FOR CAM SHAFTS, CONN. RODS, CRANK SHAFTS, AUTO AND BICYCLE PARTS, HEAVY GEARS, AND BOLTS & NUTS OF P-10.9 AND ABOVE.	1200	840-880	830-870 OIL	550-650	840-880 COOL IN AIR
H15481	15B48H	0.43-0.53	1.00-1.50	0.15-0.35	0.05	0.04	---	---	---	0.0005-0.003 BORON	85 122	60 110	14 9	250 400	HR & N H& T	SAME AS ABOVE BUT BOLTS & NUTS OF P-12.9 AND ABOVE.	1200	840-880	830-870 OIL	550-650	840-880 COOL IN AIR
H15621	15B62H	0.54-0.67	1.00-1.50	0.40-0.60	0.05	0.04	---	---	---	0.0005-0.003 BORON	90 130	65 115	12 8	290 450	HR & N H& T	SAME AS ABOVE BUT BOLTS & NUTS OF P-12.9 AND ABOVE. AND HIGH QUALITY SPRINGS.	1200	840-880	830-870 OIL	550-650	840-880 COOL IN AIR
ALLOYED "H" STEELS (STEELS WITH SCATTER BAND OF HARDENABILITY)																					
H13300	1330H	0.27-0.33	1.45-2.05	0.15-0.35	0.04	0.035	---	---	---	---	55 64	32 45	20 18	180 187/45	HR & N H & T	SUITABLE FOR LOW DUTY PINIONS, AUTO-MOBILE AND BICYCLE PARTS, STUDS, BOLTS, SCREWS, NUTS TO P-6.8 AND CONNEC-TINGRODS, ETC.	1200	840-880	840-880 OIL OR WATER	550-660	840-880 COOL IN AIR
H13350	1335H	0.32-0.38	1.45-2.05	0.15-0.35	0.04	0.035	---	---	---	---	55 64	32 45	20 18	180 187/45	HR & N H & T	SAME AS ABOVE	1200	840-880	840-880 OIL OR WATER	550-660	840-880 COOL IN AIR
H13400	1340H	0.37-0.44	1.45-2.05	0.15-0.35	0.04	0.035	---	---	---	---	65 82	36 65	16 14	187 230/43	HR & N H & T	SAME AS ABOVE BUT BOLTS, NUTS TO P-8.8, GR. S	1200	840-880	840-880 OIL OR WATER	550-660	840-880 COOL IN AIR
H13450	1345H	0.42-0.49	1.45-2.05	0.15-0.35	0.04	0.035	---	---	---	---	68 88	38 68	16 13	197 250/43	HR & N H & T	SAME AS ABOVE BUT BOLTS, NUTS TO P-8.8, GR.S	1200	840-880	840-880 OIL OR WATER	550-660	840-880 COOL IN AIR
H40270	4027H	0.24-0.30	0.60-1.00	0.15-0.35	0.04	0.035	---	---	0.20-0.30	---	65 74	40 52	18 16	180 210/43	HR & N H & T	SAME AS ABOVE	1200	840-880	840-880 OIL OR WATER	550-660	840-880 COOL IN AIR
H40280	4028H	0.24-0.30	0.60-1.00	0.15-0.35	0.035-0.050	0.035	---	---	0.20-0.30	---	65 74	40 52	18 16	180 210/43	HR & N H & T	SAME AS ABOVE	1200	840-880	840-880 OIL OR WATER	550-660	840-880 COOL IN AIR
H40320	4032H	0.29-0.35	0.60-1.00	0.15-0.35	0.04	0.035	---	---	0.20-0.30	---	68 78	42 55	17 14	190 220/41	HR & N H & T	SAME AS ABOVE	1200	840-880	840-880 OIL OR WATER	550-660	840-880 COOL IN AIR
H40370	4037H	0.34-0.41	0.60-1.00	0.15-0.35	0.04	0.035	---	---	0.20-0.30	---	70 82	45 65	16 14	200 230/35	HR & N H & T	SAME AS ABOVE BUT BOLTS, NUTS TO P-8.8/ GR S	1200	840-880	840-880 OIL OR WATER	550-660	840-880 COOL IN AIR

*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED.

ABBREVATIONS USED:
AR -- AS ROLLED
HR -- HOT ROLLED
CR -- COLD ROLLED
AC -- AIR COOLED
CD -- COLD DRAWN
OH -- OIL HARDENED
N -- NORMALISED
AH -- AIR HARDENED
A -- ANNEALED
H & T -- HARDENED & TEMPERED
S -- SOFTENED
CS - CASE HARDENED
F & I -- FLAME & INDUCTION HARDENING
FC -- FURNACE COOL

CONTD.

CHEMISTRY & MECH.PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - US STANDARD GRADES

DESIGNATION UNS NO.	SAE/AISI NO.	* CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
		C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J)			HOT FORMING TEMP. °C	ANNEALING TEMP. °C (FC)	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZING TEMP. °C
ALLOYED "H" STEELS (STEELS WITH SCATTER BAND OF HARDENABILITY) - CONTD.																					
H40420	4042H	0.39 0.46	0.60- 1.00	0.15- 0.35	0.04	0.035	---	---	0.20- 0.30	---	72 85	48 66	16 13	210 235/35	HR & N H & T	SUITABLE FOR NUTS, BOLTS, STUDS TO P-8.8/ GR. S, CONN. RODS, GEARS, AUTO AND BICYCLE PARTS	1200	840-880	840-880 OIL OR WATER	550-660	840-880 COOL IN AIR
H40470	4047H	0.44 0.51	0.60- 1.00	0.15- 0.35	0.04	0.035	---	---	0.20- 0.30	---	75 90	50 70	14 12	230 245/31	HR & N H & T	SAME AS ABOVE BUT BOLTS, NUTS TO P-8.8/ GR. S,T	1200	840-880	840-880 OIL OR WATER	550-660	840-880 COOL IN AIR
H41180	4118H	0.17 0.23	0.60- 1.00	0.15- 0.35	0.04	0.035	0.30- 0.70	---	0.08- 0.15	---	65 68	40 50	18 16	180 200/43	HR & N H & T	MOSTLY USED FOR AUTOMOBILE PARTS SUCH AS GUDGEON PINS, ROLLER BEARINGS, MEASURING INSTRUMENTS, ETC.	1200	890-940	870-930 OIL OR WATER	550-660	890-940 COOL IN AIR
H41300	4130H	0.27 0.33	0.30- 0.70	0.15- 0.35	0.04	0.035	0.75- 1.20	---	0.15- 0.25	---	70 82	42 65	16 14	200 230/41	HR & N H & T	SAME AS ABOVE AND ALSO FOR BOLTS, NUTS TO P-8.8, GR.S	1200	890-940	870-930 OIL OR WATER	550-660	890-940 COOL IN AIR
G41350	4135H	0.32 0.38	0.60- 1.00	0.15- 0.35	0.04	0.035	0.80- 1.10	---	0.15- 0.25	---	75 92	45 68	15 13	210 250/37	HR & N H & T	SAME AS ABOVE AND ALSO FOR BOLTS, NUTS TO P-8.8, GR.S	1200	890-940	870-930 OIL OR WATER	550-660	890-940 COOL IN AIR
H41370	4137H	0.34 0.41	0.60- 1.00	0.15- 0.35	0.04	0.035	0.75- 1.20	---	0.15- 0.25	---	80 102	50 92	14 11	240 350	HR & N H & T	SAME AS ABOVE AND ALSO FOR BOLTS, NUTS TO P-10.9/GR. T	1200	890-940	870-930 OIL OR WATER	550-660	890-940 COOL IN AIR
H41400	4140H	0.37 0.44	0.65- 1.10	0.15- 0.35	0.04	0.035	0.75- 1.20	---	0.15- 0.25	---	80 102	50 92	14 11	240 350	HR & N H & T	MOSTLY USED FOR TRACK PINS, CRANK-SHAFTS, CONNECTING RODS, AXLES, SHAFTS, GEARS, BOLTS, NUTS, STUDS, TO P-10.9, GR.T, A193-B7, ETC.	1200	850-880	850-880 OIL OR WATER	550-720	850-880 COOL IN AIR
H41420	4142H	0.39- 0.48	0.65- 1.10	0.15- 0.35	0.04	0.035	0.75- 1.20	---	0.15- 0.25	---	80 102	50 92	14 11	240 350	HR & N H & T	SAME AS ABOVE	1200	850-880	850-880 OIL OR WATER	550-720	850-880 COOL IN AIR
H41450	4145H	0.42- 0.49	0.65- 1.10	0.15- 0.35	0.04	0.035	0.75- 1.20	---	0.15- 0.25	---	80 102	50 92	14 11	240 350	HR & N H & T	SAME AS ABOVE AND AUTOMOBILE PARTS SUCH AS GUDGEON PINS, ROLLER BEARINGS, MEASURING INSTRUMENTS, ETC.	1200	850-880	850-880 OIL OR WATER	550-720	850-880 COOL IN AIR
H41470	4147H	0.44- 0.51	0.65- 1.10	0.15- 0.35	0.04	0.035	0.75- 1.20	---	0.15- 0.25	---	80 118	50 100	14 10	240 370	HR & N H & T	SAME AS ABOVE	1200	850-880	850-880 OIL OR WATER	550-720	850-880 COOL IN AIR

*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED.

CONTD.

ABBREVATIONS USED:
HR – HOT ROLLED, AC – AIR COOLED, OH – OIL HARDENED, AH – AIR HARDENED, H & T – HARDENED & TEMPERED, S – SOFTENED, CS - CASE HARDENED
AR – AS ROLLED, CR – COLD ROLLED, CD – COLD DRAWN, N – NORMALISED, A – ANNEALED, F & I – FLAME & INDUCTION HARDENING, FC – FURNACE COOL

CHEMISTRY & MECH.PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - US STANDARD GRADES

UNS NO.	DESIGNATION SAE/AISI NO.	* CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
ALLOYED "H" STEELS (STEELS WITH SCATTER BAND OF HARDENABILITY) - CONTD.		C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J			HOT FORMING TEMP. °C	ANNEALING TEMP. °C (FC)	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZING TEMP °C
H41500	4150H	0.47-0.54	0.65-1.10	0.15-0.35	0.04	0.035	0.75-1.20	---	0.15-0.25	---	80 118	50 100	14 10	240 370	HR & N H & T	SAME AS 4147H	1200	850-880	850-880 OIL OR WATER	550-720	850-880 COOL IN AIR
H41610	4161H	0.55-0.65	0.65-1.10	0.15-0.35	0.04	0.035	0.65-0.95	---	0.25-0.35	---	85 122	55 109	14 9	250 400	HR & N H & T	SAME AS ABOVE AND ALSO FOR BOLTS, NUTS TO P-12.9 GR.U,V	1200	850-880	850-880 OIL OR WATER	550-720	850-880 COOL IN AIR
H43200	4320H	0.17-0.23	0.40-0.70	0.15-0.35	0.04	0.035	0.35-0.65	1.55-2.00	0.20-0.30	---	70 102	50 91	16 11	210 350	HR & N H & T	SAME AS ABOVE AND ALSO FOR BOLTS, NUTS TO P-10.9 GR.S,T	1200	850-880	850-880 OIL OR WATER	550-720	850-880 COOL IN AIR
H43400	4340H	0.37-0.44	0.55-0.90	0.15-0.35	0.04	0.035	0.65-0.95	1.55-2.00	0.20-0.30	---	80 122	55 109	14 9	250 400	HR & N H & T	MOSTLY USED FOR GEARS, AXLES, SHAFTS, CONNECTING RODS, BOLTS, NUTS TO P-12.9, GR. U,V.	1200	820-850	820-850 OIL	550-660	820-850 COOL IN AIR
H43406 E - ELECTRIC FURNACE STEEL	E4340H	0.37-0.44	0.60-0.95	0.15-0.35	0.04	0.035	0.65-0.95	1.55-2.00	0.20-0.30	---	80 122	55 109	14 9	250 400	HR & N H & T	SAME AS ABOVE	1200	820-850	820-850 OIL	550-660	820-850 COOL IN AIR
H46200	4620H	0.17-0.23	0.35-0.75	0.15-0.35	0.04	0.035	---	1.55-2.00	0.20-0.30	---	75 134	45 ---	16 9	200 400/35	HR & N CH & T	CASE HARDENING STEEL SUITABLE FOR HEAVY DUTY GEARS FOR ALL TYPES OF MACHINERY, PINIONS, CAMSHAFTS, CLUTCH PLATES, ETC.	1200	840-880	CARBURIZING AT 880-930 FOLLOWED BY HARDENING AT 780-820 OIL	200-250	840-880 COOL IN AIR
H47180	4718H	0.15-0.21	0.60-0.95	0.15-0.35	0.04	0.035	0.30-0.60	0.85-1.25	0.30-0.40	---	70 124	40 ---	18 10	180 380/37	HR & N CH & T	SAME AS ABOVE	1200	840-880	SAME AS ABOVE	200-250	840-880 COOL IN AIR
H47200	4720H	0.17-0.23	0.45-0.75	0.15-0.35	0.04	0.035	0.30-0.60	0.85-1.25	0.15-0.25	---	70 124	40 ---	18 10	180 380/37	HR & N CH & T	SAME AS ABOVE	1200	840-880	SAME AS ABOVE	200-250	840-880 COOL IN AIR
H48150	4815H	0.12-0.18	0.30-0.70	0.15-0.35	0.04	0.035	---	3.20-3.80	0.20-0.30	---	70 134	40 ---	18 9	180 400/35	HR & N CH & T	SAME AS ABOVE	1200	840-880	SAME AS ABOVE	200-250	840-880 COOL IN AIR
H48170	4817H	0.14-0.20	0.30-0.70	0.15-0.35	0.04	0.035	---	3.20-3.80	0.20-0.30	---	70 134	40 ---	18 9	180 400/35	HR & N CH & T	SAME AS ABOVE	1200	840-880	SAME AS ABOVE	200-250	840-880 COOL IN AIR
H48200	4820H	0.17-0.23	0.40-0.80	0.15-0.35	0.04	0.035	---	3.20-3.80	0.20-0.30	---	70 134	40 ---	18 9	180 400/35	HR & N CH & T	SAME AS ABOVE	1200	840-880	SAME AS ABOVE	200-250	840-880 COOL IN AIR

*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED.

ABBREVATIONS USED:
AR -- AS ROLLED
HR -- HOT ROLLED
CR -- COLD ROLLED
AC -- AIR COOLED
CD -- COLD DRAWN
OH -- OIL HARDENED
N -- NORMALISED
AH -- AIR HARDENED
A -- ANNEALED
H & T -- HARDENED & TEMPERED
S -- SOFTENED
CS - CASE HARDENED
F & I -- FLAME & INDUCTION HARDENING
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CONTD.

CHEMISTRY & MECH.PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - US STANDARD GRADES

DESIGNATION UNS NO.	SAE/AISI NO.	* CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
ALLOYED "H" STEELS (STEELS WITH SCATTER BAND OF HARDENABILITY) - CONTD.		C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J)			HOT FORMING TEMP. °C	ANNEALING TEMP. °C (FC)	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZING TEMP. °C
H50401	50B40H	0.37-0.44	0.65-1.10	0.15-0.35	0.04	0.035	0.30-0.70	---	---	0.0005-0.003 BORON	70 100	40 90	18 10	180 340	HR & N H & T	HIGH QUALITY ALLOY STEEL SUITABLE FOR GEARS, CONN. RODS, SPINDLES, AXLES, CRANK SHAFTS, HIGH TENSILE FASTENERS OF P-8.8 AND ABOVE.	1200	840-880	840-880 OIL	550-700	840-880 COOL IN AIR
H50441	50B44H	0.42-0.49	0.65-1.10	0.15-0.35	0.04	0.035	0.30-0.70	---	---	0.0005-0.003 BORON	70 100	40 90	18 10	180 340	HR & N H & T	SAME AS ABOVE BUT FOR HIGH TENSILE FASTENERS OF P-10.9 & ABOVE	1200	840-880	840-880 OIL	550-700	840-880 COOL IN AIR
H50460	5046H	0.43-0.50	0.65-1.10	0.15-0.35	0.04	0.035	0.13-0.43	---	---	---	65 80	36 64	18 14	160 230	HR & N H & T	SAME AS ABOVE BUT FOR HIGH TENSILE FASTENERS OF P-8.8	1200	840-880	840-880 OIL	550-700	840-880 COOL IN AIR
H50461	50B46H	0.43-0.50	0.65-1.10	0.15-0.35	0.04	0.035	0.13-0.43	---	---	0.0005-0.003 BORON	70 90	40 70	18 14	180 250	HR & N H & T	SAME AS ABOVE BUT FOR HIGH TENSILE FASTENERS OF P-8.8 & ABOVE	1200	840-880	840-880 OIL	550-700	840-880 COOL IN AIR
H50501	50B50H	0.47-0.54	0.65-1.10	0.15-0.35	0.04	0.035	0.30-0.70	---	---	0.0005-0.003 BORON	70 100	40 90	18 10	180 340	HR & N H & T	SAME AS ABOVE BUT FOR HIGH TENSILE FASTENERS OF P-10.9 & ABOVE	1200	840-880	840-880 OIL	550-700	840-880 COOL IN AIR
H50601	50B60H	0.55-0.65	0.65-1.10	0.15-0.35	0.04	0.035	0.30-0.70	---	---	0.0005-0.003 BORON	70 100	40 90	18 10	180 340	HR & N H & T	SAME AS ABOVE BUT FOR HIGH TENSILE FASTENERS OF P-10.9 & ABOVE	1200	840-880	840-880 OIL	550-700	840-880 COOL IN AIR
H51200	5120H	0.17-0.23	0.60-1.00	0.15-0.35	0.04	0.035	0.60-1.00	---	---	---	50 80	30 ---	22 9	160 400	HR & N CH & T	SAME AS ABOVE	1200	840-880	SAME AS ABOVE	200-250	840-880 COOL IN AIR
H51300	5130H	0.27-0.33	0.60-1.00	0.15-0.35	0.04	0.035	0.75-1.20	---	---	---	70 82	42 65	16 14	200 230/41	HR & N H & T	SAME AS ABOVE AND ALSO FOR BOLTS, NUTS TO P-8.8, GR.S	1200	890-940	870-930 OIL OR WATER	550-660	890-940 COOL IN AIR
H51320	5132H	0.29-0.35	0.50-0.90	0.15-0.35	0.04	0.035	0.65-1.10	---	---	---	70 82	42 65	16 14	200 230/41	HR & N H & T	SAME AS ABOVE AND ALSO FOR BOLTS, NUTS TO P-8.8, GR.S	1200	890-940	870-930 OIL OR WATER	550-660	890-940 COOL IN AIR
H51350	5135H	0.32-0.38	0.50-0.90	0.15-0.35	0.04	0.035	0.70-1.15	---	---	---	75 92	45 68	15 13	210 250/37	HR & N H & T	SAME AS ABOVE AND ALSO FOR BOLTS, NUTS TO P-8.8, GR.S	1200	890-940	870-930 OIL OR WATER	550-660	890-940 COOL IN AIR
H51400	5140H	0.37-0.44	0.60-1.00	0.15-0.35	0.04	0.035	0.60-1.00	---	---	---	75 92	45 68	15 13	210 250/37	HR & N H & T	SAME AS ABOVE AND ALSO FOR BOLTS, NUTS TO P-8.8, GR.S	1200	890-940	870-930 OIL OR WATER	550-660	890-940 COOL IN AIR

*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED.

ABBREVATIONS USED:
AR -- AS ROLLED; HR -- HOT ROLLED; CR -- COLD ROLLED; AC -- AIR COOLED; CD -- COLD DRAWN; OH -- OIL HARDENED; N -- NORMALISED; AH -- AIR HARDENED; A -- ANNEALED; H & T -- HARDENED & TEMPERED; F & I -- FLAME & INDUCTION HARDENING; S -- SOFTENED; FC -- FURNACE COOL; CS CASE HARDENED

CONTD

CHEMISTRY & MECH.PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - US STANDARD GRADES

DESIGNATION UNS NO.	SAE/AISI NO.	* CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
ALLOYED "H" STEELS (STEELS WITH SCATTER BAND OF HARDENABILITY) - CONTD.		C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J)			HOT FORMING TEMP. °C	ANNEALING TEMP. °C (FC)	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZING TEMP °C
H51470	5147H	0.45-0.52	0.60-1.05	0.15-0.35	0.04	0.035	0.80-1.25	---	---	---	80 102	50 92	14 11	240 350	HR & N H & T	SAME AS 5140H AND ALSO FOR BOLTS, NUTS TO P-10.9/GR.T	1200	890-940	870-930 OIL OR WATER	550-660	890-940 COOL IN AIR
H51500	5150H	0.47-0.54	0.60-1.00	0.15-0.35	0.04	0.035	0.60-1.00	---	---	---	80 102	50 92	14 11	240 350	HR & N H & T	MOSTLY USED FOR TRACK PINS, CRANK-SHAFTS, CONNEC-TING RODS, AXLES, SHAFTS, GEARS, BOLTS, NUTS, STUDS, TO P-10.9, GR.T, ETC.	1200	830-880	830-880 OIL	550-720	830-880 COOL IN AIR
H51550	5155H	0.50-0.60	0.60-1.00	0.15-0.35	0.04	0.035	0.60-1.00	---	---	---	85 112	55 100	14 10	250 380	HR & N H & T	SAME AS ABOVE	1200	830-880	830-880 OIL	550-720	830-880 COOL IN AIR
H51600	5160H	0.55-0.65	0.65-1.10	0.15-0.35	0.04	0.035	0.60-1.00	---	---	---	85 122	55 110	14 9	250 400	HR & N H & T	SAME AS ABOVE AND ALSO FOR FASTENERS TO P-12.9 AND LAMI-NATED SPRINGS, ETC.	1200	830-880	830-880 OIL	550-720	830-880 COOL IN AIR
H51601	51B60H	0.55-0.65	0.65-1.10	0.15-0.35	0.04	0.035	0.60-1.00	---	---	0.0005-0.003 BORON	80 120	50 108	14 9	240 400	HR & N H & T	MOSTLY USED FOR TRACK PINS, CRANK-SHAFTS, CONNEC-TING RODS, AXLES, SHAFTS, GEARS, BOLTS, NUTS, STUDS, TO P-10.9, GR.T AND ABOVE.	1200	830-880	830-880 OIL	550-720	830-880 COOL IN AIR
H61180	6118H	0.15-0.21	0.40-0.80	0.15-0.35	0.040	0.035	0.40-0.80	---	---	0.10-0.15V	55 80	30 ---	20 14	170 400/48	HR & N CH & T	CASE HARDENING STEEL SUITABLE FOR CAMSHAFTS, BUSH-INGS, AUTOMOBILE TRANSMISSION COMPONENTS, GUDGEON PINS, CLUTCH PLATES.	1150	840-880	CARBURI-ZING AT 880-930 FOLLOWED BY HARDEN-ING AT 780-820 OIL	200-250	840-880 COOL IN AIR
H61500	6150H	0.47-0.54	0.60-1.00	0.15-0.35	0.04	0.035	0.75-1.20	---	---	0.15V	80 122	50 110	14 9	240 450	HR & N H & T	MOSTLY USED FOR TRACK PINS, CRANK-SHAFTS, CONNEC-TING RODS, AXLES, SHAFTS, GEARS, BOLTS, NUTS, STUDS, TO P-12.9, GR.U, ETC.	1200	830-880	830-880 OIL	550-720	830-880 COOL IN AIR
H61451	81B45H	0.42-0.49	0.70-1.05	0.15-0.35	0.04	0.035	0.30-0.60	0.15-0.45	0.08-0.15	0.0005-0.003 BORON	80 120	50 108	14 9	240 400	HR & N H & T	MOSTLY USED FOR TRACK PINS, CRANK-SHAFTS, AXLES, BOLTS, NUTS, STUDS, TO P-10.9, GR.T AND ABOVE.	1200	830-880	830-880 OIL	550-720	830-880 COOL IN AIR

*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED.

ABBREVATIONS USED:
AR -- AS ROLLED; HR -- HOT ROLLED; CR -- COLD ROLLED; AC -- AIR COOLED; CD -- COLD DRAWN; OH -- OIL HARDENED; N -- NORMALISED; AH -- AIR HARDENED; A -- ANNEALED; H & T -- HARDENED & TEMPERED; F & I -- FLAME & INDUCTION HARDENING; S -- SOFTENED; FC -- FURNACE COOL; CS - CASE HARDENED

CONTD

CHEMISTRY & MECH.PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - US STANDARD GRADES

DESIGNATION UNS NO.	SAE/AISI NO.	* CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
ALLOYED "H" STEELS (STEELS WITH SCATTER BAND OF HARDENABILITY) - CONTD.		C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J)			HOT FORMING TEMP. °C	ANNEALING TEMP. °C (FC)	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZING TEMP. °C
H86170	8617H	0.14-0.20	0.60-0.95	0.15-0.35	0.04	0.035	0.35-0.65	0.35-0.75	0.15-0.25	---	62 90	36 ---	19 8	180 450	HR & N CH & T	CASE HARDENING STEEL SUITABLE FOR CAMSHAFTS, BUSHINGS, AUTOMOBILE TRANSMISSION COMPONENTS, GUDGEON PINS.	1150	840-880	CARBURIZING AT 880-930 FOLLOWED BY HARDENING AT 780-820 OIL	200-250	840-880 COOL IN AIR
H86200	8620H	0.17-0.23	0.60-0.95	0.15-0.35	0.04	0.035	0.35-0.65	0.35-0.75	0.15-0.25	---	65 100	40 ---	18 8	190 480	HR & N CH & T	SAME AS ABOVE	1150	840-880	SAME AS ABOVE	200-250	840-880 COOL IN AIR
H86220	8622H	0.19-0.25	0.60-0.95	0.15-0.35	0.04	0.035	0.35-0.65	0.35-0.75	0.15-0.25	---	70 82	42 65	16 14	200 230/41	HR & N H & T	LOW ALLOY STEEL SUITABLE FOR BOLTS, NUTS TO P-8.8, GR.S	1200	890-940	870-930 OIL OR WATER	550-660	890-940 COOL IN AIR
H86250	8625H	0.22-0.28	0.60-0.95	0.15-0.35	0.04	0.035	0.35-0.65	0.35-0.75	0.15-0.25	---	70 86	42 68	16 14	200 250/41	HR & N H & T	LOW ALLOY STEEL SUITABLE FOR BOLTS, NUTS TO P-8.8, GR.S	1200	890-940	870-930 OIL OR WATER	550-660	890-940 COOL IN AIR
H86270	8627H	0.24-0.30	0.60-0.95	0.15-0.35	0.04	0.035	0.35-0.65	0.35-0.75	0.15-0.25	---	70 90	42 70	16 12	200 270/38	HR & N H & T	LOW ALLOY STEEL SUITABLE FOR BOLTS, NUTS TO P-8.8, GR.S, AND HIGH TEMP. PIPE FITTINGS, ETC.	1200	870-930	870-930 OIL OR WATER	550-660	870-930 COOL IN AIR
H86300	8630H	0.27-0.33	0.60-0.95	0.15-0.35	0.04	0.035	0.35-0.65	0.35-0.75	0.15-0.25	---	72 95	42 72	15 12	210 280/36	HR & N H & T	SAME AS ABOVE	1200	870-930	870-930 OIL OR WATER	550-660	870-930 COOL IN AIR
H86301	86B30H	0.27-0.33	0.60-0.95	0.15-0.35	0.04	0.035	0.35-0.65	0.35-0.75	0.15-0.25	0.0005-0.003 BORON	80 102	48 92	14 10	250 380/30	HR & N H & T	SAME AS ABOVE AND BOLT, NUT OF P-10.9, GR.T, ETC.	1200	870-930	870-930 OIL OR WATER	550-660	870-930 COOL IN AIR
H86370	8637H	0.34-0.41	0.70-1.05	0.15-0.35	0.04	0.035	0.35-0.65	0.35-0.75	0.15-0.25	---	72 95	42 72	15 12	210 280/36	HR & N H & T	SUITABLE FOR HIGH TEMP. BOLTING MAT.	1200	870-930	870-930 OIL	550-660	870-930 COOL IN AIR
H86400	8640H	0.37-0.44	0.70-1.05	0.15-0.35	0.04	0.035	0.35-0.65	0.35-0.75	0.15-0.25	---	75 100	45 90	14 10	220 320/31	HR & N H & T	SUITABLE FOR HIGH TEMP. BOLTING MAT.	1150	820-870	830-870 OIL	550-660	830-870 COOL IN AIR
H86420	8642H	0.39-0.46	0.70-1.05	0.15-0.35	0.04	0.035	0.35-0.65	0.35-0.75	0.15-0.25	---	75 105	45 92	14 10	220 330/31	HR & N H & T	SUITABLE FOR HIGH TEMP. BOLTING MAT. AND P-10.9	1150	820-870	830-870 OIL	550-660	830-870 COOL IN AIR
H86450	8645H	0.42-0.49	0.70-1.05	0.15-0.35	0.04	0.035	0.35-0.65	0.35-0.75	0.15-0.25	---	75 105	45 92	14 10	220 330/31	HR & N H & T	SAME AS ABOVE	1150	820-870	830-870 OIL	550-660	830-870 COOL IN AIR

*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED.

ABBREVATIONS USED.
AR -- AS ROLLED
HR -- HOT ROLLED
CR -- COLD ROLLED
AC -- AIR COOLED
CD -- COLD DRAWN
OH -- OIL HARDENED
N -- NORMALISED
AH -- AIR HARDENED
A -- ANNEALED
H & T -- HARDENED & TEMPERED
S -- SOFTENED
CS - CASE HARDENED
F & I -- FLAME & INDUCTION HARDENING
FC -- FURNACE COOL

CONTD.

CHEMISTRY & MECH.PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - US STANDARD GRADES

DESIGNATION UNS NO. SAE/AISI NO.	* CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
ALLOYED "H" STEELS (STEELS WITH SCATTER BAND OF HARDENABILITY) - CONTD.	C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J)			HOT FORMING TEMP. °C	ANNEALING TEMP °C (FC)	HARDENING TEMP °C QUENCHANT	TEMPERING TEMP °C	NORMALIZING TEMP °C
H86451 86B45H	0.42-0.49	0.70-1.05	0.15-0.35	0.04	0.035	0.35-0.65	0.35-0.75	0.15-0.25	0.0005-0.003 BORON	80 125	50 110	12 9	250 430/25	HR & N H & T	SUITABLE FOR HIGH TEMP BOLTING MAT AND P-12.9	1150	820-870	830-870 OIL	550-660	930-970 COOL IN AIR
H86500 8650H	0.47-0.54	0.70-1.05	0.15-0.35	0.04	0.035	0.35-0.65	0.35-0.75	0.15-0.25	---	80 120	48 108	13 9	240 410/28	HR & N H & T	SAME AS ABOVE AND ALSO FOR CAM AND GEAR SHAFTS, ETC.	1150	820-870	830-870 OIL	550-660	930-970 COOL IN AIR
H86550 8655H	0.50-0.60	0.70-1.05	0.15-0.35	0.04	0.035	0.35-0.65	0.35-0.75	0.15-0.25	---	80 125	48 110	13 9	240 440/24	HR & N H & T	SAME AS ABOVE AND ALSO FOR AUTOMOTIVE SPRINGS, ETC.	1150	820-870	830-870 OIL	350-660	930-970 COOL IN AIR
H86600 8660H	0.55-0.65	0.70-1.05	0.15-0.35	0.04	0.035	0.35-0.65	0.35-0.75	0.15-0.25	---	80 125	48 110	13 9	240 440/24	HR & N H & T	SAME AS ABOVE AND ALSO FOR AUTOMOTIVE SPRINGS, ETC.	1150	820-870	830-870 OIL	350-660	930-970 COOL IN AIR
H87200 8720H	0.17-0.23	0.60-0.95	0.15-0.35	0.04	0.035	0.35-0.65	0.35-0.75	0.20-0.30	---	60 90	36 ---	20 8	170 450	HR & N CH & T	CASE HARDENING STEEL SUITABLE FOR CAMSHAFTS, BUSHINGS, AUTOMOBILE TRANSMISSION COMPONENTS, GUDGEON PINS.	1150	840-880	CARBURIZING AT 880-930 FOLLOWED BY HARDENING AT 780-820 OIL	200-250	840-880 COOL IN AIR
H87400 8740H	0.37-0.44	0.70-1.05	0.15-0.35	0.04	0.035	0.35-0.65	0.35-0.75	0.20-0.30		75 100	45 90	14 10	220 320/31	HR & N H & T	SUITABLE FOR HIGH TEMP. BOLTING MAT.	1150	820-870	830-870 OIL	550-660	930-970 COOL IN AIR
H88220 8822H	0.19-0.25	0.70-1.05	0.15-0.35	0.04	0.035	0.35-0.65	0.35-0.75	0.30-0.40	---	75 85	45 70	16 12	200 240/40	HR & N H & T	LOW ALLOY STEEL SUITABLE FOR BOLTS, NUTS TO P-8.8, GR.S	1200	890-940	870-930 OIL OR WATER	550-660	890-940 COOL IN AIR
H92600 9260H	0.55-0.65	0.65-1.10	1.70-2.20	0.04	0.035	---	---	---	---	80 125	48 110	13 9	240 440/24	HR & N H & T	SAME AS ABOVE AND ALSO FOR AUTOMOTIVE SPRINGS, ETC.	1150	820-870	830-870 OIL	350-660	930-970 COOL IN AIR
H93100 E9310H E - ELECTRIC FURNACE STEEL	0.07-0.13	0.40-0.70	0.15-0.35	0.04	0.035	1.00-1.45	2.95-3.55	0.08-0.15	---	60 90	36 ---	20 12	170 400/48	HR & N CH & T	CASE HARDENING STEEL SUITABLE FOR CAMSHAFTS, BUSHINGS, AUTOMOBILE TRANSMISSION COMPONENTS, GUDGEON PINS	1150	840-880	CARBURIZING AT 880-930 FOLLOWED BY HARDENING AT 760-780 OIL	200-250	840-880 COOL IN AIR
H94151 94B15H	0.12-0.18	0.70-1.05	0.15-0.35	0.04	0.035	0.25-0.55	0.25-0.65	0.08-0.15	0.0005--0.003 BORON	80 102	48 92	14 10	250 380/30	HR & N H & T	BORON STEEL FOR BOLT, NUT OF P-8.8, GR.S. ETC	1200	870-930	870-930 OIL	550-660	870-930 COOL IN AIR

*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED.

ABBREVATIONS USED:
AR -- AS ROLLED
HR -- HOT ROLLED
CR -- COLD ROLLED
AC -- AIR COOLED
CD -- COLD DRAWN
OH -- OIL HARDENED
N -- NORMALISED
AH -- AIR HARDENED
A -- ANNEALED
H & T -- HARDENED & TEMPERED
S -- SOFTENED
CS - CASE HARDENED
F & I -- FLAME & INDUCTION HARDENING
FC -- FURNACE COOL

CONTD.

CHEMISTRY & MECH.PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - US STANDARD GRADES

DESIGNATION UNS NO.	SAE/AISI NO.	* CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
ALLOYED "H" STEELS (STEELS WITH SCATTER BAND OF HARDENABILITY) - CONTD.		C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J)			HOT FORMING TEMP. °C	ANNEALING TEMP. °C (FC)	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZ-ING TEMP °C
H94171	94B17H	0.14-0.20	0.70-1.05	0.15-0.35	0.04	0.035	0.25-0.55	0.25-0.65	0.08-0.15	0.0005-0.003 BORON	80 102	48 92	14 10	250 380/30	HR & N H & T	BORON STEEL FOR BOLT, NUT OF P-8.8, GR.S, ETC.	1200	870-930	870-930 OIL	550-660	870-930 COOL IN AIR
H94301	94B30H	0.27-0.33	0.70-1.05	0.15-0.35	0.04	0.035	0.25-0.55	0.25-0.65	0.08-0.15	0.0005-0.003 BORON	80 102	48 92	14 10	250 380/30	HR & N H & T	BORON STEEL FOR BOLT, NUT OF P-10.9, GR.T, ETC.	1200	870-930	870-930 OIL	550-660	870-930 COOL IN AIR
PLEASE SEE NOTES AT PAGE 110 FOR VARIATION IN % CONTENT OF VARIOUS TYPES OF STEELS																					
AUSTENITIC STAINLESS CHROMIUM -NICKEL STEELS (NON MAGNETIC STAINLESS STEELS)																					
S20100	30201/201	0.15	5.50-7.50	1.00	0.03	0.06	16.00-18.00	3.50-5.50	---	0.25 N	65	33	40	217 MAX	S	SUITABLE FOR FLATWARE, AUTO WHEEL COVERS, TRIM. ETC.	1200	---	950-1150 SOFTEN AIR,OIL OR WATER	---	---
S20200	30202/202	0.15	7.50-10.00	1.00	0.03	0.06	17.00-19.00	4.00-6.00	---	0.25 N	65	33	40	217 MAX	S	SUITABLE FOR FLATWARE, AUTO WHEEL COVERS, TRIM, FASTENERS TO GR. A2-70, ETC.	1200	---	950-1150 SOFTEN AIR, OIL OR WATER	---	---
S20300	--- / XM1	0.08	5.00-6.50	1.00	0.18-0.35	0.04	16.00-18.00	5.00-6.50	0.50	1.75-2.25 Cu											
S30100	30301/301	0.15	2.00	1.00	0.03	0.045	16.00-18.00	6.00-8.00	---	---	60 85	22 59	40 12	212 MAX	S CD	SUITABLE FOR FLATWARE, AUTO WHEEL COVERS, TRIM., FASTENERS. AIRCRAFT STRUCT-URALS, ETC.	1200	---	950-1150 SOFTEN AIR,OIL OR WATER	---	---
S30200	30302/302	0.15	2.00	1.00	0.03	0.045	17.00-19.00	8.00-10.00	---	---	60 85	22 59	40 12	212 MAX	S CD	SUITABLE FOR HOUSE HOLDWARE, AUTO. COVERS, REFRIGERATOR TRAYS, FASTENERS, PHOTOGRAPHIC EQUIPMENTS, ETC.	1200	---	950-1150 SOFTEN AIR,OIL OR WATER	---	---
S30200	30302/302	0.15	2.00	1.00	0.03	0.045	17.00-19.00	8.00-10.00	---	---	60 85	22 59	40 12	212 MAX	S CD	SUITABLE FOR HOUSE HOLDWARE, AUTO. COVERS, REFRIGERATOR TRAYS, FASTENERS, PHOTOGRAPHIC EQUIPMENTS, ETC.	1200	---	950-1150 SOFTEN AIR,OIL OR WATER	---	---
S30215	30302B/302B	0.15	2.00	2.00-3.00	0.03	0.045	17.00-19.00	8.00-10.00	---	---	60	22	40	212 MAX	S	SUITABLE FOR FURNACE CONVEY-ERS, CARBURISING BOXES.	1200	---	950-1150 SOFTEN AIR,OIL OR WATER	---	---
S30300	30303/303	0.15	2.00	1.00	0.15	0.20 MIN.	17.00-19.00	8.00-10.00	---	0.60 Zr OR Mo	60 85	22 59	40 12	212 MAX	S CD	SUITABLE FOR HOUSE HOLDWARE, FASTENERS (A1-50/ A1-70)SMALL MACHI-NED COMPONENTS.	1200	---	950-1150 SOFTEN AIR,OIL OR WATER	---	---
		*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED.																			

ABBREVATIONS USED: HR -- HOT ROLLED; AR -- AS ROLLED; CR -- COLD ROLLED; AC -- AIR COOLED; CD -- COLD DRAWN; OH -- OIL HARDENED; N -- NORMALISED; AH -- AIR HARDENED; A -- ANNEALED; H & T -- HARDENED & TEMPERED; F & I -- FLAME & INDUCTION HARDENING; S -- SOFTENED; FC -- FURNACE COOL; CS - CASE HARDENED

CONTD.

CHEMISTRY & MECH.PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - US STANDARD GRADES

UNS NO.	DESIGNATION SAE/AISI NO.	*CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
AUSTENITIC STAINLESS CHROMIUM -NICKEL STEELS (NON MAGNETIC STAINLESS STEELS) CONTD.		C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J)			HOT FORMING TEMP. °C	ANNEALING TEMP. °C (FC)	HARDENING TEMP °C QUENCHANT	TEMPERING TEMP °C	NORMALIZING TEMP °C
S30400	30304 / 304	0.08	2.00	1.00	0.03	0.045	18.00-20.00	8.00-10.50	---	---	50 85	21 50	40 12	192 MAX / 69	S CD	SUITABLE FOR KITCHENWARE, FASTENERS TO GR. A1-50, A1-70 CHEMICAL HANDLING EQUIPMENTS, HOSPITAL BOWLS, WASHING MACHINES, DOOR BARS, ETC.	1200	---	950-1150 SOFTEN AIR,OIL OR WATER	---	---
S30430	--- / XM7	0.10	2.00	1.00	0.03	0.045	17.00-19.00	8.00-10.00	---	3.00-4.00 Cu											
S38900	--- / 384	0.07	1.00	1.00	0.03	0.045	15.00-17.00	17.00-19.00	---	---											
S30403	30304L / 304L	0.03	2.00	1.00	0.03	0.045	18.00-20.00	8.00-12.00	---	---	45 80	18 48	40 14	192 MAX / 69	S CD	SAME AS ABOVE WITH LOW CARBON CONTENT, USABLE IN CHEMICAL, TEXTILE, PETROLEUM, PHARMACEUTICAL AND FOOD INDUSTRIES.	1200	---	950-1150 SOFTEN AIR,OIL OR WATER	---	---
S30500	30305 / 305	0.12	2.00	1.00	0.03	0.045	17.00-19.00	10.50-13.50	---	---	50 85	21 50	40 12	192 MAX / 69	S CD	SAME AS AISI-304 WITH HIGHER Ni AND C CONTENT.	1200	---	950-1150 SOFTEN AIR,OIL OR WATER	---	---
S30800	30308 / 308	0.08	2.00	1.00	0.03	0.045	19.00-21.00	10.00-12.00	---	---	50 85	21 50	40 12	192 MAX / 69	S CD	SAME AS AISI-304 WITH HIGHER Cr CONTENT.	1200	---	950-1150 SOFTEN AIR,OIL OR WATER	---	---
S30900	30309 / 309	0.20	2.00	1.00	0.03	0.045	22.00-24.00	12.00-15.00	---	---	55	23	30	223 MAX / 59	S	MOSTLY USED FOR HEAT TREATING FURNACE PARTS, ANNEALING TRAYS, EXHAUST MANIFOLD, RADIANT TUBES,ETC.	1200	---	950-1150 SOFTEN AIR,OIL OR WATER	---	---
S30908	30309S / 309S	0.08	2.00	1.00	0.03	0.045	22.00-24.00	12.00-15.00	---	---	50	21	40	192 MAX / 69	S	SAME AS ABOVE CAN BE WELDED TO GIVE DIFFERENT SHAPES.	1200	---	950-1150 SOFTEN AIR,OIL OR WATER	---	---
S31000	30310 / 310	0.25	2.00	1.50	0.03	0.045	24.00-26.00	19.00-22.00	---	---	55	23	30	223 MAX / 59	S	MOSTLY USED FOR HYDROGENATION TUBES, HEAT EXCHANGERS, ANNEALING BOXES, FURNACE DOORS, RETORTS, ETC.	1200	---	950-1150 SOFTEN AIR,OIL OR WATER	---	---
S31008	30310S / 310S	0.08	2.00	1.50	0.03	0.045	24.00-26.00	19.00-22.00	---	---	55	23	30	223 MAX / 59	S	SAME AS ABOVE WELDABLE QUALITY CORROSION RESISTANT AT HIGHER TEMP.	1200	---	950-1150 SOFTEN AIR,OIL OR WATER	---	---

*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED.

ABBREVATIONS USED:
AR -- AS ROLLED HR -- HOT ROLLED AC -- AIR COOLED OH -- OIL HARDENED AH -- AIR HARDENED H&T -- HARDENED & TEMPERED S -- SOFTENED CS - CASE HARDENED
CR -- COLD ROLLED CD -- COLD DRAWN N -- NORMALISED A -- ANNEALED F&I -- FLAME & INDUCTION HARDENING FC -- FURNACE COOL

CONTD

CHEMISTRY & MECH.PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - US STANDARD GRADES

DESIGNATION UNS NO.	SAE/AISI NO.	* CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
		C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J			HOT FORMING TEMP. °C	ANNEALING TEMP. °C (FC)	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZ-ING TEMP.°C
AUSTENITIC STAINLESS CHROMIUM -NICKEL STEELS (NON MAGNETIC STAINLESS STEELS) CONTD.																					
S31400	30314 / 314	0.25	2.00	1.50-3.00	0.03	0.045	23.00-26.00	19.00-22.00	---	---	55	23	35	223 MAX / 59	S	MOSTLY USED FOR HYDROGENATION TUBES, HEAT EX-CHANGERS, ANN-EALING BOXES, FURNACE DOORS, RETORTS, ETC.	1200	---	950-1150 SOFTEN AIR,OIL OR WATER	---	---
S31600	30316/316	0.08	2.00	1.00	0.03	0.045	16.00-18.00	10.00-14.00	2.00-3.00	---	50 85	21 50	40 12	192 MAX / 69	S CD	SUITABLE FOR FASTENERS TO GR. A4-50, A4-70 CHEMICAL HAND-LING EQUIPMENTS, PAPER, TEXTILE & PRINTING INDUS-TRIES, DYE VATS, MARINE PARTS,ETC.	1200	---	950-1150 SOFTEN AIR,OIL OR WATER	---	---
S31603	30316L / 316L	0.03	2.00	1.00	0.03	0.045	16.00-18.00	10.00-14.00	2.00-3.00	---	45 85	20 50	40 12	192 MAX / 69	S CD	SUITABLE FOR FASTENERS TO GR. A4-50, A4-70,80 HIGH TEMP. CHEMI-CAL, RAYON, RUBBER, AND MARINE , PHOTO-GRAPHIC DEVELOP-MENT, PULP HAND-LING, FOOD PROCE-SSING EQUIPMENTS, STEAM JACKETED KETTLES, ETC.	1200	---	950-1150 SOFTEN AIR,OIL OR W. TER	---	---
S31700	30317 / 317	0.08	2.00	1.00	0.03	0.045	18.00-20.00	11.00-15.00	3.00-4.00	---	55	24	35	213 MAX	S	SAME AS AISI-316	1200	---	950-1150 SOFTEN AIR,OIL OR WATER	---	---
S32100	30321 / 321	0.08	2.00	1.00	0.03	0.045	17.00-19.00	9.00-12.00	---	5 X C Ti MIN.	50	21	35	192 MAX	S	AIRCRAFT ENGINE EXHAUST MANIFO-LDS, BOILER SHIELDS, COLLECTOR RINGS, HIGH TEMP. CHEMI-CAL HANDLING EQUIPMENTS, FASTENERS TO GR. A2-50,70,80	1200	---	950-1150 SOFTEN AIR,OIL OR WATER	---	---
---	30330 / ---	0.15	2.00	1.00	0.03	0.045	14.00-17.00	33.00-37.00	---	---	60	26	30	223 MAX	S	MOSTLY USED FOR COMPONENTS SUBJECT TO VERY HIGH CORROSION AND HIGH TEMP	1200	---	950-1150 SOFTEN AIR,OIL OR WATER	---	---

*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED

CONTD

ABBREVATIONS USED
AR -- AS ROLLED
HR -- HOT ROLLED
CR -- COLD ROLLED
AC -- AIR COOLED
CD -- COLD DRAWN
OH -- OIL HARDENED
N -- NORMALISED
AH -- AIR HARDENED
A -- ANNEALED
H & T -- HARDENED & TEMPERED
F & I -- FLAME & INDUCTION HARDENING
S -- SOFTENED
CS - CASE HARDENED
FC -- FURNACE COOL

CHEMISTRY & MECH.PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - US STANDARD GRADES

DESIGNATION UNS NO.	SAE/AISI NO.	*CHEMICAL COMPOSITION, PERCENT									MECHANICAL PROPERTIES (MIN.)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
		C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J)			HOT FORMING TEMP. °C	ANNEALING TEMP. °C (FC)	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZING TEMP. °C
AUSTENITIC STAINLESS CHROMIUM -NICKEL STEELS (NON MAGNETIC STAINLESS STEELS) CONTD.																					
S34700	30347 / 347	0.08	2.00	1.00	0.03	0.045	17.00-19.00	9.00-13.00	---	10 X C MIN. Nb - Ti	50	21	35	192 MAX	S	AIRCRAFT ENGINE EXHAUST MANIFOLDS, BOILER SHIELDS, COLLECTOR RINGS, HIGH TEMP. CHEMICAL HANDLING EQUIPMENTS, FASTENERS TO GR. A2-50,70,80	1200	---	950-1150 SOFTEN AIR,OIL OR WATER	---	---
S34800	30348 / 348	0.08	2.00	1.00	0.03	0.045	17.00-19.00	9.00-13.00	---	10 X C MIN. Nb - Ti 0.10 Ta	50	21	35	192 MAX	S	SAME AS ABOVE	1200	---	950-1150 SOFTEN AIR,OIL OR WATER	---	---
S38400	30384 / 384	0.08	2.00	1.00	0.03	0.045	15.00-17.00	17.00-19.00	---	---	55	23	35	213 MAX	S	SAME AS AISI-304	1200	---	950-1150 SOFTEN AIR,OIL OR WATER	---	---
S38500	30385 / 385	0.08	2.00	1.00	0.03	0.045	11.50-13.50	14.00-16.00	---	---	50	21	40	192 MAX	S	HIGH GRADE CORROSION RESISTANCE STEEL FOR SPECIAL APPLICATIONS ONLY	1200	---	950-1150 SOFTEN AIR,OIL OR WATER	---	---
MARTENSITIC STAINLESS CHROMIUM STEELS (MAGNETIC STAINLESS STEELS)																					
S40300	51403/403	0.15	1.00	0.50	0.03	0.040	11.50-13.00	---	---	---	45 60	25 45	20 15	200 MAX 230	A H & T	MOSTLY USED FOR AIRCRAFT RIVETS, STEAM TURBINE BLADES, STRUCTURAL PARTS, HOUSEHOLD ARTICLES, BLDG. FITTINGS,ETC.	1100	750-800 FURNACE COOL	950-1000 OIL/AIR	680-780 AIR COOL	---
S41000	51410/410	0.15	1.00	1.00	0.03	0.040	11.50-13.50	---	---	---	45 60	25 45	20 15	200 MAX 230	A H & T	SAME AS ABOVE AND ALSO FOR FASTENERS TO GR. C1-50, 70, 110	1100	750-800 FURNACE COOL	950-1000 OIL/AIR	680-780 AIR COOL	---
S41400	51414/414	0.15	1.00	1.00	0.03	0.040	11.50-13.50	1.25-2.50	---	---	50 65	28 50	18 14	213 MAX 240	A H & T	SAME AS ABOVE AND ALSO FOR PAPER MACHINE PARTS.	1100	750-800 FURNACE COOL	950-1000 OIL/AIR	680-780 AIR COOL	---
S41400	51414/414	0.15	1.00	1.00	0.03	0.040	11.50-13.50	1.25-2.50	---	---	50 65	28 50	18 14	213 MAX 240	A H & T	SAME AS ABOVE AND ALSO FOR PAPER MACHINE PARTS	1100	750-800 FURNACE COOL	950-1000 OIL/AIR	680-780 AIR COOL	---

*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED.

ABBREVATIONS USED:
AR -- AS ROLLED
HR -- HOT ROLLED
CR -- COLD ROLLED
AC -- AIR COOLED
CD -- COLD DRAWN
OH -- OIL HARDENED
N -- NORMALISED
AH -- AIR HARDENED
A -- ANNEALED
H & T -- HARDENED & TEMPERED
F & I -- FLAME & INDUCTION HARDENING
S -- SOFTENED
CS : CASE HARDENED
FC -- FURNACE COOL

CONTD

CHEMISTRY & MECH.PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - US STANDARD GRADES

DESIGNATION UNS NO.	SAE/AISI NO.	* CHEMICAL COMPOSITION, PERCENT									MECH ANICAL PROPERTIES (MIN.)				CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
		C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J)			HOT FORMING TEMP. °C	ANNEALING TEMP °C (FC)	HARDENING TEMP °C QUENCHANT	TEMPERING TEMP °C	NORMALIZING TEMP °C
MARTENSITIC STAINLESS CHROMIUM STEELS (MAGNETIC STAINLESS STEELS) (CONTD.)																					
S41600	51416/416	0.15	1.25	1.00	0.15 MIN.	0.060	12.00-14.00	---	---	0.60 Zr OR Mo	45 60	25 45	20 15	200 MAX 230	A H & T	FREE CUTTING STEEL FOR COMPONENTS SUBJECT TO CORROSION IN MECHANICAL AND PLANT ENGG. & FASTENERS TO GR. C4-50. 70. 110	1100	750-800 FURNACE COOL	950-1000 OIL/AIR	680-780 AIR COOL	---
S41623	51416 Se/416 Se	0.15	1.25	1.00	0.06	0.060	12.00-14.00	---	---	0.15 MIN. Se	45 60	25 45	20 15	200 MAX 230	A H & T	SAME AS ABOVE	1100	750-800 FURNACE COOL	950-1000 OIL/AIR	680-780 AIR COOL	---
S42000	41420/420	>0.15	1.00	1.00	0.03	0.04	12.00-14.00	---	---	---	45 79	25 60	20 10	200 MAX 500	A H & T	MOSTLY USED FOR FOR HOUSE HOLD CUTLERY, HAND & MACHINE KNIVES, SPRINGS, SURGICAL AND DENTAL INSTRUMENTS, ETC.	1100	700-750 FURNACE COOL	950-1020 OIL/AIR	150-450 AIR OR WATER COOL	---
S42020	51420F/420F	0.15 MIN.	1.25	1.00	0.15 MIN.	0.06	12.00-14.00	---	---	0.60	45 79	25 60	20 10	200 MAX 500	A H & T	SAME AS AISI-416	1100	750-800 FURNACE COOL	950-1020 OIL/AIR	680-780 AIR COOL	---
S43100	51431/431	0.20	1.00	1.00	0.03	0.04	15.00-17.00	1.25-2.50	---	---	65 85	35 65	18 10	230 MAX 250	A H & T	USABLE FOR AIR-CRAFT & GENERAL ENGG. FITTINGS, PAPER MACHINE PARTS, WIND SHIELD WIPER ARMS, FASTENERS TO GR C3-80. 120	1100	750-800 FURNACE COOL	950-1000 OIL/AIR	680-780 AIR COOL	---
S44002	51440A/440A	0.60-0.75	1.00	1.00	0.03	0.04	16.00-18.00	---	---	0.75 Mo				265 570-590	A H & T	USABLE FOR KNIFE BLADES, CUTTING SURGICAL INSTRUMENTS, CORROSION RESISTANCE ROLLER BEARINGS, VALVE NEEDLES, PISTONS FOR REFRIGERATING MACHINES, ETC.	1100	800-850 FURNACE COOL	1000-1050 OIL	100-200 AIR COOL	---
S44003	51440B/440B	0.75-0.95	1.00	1.00	0.03	0.04	16.00-18.00	---	---	0.75 Mo				265 570-590	A H & T	SAME AS ABOVE	1100	800-850 FURNACE COOL	1000-1050 OIL	100-200 AIR COOL	---
S44004	51440C/440C	0.95-1.20	1.00	1.00	0.03	0.04	16.00-18.00	---	---	0.75 Mo				265 570-600	A H & T	SAME AS ABOVE	1100	800-850 FURNACE COOL	1000-1050 OIL	100-200 AIR COOL	---
---	51440F/—	0.95-1.20	1.25	1.00	0.15 MIN.	0.06	16.00-8.00	---	---	0.75 Zr. OR Mo				265 570-600	A H & T	SAME AS ABOVE FREE CUTTING STEEL	1100	800-850 FURNACE COOL	1000-1050 OIL	100-200 AIR COOL	---

*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED.

ABBREVATIONS USED:
HR -- HOT ROLLED, AC -- AIR COOLED, OH -- OIL HARDENED, AH -- AIR HARDENED, H & T -- HARDENED & TEMPERED, S - SOFTENED, CS - CASE HARDENED
AR -- AS ROLLED, CR -- COLD ROLLED, CD -- COLD DRAWN, N -- NORMALISED, A -- ANNEALED, F & I -- FLAME & INDUCTION HARDENING, FC -- FURNACE COOL

CONTD.

CHEMISTRY & MECH.PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - US STANDARD GRADES

DESIGNATION UNS NO.	SAE/AISI NO.	*CHEMICAL COMPOSITION, PERCENT C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	MECH ANICAL PROPERTIES (MIN.) UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J)	CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME HOT FORMING TEMP. °C	ANNEALING TEMP. °C (FC)	HARDENING TEMP °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZ-ING TEMP °C
MARTENSITIC STAINLESS CHROMIUM STEELS (MAGNETIC STAINLESS STEELS) (CONTD.)																					
---	51440F /--	0.95- 1.20	1.25	1.00	0.06	0.06	16.00- 18.00	---	---	0.15 Se MIN				285 570- 600	A H & T	USABLE FOR KNIFE BLADES, CUTTING SURGICAL INSTRUMENTS, CORROSION RESIS-TANCE ROLLER BEARINGS, VALVE NEEDLES, PISTONS FOR REFRIGERATING MACHINES, ETC.	1100	800-850 FURNACE COOL	1000-1050 OIL	100-200 AIR COOL	---
S50100	51501/501	>0.10	1.00	1.00	0.03	0.04	4.00- 6.00	---	---	0.40- 0.65 Mo	45 70	25 54	22 16	200 MAX 230	A H & T	MOSTLY USED FOR TUBES, PIPE FITTINGS, PLATES AND BOLT-ING MATERIAL FOR CHEMICAL AND PETROLEUM INDUS-TRIES, ETC.	1150	920-980 FURNACE COOL	920-980 OIL/AIR	680-700 AIR OR WATER COOL	---
S50200	51502/502	0.10	1.00	1.00	0.03	0.04	4.00- 6.00	---	---	0.40- 0.65 Mo	45 70	25 54	22 16	200 MAX 230	A H & T	SAME AS ABOVE	1150	920-980 FURNACE COOL	920-980 OIL/AIR	680-700 AIR OR WATER COOL	---
FERRITIC STAINLESS CHROMIUM STEELS (MAGNETIC STAINLESS STEELS)																					
S40500	51405/405	0.08	1.00	1.00	0.03	0.040	11.50- 14.50	---	---	0.10- 0.30 Al	45 60	25 45	20 15	200 MAX 230	A H & T	MOSTLY USED FOR AIRCRAFT RIVETS, STEAM TURBINE BLADES, STRUCTU-RAL PARTS, HOUSE-HOLD ARTICLES, BLDG. FITTINGS,ETC.	1100	750-800 FURNACE COOL	950-1000 OIL/AIR	680-780 AIR COOL	---
S40900	51409/409	0.08	1.00	1.00	0.03	0.040	10.50- 11.75	0.50	---	6 X C 0.75 Ti	45 60	25 45	20 15	200 MAX 230	A H & T	SAME AS ABOVE	1100	750-800 FURNACE COOL	950-1000 OIL/AIR	680-780 AIR COOL	---
S42900	51429/429	0.12	1.00	1.00	0.03	0.040	14.00- 16.00	---	---	---	45 60	25 45	20 15	200 MAX 230	A H & T	SAME AS ABOVE ALSO FOR FASTEN-ERS TO GR. F1-45, 60	1100	750-800 FURNACE COOL	950-1000 OIL/AIR	680-780 AIR COOL	---
S43000	51430/430	0.12	1.00	1.00	0.03	0.040	16.00- 18.00	---	---	---	45 60	25 45	20 15	200 MAX 230	A H & T	SAME AS ABOVE ALSO FOR FASTEN-ERS TO GR. F1-45, 60	1100	750-800 FURNACE COOL	950-1000 OIL/AIR	680-780 AIR COOL	---
S43020	51430F/430F	0.12	1.25	1.00	0.15 MIN	0.06	16.00- 18.00	0.60	---	---	45 65	25 45	20 14	200 MAX 230	A H & T	SAME AS ABOVE ALSO FOR FASTEN-ERS TO GR. F1-45, 60 (FREE CUTTING STEEL)	1100	750-800 FURNACE COOL	950-1030 OIL/AIR	680-780 AIR COOL	---
S43023	51430FSe/430F Se	0.12	1.25	1.00	0.06	0.06	16.00- 18.00	---	---	0.15 MIN	45 65	25 45	20 14	200 MAX 230	A H & T	SAME AS ABOVE	1100	750-800 FURNACE COOL	950-1030 OIL/AIR	680-780 AIR COOL	---

*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED

ABBREVATIONS USED:
AR -- AS ROLLED
HR -- HOT ROLLED
CR -- COLD ROLLED
AC -- AIR COOLED
CD -- COLD DRAWN
OH -- OIL HARDENED
N -- NORMALISED
AH -- AIR HARDENED
A -- ANNEALED
H & T -- HARDENED & TEMPERED
F & I -- FLAME & INDUCTION HARDENING
S -- SOFTENED
CS -- CASE HARDENED
FC -- FURNACE COOL

CONTD

CHEMISTRY & MECH.PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - US STANDARD GRADES

DESIGNATION UNS NO.	SAE/AISI NO.	* CHEMICAL COMPOSITION, PERCENT C	Mn	Si	S	P	Cr	Ni	Mo	OTHERS	MECH ANICAL PROPERTIES (MIN.) UTS Kg/mm²	YS	E	% HARDNESS (BHN) / IZOD(J)	CONDITION / NOMINAL THK. / DIA MIN.	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME HOT FORMING TEMP. °C	ANNEALING TEMP. °C (FC)	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZ-ING TEMP.°C
FERRITIC STAINLESS CHROMIUM STEELS (MAGNETIC STAINLESS STEELS) (CONTD.)																					
S43400	51434/434	0.12	1.00	1.00	0.03	0.040	16.00-18.00	---	0.75-1.25	---	45 70	25 50	20 14	200 MAX 250	A H & T	MOSTLY USED FOR AIRCRAFT RIVETS, STEAM TURBINE BLADES, HIGH TENSILE BOLTS, NUTS, ETC.	1100	750-800 FURNACE COOL	950-1000 OIL/AIR	680-780 AIR COOL	---
S43600	51436/436	0.12	1.00	1.00	0.03	0.040	16.00-18.00	---	0.75-1.25	5 X C-0.70 Nb+Ta	45 70	25 50	20 14	200 MAX 250	A H & T	SAME AS ABOVE AND FOR SPECIAL APPLICATIONS	1100	750-800 FURNACE COOL	950-1000 OIL/AIR	680-780 AIR COOL	---
S43900	51439/439	0.07	1.00	1.00	0.03	0.040	17.00-19.00	0.50	0.75-1.25	0,20+ 4(C+N)-1.10 Ti 0.15 Al 0.04 N	45 70	25 50	20 14	200 MAX 250	A H & T	SAME AS ABOVE AND FOR SPECIAL APPLICATIONS	1100	750-800 FURNACE COOL	950-1000 OIL/AIR	680-780 AIR COOL	---
S43903	51439LL/439LL	0.014	1.00	1.00	0.03	0.040	17.00-19.00	0.50	0.75-1.25	0,20+ 4(C+N)-1.10 Ti 0.15 Al 0.04 N	45 70	25 50	20 14	200 MAX 250	A H & T	SAME AS ABOVE AND FOR SPECIAL APPLICATIONS	1100	750-800 FURNACE COOL	950-1000 OIL/AIR	680-780 AIR COOL	---
S44200	51442/—	0.20	1.00	1.00	0.035	0.040	18.00-23.00	---	---	---	45 70	25 50	20 14	200 MAX 250	A H & T	MOSTLY USED FOR FURNACE PARTS, STEAM TURBINE BLADES, NOZZLES, COMBUSTION CHAMBERS & OTHER SPECIAL APPLICA-TIONS, ETC.	1100	750-800 FURNACE COOL	950-1000 OIL/AIR	680-780 AIR COOL	---
S44600	51446/446	0.20	1.00	1.00	0.03	0.040	23.00-27.00	---	---	0.25 N	45 70	25 50	20 14	200 MAX 250	A H & T	MOSTLY USED FOR FURNACE PARTS, STEAM TURBINE BLADES, NOZZLES, COMBUSTION CHAMBERS & OTHER SPECIAL APPLICA-TIONS, ETC.	1100	750-800 FURNACE COOL	950-1000 OIL/AIR	680-780 AIR COOL	---
17400	—/630	0.07	1.00	1.00	0.03	0.040	15.00-17.50	3.00-5.00	---	3.00-5.00 Cu 0.15-0.50 Cb+Ta	55 90	35 70	18 16	210 MAX 285	A H & T	PRECIPITATION HARDENING CORO-SSION RESISTANCE STEEL SUITABLE FOR FASTENERS TO GR P1-90 AND OTHER SPECIAL APPLICA-TIONS, ETC.	1100	750-800 FURNACE COOL	950-1000 OIL/AIR	680-780 AIR COOL	---

*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED

ABBREVATIONS USED:
AR -- AS ROLLED
HR -- HOT ROLLED
CR -- COLD ROLLED
AC -- AIR COOLED
CD -- COLD DRAWN
OH -- OIL HARDENED
N -- NORMALISED
AH -- AIR HARDENED
A -- ANNEALED
H & T -- HARDENED & TEMPERED
F & I -- FLAME & INDUCTION HARDENING
S -- SOFTENED
FC -- FURNACE COOL
CS -- CASE HARDENED

CONTD

CHEMISTRY & MECH.PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - US STANDARD GRADES

UNS NO.	DESIGNATION TYPE	* CHEMICAL COMPOSITION, PERCENT												HARDNESS (MIN.) HRC	CONDITION	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
		C	Mn	Si	S	P	Cr	Ni	Mo	V	W	Co	OTHERS				HOT FORMING TEMP. °C	ANNEALING TEMP. °C (FC)	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZING TEMP °C
HIGH SPEED STEELS (TUNGSTEN ALLOYED) - ASTM A600-86																					
T12001	T1	0.65-0.80	0.10-0.40	0.20-0.40	0.03	0.03	3.75-4.50	---	---	0.90-1.30	17.25-18.75	---	---	27 MAX 62	A H & T	SUITABLE FOR MILLING AND WORM CUTTERS, REAMERS, SLITTING SAWS, DRILLS, HACKSAWS, ETC.	950-1150	850 FURNACE COOL	1260 OIL/AIR.	570(DT) AIR COOL	---
T12002	T2	0.80-0.90	0.20-0.40	0.20-0.40	0.03	0.03	3.75-4.50	---	1.00	1.80-2.40	17.50-19.00	---	---	29MAX 62	A H & T	SAME AS ABOVE AND ALSO FOR BROACHES.	950-1150	850 FURNACE COOL	1260 OIL/AIR	570(DT) AIR COOL	---
T12004	T4	0.70-0.80	0.10-0.40	0.20-0.40	0.03	0.03	3.75-4.50	---	0.40-1.00	0.80-1.20	17.50-19.00	4.25-5.75	---	29MAX 62	A H & T	SAME AS ABOVE	950-1150	850 FURNACE COOL	1300 OIL/AIR	570(DT) AIR COOL	---
T12005	T5	0.75-0.85	0.20-0.40	0.20-0.40	0.03	0.03	3.75-5.00	---	0.50-1.25	1.80-2.40	17.50-19.00	7.00-9.50	---	29MAX 62	A H & T	SAME AS ABOVE	950-1150	850 FURNACE COOL	1310 OIL/AIR	570(DT) AIR COOL	---
T12006	T6	0.75-0.85	0.20-0.40	0.20-0.40	0.03	0.03	4.00-4.75	---	0.40-1.00	1.50-2.10	18.50-21.00	11.00-13.00	---	29MAX 62	A H & T	SAME AS ABOVE	950-1150	850 FURNACE COOL	1310 OIL/AIR	570(DT) AIR COOL	---
T12008	T8	0.75-0.85	0.20-0.40	0.20-0.40	0.03	0.03	3.75-4.50	---	0.40-1.00	1.80-2.40	13.25-14.75	4.25-5.75	---	29MAX 62	A H & T	SAME AS T1	950-1150	850 FURNACE COOL	1310 OIL/AIR	570(DT) AIR COOL	---
T12015	T15	1.50-1.60	0.15-0.40	0.15-0.40	0.03	0.03	3.75-5.00	---	1.00	4.50-5.25	11.75-13.00	4.75-5.25	---	29MAX 62	A H & T	SAME AS ABOVE	950-1150	850 FURNACE COOL	1310 OIL/AIR	570(DT) AIR COOL	---
HIGH SPEED STEELS (MOLYBDENUM ALLOYED) - ASTM A600-86																					
T11301	M1	0.78-0.88	0.15-0.40	0.20-0.50	0.03	0.03	3.50-4.00	---	8.20-9.20	1.00-1.35	1.40-2.10	---	---	27MAX 62	A H & T	SAME AS ABOVE	950-1150	850 FURNACE COOL	1210 OIL/AIR	570(DT) AIR COOL	---
T11302	M2(REGULAR C)	0.78-0.88	0.15-0.40	0.20-0.45	0.03	0.03	3.75-4.50	---	4.50-5.50	1.75-2.20	5.50-6.75	---	---	27MAX 62	A H & T	SAME AS ABOVE	950-1150	850 FURNACE COOL	1210 OIL/AIR	570(DT) AIR COOL	---
T11302	M2(HIGH C)	0.95-1.05	0.15-0.40	0.20-0.45	0.03	0.03	3.75-4.50	---	4.50-5.50	1.75-2.20	5.50-6.75	---	---	29MAX 62	A H & T	SAME AS ABOVE	950-1150	850 FURNACE COOL	1210 OIL/AIR	570(DT) AIR COOL	---
T11313	M3(CLASS 1)	1.00-1.10	0.15-0.40	0.20-0.45	0.03	0.03	3.75-4.50	---	4.75-6.50	2.25-2.75	5.00-6.75	---	---	29MAX 62	A H & T	SAME AS ABOVE	950-1150 850	1210 FURNACE COOL	570(DT) OIL/AIR	--- AIR COOL	
T11323	M3(CLASS 2)	1.15-1.25	0.15-0.40	0.20-0.45	0.03	0.03	3.75-4.50	---	4.75-6.50	2.75-3.25	5.00-6.75	---	---	29MAX 62	A H & T	SAME AS ABOVE	950-1150	850 FURNACE COOL	1210 OIL/AIR	570(DT) AIR COOL	---
T11304	M4	1.25-1.40	0.15-0.40	0.20-0.45	0.03	0.03	3.75-4.75	---	4.25-5.50	3.75-4.50	5.25-6.50	---	---	29MAX 62	A H & T	SAME AS ABOVE	950-1150	850 FURNACE COOL	1210 OIL/AIR	570(DT) AIR COOL	---

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ABBREVATIONS USED:
AR -- AS ROLLED
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AH -- AIR HARDENED
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S -- SOFTENED
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F & I -- FLAME & INDUCTION HARDENING
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CONTD.

CHEMISTRY & MECH.PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - US STANDARD GRADES

UNS NO.	DESIGNATION TYPE	* CHEMICAL COMPOSITION, PERCENT												HARDNESS (MIN.) HRC	CONDITION	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
		C	Mn	Si	S	P	Cr	Ni	Mo	V	W	Co	OTHERS				HOT FORMING TEMP. °C	ANNEALING TEMP. °C (FC)	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZING TEMP. °C
HIGH SPEED STEELS (MOLYBDENUM ALLOYED) - ASTM A600-86 - CONTD.																					
T11306	M6	0.75-0.85	0.15-0.40	0.20-0.45	0.03	0.03	3.75-4.50	---	4.50-5.50	1.30-1.70	3.75-4.75	11.00-13.00	---	27 MAX 62	A H & T	SUITABLE FOR MILLING AND WORM CUTTERS, REAMERS, SLITTING SAWS, DRILLS, HACKSAWS, ETC.	950-1150	850 FURNACE COOL	1260 OIL/AIR	570(DT) AIR COOL	---
T11307	M7	0.97-1.05	0.15-0.40	0.20-0.55	0.03	0.03	3.50-4.00	---	8.20-9.20	1.75-2.25	1.40-2.10	---	---	29MAX 62	A H & T	SAME AS ABOVE	950-1150	850 FURNACE COOL	1210 OIL/AIR	570(DT) AIR COOL	---
T11310	M10 (REGULAR C)	0.84-0.94	0.10-0.40	0.20-0.45	0.03	0.03	3.75-4.50	---	7.75-8.50	1.80-2.20	---	---	---	29MAX 62	A H & T	SAME AS ABOVE	950-1150	850 FURNACE COOL	1210 OIL/AIR	570(DT) AIR COOL	---
T11310	M10 (HIGH C)	0.95-1.05	0.10-0.40	0.20-0.45	0.03	0.03	3.75-4.50	---	7.75-8.50	1.80-2.20	---	---	---	29MAX 62	A H & T	SAME AS ABOVE	950-1150	850 FURNACE COOL	1210 OIL/AIR	570(DT) AIR COOL	---
T11330	M30	0.75-0.85	0.15-0.40	0.20-0.45	0.03	0.03	3.50-4.25	---	7.75-9.00	1.00-1.40	1.30-2.30	4.50-5.50	---	29MAX 62	A H & T	SAME AS ABOVE	950-1150	850 FURNACE COOL	1210 OIL/AIR	570(DT) AIR COOL	---
T11333	M33	0.85-0.92	0.15-0.40	0.15-0.50	0.03	0.03	3.50-4.00	---	9.00-10.00	1.00-1.35	1.30-2.10	7.75-8.75	---	29MAX 62	A H & T	SAME AS ABOVE	950-1150	850 FURNACE COOL	1210 OIL/AIR	570(DT) AIR COOL	---
T11334	M34	0.85-0.92	0.15-0.40	0.20-0.45	0.03	0.03	3.50-4.00	---	7.75-9.20	1.90-2.30	1.40-2.10	7.75-8.75	---	29MAX 62	A H & T	SAME AS ABOVE	950-1150	850 FURNACE COOL	1210 OIL/AIR	570(DT) AIR COOL	---
T11336	M36	0.80-0.90	0.15-0.40	0.20-0.45	0.03	0.03	3.75-4.50	---	4.50-5.50	1.75-2.25	5.50-6.50	7.75-8.75	---	29MAX 62	A H & T	SAME AS ABOVE	950-1150	850 FURNACE COOL	1210 OIL/AIR	570(DT) AIR COOL	---
T11341	M41	1.05-1.15	0.20-0.60	0.15-0.50	0.03	0.03	3.75-4.50	---	3.25-4.25	1.75-2.25	6.25-7.00	4.75-5.75	---	29MAX 62	A H & T	SAME AS ABOVE	950-1150	850 FURNACE COOL	1210 OIL/AIR	570(DT) AIR COOL	---
T11342	M42	1.05-1.15	0.15-0.60	0.15-0.65	0.03	0.03	3.50-4.25	---	9.00-10.00	0.95-1.35	1.15-1.85	7.75-8.75	---	29MAX 62	A H & T	SAME AS ABOVE	950-1150	850 FURNACE COOL	1210 OIL/AIR	570(DT) AIR COOL	---
T11343	M43	1.15-1.25	0.20-0.60	0.15-0.65	0.03	0.03	3.50-4.25	---	7.50-8.50	1.50-1.75	2.25-3.00	7.75-8.75	---	29MAX 62	A H & T	SAME AS ABOVE	950-1150	850 FURNACE COOL	1210 OIL/AIR	570(DT) AIR COOL	---
T11344	M44	1.10-1.20	0.20-0.60	0.30-0.55	0.03	0.03	4.00-4.75	---	6.00-7.00	1.85-2.20	5.00-5.75	11.00-12.25	---	29MAX 62	A H & T	SAME AS ABOVE	950-1150	850 FURNACE COOL	1210 OIL/AIR	570(DT) AIR COOL	---
T11346	M46	1.22-1.30	0.20-0.60	0.40-0.65	0.03	0.03	3.70-4.20	---	8.00-8.50	3.00-3.30	1.90-2.20	11.00-12.25	---	29MAX 62	A H & T	SAME AS ABOVE	950-1150	850 FURNACE COOL	1210 OIL/AIR	570(DT) AIR COOL	---

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S -- SOFTENED
DT - DOUBLE TEMPERED
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CONTD

CHEMISTRY & MECH.PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - US STANDARD GRADES

DESIGNATION UNS NO.	TYPE	* CHEMICAL COMPOSITION, PERCENT												HARDNESS (MIN.) HRC	CONDITION	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
		C	Mn	Si	S	P	Cr	Ni	Mo	V	W	Co	OTHERS				HOT FORMING TEMP. °C	ANNEALING TEMP. °C (FC)	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZING TEMP. °C
HIGH SPEED STEELS (MOLYBDENUM ALLOYED) - ASTM A600-86 - CONTD.																					
T11347	M47	1.05-1.15	0.15-0.40	0.20-0.45	0.03	0.03	3.50-4.00	---	9.25-10.00	1.15-1.35	1.30-1.80	4.75-5.25	---	27 MAX 62	A H & T	SUITABLE FOR MILLING AND WORM CUTTERS, REAMERS, SLITTING SAWS, DRILLS, HACKSAWS, ETC.	950-1150	850 FURNACE COOL	1260 OIL/AIR	570(DT) AIR COOL	---
---	M48	1.42-1.52	0.15-0.40	0.15-0.40	0.07	0.03	3.50-4.00	---	4.75-5.50	2.75-3.25	9.50-10.50	8.00-10.00	---	27 MAX 62	A H & T	SAME AS ABOVE (FREE CUTTING STEEL)	950-1150	850 FURNACE COOL	1260 OIL/AIR	570(DT) AIR COOL	---
---	M61	1.75-1.85	0.25-0.50	0.20-0.45	0.07	0.03	3.50-4.25	---	6.00-6.75	4.50-5.25	11.75-13.00	---	---	27 MAX 62	A H & T	SAME AS ABOVE (FREE CUTTING STEEL)	950-1150	850 FURNACE COOL	1260 OIL/AIR	570(DT) AIR COOL	---
---	M62	1.25-1.35	0.15-0.40	0.15-0.40	0.07	0.03	3.50-4.00	---	10.00-11.00	1.80-2.10	5.75-6.50	---	---	27 MAX 62	A H & T	SAME AS ABOVE (FREE CUTTING STEEL)	950-1150	850 FURNACE COOL	1260 OIL/AIR	570(DT) AIR COOL	---
ALLOYED TOOL STEELS - (ASTM A681-86)																					
T20810	H10	0.35-0.45	0.25-0.70	0.80-1.20	0.03	0.03	3.00-3.75	---	2.00-3.00	0.25-0.75	---	---	---	23 MAX 45	A H & T	SUITABLE FOR ALL ALL TYPES OF DIES FOR EXTRUSION, FORGING, STAMPING, CASTING, HOT HEADING, RIVETING, ETC.	850-1150	850 FURNACE COOL	1100 OIL/AIR	600-650 AIR COOL	---
T20811	H11	0.33-0.43	0.20-0.50	0.80-1.20	0.03	0.03	4.75-5.50	---	1.10-1.60	030-0.60	---	---	---	23 MAX 45	A H & T	SAME AS ABOVE	850-1150	850 FURNACE COOL	1100 OIL/AIR	600-650 AIR COOL	---
T20812	H12	0.30-0.40	0.20-0.50	0.80-1.20	0.03	0.03	4.75-5.50	---	1.25-1.75	050	1.00-1.70	---	---	23 MAX 34	A H & T	SAME AS ABOVE	850-1150	850 FURNACE COOL	1010 OIL/AIR	500-650 AIR COOL	---
T20813	H13	0.32-0.45	0.20-0.50	0.80-1.20	0.03	0.03	4.75-5.50	---	1.10-1.75	080-1.20	---	---	---	23 MAX 35	A H & T	SAME AS ABOVE	850-1150	850 FURNACE COOL	1000 OIL/AIR	550-650 AIR COOL	---
T20814	H14	0.35-0.45	0.20-0.50	0.80-1.20	0.03	0.03	4.75-5.50	---	---	---	4.00-5.25	---	---	23 MAX 45	A H & T	SAME AS ABOVE	850-1150	850 FURNACE COOL	1100 OIL/AIR	550-650 AIR COOL	---
T20819	H19	0.32-0.45	0.20-0.50	0.20-0.50	0.03	0.03	4.00-4.75	---	030-0.55	1.75-2.20	3.75-4.50	4.00-4.50	---	23 MAX A 45	H & T	SAME AS ABOVE	850-1150	850 FURNACE COOL	1100 OIL/AIR	550-650 AIR COOL	---
T20821	H21	0.26-0.36	0.15-0.40	015-0.50	0.03	0.03	3.00-3.75	---	---	0.30-0.60	8.50-10.00	---	---	23 MAX 45	A H & T	SAME AS ABOVE	850-1150	850 FURNACE COOL	1100 OIL/AIR	550-650 AIR COOL	---
T20822	H22	0.30-0.40	0.15-0.40	015-0.40	0.03	0.03	1.75-3.75	---	---	0.25-0.50	10.00-11.75	---	---	23 MAX 45	A H & T	SAME AS ABOVE	850-1150	850 FURNACE COOL	1100 OIL/AIR	550-650 AIR COOL	---

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F & I -- FLAME & INDUCTION HARDENING
S -- SOFTENED
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CONTD.

CHEMISTRY & MECH.PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - US STANDARD GRADES

UNS NO.	DESIGNATION TYPE	* CHEMICAL COMPOSITION, PERCENT												HARDNESS (MIN.) HRC	CONDITION	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
ALLOYED TOOL STEELS (ASTM A681-86) - - CONTD.		C	Mn	Si	S	P	Cr	Ni	Mo	V	W	Co	OTHERS				HOT FORMING TEMP. °C (FC)	ANNEALING TEMP. °C	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZ-ING TEMP.°C
T20822	H22	0.30-0.40	0.15-0.40	0.15-0.40	0.03	0.03	1.75-3.75	---	---	0.25-0.75	10.00-11.75	---	---	23 MAX 45	A H & T	SUITABLE FOR ALL ALL TYPES OF DIES FOR EXTRUSION, FORGING, STAMPING, CASTING, HOT HEAD-iNG, RIVETING, ETC	850-1150	850 FURNACE COOL	1100 OIL/AIR	600-650 AIR COOL	---
T20823	H23	0.25-0.35	0.15-0.40	0.15-0.60	0.03	0.03	11.00-12.75	---	---	0.75-1.25	11.00-12.75	---	---	23 MAX 45	A H & T	COROSSION RESIS-TANCE TOOL STEEL FOR ALL TYPES OF DIES AS ABOVE.	850-1150	850 FURNACE COOL	1100 OIL/AIR	600-650 AIR COOL	---
T20824	H24	0.42-0.53	0.15-0.40	0.15-0.40	0.03	0.03	2.50-3.50	---	---	0.40-0.60	14.00-16.00	---	---	23 MAX 45	A H & T	HIGH CARBON-TUNGSTEN TOOL STEEL FOR ALL TYPES OF DIES AS ABOVE.	850-1150	850 FURNACE COOL	1100 OIL/AIR	600-650 AIR COOL	---
T20825	H25	0.22-0.32	0.15-0.40	0.15-0.40	0.03	0.03	3.75-4.50	---	---	0.40-0.60	14.00-16.00	---	---	23 MAX 45	A H & T	LOW CARBON-TUNGSTEN TOOL STEEL FOR ALL TYPES OF DIES AS ABOVE.	850-1150	850 FURNACE COOL	1100 OIL/AIR	600-650 AIR COOL	---
T20826	H26	0.42-0.53	0.15-0.40	0.15-0.40	0.03	0.03	3.75-4.50	---	---	0.75-1.25	17.25-19.00	---	---	23 MAX 45	A H & T	HIGH CARBON-TUNGSTEN TOOL STEEL FOR ALL TYPES OF DIES AS ABOVE.	850-1150	850 FURNACE COOL	1100 OIL/AIR	600-650 AIR COOL	---
T20841	H41	0.60-0.75	0.15-0.40	0.20-0.45	0.03	0.03	3.50-4.00	---	8.20-9.20	1.00-1.30	1.40-2.10	---	---	23 MAX 45	A H & T	HIGH CARBON-MOLYBDENUM TOOL STEEL FOR ALL TYPES OF DIES AS ABOVE.	850-1150	850 FURNACE COOL	1100 OIL/AIR	600-650 AIR COOL	---
T20842	H42	0.55-0.70	0.15-0.40	0.20-0.45	0.03	0.03	3.75-4.50	---	4.50-5.50	1.75-2.20	5.50-6.75	---	---	23 MAX 45	A H & T	HIGH CARBON-MOLYBDENUM TOOL STEEL FOR ALL TYPES OF DIES AS ABOVE.	850-1150	850 FURNACE COOL	1100 OIL/AIR	600-650 AIR COOL	---
T20843	H43	0.50-0.65	0.15-0.40	0.20-0.45	0.03	0.03	3.75-4.50	---	7.75-8.50	1.80-2.20	5.50-6.75	---	---	23 MAX 45	A H & T	HIGH CARBON-MOLYBDENUM TOOL STEEL FOR ALL TYPES OF DIES AS ABOVE.	850-1150	850 FURNACE COOL	1100 OIL/AIR	600-650 AIR COOL	---
T30102	A2	0.95-1.05	1.00	0.50	0.03	0.03	4.75-5.50	---	0.90-1.40	0.15-0.50	---	---	---	23 MAX 45	A H & T	HIGH CARBON-CHROMIUM TOOL STEEL FOR ALL TYPES OF DIES AS ABOVE.	850-1150	850 FURNACE COOL	1100 OIL/AIR	600-650 AIR COOL	---
T30103	A3	1.20-1.30	0.40-0.60	0.50	0.03	0.03	4.75-5.50	---	0.90-1.40	0.80-1.40	---	---	---	23 MAX 45	A H & T	HIGH CARBON-Cr - V TOOL STEEL FOR ALL TYPES OF DIES AS ABOVE.	850-1150	850 FURNACE COOL	1100 OIL/AIR	600-650 AIR COOL	---
T30104	A4	0.95-1.05	1.80-2.20	0.50	0.03	0.03	0.90-2.20	---	0.90-1.40	---	---	---	---	23 MAX 45	A H & T	HIGH CARBON-Cr - Mo TOOL STEEL FOR ALL TYPES OF DIES AS ABOVE	850-1150	850 FURNACE COOL	1100 OIL/AIR	600-650 AIR COOL	---

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AR -- AS ROLLED; CR -- COLD ROLLED; CD -- COLD DRAWN; N -- NORMALISED; A -- ANNEALED; F & I -- FLAME & INDUCTION HARDENING; FC -- FURNACE COOL

CHEMISTRY & MECH.PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - US STANDARD GRADES

DESIGNATION UNS NO.	TYPE	* CHEMICAL COMPOSITION, PERCENT												HARDNESS (MIN.) HRC	CONDITION	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
ALLOYED TOOL STEELS (ASTM A681-86) - - CONTD.		C	Mn	Si	S	P	Cr	Ni	Mo	V	W	Co	OTHERS				HOT FORMING TEMP. °C (FC)	ANNEALING TEMP. °C	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZING TEMP °C
T30105	A5	0.95-1.05	2.80-3.20	0.50	0.03	0.03	0.90-1.20	---	0.90-1.40	---	---	---	---	20 MAX 35	A H & T	SUITABLE FOR ALL ALL TYPES OF DIES FOR EXTRUSION, FORGING, STAMPING, CASTING, HOT HEAD-	850-1150	850 FURNACE COOL	1100 OIL/AIR	600-650 AIR COOL	---
T30106	A6	0.65-0.75	1.80-2.50	0.50	0.03	0.03	0.90-1.20	---	0.90-1.40	---	---	---	---	20 MAX 35	A H & T	SAME AS ABOVE	850-1150	850 FURNACE COOL	1100 OIL/AIR	600-650 AIR COOL	---
T30107	A7	2.00-2.85	0.80	0.50	0.03	0.03	5.00-5.75	---	0.90-1.40	3.90-5.15	0.50-1.50	---	---	22 MAX 45	A H & T	SAME AS ABOVE	850-1150	850 FURNACE COOL	1100 OIL/AIR	600-650 AIR COOL	---
T30108	A8	0.50-0.60	0.50	0.75-1.10	0.03	0.03	4.75-5.50	---	1.15-1.65	---	1.00-1.50	---	---	22 MAX 35	A H & T	SAME AS ABOVE	850-1150	850 FURNACE COOL	1100 OIL/AIR	600-650 AIR COOL	---
T30109	A9	0.45-0.55	0.50	0.95-1.15	0.03	0.03	4.75-5.50	1.25-1.75	1.30-1.80	0.80-1.40	---	---	---	22 MAX 40	A H & T	SAME AS ABOVE	850-1150	850 FURNACE COOL	1100 OIL/AIR	600-650 AIR COOL	---
T30402	D2	1.40-1.60	0.60	0.60	0.03	0.03	11.00-13.00	---	0.70-1.20	1.10	---	1.00	---	26 MAX 58	A H & T	SUITABLE FOR HIGH QUALITY PRESS TOOLS, DRAWING AND CUTTER DIES, SHEAR BLADES, THREAD ROLLERS.	900-1050	850 FURNACE COOL	1020 OIL/AIR	150-400 AIR COOL	---
T30403	D3	2.00-2.30	0.60	0.60	0.03	0.03	11.00-13.50	---	0.70-1.20	1.00	1.00	---	---	26 MAX 58	H & T	SAME AS ABOVE BUT WITH HIGHER CARBON	900-1050	850 FURNACE COOL	980 OIL/AIR	150-400 AIR COOL	---
T30404	D4	2.05-2.40	0.60	0.60	0.03	0.03	11.00-13.00	---	0.70-1.20	1.00	---	---	---	26 MAX 58	A H & T	SAME AS ABOVE BUT WITH HIGHER CARBON	900-1050	850 FURNACE COOL	980 OIL/AIR	150-400 AIR COOL	---
T30405	D5	1.40-1.60	0.60	0.60	0.03	0.03	11.00-13.00	---	0.70-1.20	1.00	---	2.50-3.50	---	26 MAX 58	A H & T	SAME AS ABOVE BUT WITH COBALT	900-1050	850 FURNACE COOL	1000 OIL/AIR	150-400 AIR COOL	---
T30407	D7	2.15-2.50	0.60	0.60	0.03	0.03	11.50-13.50	---	0.70-1.20	3.80-4.40	---	---	---	26 MAX 58	A H & T	SAME AS ABOVE BUT WITH HIGH C-V	900-1050	850 FURNACE COOL	1000 OIL/AIR	150-400 AIR COOL	---
T31501	O1	0.85-1.00	1.00-1.40	0.50	0.03	0.03	0.40-0.60	---	---	0.30	0.40-0.60	---	---	23 MAX 50	A H & T	SUITABLE FOR ENGRAVING TOOLS, DRILLS, MILLING CUTTERS, COLD PUNCHES, ETC.	850-1050	780 FURNACE COOL	790 OIL/AIR	150-400 AIR COOL	---
T31502	O2	0.85-0.95	1.40-1.80	0.50	0.03	0.03	0.35	---	0.30	0.30	---	---	---	23 MAX 50	A H & T	SAME AS ABOVE	850-1050	780 FURNACE COOL	790 OIL/AIR	150-400 AIR COOL	---
T31506	O6	1.25-1.55	0.30-1.10	0.55-1.50	0.03	0.03	0.30	---	0.20-0.30	---	---	---	---	23 MAX 50	A H & T	SAME AS ABOVE	850-1050	780 FURNACE COOL	790 OIL/AIR	150-400 AIR COOL	---

*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED

ABBREVATIONS USED:
AR -- AS ROLLED
HR -- HOT ROLLED
CR -- COLD ROLLED
AC -- AIR COOLED
CD -- COLD DRAWN
OH -- OIL HARDENED
N -- NORMALISED
AH -- AIR HARDENED
A -- ANNEALED
H & T -- HARDENED & TEMPERED
F & I -- FLAME & INDUCTION HARDENING
S -- SOFTENED
DT - DOUBLE TEMPERED
FC -- FURNACE COOL

CONTD

CHEMISTRY & MECH.PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - US STANDARD GRADES

UNS NO.	DESIGNATION TYPE	* CHEMICAL COMPOSITION, PERCENT												HARDNESS (MIN.) HRC	CONDITION	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
ALLOYED TOOL STEELS (ASTM A681-86) - - CONTD.		C	Mn	Si	S	P	Cr	Ni	Mo	V	W	Co	OTHERS				HOT FORMING TEMP. °C (FC)	ANNEALING TEMP. °C	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZING TEMP. °C
T31507	O7	1.10-1.30	1.00	0.60	0.03	0.03	0.35-0.65	---	0.30	0.40	1.00-2.00	---	---	23 MAX 50	A H & T	SUITABLE FOR ENGRAVING TOOLS, DRILLS, MILLING CUTTERS, COLD PUNCHES, ETC.	850-1050	780 FURNACE COOL	790 OIL/AIR	150-400 AIR COOL	---
T41901	S1	0.40-0.55	0.10-0.40	0.15-1.20	0.03	0.03	1.00-1.80	---	0.50	0.15-0.30	1.50-3.00	---	---	26 MAX 42	A H & T	SHOCK RESISTANCE TOOL STTEL SUITABLE FOR HEAVY DUTY PUNCHES, DIES, SHEAR BLADES, PNEUMATIC CHISELS, ETC.	800-1100	800 FURNACE COOL	880 OIL/AIR	150-400 AIR COOL	---
T41902	S2	0.40-0.55	0.30-0.50	0.90-1.20	0.03	0.03	---	---	0.30-0.60	0.50	---	---	---	26 MAX 42	A H & T	SAME AS ABOVE	800-1100	800 FURNACE COOL	880 OIL/AIR	150-400 AIR COOL	---
T41904	S4	0.50-0.65	0.60-0.95	1.75-2.25	0.03	0.03	0.35	---	---	0.35	---	---	---	26 MAX 42	A H & T	SAME AS ABOVE	850-1100	740 FURNACE COOL	880 OIL/AIR	150-400 AIR COOL	---
T41905	S5	0.50-0.65	0.60-1.00	1.75-2.25	0.03	0.03	0.35	---	0.20-1.35	0.35	---	---	---	26 MAX 42	A H & T	SAME AS ABOVE	850-1100	740 FURNACE COOL	880 OIL/AIR	150-400 AIR COOL	---
T41906	S6	0.40-0.50	1.20-1.50	2.00-2.50	0.03	0.03	1.20-1.50	---	0.30-0.50	0.20-0.40	---	---	---	26 MAX 42	A H & T	SAME AS ABOVE	850-1100	740 FURNACE COOL	880 OIL/AIR	150-400 AIR COOL	---
T41907	S7	0.45-0.55	0.20-0.80	0.20-1.00	0.03	0.03	3.00-3.50	---	1.30-1.80	0.35	---	---	---	26 MAX 42	A H & T	SAME AS ABOVE	850-1100	740 FURNACE COOL	880 OIL/AIR	150-400 AIR COOL	---
T61202	L2	0.45-1.00	0.10-0.90	0.50	0.03	0.03	0.70-1.20	---	0.25	0.10-0.30	---	---	---	26 MAX 42	A H & T	SAME AS ABOVE	850-1100	740 FURNACE COOL	880 OIL/AIR	150-400 AIR COOL	---
T61203	L3	0.95-1.10	0.25-0.80	0.50	0.03	0.03	1.30-1.70	---	---	0.10-0.30	---	---	---	26 MAX 42	A H & T	SAME AS ABOVE	850-1100	740 FURNACE COOL	880 OIL/AIR	150-400 AIR COOL	---
T61206	L6	0.65-0.75	0.25-0.80	0.50	0.03	0.03	0.60-1.20	1.25-2.00	0.50	--	---	---	---	26 MAX 42	A H & T	SAME AS ABOVE	850-1100	740 FURNACE COOL	880 OIL/AIR	150-400 AIR COOL	---
T60601	F1	0.95-1.25	0.50	0.50	0.03	0.03	---	---	---	---	1.00-1.75	---	---	23 MAX 50	A H & T	SUITABLE FOR GAUGES, ENGRAVING TOOLS, DIES, TAPS, MILLING CUTTERS, COLD PUNCHES, KNIVES. ETC.	850-1050	780 FURNACE COOL	850 OIL/AIR	150-400 AIR COOL	---
T60602	F2	1.20-1.40	0.50	0.50	0.03	0.03	0.20-0.40	---	---	---	3.00-4.50	---	---	23 MAX 50	A H & T	SAME AS ABOVE	850-1050	780 FURNACE COOL	850 OIL/AIR	150-400 AIR COOL	---

*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED.

ABBREVATIONS USED
AR -- AS ROLLED
HR -- HOT ROLLED
CR -- COLD ROLLED
AC -- AIR COOLED
CD -- COLD DRAWN
OH -- OIL HARDENED
N -- NORMALISED
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A -- ANNEALED
H & T -- HARDENED & TEMPERED
F & I -- FLAME & INDUCTION HARDENING
S -- SOFTENED
DT - DOUBLE TEMPERED
FC -- FURNACE COOL

CONTD.

CHEMISTRY & MECH.PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - US STANDARD GRADES

DESIGNATION UNS NO.	TYPE	* CHEMICAL COMPOSITION, PERCENT												HARDNESS (MIN.) HRC	CONDITION	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
ALLOYED TOOL STEELS (ASTM A681-86) - - CONTD.		C	Mn	Si	S	P	Cr	Ni	Mo	V	W	Co	OTHERS				HOT FORMING TEMP. °C (FC)	ANNEALING TEMP. °C	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZING TEMP. °C
T51602	P2	0.10	0.10-0.40	0.10-0.40	0.03	0.03	0.75-1.25	0.10-0.50	0.15-0.40	---	---	---	---	20 MAX 50	A H & T	CASE HARDENING ALLOY TOOL STEEL SUITABLE FOR PINS FOR DIES, SMALL DIES NOT SUBJECT TO HEAVY ABRASION LIKE SUPPORT PLATES FOR MAJOR DIES.	1150	850 FURNACE COOL	CARBURISING AT 880-930 FOLLOWED BY HARDENING AT 780-820 OIL	180-250 AIR COOL	---
T51603	P3	0.10	0.20-0.60	0.40	0.03	0.03	0.40-0.75	1.00-1.50	---	---	---	---	---	20 MAX 50	A H & T	SAME AS ABOVE	1150	850 FURNACE COOL	SAME AS ABOVE	180-250 AIR COOL	---
T51604	P4	0.12	0.20-0.60	0.10-0.40	0.03	0.03	4.00-5.25	---	0.40-1.00	---	---	---	---	20 MAX 50	A H & T	SAME AS ABOVE AND ALSO FOR PRESS AND PLASTIC MOULDS. CAN ALSO BE NITRIDED.	1150	850 FURNACE COOL	SAME AS ABOVE	180-250 AIR COOL	---
T51605	P5	0.10	0.20-0.60	0.40	0.03	0.03	2.00-2.50	0.35	---	---	---	---	---	20 MAX 50	A H & T	SAME AS ABOVE	1150	850 FURNACE COOL	SAME AS ABOVE	180-250 AIR COOL	---
T51606	P6	0.05-0.15	0.35-0.70	0.10-0.40	0.03	0.03	1.25-1.75	3.25-3.75	---	---	---	---	---	20 MAX 50	A H & T	SAME AS ABOVE CAN ALSO BE USED FOR HEAVY CROWN WHEELS , WORMS, GEARS, ETC,	1150	850 FURNACE COOL	SAME AS ABOVE	180-250 AIR COOL	---
T51620	P20	0.28-0.40	0.60-1.00	0.20-0.80	0.03	0.03	1.40-2.00	---	0.30-0.55	---	---	---	---	20 MAX 50	A H & T	SAME AS ABOVE CAN ALSO BE USED FOR HEAVY CROWN WHEELS , WORMS, GEARS, ETC, CAN ALSO BE DIRECTLY HARDENED.	1150	850 FURNACE COOL	SAME AS ABOVE	180-250 AIR COOL	---
T51620	P20	0.28-0.40	0.60-1.00	0.20-0.80	0.03	0.03	1.40-2.00	---	0.30-0.55	---	---	---	---	20 MAX 50	A H & T	SAME AS ABOVE CAN ALSO BE USED FOR HEAVY CROWN WHEELS , WORMS, GEARS, ETC, CAN ALSO BE DIRECTLY HARDENED.	1150	850 FURNACE COOL	SAME AS ABOVE	180-250 AIR COOL	---
T51621	P21	0.18-0.22	0.20-0.40	0.20-0.40	0.03	0.03	0.20 0.30	3.90-4.25	---	0.15-0.25	---	---	1.05-1.25 Al	20 MAX 50	A H & T	SAME AS ABOVE CAN ALSO BE USED FOR HEAVY CROWN WHEELS , WORMS, GEARS, ETC, CAN ALSO BE DIRECTLY HARDENED.	1150	850 FURNACE COOL	SAME AS ABOVE	180-250 AIR COOL	---

UNLESS OTHERWISE SPECIFIED, Cu 0.25% MAX., Ni 0.50% MAX. FOR ALL TYPES FOR FREE CUTTING STEELS, S 0.06-0.15%

*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED

ABBREVATIONS USED:
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OH -- OIL HARDENED
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AH -- AIR HARDENED
A -- ANNEALED
H & T -- HARDENED & TEMPERED
F & I -- FLAME & INDUCTION HARDENING
S -- SOFTENED
DT - DOUBLE TEMPERED
FC -- FURNACE COOL

CONTD.

CHEMISTRY & MECH.PROPERTIES OF IMPORTANT STEELS WITH HEAT TREATMENT REGIMES - US STANDARD GRADES

DESIGNATION UNS NO.	TYPE	* CHEMICAL COMPOSITION, PERCENT												HARDNESS (MIN.) HRC	CONDITION	TYPE OF STEEL AND APPLICATIONS	HEAT TREATMENT REGIME				
UNALLOYED TOOL STEELS (ASTM A686-85)		C	Mn	Si	S	P	Cr	Ni	Mo	V	W	Co	OTHERS				HOT FORMING TEMP. °C (FC)	ANNEALING TEMP. °C	HARDENING TEMP. °C QUENCHANT	TEMPERING TEMP. °C	NORMALIZING TEMP °C
T72301	W1 - GRADE A	0.80-1.25	0.10-0.40	0.10-0.40	0.03	0.03	0.15	0.20	0.10	0.10	0.15	---	0.20 Cu	20 MAX 48	A H & T	PLAIN CARBON WATER HARDENING TOOL STEEL SUITABLE FOR WOOD WORKING, ENGRAVING, PRESS TOOLS, PUNCHES, SHEARING BLADES, RAZORS,ETC.	850-1050	750 FURNACE COOL	800 WATER	150-400 AIR	---
T72301	W1 - GRADE C	0.80-1.25	0.10-0.40	0.10-0.40	0.03	0.03	0.30	0.20	0.10	0.10	0.15	---	0.20 Cu	20 MAX 48	A H & T	SAME AS ABOVE	850-1050	750 FURNACE COOL	800 WATER	150-400 AIR	---
T72302	W2 - GRADE A	0.85-1.50	0.10-0.40	0.10-0.40	0.03	0.03	0.15	0.20	0.10	0.15-0.35	0.15	---	0.20 Cu	20 MAX 48	A H & T	SAME AS ABOVE IMPROVED CARBON TOOL STEEL	850-1050	750 FURNACE COOL	800 WATER	150-400 AIR	---
T72302	W2 - GRADE C	0.85-1.50	0.10-0.40	0.10-0.40	0.03	0.03	0.30	0.20	0.10	0.15-0.35	0.15	---	0.20 Cu	20 MAX 48	A H & T	SAME AS ABOVE IMPROVED CARBON TOOL STEEL	850-1050	750 FURNACE COOL	800 WATER	150-400 AIR	---
T72305	W5	1.05-1.15	0.10-0.40	0.10-0.40	0.03	0.03	0.40-0.60	0.20	0.10	0.10	0.15	---	0.20 Cu	20 MAX 48	A H & T	SAME AS ABOVE IMPROVED CARBON TOOL STEEL	850-1050	750 FURNACE COOL	800 WATER	150-400 AIR	---

*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED.

ABBREVATIONS USED:
AR – AS ROLLED
HR – HOT ROLLED
CR – COLD ROLLED
AC – AIR COOLED
CD – COLD DRAWN
OH – OIL HARDENED
N – NORMALISED
AH – AIR HARDENED
A – ANNEALED
H & T – HARDENED & TEMPERED
S – SOFTENED
DT - DOUBLE TEMPERED
F & I – FLAME & INDUCTION HARDENING
FC – FURNACE COOL

CONTD.

NOTE 1 - Grades shown in the above Tables with prefix letter E are normally made only by the basic electric furnace process. All others are normally manufactured by the basic open-hearth or baisc-oxygen processes but may be manufactured by the baisc electric furnace with adjustments in phosphorus and sulphur.
NOTE 2 - The phosphorus and sulphur limitations for each process are as follows.

	Maximum, percent	
	Phosphorus	Sulphur
Basic electric	0.025	0.025
Baisc open-hearth or basic oxygen	0.035	0.040
Acid Electric	0.050	0.050
Acid open-hearth	0.050	0.050

NOTE 3 - Minimum silicon limit for acid open hearth or acid electric furnace alloy steels is 0.15%.
NOTE 4 - Small quantities of certain elements are present in alloy steels which are not specified or required. These elements are considered as incidental and may be present to the following maximum amounts Copper, 0.35% ; Nickel, 0.25% ; Chromium, 0.20% and Molybdenum, 0.06%.
NOTE 5 - Where minimum and maximum sulphur content is shown, it is indicative of resulphurized steel.
NOTE 6 - Standard alloy steels can be produced with a lead range of 0.15-0.35%. Such steels are identified by inserting letter "L" between the socond and third numerals of the AISI number, e.g. 41L40. A heat analysis for lead is not determinable, since lead is added to the ladle stream which each ingot is poured.
NOTE 7 - The chemical ranges and limits shown in thease Tables are produced to product analysis tolerances shown in Page - 145.

HEAT TREATMENT OF TOOL STEELS

IMPORTANT POINTS FOR HEAT TREATMENT OF TOOL STEELS :

1. Step heating should be employed as a safeguard against distortion and cracking due to thermal shocks. This practice is specially beneficial for bigger and complicated shapes and for highly alloyed hot work and high speed steels.

2. The work piece should always be protected against decarburization during heating.

3. Precision tools usually are stress relieved after machining and before hardening, it is often desirable to stress relieve after rough machining, but before finish machining.

4. WATER HARDENING TOOL STEELS: These steels have relatively low hardenability, allowing tough core with high surface hardness. Oil hardening of very thin sections to avoid distortion and cracking is an exception. Quenched tools should not be allowed to stand at room temperature to avoid cracking. Instead, they should either be immediately tempered or should be stocked in warm furnaces for subsequent tempering.

5. SHOCK RESISTANT TOOL STEELS:
5.1 Normalizing is not recommended.
5.2 For high silicon type, annealing temperature should be low in order to avoid decarburization and graphitization.These tools should be immediately quenched as soon as the austenitizing temperature is reached.

6. OIL HARDENING TOOL STEELS: Such tool steels for which dimensional stability is very critical (gauges, etc.), multiple tempering or subzero treatment is desirable. Before each re-tempering, the piece should be cooled below 70° C.

7. MEDIUM ALLOY AIR HARDENING COLD WORK TOOL STEELS:
7.1 Normalizing is not recommended.
7.2 Depending on hardenability, section, size and complexity of shape, the tools can be hardened in still air, forced air, oil quench to black, followed by air cooling and conventional oil quenching.

8. HIGH CARBON HIGH CHROMIUM TOOL STEELS:
8.1 Same as 7.1 and 7.2.
8.2 These steels can be nitrided after hardening and tempering.

9. HOT WORK TOOL STEELS:
9.1 Normalizing generally not recommended.
9.2 These steels range from high to extremely high hardenability.
9.3 Depending on the piece, air cooling, blast cooling or oil quenching can be used.
9.4 Multiple tempering is advantageous for large and sharp cornered die blocks.They are not allowed to reach room temperature before the first tempering operation.
9.5 Most of these steels have secondary hardness characteristics. These steels can be carburized with low carbon.
9.6 To increase the resistance to heat and wear, nitriding can be applied.

10. HIGH SPEED TOOL STEELS:
10.1 As the complex carbides do not dissolve to an appreciable extent unless heated near the melting point, extremely accurate temperature control to austenitizing is required in order to achieve high resistance quality and quality and cutting ability.
10.2 With lower temperature, full hardness cannot be developed because of insufficient carbide solution. On the other hand, higher temperature may introduce partial melting or excessive amount of retained austenite due to too much carbon and alloy solution.
10.3 Red hardness property is mainly due to the highly alloyed tempered martensitic matrix.
10.4 These steels show secondary hardness characteristics.
10.5 Multiple tempering is needed to convert all the retained austenite into martensite and to relieve internal stresses set up due to martensite formation.
10.6 Liquid nitriding is preferred to gas nitriding in high speed tool steels.

CHEMISTRY AND MECHANICAL PROPERTIES

OF

IMPORTANT STEELS

OF

RUSSIAN STANDARDS

AT A GLANCE

CHEMICAL AND MECHANICAL PROPERTIES OF IMPORTANT STEELS - RUSSIAN GRADES

DESIGNATION	* CHEMICAL COMPOSITION, PERCENT												MECH. PROPERTIES MIN.		
GRADE OF STEEL OR ALLOY	C	Mn	Si	S	P	Cr	Ni	Mo	V	W	Co	OTHERS	UTS Kg/mm²	ELON. %	HARDNESS MAX.,BHN
PLAIN CARBON STEELS															
MCT.0	≤0.23	---	---	0.06	0.07										
MCT.1 КП	0.06-0.12	≤0.05	0.25-0.50	0.06	0.045										
MCT.2 КП	0.09-0.15	≤0.07	0.25-0.50	0.06	0.045										
MCT.3 КП	014-0.22	≤0.07	0.30-0.50	0.06	0.045										
MCT.3	014-0.22	0.12-0.30	0.45-0.65	0.06	0.045										
MCT.4 КП	018-0.27	≤0.07	0.45-0.70	0.06	0.045										
MCT.4	018-0.27	0.12-0.30	0.40-0.70	0.06	0.045										
MCT.5	028-0.37	0.15-0.35	0.50-0.80	0.06	0.045										
MCT.6	038-0.49	0.15-0.35	0.50-0.80	0.06	0.045										
MCT.7	0.50-0.62	0.15-0.35	0.50-0.80	0.06	0.045										
FREE CUTTING STEELS															
A12	0.08-0.16	0.60-0.90	0.15-0.35	0.08-0.20	0.08-0.15										167-247
A20	0.15-0.25	0.60-0.90	0.15-0.35	0.08-0.15	<0.06										167-217
A30	0.25-0.35	0.70-1.00	0.15-0.35	0.08-0.15	<0.06										174-223
A40Г	0.35-0.45	1.20-1.55	0.15-0.35	0.18-0.30	<0.06										179-229

*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED

CHEMICAL AND MECHANICAL PROPERTIES OF IMPORTANT STEELS - RUSSIAN GRADES

DESIGNATION	* CHEMICAL COMPOSITION, PERCENT												MECH. PROPERTIES MIN.		
GRADE OF STEEL OR ALLOY	C	Mn	Si	S	P	Cr	Ni	Mo	V	W	Co	OTHERS	UTS Kg/mm^2	ELON. %	HARDNESS MAX.,BHN
QUALITY CARBON STEELS - GROUP I															
05КП	≤0.06	≤0.04	≤0.30	0.04	0.035										
08КП	0.05-0.11	0.25-0.50	≤0.30	0.04	0.04								30	35	131
08	0.05-0.12	0.35-0.65	0.17-0.37	0.04	0.035								33	33	131
10КП	0.07-0.141	0.25-0.50	0.07	0.04	0.035								32	33	137
10КП	0.07-0.141	0.35-0.65	0.17-0.37	0.04	0.035								34	31	137
15КП	0.12-0.19	0.25-0.50	≤0.07	0.04	0.035								36	29	143
15	0.12-0.19	0.35-0.65	0.17-0.37	0.04	0.035								38	27	143
20КП	0.17-0.24	0.25-0.50	≤0.07	0.04	0.035								39	27	156
20	0.17-0.24	0.35-0.65	0.17-0.37	0.04	0.035								42	25	156
25	0.22-0.30	0.50-0.80	0.17-0.37	0.04	0.035								46	23	170
30	0.27-0.35	0.50-0.80	0.17-0.37	0.04	0.035								50	21	179
35	0.32-0.40	0.50-0.80	0.17-0.37	0.04	0.04								54	20	187
40	0.37-0.45	0.50-0.80	0.17-0.37	0.04	0.04								58	19	187
45	0.42-0.50	0.50-0.80	0.17-0.37	0.04	0.04								61	16	197
50	0.47-0.55	0.50-0.80	0.17-0.37	0.04	0.04								64	14	207

*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED

CHEMICAL AND MECHANICAL PROPERTIES OF IMPORTANT STEELS - RUSSIAN GRADES

DESIGNATION	* CHEMICAL COMPOSITION, PERCENT												MECH. PROPERTIES MIN.		
GRADE OF STEEL OR ALLOY	C	Mn	Si	S	P	Cr	Ni	Mo	V	W	Co	OTHERS	UTS Kg/mm^2	ELON. %	HARDNESS MAX.,BHN
QUALITY CARBON STEELS - GROUP - I (CONTD.)															
55	0.52-0.60	0.50-0.80	0.17-0.37	0.04	0.04								66	13	217
60	0.57-0.65	0.50-0.80	0.17-0.37	0.04	0.04								69	12	229
65	0.62-0.70	0.50-0.80	0.17-0.37	0.04	0.04								71	10	229
70	0.67-0.75	0.50-0.80	0.17-0.37	0.04	0.04								73	9	229
75	0.72-0.80	0.50-0.80	0.17-0.37	0.04	0.04								110	7	241
80	0.77-0.85	0.50-0.80	0.17-0.37	0.04	0.04								110	6	241
85	0.82-0.90	0.50-0.80	0.17-0.37	0.04	0.04								115	6	255
QUALITY CARBON STEELS - GROUP II															
15Г	0.12-0.19	0.70-1.00	0.17-0.37	0.04	0.04								42	26	163
20Г	0.17-0.24	0.70-1.00	0.17-0.37	0.04	0.04								46	24	197
25Г	0.22-0.30	0.70-1.00	0.17-0.37	0.04	0.04								50	22	197
30Г	0.27-0.35	0.70-1.00	0.17-0.37	0.04	0.04								55	20	197
35Г	0.32-0.40	0.70-1.00	0.17-0.37	0.04	0.04								57	18	197
40Г	0.37-0.45	0.70-1.00	0.17-0.37	0.04	0.04								60	17	207
45Г	0.42-0.50	0.70-1.00	0.17-0.37	0.04	0.04								63	15	217

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CHEMICAL AND MECHANICAL PROPERTIES OF IMPORTANT STEELS - RUSSIAN GRADES

DESIGNATION	* CHEMICAL COMPOSITION, PERCENT												MECH. PROPERTIES MIN.		
GRADE OF STEEL OR ALLOY	C	Mn	Si	S	P	Cr	Ni	Mo	V	W	Co	OTHERS	UTS Kg/mm²	ELON. %	HARDNESS MAX.,BHN
QUALITY CARBON STEELS - GROUP - II (CONTD.)															
50Г	0.48-0.56	0.70-1.00	0.17-0.37	0.04	0.04								66	13	217
60Г	0.57-0.65	0.70-1.00	0.17-0.37	0.04	0.04								71	11	229
65Г	0.62-0.70	0.90-1.20	0.17-0.37	0.04	0.04								75	9	229
70Г	0.67-0.75	0.90-1.20	0.17-0.37	0.04	0.04								80	8	229
PLOUGH SHARE ROLLED STEEL															
П53	0.47-0.57	0.50-0.80	0.15-0.40	0.04	0.04										255
ALLOY STRUCTURAL STEELS															
35 Г 2	0.31-0.39	1.40-1.80	0.10-0.35	0.04	0.04										207
40 Г 2	0.36-0.44	1.40-1.80	0.10-0.35	0.04	0.04										217
45 Г 2	0.41-0.49	1.40-1.80	0.10-0.35	0.04	0.04										229
50 Г 2	0.46-0.55	1.40-1.80	0.10-0.35	0.04	0.04										229
15X	0.12-0.18	0.40-0.70	0.10-0.35	0.04	0.04	0.70-1.00									179
15XP	0.12-0.18	0.40-0.70	0.10-0.35	0.04	0.04	0.70-1.00						0.002 B			177
20X	0.17-0.23	0.50-0.80	0.10-0.35	0.04	0.04	0.70-1.00									179
30X	0.25-0.33	0.50-0.80	0.10-0.35	0.04	0.04	0.80-1.10									187
30XPA	0.27-0.33	0.50-0.80	0.10-0.35	0.04	0.04	1.00-1.30						0.002 B			197

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CHEMICAL AND MECHANICAL PROPERTIES OF IMPORTANT STEELS - RUSSIAN GRADES

DESIGNATION	* CHEMICAL COMPOSITION, PERCENT												MECH. PROPERTIES MIN.		
GRADE OF STEEL OR ALLOY	C	Mn	Si	S	P	Cr	Ni	Mo	V	W O	Co	OTHERS	UTS Kg/mm²	ELON. %	HARDNESS MAX.,BHN
ALLOY STRUCTURAL STEELS (CONTD.)															
35X	0.31-0.39	0.50-0.80	0.10-0.35	0.04	0.04	0.80-1.00									197
35XPA	0.33-0.40	0.50-0.80	0.10-0.35	0.04	0.04	0.80-1.00						0.002 B			217
38XA	0.34-0.42	0.50-0.80	0.10-0.35	0.04	0.04	0.80-1.10									207
40X	0.36-0.44	0.50-0.80	0.10-0.35	0.04	0.04	0.80-1.10									217
40XP	0.37-0.45	0.50-0.80	0.10-0.35	0.04	0.04	0.80-1.10						0.002 B			229
45X	0.41-0.49	0.50-0.80	0.10-0.35	0.04	0.04	0.80-1.10									229
45XII	0.41-0.49	0.50-0.80	0.10-0.35	0.04	0.04	0.80-1.10						0.50 Al 0.25 Zr			217
50X	0.46-0.54	0.50-0.80	0.10-0.35	0.04	0.04	0.80-1.10									229
15XГ	0.12-0.20	1.10-1.40	0.10-0.35	0.04	0.04	0.40-0.70									187
18XГ	0.15-0.21	0.90-1.20	0.10-0.35	0.04	0.04	0.90-1.20									187
18XГT	0.17-0.23	0.80-1.10	0.10-0.35	0.04	0.04	1.00-1.30						0.06-0.12 Ti			229
20XГ	0.15-0.25	0.90-1.20	0.10-0.35	0.04	0.04	0.90-1.20									187
20XГP	0.18-0.24	0.70-1.10	0.10-0.35	0.04	0.04	0.80-1.10						0.002-0.005 B			197
30XГT	0.24-0.32	0.80-1.10	0.10-0.35	0.04	0.04	1.00-1.30						0.06-0.12 Ti			229
40XГ	0.37-0.45	0.90-1.20	0.10-0.35	0.04	0.04	0.90-1.20									229

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CHEMICAL AND MECHANICAL PROPERTIES OF IMPORTANT STEELS - RUSSIAN GRADES

DESIGNATION	* CHEMICAL COMPOSITION, PERCENT												MECH. PROPERTIES MIN.		
GRADE OF STEEL OR ALLOY	C	Mn	Si	S	P	Cr	Ni	Mo	V	W	Co	OTHERS	UTS	ELON.	HARDNESS
													Kg/mm^2	%	MAX.,BHN
ALLOY STRUCTURAL STEELS (CONTD.)															
40ХГР	0.37-0.45	0.70-1.00	0.10-0.35	0.04	0.04	0.80-1.10						0.002-0.005 B			241
35ХГ2	0.32-0.40	1.60-1.90	0.10-0.35	0.04	0.04	0.40-0.70									229
40ХГМ	0.37-0.45	0.90-1.20	0.10-0.35	0.04	0.04	0.90-1.20		0.20-0.30							241
33ХС	0.29-0.37	0.30-0.60	1.00-1.30	0.04	0.04	1.30-1.60									241
38ХС	0.34-0.42	0.30-0.60	1.00-1.30	0.04	0.04	1.30-1.60									255
40ХС	0.37-0.45	0.30-0.60	1.20-1.60	0.04	0.04	1.30-1.60									255
27СГ	0.23-0.31	1.10-1.40	1.10-1.40	0.04	0.04	≤0.25									217
35СГ	0.31-0.39	1.10-1.40	1.10-1.40	0.04	0.04	≤0.25									229
36Г2С	0.32-0.40	0.40-0.70	1.10-1.40	0.04	0.04	≤0.25									229
15ХМ	0.11-0.18	0.40-0.70	0.10-0.35	0.04	0.04	0.80-1.10		0.40-0.55							179
20ХМ	0.15-0.25	0.40-0.70	0.10-0.35	0.04	0.04	≤0.30		0.40-0.55							197
30ХМ	0.26-0.34	0.40-0.70	0.10-0.35	0.04	0.04	0.80-1.10		0.15-0.25							229
35ХМ	0.32-0.40	0.40 0.70	0.10-0.35	0.04	0.04	0.80-1.10		0.15-0.25							241
35Х2МА	0.32-0.40	0.40-0.70	0.10-0.35	0.04	0.04	1.60-1.90		0.15-0.25							241
38ХВА	0.35-0.42	0.25-0.50	0.10-0.35	0.04	0.04	0.90-1.30				0.50-0.80					229

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CHEMICAL AND MECHANICAL PROPERTIES OF IMPORTANT STEELS - RUSSIAN GRADES

DESIGNATION	* CHEMICAL COMPOSITION, PERCENT												MECH. PROPERTIES MIN.		
GRADE OF STEEL OR ALLOY	C	Mn	Si	S	P	Cr	Ni	Mo	V	W	Co	OTHERS	UTS Kg/mm^2	ELON. %	HARDNESS MAX.,BHN
ALLOY STRUCTURAL STEELS (CONTD.)															
15XΦ	0.12-0.18	0.40-0.70	0.10-0.35	0.04	0.04	0.80-1.10			0.10-0.20						187
20XΦ	0.17-0.23	0.50-0.80	0.10-0.35	0.04	0.04	0.80-1.10			0.10-0.20						197
40XΦ	0.37-0.44	0.50-0.80	0.10-0.35	0.04	0.04	0.80-1.10			0.10-0.20						241
15HM	0.10-0.18	0.40-0.70	0.10-0.35	0.04	0.04	≤0.30	1.50-2.00	0.20-0.30							197
20HM	0.17-0.25	0.40-0.70	0.10-0.35	0.04	0.04	≤0.30	1.50-2.00	0.20-0.30							197
20XH	0.17-0.23	0.40-0.70	0.10-0.35	0.04	0.04	0.45-0.75	1.00-1.40								197
40XH	0.36-0.45	0.50-0.80	0.10-0.35	0.04	0.04	0.45-0.75	1.00-1.40								217
45XH	0.41-0.45	0.50-0.80	0.10-0.35	0.04	0.04	0.45-0.75	1.00-1.40								227
50XH	0.46-0.54	0.50-0.80	0.10-0.35	0.04	0.04	0.45-0.75	1.00-1.40								227
12XH2	0.09-0.16	0.30-0.60	0.10-0.35	0.04	0.04	0.60-0.90	1.50-1.90								207
12XH3A	0.09-0.16	0.30-0.60	0.10-0.35	0.04	0.04	0.60-0.90	2.80-3.20								217
12X2H4A	0.09-0.16	0.30-0.60	0.10-0.35	0.04	0.04	1.25-1.65	3.30-3.70								269
20XH3A	0.17-0.24	0.30-0.60	0.10-0.35	0.04	0.04	0.60-0.90	2.80-3.20								241
20X2H4A	0.16-0.22	0.30-0.60	0.10-0.35	0.04	0.04	1.25-1.65	3.30-3.70								269
30XH3A	0.27-0.34	0.30-0.60	0.10-0.35	0.04	0.04	0.60-0.90	2.80-3.20								241

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CHEMICAL AND MECHANICAL PROPERTIES OF IMPORTANT STEELS - RUSSIAN GRADES

DESIGNATION	* CHEMICAL COMPOSITION, PERCENT												MECH. PROPERTIES MIN.		
GRADE OF STEEL OR ALLOY	C	Mn	Si	S	P	Cr	Ni	Mo	V	W	Co	OTHERS	UTS Kg/mm²	ELON. %	HARDNESS MAX.,BHN
ALLOY STRUCTURAL STEELS (CONTD.)															
37XH3A	0.33-0.41	0.25-0.55	0.10-0.35	0.04	0.04	1.20-1.60	3.00-3.50								269
34XH1M	0.30-0.40	0.50-0.80	0.10-0.35	0.04	0.04	1.30-1.70	1.30-1.70	0.20-0.30							269
34XH2M	0.30-0.40	0.50-0.80	0.10-0.35	0.04	0.04	0.80-1.20	1.75-2.25	0.25-0.40							269
34XH3M	0.30-0.40	0.50-0.80	0.10-0.35	0.04	0.04	0.70-1.10	2.75-3.25	0.25-0.40							289
36XH1MΦA	0.32-0.40	0.30-0.50	0.10-0.35	0.04	0.04	1.30-1.70	1.30-1.70	0.30-0.40	0.08-0.15						289
45XHMΦA	0.42-0.50	0.50-0.80	0.10-0.35	0.04	0.04	0.80-1.10	1.30-1.80	0.20-0.30	0.10-0.20						300
30XHBA	0.27-0.34	0.30-0.60	0.10-0.35	0.04	0.04	0.60-0.80	1.27-1.65			0.50-0.80					241
38XHBA	0.34-0.42	0.30-0.60	0.10-0.35	0.04	0.04	1.30-1.70	1.25-1.65			0.50-0.80					269
40XHBA	0.37-0.44	0.50-0.80	0.10-0.35	0.04	0.04	0.60-0.90	1.25-1.65			0.80-1.20					269
40XHMA	0.37-0.44	0.50-0.80	0.10-0.35	0.04	0.04	0.60-0.90	1.25-1.65		0.15-0.25						269
30X2HBA	0.27-0.34	0.30-0.60	0.10-0.35	0.04	0.04	1.60-2.00	1.40-1.80			1.20-1.50					269
38XH3BA	0.34-0.42	0.25-0.55	0.10-0.35	0.04	0.04	0.80-1.20	2.75-3.25			0.50-0.80					269
18X2H4BA	0.14-2.00	0.25-0.55	0.10-0.35	0.04	0.04	1.35-1.65	4.00-4.50			0.80-1.20					269
25X2H4BA	0.21-0.28	0.25-0.55	0.10-0.35	0.04	0.04	1.35-1.65	4.00-4.50			0.80-1.20					269
30XHBΦA	0.27-0.34	0.30-0.60	0.10-0.35	0.04	0.04	0.60-0.90	2.00-2.50		0.15-3.00	0.50-0.80					269

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CHEMICAL AND MECHANICAL PROPERTIES OF IMPORTANT STEELS - RUSSIAN GRADES

DESIGNATION	* CHEMICAL COMPOSITION, PERCENT												MECH. PROPERTIES MIN.		
GRADE OF STEEL OR ALLOY	C	Mn	Si	S	P	Cr	Ni	Mo	V	W	Co	OTHERS	UTS Kg/mm^2	ELON. %	HARDNESS MAX.,BHN
ALLOY STRUCTURAL STEELS (CONTD.)															
30Х2НВФА	0.27-0.34	0.30-0.60	0.10-0.35	0.04	0.04	1.60-2.00	1.40-1.80		0.18-0.28	1.20-1.50					269
30ХН3ВФА	0.34-0.42	0.25-0.55	0.10-0.35	0.04	0.04	1.00-1.40	3.00-3.50		0.10-0.20	0.50-0.80					269
20ХН4ФА	0.17-2.00	0.25-0.55	0.10-0.35	0.04	0.04	0.70-1.10	3.75-4.25		0.15-0.30						269
38ХН3МФА	0.34-0.42	0.25-0.55	0.10-0.35	0.04	0.04	1.20-1.50	3.00-3.50	0.35-0.40	0.10-0.20						269
15ХГНТ	0.12-0.18	0.70-1.00	0.10-0.35	0.04	0.04	0.70-1.00						0.06-0.12 Ti			229
15Х2ГН2	0.12-0.18	0.70-1.00	0.10-0.35	0.04	0.04	1.40-1.80						0.06-0.12 Ti			269
15Х2ГН2ТРА	0.12-0.18	0.70-1.00	0.10-0.35	0.04	0.04	1.40-1.80						0.06-0.12 Ti	0.002-0.005 B		269
18ХГН	0.16-0.22	0.80-1.10	0.10-0.35	0.04	0.04	0.40-0.70	0.40-0.70								255
25Х2ГНТА	0.22-0.29	0.80-1.10	0.10-0.35	0.04	0.04	1.30-1.70	0.90-1.30					0.06-0.12 Ti			269
30ХГНА	0.28-0.35	0.60-0.90	0.10-0.35	0.04	0.04	0.90-1.20	0.30-0.60								229
38ХГН	0.35-0.43	0.80-1.10	0.10-0.35	0.04	0.04	0.50-0.80	0.70-1.10								229
30Х2ГН2	0.26-0.34	0.80-1.10	0.10-0.35	0.04	0.04	1.40-1.70	1.40-1.80								255
16ХСН	0.13-0.20	0.30-0.60	0.60-0.90	0.04	0.04	0.80-1.10	0.60-0.90								255
18ХСНРА	0.16-0.21	0.60-0.90	0.60-0.90	0.04	0.04	0.80-1.10	0.80-1.10					0.002-0.005 B			197
20ХГСА	0.17-0.23	0.80-1.10	0.90-1.20	0.04	0.04	0.80-1.10									207

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CHEMICAL AND MECHANICAL PROPERTIES OF IMPORTANT STEELS - RUSSIAN GRADES

DESIGNATION	* CHEMICAL COMPOSITION, PERCENT												MECH. PROPERTIES MIN.		
GRADE OF STEEL OR ALLOY	C	Mn	Si	S	P	Cr	Ni	Mo	V	W	Co	OTHERS	UTS Kg/mm²	ELON. %	HARDNESS MAX.,BHN
ALLOY STRUCTURAL STEELS (CONTD.)															
25ХГСА	0.22-0.28	0.80-1.10	0.90-1.20	0.04	0.04	0.80-1.10									217
30ХГСА	0.28-0.34	0.80-1.10	0.90-1.20	0.04	0.04	0.80-1.10									229
30ХГСНА	0.27-0.34	1.00-1.30	0.90-1.20	0.04	0.04	0.90-1.20	1.40-1.80								255
35ХГСА	0.32-0.39	0.80-1.10	1.10-1.40	0.04	0.04	1.10-1.40									241
38Х Ю	0.35-0.43	0.20-0.50	0.10-0.35	0.04	0.04	1.50-1.80						0.50-0.80 Al.			229
38ХМ Ю А	0.35-0.42	0.30-0.60	0.10-0.35	0.04	0.04	1.35-1.65		0.15-0.25				0.50-0.80 Al.			229
38ХВФЮ	0.35-0.43	0.20-0.40	0.10-0.35	0.04	0.04	1.50-1.80			0.10-0.2	0.20-0.40		0.40-0.70 Al.			229
13Н2ХА	0.09-0.12	0.30-0.60	0.10-0.35	0.04	0.04	0.20-0.50	1.70-2.10								269
NON - ANNEALED SPRING STEELS															
65	0.60-0.70	0.50-0.80	0.17-0.37	0.04	0.04										255
70	0.65-0.75	0.50-0.80	0.17-0.37	0.04	0.04										269
75	0.72-0.80	0.50-0.80	0.17-0.37	0.04	0.04										285
85	0.82-0.90	0.50-0.80	0.17-0.37	0.04	0.04										302
55ГС	0.52-0.60	0.60-0.90	0.50-0.8	0.04	0.04										285
65Г	0.60-0.70	0.90-1.20	0.17-0.37	0.04	0.04										269

*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED

CHEMICAL AND MECHANICAL PROPERTIES OF IMPORTANT STEELS - RUSSIAN GRADES

DESIGNATION	* CHEMICAL COMPOSITION, PERCENT												MECH. PROPERTIES MIN.		
GRADE OF STEEL OR ALLOY	C	Mn	Si	S	P	Cr	Ni	Mo	V	W	Co	OTHERS	UTS Kg/mm^2	ELON. %	HARDNESS MAX.,BHN
NON - ANNEALED SPRING STEELS (CONTD.)															
50C2	0.47-0.55	0.60-0.90	1.50-2.00	0.04	0.04										285
55C2	0.52-0.60	0.60-0.90	1.50-2.00	0.04	0.04										285
60C2	0.57-0.65	0.60-0.90	1.50-2.00	0.04	0.04										302
60C2A	0.56-0.64	0.60-0.90	1.60-2.00	0.04	0.04										302
63C2A	0.60-0.65	0.60-0.90	1.80-2.20	0.04	0.04										302
70C3A	0.66-0.74	0.60-0.90	2.40-2.80	0.04	0.04										302
70C2XA	0.65-0.75	0.40-0.60	1.40-1.70	0.04	0.04	0.20-0.40									329
50ХГ	0.46-0.54	0.70-1.00	0.17-0.37	0.04	0.04	0.90-1.20									302
50ХГА	0.46-0.54	0.80-1.00	0.17-0.37	0.04	0.04	0.95-1.20									302
50ХФА	0.46-0.54	0.50-0.80	0.17-0.37	0.04	0.04	0.80-1.10			0.15-0.25						302
50ХГФА	0.48-0.55	0.80-1.00	0.17-0.37	0.04	0.04	0.95-1.20			0.15-0.25						321
60С2ХФА	0.56-0.64	0.40-0.70	1.40-1.80	0.04	0.04	0.90-1.20			0.10-0.20						302
60С2ХА	0.56-0.64	0.40-0.70	1.40-1.80	0.04	0.04	0.70-1.00									321
65С2ВА	0.61-0.69	0.70-1.00	1.50-2.00	0.04	0.04					0.80-1.20					302
60С2Н2А	0.56-0.64	0.40-0.70	1.40-1.80	0.04	0.04		1.40-1.70								302

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CHEMICAL AND MECHANICAL PROPERTIES OF IMPORTANT STEELS - RUSSIAN GRADES

DESIGNATION	* CHEMICAL COMPOSITION, PERCENT												MECH. PROPERTIES MIN.		
GRADE OF STEEL OR ALLOY	C	Mn	Si	S	P	Cr	Ni	Mo	V	W	Co	OTHERS	UTS Kg/mm^2	ELON. %	HARDNESS MAX.,BHN
NON - ANNEALED SPRING STEELS (CONTD.)															
55СГ	0.50-0.60	0.80-1.00	1.30-1.80	0.04	0.04										285
60СГ	0.55-0.65	0.80-1.00	1.30-1.80	0.04	0.04										285
60СГА	0.56-0.64	0.80-1.00	1.30-1.80	0.04	0.04										285
CORROSION RESISTING, NON SCALING AND HEAT RESISTING STEELS AND ALLOYS															
X5	≤0.15	≤0.50	0.5	0.04	0.04	4.50-6.00									
X5M	≤0.15	≤0.50	0.5	0.04	0.04	4.50-6.00		0.45-0.60							
X5BΦ	≤0.15	≤0.50	0.30-0.60	0.04	0.04	4.50-6.00				0.40-0.70					
X6CM	≤0.15	≤0.70	1.50-2.00	0.04	0.04	5.00-6.00		0.45-0.60							
1X8BΦ	0.08-0.15	≤0.50	≤0.60	0.04	0.04	7.00-8.50			0.30-0.50	0.60-1.00					
4X9C2	0.35-0.45	≤0.70	2.00-3.00	0.04	0.04	8.00-10.00									
4X10C2M	0.35-0.45	≤0.70	1.90-2.60	0.04	0.04	9.00-10.50		0.70-0.90							
1X12H2BMΦ	0.10-0.16	≤0.60	0.60	0.04	0.04	10.50-12.00	1.50-1.8	0.35-0.50	0.18-0.30	1.60-2.00					
2X13	0.16-0.24	≤0.60	≤0.60	0.04	0.04	12.00-14.00									126-197
3X13	0.25-0.34	≤0.60	≤0.60	0.04	0.04	12.00-14.00									131-207
4X13	0.35-0.44	≤0.60	≤0.60	0.04	0.04	12.00-14.00									143-229

*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED

CHEMICAL AND MECHANICAL PROPERTIES OF IMPORTANT STEELS - RUSSIAN GRADES

DESIGNATION	* CHEMICAL COMPOSITION, PERCENT												MECH. PROPERTIES MIN.		
GRADE OF STEEL OR ALLOY	C	Mn	Si	S	P	Cr	Ni	Mo	V	W	Co	OTHERS	UTS Kg/mm^2	ELON. %	HARDNESS MAX.,BHN
CORROSION RESISTING, NON SCALING AND HEAT RESISTING STEELS AND ALLOYS (CONTD.)															
3X13H7C2	0.25-0.34	≤0.70	2.00-3.00	0.04	0.04	12.00-14.00	6.00-7.50								143-229
1X13H3	0.08-0.15	≤0.60	≤0.60	0.04	0.04	12.00-14.50	2.20-3.00								126-197
1X17H2	0.11-0.17	≤0.80	≤0.80	0.04	0.04	16.00-18.00	1.50-2.50								126-197
9X18	0.90-1.00	≤0.70	≤0.80	0.04	0.04	17.00-19.00									143-229
MARTENSITO - FERRITIC STEELS												OTHER ELEMENTS			
X6Ю	≤0.15	≤0.50	1.20-1.80	0.04	0.04	5.50-7.00						0.70-1.10 Al.			
1X11MΦ	0.12-0.19	≤0.70	≤0.50	0.04	0.04	10.00-11.50		0.60-0.80	0.25-4.00						
1X12BHMΦ	0.12-0.18	0.50-0.90	≤0.40	0.04	0.04	11.00-13.00	0.40-0.80	0.50-0.70	0.15-0.30	0.70-2.20					
2X12BMБΦP	0.15-0.22	11.00-13.00	≤0.50	0.04	0.04	11.00-13.00		0.40-0.60	0.15-0.30	0.40-0.70		0.20-0.40 Nb	0.003 B		
1X12B2MΦ	0.10-0.17	0.50-0.80	≤0.50	0.04	0.04	11.00-13.00		0.60-0.90	0.15-0.30	1.70-2.20					
1X13	0.09-0.15	≤0.60	≤0.60	0.04	0.04	12.00-14.00									
FERRITIC STEELS															
1X12CЮ	0.07-0.12	1.20-2.00	≤0.70	0.04	0.04	12.00-14.00									
0X13	≤0.80	≤0.60	≤0.60	0.04	0.04	11.00-13.00									
X14	≤0.15	≤0.70	≤0.70	0.04	0.04	13.00-15.00									
X17	≤0.12	≤0.70	≤0.80	0.04	0.04	16.00-18.00									

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CHEMICAL AND MECHANICAL PROPERTIES OF IMPORTANT STEELS - RUSSIAN GRADES

DESIGNATION	* CHEMICAL COMPOSITION, PERCENT												MECH. PROPERTIES MIN.		
GRADE OF STEEL OR ALLOY	C	Mn	Si	S	P	Cr	Ni	Mo	V	W	Co	OTHERS	UTS Kg/mm²	ELON. %	HARDNESS MAX.,BHN
FERRITIC STEELS (CONTD.)															
0X17T	≤0.08	≤0.70	≤0.80	0.04	0.04	16.00-18.00						5C-0.80 Ti			
X18CЮ	≤0.15	≤0.50	1.00-1.50	0.04	0.04	17.00-20.00						0.70-1.20 Al.			
X25T	≤0.15	≤0.80	≤1.00	0.04	0.04	24.00--27.00						5C-0.80 Ti			
X28	≤0.15	≤0.80	≤1.00	0.04	0.04	27.00-30.00									
ASTENO - MARTENSITIC STEELS															
2X13H4Г9	0.15-0.30	≤0.80	≤0.80	0.04	0.04	12.00-14.00	3.70-4.70								
X15H9Ю	≤0.09	≤0.80	≤0.80	0.04	0.04	14.00-16.00	7.00-9.40					0.70-1.30 Al.			
X17H7Ю	≤0.09	≤0.80	≤0.80	0.04	0.04	16.00-18.00	6.50-7.50					0.70-1.30 Al.			
2X17H2	0.22-0.28	≤0.80	≤0.80	0.04	0.04	16.00-18.00	1.50-2.50								
ASTENO - FERRITIC STEELS															
0X20H14C2	≤0.80	≤1.50	2.00-3.00	0.04	0.04	19.00-22.00	12.00-15.00								143-229
X20H14C2	≤0.20	≤1.50	2.00-3.00	0.04	0.04	19.00-22.00	12.00-15.00								143-229
0X21H5T	≤0.80	≤0.80	≤0.80	0.04	0.04	20.00-22.00	4.80-5.80					0.30-0.60 Ti			
0X21H6M2T	≤0.80	≤0.80	≤0.80	0.04	0.04	20.00-22.00	5.50-6.50	1.80-2.50				0.20-0.40 Ti			
1X21H5T	0.09-0.14	≤0.80	≤0.80	0.04	0.04	20.00-22.00	4.80-5.80					(C-0.02)5-0.80 Ti			
X23H13	≤0.20	≤2.00	≤1.00	0.04	0.04	22.00-25.00	12.00-15.00								

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CHEMICAL AND MECHANICAL PROPERTIES OF IMPORTANT STEELS - RUSSIAN GRADES

DESIGNATION	* CHEMICAL COMPOSITION, PERCENT												OTHER ELEMENTS		
GRADE OF STEEL OR ALLOY	C	Mn	Si	S	P	Cr	Ni	Mo	V	W	Co	OTHERS			
ASTENO - FERRITIC STEELS (CONTD.)															
X28AH	≤0.15	≤1.50	≤1.00	0.04	0.04	25.00-28.00	1.00-1.70								
AUSTENITIC STEELS															
4X12H8Г8МФБ	0.34-0.40	7.50-9.50	0.30-0.80	0.04	0.04	11.50-13.50	7.00-9.00	1.10-1.40	1.25-1.55			0.25-0.45 Nb			
0X10H20T2	≤0.80	≤2.00	≤0.80	0.04	0.04	10.00-12.00	18.00-20.00					1.50-2.50 Ti,Al≤1.0			
X12H20T3P	≤1.00	≤1.00	≤1.00	0.04	0.04	10.00-12.50	18.00-21.00					2.60-3.2 Ti,Al≤0.8			
X12H22T3MP	≤1.00	≤0.60	≤0.60	0.04	0.04	10.00-12.50	21.00-25.00	1.00-1.60				2.60-3.2 Ti,Al≤0.8	0.02 B		
1X14H16Б	0.07-0.12	1.00-2.00	≤0.60	0.04	0.04	13.0-15.00	14.00-17.00					0.90-1.30 Nb			
1X14H16БР	0.07-0.12	1.00-2.00	≤0.60	0.04	0.04	13.0-15.00	14.00-17.00					0.90-1.30 Nb.	0.005 B		
1X14H18B2Б	0.07-0.12	1.00-2.00	≤0.60	0.04	0.04	13.0-15.00	18.00-20.00			2.00-2.75		0.90-1.30 Nb			
1X14H18B2БР	0.07-0.12	1.00-2.00	≤0.60	0.04	0.04	13.0-15.00	18.00-20.00			2.00-2.75		0.90-1.30 Nb	≤0.005B ≤0.02Ce		
1X14H18B2БР1	0.07-0.12	1.00-2.00	≤0.60	0.04	0.04	13.0-15.00	18.00-20.00			2.00-2.75		0.90-1.30 Nb	≤0.025B ≤0.02Ce		
X14Г14H	≤0.12	13.00-15.00	≤0.80	0.04	0.04	13.00-15.00	1.00-1.50								
X14Г14H3T	≤0.10	13.00-15.00	≤0.80	0.04	0.04	13.00-15.00	2.50-3.50						(C-0.02)5-0.60 Ti		
4X14H14B2M	0.40-0.50	0.70	≤0.80	0.04	0.04	13.00-15.00	13.00-15.00	0.25-0.40		2.00-2.75					
4X15H7Г7Ф2MC	0.38-0.47	6.00-8.00	0.90-1.40	0.04	0.04	14.00-16.00	6.00-8.00	0.65-0.95	1.50-1.90						

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CHEMICAL AND MECHANICAL PROPERTIES OF IMPORTANT STEELS - RUSSIAN GRADES

DESIGNATION	* CHEMICAL COMPOSITION, PERCENT												MECH. PROPERTIES MIN.		
GRADE OF STEEL OR ALLOY	C	Mn	Si	S	P	Cr	Ni	Mo	V	W	Co	OTHERS	UTS Kg/mm²	ELON. %	HARDNESS MAX.,BHN
AUSTENITIC STEELS (CONTD.)															
0X14H28B3T3ЮР	≤0.80	≤0.60	≤0.60	0.04	0.04	13.00-15.00	26.00-29.00					0.5-1.2Al. 2.4-3.2Ti	≤0.02B		
1X16H13M2Б	0.06-0.12	≤1.00	≤0.80	0.04	0.04	15.00-17.00	12.50-14.50	2.00-2.50				0.90-1.30 Nb.			
X16H15M3Б	0.09	≤0.60	≤0.80	0.04	0.04	15.00-17.00	14.00-16.00	2.50-3.00				0.60-0.90 Nb.			
0X17Г9АН4	≤0.12	8.00-10.50	≤0.80	0.04	0.04	16.00-18.00	3.50-4.50					0.15-0.25 N			143-229
X17H13M2T	≤0.10	1.00-2.00	≤0.80	0.04	0.04	16.00-18.00	12.00-14.00	1.80-2.50				0.30-0.60 Ti			143-229
X17H13M3T	≤0.10	1.00-2.00	≤0.80	0.04	0.04	16.00-18.00	12.00-14.00	3.00-4.00				0.30-0.60 Ti			
0X17H16M3T	≤0.08	1.00-2.00	≤0.80	0.04	0.04	16.00-18.00	15.00-17.00	2.00-3.50				0.30-0.60 Ti			
X17АГ14	≤0.15	13.50-15.50	≤0.80	0.04	0.04	16.00-18.00	0.60					0.30-0.40 N			
00X18H10	≤0.04	1.00-2.00	≤0.80	0.04	0.04	17.00-19.00	9.00-11.00								
0X18H10	≤0.08	1.00-2.00	≤0.80	0.04	0.04	17.00-19.00	9.00-11.00								
X18H9	≤0.12	1.00-2.00	≤0.80	0.04	0.04	17.00-19.00	8.00-10.00								
2X18H9	0.13-0.21	1.00-2.00	≤0.80	0.04	0.04	17.00-19.00	8.00-10.00								
X18H10E	≤0.12	1.00-2.00	≤0.80	0.04	0.04	17.00-19.00	9.00-11.00					0.18-0.35 Se or Te			
0X18H10T	≤0.08	1.00-2.00	≤0.80	0.04	0.04	17.00-19.00	9.00-11.00					Ti ≤0.06			
X18H10T	≤0.12	1.00-2.00	≤0.80	0.04	0.04	17.00-19.00	9.00-11.00					(C-0.02) 5-0.70 Ti			

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CHEMICAL AND MECHANICAL PROPERTIES OF IMPORTANT STEELS - RUSSIAN GRADES

DESIGNATION	* CHEMICAL COMPOSITION, PERCENT												MECH. PROPERTIES MIN.		
GRADE OF STEEL OR ALLOY	C	Mn	Si	S	P	Cr	Ni	Mo	V	W	Co	OTHERS	UTS Kg/mm²	ELON. %	HARDNESS MAX.,BHN
AUSTENITIC STEELS (CONTD.)															
X18H9T	≤0.12	1.00-2.00	≤0.80	0.04	0.04	17.00-19.00	8.00-9.50					(C-0.02) 5-0.70 Ti			
0X18H11	≤0.06	1.00-2.00	≤0.80	0.04	0.04	17.00-19.00	10.00-12.00								
0X18H12T	≤0.08	1.00-2.00	≤0.80	0.04	0.04	17.00-19.00	11.00-13.00					5C-0.80 Ti			
X18H12T	≤0.12	1.00-2.00	≤0.80	0.04	0.04	17.00-19.00	11.00-13.00					(C-0.02) 5-0.70 Ti			
0X18H12Б	≤0.08	1.00-2.00	≤0.80	0.04	0.04	17.00-19.00	11.00-12.00					8C-1.20 Nb			
4X18H25C2	0.32-0.40	≤1.50	2.00-3.00	0.04	0.04	17.00-19.00	23.00-26.00								
3X19H9MBБT	0.28-0.35	0.80-1.50	≤0.08	0.04	0.04	18.00-20.00	8.00-10.00	1.00-1.50		1.00-1.50		0.2-0.5Nb 0.2-0.5Ti			
0X23H18	≤0.10	≤2.00	≤1.00	0.04	0.04	22.00-25.00	17.00-20.00								
X23H18	≤0.20	≤2.00	≤1.00	0.04	0.04	22.00-25.00	17.00-20.00								
0X23H28M2T	≤0.06	≤0.80	≤0.80	0.04	0.04	22.00-25.00	26.00-29.00	1.80-2.50				0.4-0.7 Ti			
0X23H28M3Д3T	≤0.06	≤0.80	≤0.80	0.04	0.04	22.00-25.00	26.00-29.00	2.50-3.00				0.4-0.7Ti 2.5-3.5Cu			
X25H16Г7AP	≤0.12	5.00-7.00	≤1.00	0.04	0.04	23.00-26.00	15.00-18.00					0.3-0.45 N			
X25H20C2	≤0.20	≤1.50	2.00-3.00	0.04	0.04	24.00-27.00	18.00-21.00								
1X25H25TP	0.07-0.12	1.00-2.00	≤0.80	0.04	0.04	23.00-26.00	24.00-27.00					1.1-1.6Ti B ≤0.01			

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CHEMICAL AND MECHANICAL PROPERTIES OF IMPORTANT STEELS - RUSSIAN GRADES

DESIGNATION	* CHEMICAL COMPOSITION, PERCENT												OTHER ELEMENTS		
GRADE OF STEEL OR ALLOY	C	Mn	Si	S	P	Cr	Ni	Mo	V	W	Co	OTHERS			
IRON-NICKEL ALLOYS															
XH35BMT	≤0.12	≤1.00	≤0.60	0.04	0.04	14.00-16.00	32.00-36.00	2.00-3.00		2.30-3.00		B ≤0.02 Ce 0.025			
XH38BT	≤0.12	1.00-2.00	≤0.60	0.04	0.04	14.00-16.00	34.00-38.00			2.80-3.50		1.1-1.5Ti			
NICKEL-BASE ALLOYS															
XH60Ю	≤0.10	≤0.30	≤0.80	0.04	0.04	15.00-18.00	55.00-58.00					1.1-1.4Ti, 2.6-3.5Al	Ba ≤0.1 Ce0.003		
XH70BMЮT	0.10-0.16	≤0.50	≤0.60	0.04	0.04	14.00-16.00	THE BAL.	3.00-5.00		4.00-6.00		Fe≤3.0 1.7-2.2Al	B ≤0.01		
XH70BMTЮ	≤0.12	≤0.50	≤0.60	0.04	0.04	13.00-16.00	THE BAL.	2.00-4.00	0.10-0.50	5.00-7.00		1.8-2.3Ti 1.7-2.3Al	B ≤0.02 Ce0.02	Fe5.0	
XH80TБЮ	≤0.08	≤1.00	≤0.80	0.04	0.04	15.00-18.00	THE BAL.					1.8-2.3Ti 0.5-1.0Al	1.0-1.5 Nb	Fe≤3.0	
XH70MBTЮБ	≤0.12	≤1.00	≤0.80	0.04	0.04	15.00-18.00	THE BAL.	4.00-6.00		2.00-3.50		1.9-2.8Ti 1.0-1.7Al	0.50-1.3 Nb	Fe≤5.0	Ce≤0.02
XH67BMTЮ	≤0.08	≤0.50	≤0.60	0.04	0.04	17.00-20.00	THE BAL.	4.00-5.00		4.00-5.00		2.2-2.8Ti 1.0-1.5Al	Ce≤0.01	Fe≤4.0	
XH75MБTЮ	≤0.10	≤0.40	≤0.80	0.04	0.04	19.00-22.00	THE BAL.	1.80-2.30				.35-.75Ti .35-.75Al	0.9-1.30 Nb	Fe≤8.0	
XH78T	≤0.12	≤0.70	≤0.80	0.04	0.04	19.00-22.00	THE BAL.					.15-.35Ti ≤0.15 Al	Fe≤6.0		
XH77TЮ	≤0.06	≤0.40	≤0.60	0.04	0.04	19.00-22.00	THE BAL.					2.3-2.7Ti Fe≤4.0	0.55-0.9 Nb	≤0.01 Ce	
XH77TЮP	≤0.06	≤0.40	≤0.60	0.04	0.04	19.00-22.00	THE BAL.					2.3-2.7Ti Fe≤4.0	0.55-0.9 Nb	≤0.01 Ce/B	
XH60B	≤0.10	≤0.50	≤0.80	0.04	0.04	23.50-26.50	THE BAL.			13.0-16.00		0.3-0.7Ti Fe≤4.0	≤0.50 Al.		
XH70Ю	≤0.10	≤0.30	≤0.80	0.04	0.04	26.00-29.00	THE BAL.					Fe≤1.0 Ba≤0.1	2.6-3.5 Al.	≤0.03 Ce	

*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED

CHEMICAL AND MECHANICAL PROPERTIES OF IMPORTANT STEELS - RUSSIAN GRADES

DESIGNATION	* CHEMICAL COMPOSITION, PERCENT												MECH. PROPERTIES MIN.		
GRADE OF STEEL OR ALLOY	C	Mn	Si	S	P	Cr	Ni	Mo	V	W	Co	OTHERS	UTS Kg/mm²	ELON. %	HARDNESS MAX.,BHN
NICKEL-BASE ALLOYS (CONTD.)															
XH70	≤0.07	≤0.50	≤0.80	0.04	0.04	28.00-31.00	THE BAL.					Fe≤5.0 Al≤0.15			
CARBON TOOL STEELS															
Y7	0.65-0.74	0.20-0.40	0.15-0.35	0.04	0.04										187
Y8	0.75-0.84	0.20-0.40	0.15-0.35	0.04	0.04										187
Y8Г	0.80-0.90	0.35-0.60	0.15-0.35	0.04	0.04										187
Y9	0.85-0.94	0.15-0.35	0.15-0.35	0.03	0.04										192
Y10	0.95-1.04	0.15-0.35	0.15-0.35	0.03	0.04										197
Y11	1.05-1.14	0.15-0.35	0.15-0.35	0.03	0.04										207
Y12	1.15-1.24	0.15-0.35	0.15-0.35	0.03	0.035										207
Y13	1.25-1.35	0.15-0.35	0.15-0.35	0.03	0.035										217
ALLOY TOOL STEELS															
7X3	0.60-0.75	0.40-0.70	0.15-0.35	0.04	0.04	3.20-3.80									187-229
8X3	0.76-0.85	0.40-0.70	0.15-0.35	0.04	0.04	3.20-3.80									207-255
65X	0.60-0.70	0.40-0.70	0.15-0.35	0.04	0.04	0.50-0.75									187
7X	0.80-0.95	0.40-0.70	0.15-0.35	0.04	0.04	1.40-1.70									179-217
X	0.95-1.10	0.40-0.70	0.15-0.35	0.04	0.04	1.30-1.60									187-229

*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED

CHEMICAL AND MECHANICAL PROPERTIES OF IMPORTANT STEELS - RUSSIAN GRADES

DESIGNATION	* CHEMICAL COMPOSITION, PERCENT												MECH. PROPERTIES MIN.		
GRADE OF STEEL OR ALLOY	C	Mn	Si	S	P	Cr	Ni	Mo	V	W	Co	OTHERS	UTS Kg/mm^2	ELON. %	HARDNESS MAX.,BHN
ALLOY TOOL STEELS (CONTD.)															
X09	0.95-1.10	0.40-0.70	0.15-0.35	0.04	0.04	0.75-1.05									179-229
X05	1.25-1.40	0.40-0.70	0.15-0.35	0.04	0.04	0.40-0.60									197-241
X12	2.00-2.30	0.40-0.70	0.15-0.35	0.04	0.04	11.50-13.00									217-269
Ф	0.95-1.05	0.40-0.70	0.15-0.35	0.04	0.04			0.20-0.40							179-219
B1	1.05-1.25	0.40-0.70	0.15-0.35	0.04	0.04	0.10-0.30		0.15-0.30							187-229
XГ	1.30-1.50	0.45-0.70	0.15-0.35	0.04	0.04	1.30-1.60									197-241
4XC	0.35-0.45	0.40-0.70	1.20-1.60	0.04	0.04	1.30-1.60									170-207
6XC	0.60-0.70	0.40-0.70	0.60-1.00	0.04	0.04	0.95-1.25									187-229
XГC	0.95-1.10	0.80-1.20	0.50-1.00	0.04	0.04										207-225
9XC	0.85-0.95	0.40-0.70	1.20-1.60	0.04	0.04	0.95-1.25									197-241
8XФ	0.75-0.85	0.40-0.70	0.15-0.35	0.04	0.04	0.50-0.80			0.15-0.30						170-207
85XФ	0.80-0.90	0.35-0.60	0.15-0.35	0.04	0.04	0.45-0.70			0.15-0.30						170-207
4XB2C	0.35-0.44	0.40-0.70	0.60-0.90	0.03	0.04	1.00-1.30				2.00-2.50					179-217
5XB2C	0.45-0.54	0.40-0.70	0.50-0.80	0.03	0.04	1.00-1.30				2.00-2.50					207-255
6XB2C	0.55-0.65	0.40-0.70	0.50-0.80	0.03	0.04	1.00-1.30				2.20-2.70					229-285

*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED

CHEMICAL AND MECHANICAL PROPERTIES OF IMPORTANT STEELS - RUSSIAN GRADES

DESIGNATION	* CHEMICAL COMPOSITION, PERCENT												MECH. PROPERTIES MIN.		
GRADE OF STEEL OR ALLOY	C	Mn	Si	S	P	Cr	Ni	Mo	V	W	Co	OTHERS	UTS Kg/mm^2	ELON. %	HARDNESS MAX.,BHN
ALLOY TOOL STEELS (CONTD.)															
5XBГ	0.55-0.70	0.90-1.20	0.15-0.35	0.03	0.035	0.50-0.80				0.50-0.80					179-217
9XBГ	0.85-0.95	0.90-1.20	0.15-0.35	0.03	0.035	0.50-0.80				0.50-0.80					197-241
XBГ	0.90-1.05	0.80-1.10	0.15-0.35	0.03	0.035	0.90-1.20				1.20-1.60					207-255
3X2B8	0.30-0.40	0.40-0.70	0.15-0.35	0.03	0.035	2.20-2.70				7.50-9.00					207-255
4X8B2	0.35-0.45	0.40-0.70	0.15-0.35	0.03	0.035	7.00-9.00				2.00-3.00					207-255
XB5	1.25-1.50	0.40-0.70	0.15-0.35	0.03	0.035	0.40-0.70			0.15-0.30	4.50-5.50					229-285
X12M	1.45-1.70	0.40-0.70	0.15-0.35	0.03	0.035	11.00-12.50		0.40-0.60	0.15-0.30						207-255
5XHM	0.50-0.60	0.40-0.70	0.15-0.35	0.03	0.035	0.50-0.80	1.40-1.80	0.15-0.30							197-241
5XГM	0.50-0.60	1.20-1.60	0.15-0.35	0.03	0.035	0.60-0.90		0.15-0.30							197-241
5XHT	0.50-0.60	0.40-0.70	0.15-0.35	0.03	0.035	0.60-0.90	1.40-1.80					0.10-0.20 Ti			192-235
5XHC	0.50-0.60	0.40-0.70	0.60-0.90	0.03	0.035	1.30-1.60	0.80-1.30								207-255
5XHCB	0.50-0.60	0.40-0.70	0.60-0.90	0.03	0.035	1.30-1.60	0.80-1.30			0.40-0.60					207-255
5XHB	0.50-0.60	0.40-0.70	0.15-0.35	0.03	0.035	0.50-0.80	1.40-1.80			0.60-1.00					197-241
45XHT	0.43-0.50	0.40-0.70	0.15-0.35	0.03	0.035	1.20-1.40	1.50-2.00					0.03-0.12 Ti			241
45XHB	0.43-0.50	0.40-0.70	0.15-0.35	0.03	0.035	0.50-0.80	1.50-2.00			0.60-1.00					241

*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED

CHEMICAL AND MECHANICAL PROPERTIES OF IMPORTANT STEELS - RUSSIAN GRADES

DESIGNATION	* CHEMICAL COMPOSITION, PERCENT												MECH. PROPERTIES MIN.		
GRADE OF STEEL OR ALLOY	C	Mn	Si	S	P	Cr	Ni	Mo	V	W	Co	OTHERS	UTS	ELON.	HARDNESS
													Kg/mm^2	%	MAX.,BHN
ALLOY TOOL STEELS (CONTD.)															
X12Φ1	1.45-1.70	0.40-0.70	0.15-0.35	0.03	0.035	11.00-12.50			0.70-0.90						207-255
HIGH SPEED STEELS															
P9	0.85-0.95	0.40-0.70	0.15-0.35	0.04	0.04	3.80-4.40		≤0.30	2.00-2.60	8.50-10.00					207-255
P9Φ5	1.40-1.50	0.40-0.70	0.15-0.35	0.04	0.04	3.80-4.40		≤0.40	4.30-5.10	9.00-10.50					≤293
P10K5Φ5	1.45-1.55	0.40-0.70	0.15-0.35	0.04	0.04	4.00-4.60		≤0.30	4.30-5.10	10.00-11.50	5.00-6.00				≤293
P9K10	0.90-1.00	0.40-0.70	0.15-0.35	0.04	0.04	3.80-4.40		≤0.30	2.00-2.60	9.00-10.50	9.50-10.50				≤293
P14Φ4	1.20-1.30	0.40-0.70	0.15-0.35	0.04	0.04	4.00-4.60		≤0.40	3.80-4.10	13.00-14.50					≤293
P18	0.70-0.80	0.40-0.70	0.15-0.35	0.04	0.04	3.80-4.40		≤0.30	1.00-1.40	17.50-19.00					207-255
P18Φ2	0.85-0.95	0.40-0.70	0.15-0.35	0.04	0.04	3.80-4.40		≤0.50	1.80-2.40	17.50-19.00					≤293
P18K5Φ2	0.85-0.95	0.40-0.70	0.15-0.35	0.04	0.04	3.80-4.40		≤0.50	1.80-2.40	17.50-19.00	5.00-6.00				≤295
P18K5	0.65-0.77	0.40-0.70	0.15-0.35	0.04	0.04	3.60-4.50		0.30-0.60	1.00-1.40	17.00-18.50	4.50-5.50				217-269
P18K10	0.65-0.77	0.40-0.70	0.15-0.35	0.04	0.04	3.60-4.50		0.30-0.60	1.00-1.40	17.00-18.50	9.50-10.50				217-269

*VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED

CHEMISTRY AND MECHANICAL PROPERTIES

OF

IMPORTANT STEELS

OF

DIN SERIES

AT A GLANCE

CHEMICAL AND MECHANICAL PROPERTIES OF IMPORTANT STEELS - DIN SERIES

DESIGNATIO	* CHEMICAL COMPOSITION, PERCENT												MECH. PROPERTIES, MIN.				
GRADE OF STEEL	C	Mn	Si	S	P	Cr	Ni	Mo	V	W	Co	OTHERS	UTS Kg/mm²	YS Kg/mm²	% ELON.	%RED. IN AREA	HARDNESS MAX.,HRC
COLD DRAWN STEELS FOR COLD FORGED SCREWS																	
Cq 15	0.12-0.18	0.25-0.50	0.15-0.35	0.04	0.04								60			58	
Cq 22	0.18-0.25	0.30-0.60	0.15-0.35	0.04	0.04								65			55	
Cq 35	0.32-0.40	0.40-0.70	0.15-0.35	0.04	0.04								68			53	
Cq 45	0.42-0.50	0.50-0.80	0.15-0.35	0.04	0.04								70			50	
34Cr4	0.30-0.37	0.50-0.80	0.15-0.35	0.035	0.035	0.90-1.20							70			50	
41Cr4	0.38-0.44	0.50-0.80	0.15-0.35	0.035	0.035	0.90-1.20							70			50	
42CrV6	0.38-0.46	0.50-0.80	0.15-0.35	0.035	0.035	1.40-1.70			0.07-0.12				70			50	
42CrMo4	0.38-0.45	0.50-0.80	0.15-0.35	0.035	0.035	0.90-1.20		0.15-0.25					70			50	
24CrMo5	0.20-0.28	0.50-0.80	0.15-0.35	0.035	0.035	0.90-1.20	0.60 max.	0.20-0.30					65			50	
13CrMo44	0.10-0.18	0.40-0.70	0.15-0.35	0.040	0.040	0.70-1.00		0.40-0.50					65			50	
QUENCHED AND TEMPERED STEELS - HIGH GRADE STEELS																	
C 22	0.18-0.25	0.30-0.60	0.15-0.35	0.045	0.045								50-65	30	22	45	
C 35	0.32-0.39	0.50-0.80	0.15-0.35	0.045	0.045								59-74	37	19	40	
C 45	0.42-0.50	0.50-0.80	0.15-0.35	0.045	0.045								67-82	42	16	35	
C 55	0.52-0.60	0.60-0.90	0.15-0.35	0.045	0.045								75-90	47	14	30	
C 60	0.57-0.65	0.60-0.90	0.15-0.35	0.045	0.045								80-95	50	13	30	

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CHEMICAL AND MECHANICAL PROPERTIES OF IMPORTANT STEELS - DIN SERIES

DESIGNATIO	* CHEMICAL COMPOSITION, PERCENT												MECH. PROPERTIES, MIN.				
GRADE OF STEEL	C	Mn	Si	S	P	Cr	Ni	Mo	V	W	Co	OTHERS	UTS Kg/mm²	YS Kg/mm²	% ELON.	%RED. IN AREA	HARDNESS MAX.,HRC
COLD DRAWN STEELS FOR COLD FORGED SCREWS																	
Cq 15	0.12-0.18	0.25-0.50	0.15-0.35	0.04	0.04								60			58	
Cq 22	0.18-0.25	0.30-0.60	0.15-0.35	0.04	0.04								65			55	
Cq 35	0.32-0.40	0.40-0.70	0.15-0.35	0.04	0.04								68			53	
Cq 45	0.42-0.50	0.50-0.80	0.15-0.35	0.04	0.04								70			50	
34Cr4	0.30-0.37	0.50-0.80	0.15-0.35	0.035	0.035	0.90-1.20							70			50	
41Cr4	0.38-0.44	0.50-0.80	0.15-0.35	0.035	0.035	0.90-1.20							70			50	
42CrV6	0.38-0.46	0.50-0.80	0.15-0.35	0.035	0.035	1.40-1.70			0.07-0.12				70			50	
42CrMo4	0.38-0.45	0.50-0.80	0.15-0.35	0.035	0.035	0.90-1.20		0.15-0.25					70			50	
24CrMo5	0.20-0.28	0.50-0.80	0.15-0.35	0.035	0.035	0.90-1.20	0.60 max.	0.20-0.30					65			50	
13CrMo44	0.10-0.18	0.40-0.70	0.15-0.35	0.040	0.040	0.70-1.00		0.40-0.50					65			50	
QUENCHED AND TEMPERED STEELS - HIGH GRADE STEELS																	
C 22	0.18-0.25	0.30-0.60	0.15-0.35	0.045	0.045								50-65	30	22	45	
C 35	0.32-0.39	0.50-0.80	0.15-0.35	0.045	0.045								59-74	37	19	40	
C 45	0.42-0.50	0.50-0.80	0.15-0.35	0.045	0.045								67-82	42	16	35	
C 55	0.52-0.60	0.60-0.90	0.15-0.35	0.045	0.045								75-90	47	14	30	
C 60	0.57-0.65	0.60-0.90	0.15-0.35	0.045	0.045								80-95	50	13	30	

* VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED

CHEMICAL AND MECHANICAL PROPERTIES OF IMPORTANT STEELS - DIN SERIES

DESIGNATIO	* CHEMICAL COMPOSITION, PERCENT												MECH. PROPERTIES, MIN.				
GRADE OF STEEL	C	Mn	Si	S	P	Cr	Ni	Mo	V	W	Co	OTHERS	UTS Kg/mm²	YS Kg/mm²	% ELON.	%RED. IN AREA	HARDNESS MAX. HRC
QUENCHED AND TEMPERED STEELS - HIGH GRADE STEELS																	
C 22	0.18-0.25	0.30-0.60	0.15-0.35	0.035	0.035								50-65	30	22	50	
C 35	0.32-0.39	0.50-0.80	0.15-0.35	0.035	0.035								59-74	37	19	45	
C 45	0.42-0.50	0.50-0.80	0.15-0.35	0.035	0.035								67-82	42	16	40	
C 55	0.52-0.60	0.60-0.90	0.15-0.35	0.035	0.035								75-90	47	14	35	
C 60	0.57-0.65	0.60-0.90	0.15-0.35	0.035	0.035								80-95	50	13	35	
40Mn4	0.36-0.44	0.80-1.10	0.25-0.50	0.035	0.035								80-95	55	14	45	
28Mn6	0.25-0.32	1.30-1.65	0.15-0.40	0.035	0.035								70-85	50	15	45	31
38Cr2	0.34-0.41	0.50-0.80	0.15-0.40	0.035	0.035	0.40-0.60							70-85	45	15	45	30
46Cr2	0.42-0.50	0.50-0.80	0.15-0.40	0.035	0.035	0.40-0.60							80-95	55	14	45	33
34Cr4	0.30-0.37	0.60-0.90	0.15-0.40	0.035	0.035	0.90-1.20							80-95	60	14	45	21-37
30Mn5	0.27-0.34	1.20-1.50	0.15-0.35	0.035	0.035								70-85	45	15	45	30
42MnV7	0.38-0.45	1.60-1.90	0.15-0.35	0.035	0.035				0.07-0.12				80-95	60	14	45	21-37
15Mo3	0.12-0.20	0.50-0.70	0.15-0.35	0.040	0.040			0.25-0.35					70-85	45	15	45	30
37Cr4	0.34-0.41	0.60-0.90	0.15-0.40	0.035	0.035	0.90-1.20							85-100	64	13	45	24-39
41Cr4	0.38-0.45	0.50-0.80	0.15-0.40	0.035	0.035	0.90-1.20							85-100	64	13	45	27-42
25CrMo4	0.22-0.29	0.50-0.80	0.15-0.40	0.035	0.035	0.90-1.20		0.15-0.30					80-95	60	14	55	21-35

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CHEMICAL AND MECHANICAL PROPERTIES OF IMPORTANT STEELS - DIN SERIES

DESIGNATIO	* CHEMICAL COMPOSITION, PERCENT												MECH. PROPERTIES, MIN.				
GRADE OF STEEL	C	Mn	Si	S	P	Cr	Ni	Mo	V	W	Co	OTHERS	UTS Kg/mm²	YS Kg/mm²	% ELON.	%RED. IN AREA	HARDNESS MAX.,HRC
QUENCHED AND TEMPERED STEELS - HIGH GRADE STEELS (CONTD.)																	
34CrMo4	0.30-0.37	0.50-0.80	0.15-0.40	0.035	0.035	0.90-1.20		0.15-0.30					90-110	68	12	50	28-45
42CrMo4	0.38-0.45	0.50-0.80	0.15-0.40	0.035	0.035	0.90-1.20		0.15-0.30					100-120	78	11	45	35-53
50CrMo4	0.46-0.54	0.50-0.80	0.15-0.40	0.035	0.035	0.90-1.20		0.15-0.30					100-120	80	10	45	40-57
32CrMo12	0.28-0.35	0.40-0.70	0.15-0.40	0.035	0.035	2.80-3.30	0.30	0.30-0.50					125-145	105	9	35	46-55
36CrNiMo4	0.32-0.40	0.50-0.80	0.15-0.40	0.035	0.035	0.90-1.20	0.90-1.20	0.15-0.30					100-120	80	11	50	41-54
34CrNiMo6	0.30-0.38	0.40-0.70	0.15-0.40	0.035	0.035	1.40-1.70	1.40-1.70	0.15-0.30					100-130	90	10	45	45-56
30CrNiMo8	0.26-0.33	0.30-0.60	0.15-0.40	0.035	0.035	1.80-2.20	1.80-2.20	0.30-0.50					125-145	105	9	40	46-55
50CrV4	0.47-0.55	0.70-1.10	0.15-0.40	0.035	0.035	0.90-1.20			0.10-0.20				100-120	80	10	45	41-58
30CrMoV9	0.26-0.34	0.40-0.70	0.15-0.40	0.035	0.035	2.30-2.70		0.15-0.25	0.10-0.20				125-145	105	9	35	39-53
22CrMo5 4	0.18-0.25	0.50-0.80	0.30-0.50	0.040	0.040	0.80-1.10		0.40-0.50					90-110	68	12	50	28-45
24CrMoV55	0.20-0.28	0.30-0.60	0.15-0.35	0.035	0.035	1.20-1.50	0.60 MAX	0.50-0.60	0.15-0.25				100-120	80	10	45	41-58
21CrMoV5 11	0.20-0.28	0.30-0.50	0.30-0.60	0.035	0.035	1.20-1.50	0.60 MAX	1.00-1.20	0.25-0.35				100-120	80	10	45	41-58
QUENCHED AND TEMPERED STEELS WITH GUARANTEED SPREAD OF SULPHUR CONTENT- HIGH GRADE STEELS																	
Cm35	0.32-0.39	0.50-0.80	0.15-0.35	0.020-0.035	0.035								50--65	28	21		
Cm45	0.42-0.50	0.50-0.80	0.15-0.35	0.020-0.035	0.035								60-75	34	17		
Cm55	0.52-0.60	0.60-0.90	0.15-0.35	0.020-0.035	0.035								67-85	37	15		
Cm60	0.57-0.65	0.60-0.90	0.15-0.35	0.020-0.035	0.035								70-90	39	14		

* VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED

CHEMICAL AND MECHANICAL PROPERTIES OF IMPORTANT STEELS - DIN SERIES

DESIGNATIO	* CHEMICAL COMPOSITION, PERCENT												MECH. PROPERTIES, MIN.				
GRADE OF STEEL	C	Mn	Si	S	P	Cr	Ni	Mo	V	W	Co	OTHERS	UTS Kg/mm²	YS Kg/mm²	% ELON.	%RED. IN AREA	HARDNESS MAX.,HRC
QUENCHED AND TEMPERED STEELS WITH GUARANTEED SPREAD OF SULPHUR CONTENT- HIGH GRADE STEELS (CONTD.)																	
34CrS4	0.30-0.37	0.60-0.90	0.15-0.40	0.020-0.035	0.035	0.90-1.20							80-95	60	14	45	21-37
37CrS4	0.34-0.41	0.60-0.90	0.15-0.40	0.020-0.035	0.035	0.90-1.20							85-100	64	13	45	24-39
40CrS4	0.38-0.45	0.50-0.80	0.15-0.40	0.020-0.035	0.035	0.90-1.20							90-110	68	12	50	27-42
34CrMoS4	0.30-0.37	0.50-0.80	0.15-0.40	0.020-0.035	0.035	0.90-1.20		0.15-0.30					90-110	68	12	45	28-35
42CrMoS4	0.38-0.45	0.50-0.80	0.15-0.40	0.020-0.035	0.035	0.90-1.20		0.15-0.30					100-120	80	11	50	35-53
10CrMo9 10	0.15 MAX	0.40-0.60	0.15-0.50	0.035	0.035	2.00-2.50		0.90-1.10					100-120	80	11	50	35-53
CASE HARDENING STEELS - QUALITY STEELS																	
C10	0.07-0.13	0.30-0.60	0.15-0.35	0.045	0.045								65-80	40	13	35	
C15	0.12-0.18	0.30-0.60	0.15-0.35	0.045	0.045								75-90	45	12	30	
CASE HARDENING STEELS - HIGH GRADE STEELS																	
Ck10	0.07-0.13	0.30-0.60	0.15-0.35	0.035	0.035								65-80	40	13	35	
Ck15	0.12-0.18	0.30-0.60	0.15-0.35	0.035	0.035								75-90	45	12	30	
15Cr3	0.12-0.18	0.40-0.60	0.15-0.40	0.035	0.035	0.40-0.70							80-105	52	10	35	
16MnCr5	0.14-0.19	1.00-1.30	0.15-0.40	0.035	0.035	0.80-1.10							90-120	65	9	35	
20MnCr5	0.17-0.22	1.00-1.40	0.15-0.40	0.035	0.035	1 00-1.30							90-120	65	9	35	
20MoCr4	0.17-0.22	0.60-0.90	0.15-0.40	0.035	0.035	0.30-0.50		0.40-0.50					90-120	65	9	35	
25MoCr4	0.23-0.29	0.60-0.90	0.15-0.40	0.035	0.035	0.40-0.60		0.40-0.50					110-140	75	7	30	

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CHEMICAL AND MECHANICAL PROPERTIES OF IMPORTANT STEELS - DIN SERIES

DESIGNATIO / GRADE OF STEEL	* CHEMICAL COMPOSITION, PERCENT												MECH. PROPERTIES, MIN.				
	C	Mn	Si	S	P	Cr	Ni	Mo	V	W	Co	OTHERS	UTS Kg/mm²	YS Kg/mm²	% ELON.	%RED. IN AREA	HARDNESS MAX.,HRC
CASE HARDENING STEELS - HIGH GRADE STEELS (CONTD.)																	
20MoV5 3	0.16-0.23	0.70-0.90	0.30-0.50	0.040	0.040	0.30 MAX		0.40-0.55	0.25-0.35				80-105	52	10	35	
20MoV8 4	0.16-0.23	0.50-0.80	0.30-0.50	0.040	0.040	0.30 MAX		0.80-0.90	0.35-0.45				110-140	75	7	30	
15CrNi6	0.12-0.17	0.40-0.60	0.15-0.40	0.035	0.035	1.40-1.70	1.40-1.70						110-140	75	7	30	
18CrNi6	0.15-0.20	0.40-0.60	0.15-0.40	0.035	0.035	1.80-2.10	1.80-2.10						125-150	85	7	30	
17CrNiMo6	0.14-0.19	0.40-0.60	0.15-0.40	0.035	0.035	1.50-1.80	1.40-1.70	0.25-0.35					120-145	85	7	30	
CASE HARDENING STEELS WITH GUARANTEED SPREAD OF SULPHUR CONTENT																	
Cm15	0.12-0.18	0.30-0.60	0.15-0.35	0.02-0.035	0.035								75-90	45	12	35	
16MnCrS5	0.14-0.19	1.00-1.30	0.15-0.40	0.02-0.035	0.035	0.80-1.10							90-120	65	9	35	
20MnCrS5	0.17-0.22	1.00-1.40	0.15-0.40	0.02-0.035	0.035	1.00-1.30							90-120	65	9	35	
20MoCrS4	0.17-0.22	0.60-0.90	0.15-0.40	0.02-0.035	0.035	0.30-0.50		0.40-0.50					90-120	65	9	35	
25MoCrS4	0.23-0.29	0.60-0.90	0.15-0.40	0.02-0.035	0.035	0.40-0.60		0.40-0.50					110-140	75	7	30	
SPRING STEEL - QUALITY STEELS																	
C 53	0.50-0.57	0.40-0.70	0.25-0.50	0.045	0.045								120-145	105	6		36-43
C 60	0.57-0.65	0.50-0.80	0.25-0.50	0.045	0.045								120-160	110	6		36-47
C 67	0.65-0.72	0.60-0.80	0.25-0.50	0.045	0.045								120-160	110	6		36-47
C 75	0.70-0.80	0.60-0.80	0.25-0.50	0.045	0.045								120-160	110	6		36-47
M 75	0.70-0 79	0.30-0 60	0.10-0.25	0.045	0.045								120-160	110	6		36-47

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CHEMICAL AND MECHANICAL PROPERTIES OF IMPORTANT STEELS - DIN SERIES

DESIGNATIO	* CHEMICAL COMPOSITION, PERCENT												MECH. PROPERTIES, MIN.				
GRADE OF STEEL	C	Mn	Si	S	P	Cr	Ni	Mo	V	W	Co	OTHERS	UTS Kg/mm²	YS Kg/mm²	% ELON.	%RED IN AREA	HARDNESS MAX.,HRC
SPRING STEEL - QUALITY STEELS (CONTD.)																	
M 85	0.80-0.89	0.30-0.60	0.10-0.25	0.045	0.045								120-160	110	6		36-47
38Si6	0.35-0.42	0.50-0.80	1.40-1.60	0.05	0.05								120-160	110	6		36-47
46Si7	0.42-0.50	0.50-0.80	1.50-1.80	0.05	0.05								120-160	110	6		36-47
51Si7	0.48-0.55	0.50-0.80	1.50-1.80	0.05	0.05								160-200	150	6		47-53
55Si7	0.52-0.60	0.70-1.00	1.50-1.80	0.05	0.05								160-200	150	6		47-53
65Si7	0.60-0.68	0.70-1.00	1.50-1.80	0.05	0.05								170-220	160	5		49-56
60SiMn5	0.55-0.65	0.90-1.10	1.00-1.30	0.05	0.05								170-220	160	5		49-56
Ck 53	0.50-0.57	0.40-0.70	0.25-0.50	0.035	0.035								120-145	105	7		36-43
CK 60	0.57-0.65	0.50-0.80	0.25-0.50	0.035	0.035								130-155	120	7		39-45
Ck 67	0.65-0.72	0.60-0.80	0.25-0.50	0.035	0.035								140-165	130	6		41-48
Mk 75	0.70-0.80	0.40-0.60	0.15-0.25	0.035	0.035								160-180	150	6		47-51
Mk 101	0.98-1.05	0.35-0.45	0.15-0.25	0.035	0.035								180-240	170	5		51-58
71Si7	0.68-0.75	0.60-0.80	1.50-1.80	0.035	0.035								190-240	180	4		52-58
66Si7	0.60-0.70	0.70-1.00	1.50-1.80	0.035	0.035								180-230	170	5		51-57
67SiCr5	0.62-0.72	0.40-0.60	1.20-1.40	0.035	0.035	0.40-0.60							190-240	180	4		52-58
50CrV4	0.47-0.55	0.80-1.10	0.15-0.35	0.035	0.035	0.90-1.20			0.07-0.12				170-230	160	5		49-57

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CHEMICAL AND MECHANICAL PROPERTIES OF IMPORTANT STEELS - DIN SERIES

DESIGNATIO / GRADE OF STEEL	* CHEMICAL COMPOSITION, PERCENT												MECH. PROPERTIES, MIN.				
	C	Mn	Si	S	P	Cr	Ni	Mo	V	W	Co	OTHERS	UTS Kg/mm²	YS Kg/mm²	% ELON.	%RED. IN AREA	HARDNESS MAX.,HRC
SPRING STEEL - QUALITY STEELS (CONTD.)																	
58CrV4	0.55-0.62	0.80-1.10	0.15-0.35	0.035	0.035	0.90-1.20			0.07-0.12				190-240	180	4		52-58
50CrV4	0.47-0.55	0.80-1.10	0.15-0.35	0.04	0.04	0.90-1.20			0.07-0.12				170-220	160	5		49-56
TOOL STEELS																	
C55WS	0.50-0.60	0.30-0.50	0.08-0.15	0.03	0.03	0.20	0.20										
C60W3	0.55-0.65	0.60-0.80	0.25-0.50	0.035	0.035	0.25	0.25										
C70W2	0.65-0.74	0.15-0.35	0.15-0.35	0.03	0.035	0.20	0.25										
C75W3	0.70-0.80	0.60-0.80	0.25-0.50	0.035	0.035	0.25	0.25										
C80W2	0.75-0.84	0.15-0.35	0.15-0.35	0.035	0.035	0.20	0.25										
C85WS	0.80-0.90	0.50-0.70	0.25-0.40	0.03	0.03	0.20	0.20										
C90W2	0.85-0.94	0.15-0.35	0.15-0.35	0.03	0.035	0.20	0.25										
C100W1	0.95-1.04	0.15-0.25	0.15-0.30	0.02	0.02	0.20	0.20										
C100W2	0.95-1.04	0.15-0.35	0.15-0.35	0.03	0.035	0.20	0.25										
C110W1	1.05-1.14	0.15-0.25	0.15-0.30	0.02	0.02	0.20	0.20										
C110W2	1.05-1.14	0.15-0.35	0.15-0.50	0.03	0.035	0.20	0.25										
C120W2	1.15-1.24	0.15-0.35	0.15-0.50	0.03	0.035	0.20	0.25										
C130W2	1.25-1.45	0.15-0.35	0.15-0.50	0.03	0.035	0.20	0.25										

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CHEMICAL AND MECHANICAL PROPERTIES OF IMPORTANT STEELS - DIN SERIES

DESIGNATIO	* CHEMICAL COMPOSITION, PERCENT												MECH. PROPERTIES, MIN.				
GRADE OF STEEL	C	Mn	Si	S	P	Cr	Ni	Mo	V	W	Co	OTHERS	UTS Kg/mm²	YS Kg/mm²	% ELON.	%RED. IN AREA	HARDNESS MAX.,HRC
TOOL STEELS (CONTD.)																	
E2-50	0.90-1.10	0.80-1.20	0.40-0.60	0.04	0.04	4.50-5.50	0.25	0.70-0.90	0.10-0.20								
E3-55/60W	0.20-0.40			0.04	0.04	2.00-3.50			0.30-0.50	7.00-10.00	1.25-2.50						
E3-60	0.80-1.00			0.04	0.04	3.75-5.00			0.20-2.70	8.50-11.00							
E3-65	0.50-0.70	0.30-0.60	0.80-1.20	0.04	0.04	2.40-3.00			0.20-2.70	9.00-10.50	2.00-2.50						
E4-60/64W	1.00-1.20	0.40-0.70	1.00-1.20	0.04	0.04	4.30-4.70		2.50-3.00	2.50-3.00	15.00-18.00							
E4-60/65W	0.70-1.10			0.04	0.04	3.80-5.00			0.70-1.50	16.50-19.50							
E4-60/69W	0.90-1.20	0.40-0.60	1.10-1.40	0.04	0.04	4.50-5.00				14.00-17.00	14.00-16.00						
E4-60/70W	0.90-1.40	0.40-0.60	0.50-0.70	0.04	0.04	4.30-4.70			2.50-3.00	15.00-18.00	14.00-16.00						
X40Cr13	0.35-0.44	0.60	0.60	0.03	0.04	12.00-14.00											
X74WV18.1	0.70-0.80	0.45	0.30	0.03	0.03	3.80-4.50		0.30	1.00-1.20	17.50-18.50							
X76WCo18.10	0.72-0.82	0.45	0.30	0.03	0.03	3.80-4.50		0.50-0.80	1.40-1.70	17.50-18.50	9.00-10.00						
X79WCo18.5	0.75-0.85	0.45	0.30	0.03	0.03	3.80-4.50		0.50-0.80	1.40-1.70	17.50-18.50	4.50-5.00						
X82WMo6.5	0.78-0.88	0.45	0.30	0.03	0.03	3.80-4.50		4.80-5.30	1.70-2.00	6.00-6.70							
X82WV9.2	0.77-0.87	0.45	0.30	0.03	0.03	3.80-4.50		0.70-1.00	1.40-1.70	8.00-9.00							
X85WMoCo 6.5.5	0.82-0.90	0.45	0.30	0.03	0.03	3.80-4.50		4.80-5.30	2.00-2.30	6.00-6.70	4.50-5.00						
X86WV12.2	0.80-0.90	0.45	0.30	0.03	0.03	3.80-4.50		0.70-1.00	2.30-2.60	11.50-12.50							

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CHEMICAL AND MECHANICAL PROPERTIES OF IMPORTANT STEELS - DIN SERIES

DESIGNATIO	* CHEMICAL COMPOSITION, PERCENT												MECH. PROPERTIES, MIN.				
GRADE OF STEEL	C	Mn	Si	S	P	Cr	Ni	Mo	V	W	Co	OTHERS	UTS Kg/mm²	YS Kg/mm²	% ELON.	%RED. IN AREA	HARDNESS MAX.,HRC
TOOL STEELS (CONTD.)																	
X90CrMoV18	0.85-0.95	0.80	0.80	0.03	0.04	16.00-18.00		1.00-1.30	0.07-0.12								
X97WMo3.3	0.92-1.02	0.45	0.30	0.03	0.03	3.80-4.50		2.50-2.80	2.20-2.50	2.50-3.00							
X100WCo5.3	0.95-1.05	0.45	0.30	0.03	0.03	3.80-4.50		2.70-3.00	2.50-2.80	4.50-5.00	2.70-3.00						
X125WV12.4	1.20-1.30	0.45	0.30	0.03	0.03	3.80-4.50		0.70-1.00	3.50-4.00	11.50-12.50							
X133WCo12.5	1.25-1.40	0.45	0.30	0.03	0.03	3.80-4.50		0.70-1.00	3.50-4.00	11.50-12.50	4.00-5.00						
6CrMo18.5	0.07	0.20-0.40	0.15-0.30	0.03	0.03	4.50-5.00		0.40-0.60									
13NiCr12	0.11-0.17	0.30-0.60	0.17-0.37	0.03	0.03	0.60-0.90	2.75-3.25										
13NiCr14	0.11-0.17	0.30-0.60	0.17-0.37	0.03	0.03	1.25-1.75	3.25-3.75										
13NiCr18	0.11-0.17	0.30-0.50	0.20-0.40	0.03	0.03	0.90-1.30	4.25-4.75										
16MnCr5	0.14-0.19	1.00-1.30	0.17-0.37	0.03	0.03	0.80-1.10											
19NiCrMo15	0.17-0.22	0.30-0.50	0.13-0.30	0.03	0.03	1.01-1.40	3.50-4.00	0.15-0.25									
20MnCr5	0.17-0.22	1.00-1.40	0.17-0.37	0.03	0.03	1.00-1.30											
22MnCr4	0.19-0.26	0.50-0.80	0.20-0.40	0.03	0.03	0.90-1.20		0.20-0.30									
28CrMoV11.28	0.24-0.32	0.20-0.40	0.20-0.40	0.03	0.03	2.60-3.00		2.60-2.90	0.45-0.55								
30WCrCo34.10	0.26-0.34	0.20-0.40	0.15-0.35	0.03	0.03	2.20-2.50			0.25-0.35	7.50-9.00	1.80-2.10						
30WCrV15	0.25-0.35	0.30-0.50	0.70-1.00	0.03	0.03	0.90-1.20			0.10-0.20	3.50-4.00							

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CHEMICAL AND MECHANICAL PROPERTIES OF IMPORTANT STEELS - DIN SERIES

DESIGNATIO	* CHEMICAL COMPOSITION, PERCENT												MECH. PROPERTIES, MIN.				
GRADE OF STEEL	C	Mn	Si	S	P	Cr	Ni	Mo	V	W	Co	OTHERS	UTS Kg/mm²	YS Kg/mm²	% ELON.	%RED. IN AREA	HARDNESS MAX.,HRC
TOOL STEELS (CONTD.)																	
30WCrV17.9	0.25-0.35	0.20-0.40	0.15-0.35	0.03	0.03	2.20-2.70			0.45-0.60	3.80-4.50							
30WCrV34.11	0.25-0.35	0.20-0.40	0.15-0.35	0.03	0.03	2.20-2.70			0.30-0.45	7.50-9.00							
31CrV3	0.28-0.35	0.40-0.60	0.20-0.40	0.035	0.035	0.50-0.70			0.07-0.12								
37CrMoWV 20.15	0.32-0.42	0.40-0.60	0.70-1.00	0.03	0.03	4.50-5.00		1.30-1.60	0.20-0.30	1.20-1.50							
33CrMoV21.14	0.33-0.43	0.40-0.60	0.70-1.00	0.03	0.03	5.00-5.50		1.20-1.50	0.30-0.45								
38CrSi6	0.35-0.43	0.30-0.60	1.00-1.30	0.035	0.035	1.30-1.60											
40CrMnMo7	0.35-0.45	1.10-1.50	0.15-0.35	0.03	0.03	1.70-2.00		0.15-0.25									
40NiCrMo15	0.35-0.45	0.30-0.50	0.15-0.35	0.03	0.03	1.10-1.40	3.50-4.00	0.15-0.25									
45CrMoV6.7	0.40-0.50	0.60-0.80	0.15-0.35	0.03	0.03	1.30-1.50		0.65-0.75	0.25-0.35								
45CrSiV6	0.40-0.50	0.50-0.70	1.20-1.50	0.03	0.03	1.20-1.50			0.07-12.00								
45CrVMoW5.8	0.40-0.50	0.30-0.50	0.50-0.70	0.03	0.03	1.20-1.50		0.40-0.60	0.70-0.85	0.40-0.50							
45WCrV7	0.40-0.50	0.20-0.40	0.80-1.00	0.03	0.03	0.90-1.20			0.15-0.20	1.70-2.00							
50NiCr13	0.45-0.55	0.40-0.60	0.20-0.40	0.03	0.03	0.90-1.20	3.00-3.50										
50NiCrMoV6	0.50-0.60	0.50-0.70	0.15-0.35	0.03	0.03	0.60-0.80	1.45-1.75	0.20-0.30	0.07-0.12								
55WCrV7	0.50-0.60	0.20-0.40	0.80-1.00	0.03	0.03	0.90-1.20	1.45-1.75		0.15-0.20	1.70-2.00							
56NiCrMoV7.4	0.52-0.60	0.60-0.80	0.15-0.35	0.03	0.03	1.00-1.20	1.60-1.90	0.50-0.60	0.07-0.12								

* VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED

CHEMICAL AND MECHANICAL PROPERTIES OF IMPORTANT STEELS - DIN SERIES

DESIGNATIO / GRADE OF STEEL	* CHEMICAL COMPOSITION, PERCENT												MECH. PROPERTIES, MIN.				
	C	Mn	Si	S	P	Cr	Ni	Mo	V	W	Co	OTHERS	UTS Kg/mm²	YS Kg/mm²	% ELON.	%RED. IN AREA	HARDNESS MAX.,HRC
TOOL STEELS (CONTD.)																	
76SiCr5	0.62-0.72	0.40-0.60	1.20-1.40	0.03	0.03												
80CrV3	0.75-0.85	0.20-0.40	0.15-0.35	0.03	0.03	0.50-0.80			0.15-0.30								
80WCrV8	0.75-0.85	0.30-0.50	0.40-0.60	0.03	0.03	1.00-1.20			0.25-0.35	1.80 2.00							
85Cr1	0.80-0.90	0.50-0.70	0.30-0.50	0.03	0.03	0.30-0.45											
85Cr7	0.80-0.90	0.25-0.40	0.15-0.35	0.03	0.03	1.60-1.90											
90MnV8	0.85-0.95	1.80-2.00	0.15-0.35	0.03	0.03				0.07-0.12								
100Cr6	0.95-1.10	0.20-0.40	0.15-0.35	0.03	0.03	1.30-1.65											
100V3	0.95-1.05	0.20-0.40	0.15-0.35	0.03	0.03				0.25-0.35								
105MnCr4	1.00-1.10	1.00-1.20	0.15-0.30	0.03	0.03	0.70-1.00											
105WCr6	1.00-1.10	0.80 1.10	0.15-0.35	0.03	0.03	0.90-1.20				1.20-1.60							
110MoV5	1.05-1.15	0.30-0.40	0.15-0.30	0.03	0.03			0.45-0.55	0.15-0.20								
110WCrV5	1.05-1.15	0.30-0.40	0.15-0.30	0.03	0.03	1.10-1.30			0.15-0.20	1.15-1.45							
115CrV3	1.10-1.25	0.20-0.40	0.15-0.30	0.03	0.03	0.50-0.80			0.07-0.12								
115W8	1.10-1.20	0.20-0.40	0.15-0.30	0.03	0.03	0.15-0.25				1.70-2.00							
120WV4	1.15-1.25	0.20-0.35	0.15-0.30	0.03	0.03				0.15-0.25	0.85-1.15							
125Cr1	1.20-1.30	0.20-0.40	0.15-0.30	0.03	0.03	0.20-0.35											

* VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED

CHEMICAL AND MECHANICAL PROPERTIES OF IMPORTANT STEELS - DIN SERIES

DESIGNATIO / GRADE OF STEEL	* CHEMICAL COMPOSITION, PERCENT												MECH. PROPERTIES, MIN.				
	C	Mn	Si	S	P	Cr	Ni	Mo	V	W	Co	OTHERS	UTS Kg/mm²	YS Kg/mm²	% ELON.	%RED IN AREA	HARDNESS MAX.,HRC
TOOL STEELS (CONTD.)																	
125CrSi5	1.20-1.30	0.60-0.80	1.05-1.25	0.03	0.03	1.10-1.30											
135WV4	1.30-1.40	0.30-0.40	0.15-0.30	0.03	0.03	0.30-0.40			0.20-0.30	0.80-1.10							
140Cr2	1.35-1.45	0.25-0.35	0.20-0.35	0.03	0.03	0.40-0.60											
142WV13	1.35-1.50	0.20-0.40	0.15-0.35	0.03	0.03	0.20-0.50			0.20-0.30	3.00-3.50							
145CrV6	1.40-1.50	0.45-0.70	0.15-0.30	0.03	0.03	1.30-1.60			0.08-0.15								
165CrMoWV46	1.55-1.75	0.20-0.40	0.20-0.40	0.03	0.03	11.00-12.00		0.50-0.80	0.20-0.30	0.40-0.60							
210Cr46	2.00-2.20	0.20-0.40	0.20-0.40	0.03	0.03	11.00-12.00											
210CrW46	2.00-2.20	0.20-0.40	0.20-0.40	0.03	0.03	11.00-12.00				0.60-0.80							

* VALUES ARE MAXIMUM, UNLESS OTHERWISE SPECIFIED

PERMISSIBLE VARIATION OF ELEMENTS IN CARBON STEEL

ELEMENT	LIMIT OR MAXIMUM OF SPECIFIED RANGE,PERCENT	OVER MAX. LIMIT %	UNDER MIN. LIMIT %
CARBON	0.25 AND UNDER	0.020	0.020
	OVER 0.25 TO 0.55 INCL.	0.030	0.030
	OVER 0.55	0.040	0.040
MANGANESE	0.90 AND UNDER	0.030	0.030
	OVER 0.90 TO 1.65 INCL.	0.060	0.060
PHOSPHORUS	BASIC STEELS	0.008	---
	ACID BESSEMER STEEL	0.010	0.010
SULPHUR		0.008	---
SILICON	0.35 AND UNDER	0.020	0.020
	OVER 0.35 TO 0.60 INCL.	0.050	0.050
COPPER	UNDER MINIMUM ONLY	---	0.020
LEAD	UNDER 0.15 TO 0.35 INCL.	0.030	0.030

PERMISSIBLE VARIATION OF ELEMENTS IN ALLOY STEEL

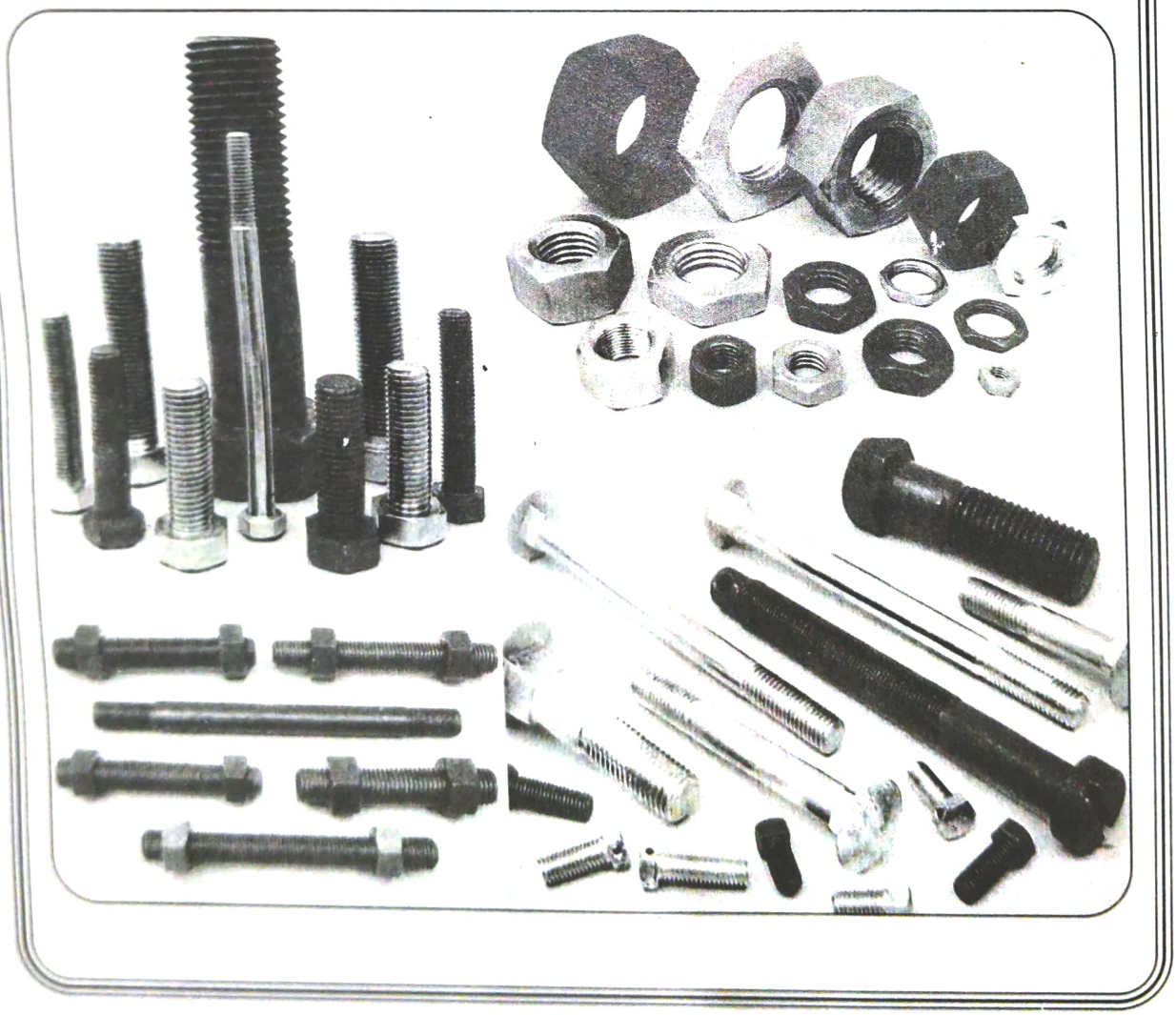
ELEMENT	LIMIT OR MAX. OF SPECIFIED RANGE, PERCENT	OVER MAX. OR UNDER MIN. LIMIT, PERCENT
CARBON	0.30 AND UNDER	0.010
	OVER 0.30 TO 0.75	0.020
	OVER 0.75	0.030
MANGANESE	0.90 AND UNDER	0.030
	OVER 0.90 TO 2.10	0.040
PHOSPHORUS	OVER MAX. ONLY	0.005
SULPHUR	0.06 AND UNDER	0.005
SILICON	0.40 AND UNDER	0.020
	OVER 0.40 TO 2.20	0.050
NICKEL	1.00 AND UNDER	0.030
	OVER 1.00 TO 2.00	0.050
	OVER 2.00 TO 5.30	0.070
	OVER 5.30 TO 10.0	0.100
CHROMIUM	0.90 AND UNDER	0.030
	OVER 0.90 TO 2.10	0.050
	OVER 2.10 TO 3.99	0.100
	4.00 AND ABOVE	0.120
COPPER	UPTO 1.00 INCL.	0.030
	OVER 1.00 TO 2.00	0.050

ELEMENT	LIMIT OR MAX. OF SPECIFIED RANGE, PERCENT	OVER MAX. OR UNDER MIN. LIMIT, PERCENT
MOLYBDENUM	0.20 AND UNDER	0.010
	OVER 0.20 TO 0.40	0.020
	OVER 0.40 TO 1.15	0.030
VANADIUM	0.10 AND UNDER	0.010
	OVER 0.10 TO 0.25	0.020
	OVER 0.25 TO 0.50	0.030
	MIN. VALUE SPECI-FIED, UNDER MIN. LIMIT ONLY	0.010
TUNGSTEN	1.00 AND UNDER	0.040
	OVER 1.00 TO 4.00	0.080
ALUMINIUM	0.10 AND UNDER	0.030
	OVER 0.10 TO 0.20	0.040
	OVER 0.20 TO 0.30	0.050
	OVER 0.30 TO 0.80	0.070
	OVER 0.80 TO 1.80	0.100
LEAD	0.15 TO 0.35 INCL.	0.030

CHAPTER - 2

INDUSTRIAL

FASTENERS



INDUSTRIAL FASTENERS

1. SCOPE:

This chapter deals with the procedure for the manufacture of various types of Industrial Fasteners alongwith test methods to be adopted in line with various product standards to ensure the product in conformance with its technical requirements. The data of various chemical and mechanical properties and acceptance criterion of different types of Industrial Fasteners has been displayed in ready to read charts which the manufacturers will find very useful for their day to day reference.

2. AN OUTLOOK AND FUTURE OF FASTENERS INDUSTRY:

Industry of Industrial Fasteners, Nuts, Bolts, Screws, Studs, Washers, etc. is firming its grip with bright future. The fasteners industry which mainly depends on the fortunes of automobile and engineering industry is riding piggy back with soaring demands for the automobiles in the country. Good days are ahead for the manufacturers of Industrial Fasteners, having performed well in the past two years. Although the fasteners industry has under gone recession earlier but now it had recovered from that stage and is on its way to boom.

Thanks to the leading manufacturers of automobiles which consume a large chunk of the produce, having future plans of expansion inclusive of global automobile giants setting up capacities in the country and the growth registered by the engineering sector in the past two years. Eventually it is the liberalization and the economic process set in by the earlier Government that the industry has and is expected to perform well in future.

The main products of fastener industry are Nuts, Bolts, Screws, Studs, Washers, etc. which act as fastener in the joints. The fasteners are classified on the basis of tensile strength as Mild steel (MS fasteners) and High Tensile or Alloy steel (HT fasteners). M.S. Fasteners are useful for general applications and are mainly produced in the unorganized and tiny/SSI sector and are not technology and capital intensive and can be manufactured by using simple Machine tools. However, HT fasteners are technology and capital extensive and require a number of Machine tools and special equipments and hence comes under the purview of the organized sector.

The application of Industrial fasteners is wide but not general purpose and are used in the manufacture and assembly of the engineering components. HT fasteners are further classified as standard; off the shelf and specific; to the design specifications. The HT fasteners come in as Hexagonal/Square and Socket according to the design of the heads. Whereas the socked head fasteners are mainly used for industrial applications in textiles, machine tools, etc., the Hexagonal/Square type are used predominantly automobile and general engineering industries. Special types of fasteners are designed and manufactured to the specific design of the customers and hence used in the special applications. The major consumption of all kinds of HT fasteners which accounts for over 50 % is in the automobile sector is also the most visible, but the uses are every where including the construction industry, etc.

3. AVAILABILITY OF RAW MATERIALS - PRESENT POSITION:

The manufacture of fasteners require various grades of steel depending upon the type of the fastener being manufactured and the purpose for which it is being manufactured. The steel required for the manufacture of MS fasteners is available in domestic market in abundant quantity and hence the difficulties are not faced. However, for HT fasteners, due to unavailability of certain grades of carbon, alloy and Cold Heading Quality (CHQ) steels, the raw material needs to be imported, hence the industry faces difficulty on this front. Initially, due to the poor quality and low availability of the required grades of steel in the domestic market, the manufacturers of the HT fasteners had to import nearly 60% of their raw material requirements. But , in the recent times, the leading steel makers like Mukand steel, Bihar Alloys, Shri Ishar Alloys, Mahindra Ugine Ltd., Steel Authority of India Ltd. (Alloy steel Plant, Durgapur), etc. getting into the production of some of the popular grades of steel, the demand on the imported inputs is on the decline, yet even today some of the manufacturers of fasteners have to import 30-40% of their raw materials. The situation is likely to improve when the domestic steel is available in the market at a lower rate.

4. GROWTH OF FASTENER INDUSTRY:

The reasons for the spectacular growth recorded by the fastener industry in the last 2-3 years can be attributed to the boom in the automobile sector (commercial vehicles recorded a 38% growth in production and the passenger car segment grew 37%) and the 24% growth in the engineering sector. With the decision of the government to

reduce import duty on CHQ steel from 85 % to 50% in 1994-95 and further to 40% in 1995-96 budget, the fastener industry is helped to reduce its raw material input cost. Though the present erosion in the value of rupee will have an inflationary impact on raw material prices, industry sources are confident of passing on the increased costs to their customers.

5. WORKING CAPITAL REQUIREMENTS:

A characteristic of the HT fastener industry is that it is highly working capital intensive. As CHQ steel was imported, consignments arrived in bunches and, as such, companies were forced to maintain high stocks, resulting in higher inventory carrying cost and, in turn, higher working capital requirements. With the availability of CHQ steel in India, and the reduction in import duty, the stress on working capital has decreased, although a bit. If this is the scene on the raw material front, it is even more constraining on the finished goods front, where the constraining factor is the varieties which run into thousands, with as many sizes. Adding to the woes, is the preference of OEMs to buy fasteners only when required. Funds blocked in receivables, due to longer credit given to OEMs, places further stress on working capital. While the OEM market is an assured one even during lean periods, the replacement market, though it offers higher margins, requires a large dealer network, necessitating higher investment.

6. EXPORT OPPORTUNITIES:

The concept of out - sourcing fasteners is undergoing a sea change globally. Auto giants around the world have identified countries to buy a particular component, depending upon the technology and cost. Quality is an important factor in exports, but not the only criterion; what is more important is timely delivery and after sales service. Though there is a vast potential to export fasteners to OEMs abroad, it has not been exploited due to difficulties in setting up service points near each of the OEMs manufacturers. Hence, the domestic producers foray abroad is limited to the replacement market. To be successful in exports, Indian companies do not require foreign collaboration, as a fastener is not a very hi-tech item. What is required is a foreign tie up for marketing and after sales service.

7. FUTURE PROSPECTS:

At present, the automobile industry which accounts for more than 50% of HT fasteners, is on roll. The fastener industry is all set to better its performance in the next couple of years. The domestic market will continue to expand rapidly as growth impulses in the automobile industry get stronger. With the continuing uptrend in the automobile industry and brisk growth in the general engineering industry, the future looks promising. Companies with higher volumes, an export thrust and good Administrative and Quality Management should be in a better position to take advantage of the expected growth. It is with this view, Quality aspects for the manufacture of Carbon steel, HT and alloy steel fasteners are covered in the foregoing sections, in line with ISO-9000 quality systems. In case, such quality systems are set up and adopted by the companies, they should be sure to take a lead over others and bag the maximum share of orders of fasteners.